

**RNA regulation in host-fungal interactions:
investigation of the host epitranscriptome and fungal small RNAs**

Dissertation

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Summary

Host-fungal interactions trigger complex transcriptomic responses in both partners, yet our understanding of RNA structure, and particularly the metabolism of double-stranded RNA (dsRNA) during infection, remains limited. My doctoral research focused on two key aspects of dsRNA regulation. First, I explored dsRNA metabolism in a fungal pathogen, *Aspergillus fumigatus*. As a largely conserved mechanism, fungi use the RNA interference (RNAi) pathway to process dsRNA for diverse functions. Analysis of mRNA-seq data revealed that *A. fumigatus* ribosome biogenesis genes are regulated by the dicer-like enzymes (DclA/B) and RNA-dependent RNA polymerases (RrpA/B) in conidia. Sequencing of small RNA (sRNA) and tRNA-derived RNA (tDR) revealed dynamic expression of tDRs in conidia and mycelium. To study canonical RNAi, an inverted-repeat transgene (IRT), which can form a 500-bp dsRNA structure, was overexpressed to induce RNAi-dependent sRNA production. This resulted in sRNAs predominantly from the 3' end of the produced hairpin, with a length of 20 nt, a 5' uridine, and dependent on DclB for biogenesis. Interestingly, not only the wild type but also the Dcl double- and single- knockout strains overexpressing the IRT were observed to exhibit growth inhibition, suggesting that excessive dsRNA itself may impose a growth restriction on *A. fumigatus*.

Second, I investigated dsRNA metabolism in the host. Adenosine deaminase acting on RNA 1 (ADAR1), a dsRNA binding protein, mediates human inflammation by deaminating adenosine to inosine (A-to-I RNA editing) in long dsRNA, but systematic characterization of extensive A-to-I RNA editing (termed hyper-editing) in response to pathogenic fungal stimulations is incomplete. To address this gap, I analyzed publicly available RNA-seq data of human peripheral blood mononuclear cells (PBMCs) exposed to the three fungal pathogens (*Candida albicans*, *Aspergillus fumigatus*, and *Rhizopus oryzae var delemar*). This analysis revealed that *R. oryzae* triggered a significant increase in A-to-I RNA hyper-editing. Additionally, *R. oryzae* stimulation differentially expressed host splicing genes and increased alternative splicing (AS) events, resulting in alternatively spliced transcripts having lower minimum free energy (MFE) on average and suggesting that disrupted splicing may lead to more dsRNA. These dsRNAs are then likely targeted by ADAR1-mediated hyper-editing. In contrast, *A. fumigatus* and *C. albicans* triggered no such response, indicating *R. oryzae* stimulation may drive host transcriptomic changes and specifically alter RNA splicing differently from the other two tested fungi. These findings highlight the unique transcriptomic response to *R. oryzae* coinubation and underscore the interplay between hyper-editing and RNA splicing.

In conclusion, this thesis enhances our understanding of dsRNA regulation in *A. fumigatus* and in the host upon fungal coinubation. In the fungus, I improved our understanding of *A. fumigatus* RNAi

biology and revealed an intriguing phenotype of dsRNA-dependent growth inhibition. Further, I elucidated an exciting form of transcriptional reprogramming in host immune cells upon *R. oryzae* stimulation, where dysregulated AS results in higher levels of hyper-editing of dsRNA. In both cases, the investigation of RNA processing has led to an exciting view of host-pathogen interactions and improved our understanding of human fungal pathogens.

Zusammenfassung

Wirt-Pilz-Interaktionen lösen komplexe transkriptomische Reaktionen in beiden Partnern aus, doch unser Verständnis der RNA-Struktur und insbesondere des Metabolismus von doppelsträngiger RNA (dsRNA) während einer Infektion bleibt begrenzt. Meine Doktorarbeit konzentrierte sich auf zwei Schlüsselaspekte der dsRNA-Regulation.

Zunächst untersuchte ich den dsRNA-Metabolismus im humanpathogenen Pilz *Aspergillus fumigatus*. Als ein konservierter Mechanismus nutzen Pilze den RNAi-Weg, um dsRNA für verschiedene Funktionen zu prozessieren. Analysen von mRNA-Seq-Daten ergaben, dass die Ribosomenbiogenese-Gene von *A. fumigatus* in Konidien durch Dicer-ähnliche Enzyme (DclA/B) und RNA-abhängige RNA-Polymerasen (RrpA/B) reguliert werden. Sequenzierungen von kleiner RNA (sRNA) und tRNA-abgeleiteter RNA (tDR) zeigten eine dynamische Expression von tDRs zwischen Konidien und Myzel. Um die kanonische RNAi zu untersuchen, wurde ein Transgen mit invertierten Wiederholungen (IRT), das eine 500 bp lange dsRNA-Struktur bilden kann, überexprimiert, um eine RNAi-abhängige sRNA-Produktion zu induzieren. Dies führte zur Entstehung von sRNAs, die überwiegend vom 3'-Ende des Haarnadelstruktur-Transkripts stammten, eine Länge von 20 Nukleotiden sowie ein 5'-Uridin aufwiesen und in ihrer Biogenese von DclB abhängig waren. Interessanterweise zeigten nicht nur der Wildtyp, sondern auch Dcl-Doppel- und Einzel-Knockout-Stämme, die das IRT überexprimierten, Wachstumshemmung. Dies legt nahe, dass überschüssige dsRNA selbst eine Wachstumsrestriktion bei *A. fumigatus* bewirken könnte.

Zweitens untersuchte ich den dsRNA-Metabolismus im Wirt. Die Adenosin-Desaminase, die auf RNA wirkt 1 (ADAR1), ein dsRNA-bindendes Protein, vermittelt die menschliche Immunantwort durch Deaminierung von Adenosin zu Inosin (A-to-I RNA editing) in langer dsRNA. Eine systematische Charakterisierung von umfangreichem A-to-I RNA-Editing (als Hyper-Editing bezeichnet) als Reaktion auf Stimulation mit pathogenen Pilzen liegt jedoch bisher nicht vollständig vor. Um diese Wissenslücke zu schließen, analysierte ich öffentlich verfügbare RNA-Seq-Daten von humanen peripheren mononukleären Blutzellen (PBMCs), die mit drei pilzlichen Krankheitserregern (*Candida albicans*, *Aspergillus fumigatus* und *Rhizopus oryzae* var. *delemar*) stimuliert wurden. Die Analyse ergab, dass *R. oryzae* einen signifikanten Anstieg des A-to-I RNA-Hyper-Editings auslöste. Zudem regulierte die *R. oryzae*-Stimulation Wirts-Spleißgene differentiell hoch und erhöhte die Anzahl von alternativen Spleiß-Ereignissen (AS). Die resultierenden alternativ gespleißten Transkripte wiesen durchschnittlich eine niedrigere minimale freie Energie (MFE) auf, was darauf hindeutet, dass eine gestörte Spleißregulation zu mehr dsRNA führen könnte, die Ziel einer ADAR1-vermittelten Hyper-Editierung

sein könnte. Im Gegensatz dazu lösten *A. fumigatus* und *C. albicans* keine solche Reaktion aus, was darauf hindeutet, dass die *R. oryzae*-Stimulation transkriptomische Veränderungen im Wirt antreibt und die RNA-Spleißung spezifisch anders verändert als die beiden anderen getesteten Pilze. Diese Ergebnisse unterstreichen die einzigartige transkriptomische Antwort auf eine *R. oryzae*-Koinkubation und heben das Wechselspiel zwischen A-to-I RNA-Editing und RNA-Spleißung hervor.

Zusammenfassend erweitert diese Arbeit unser Verständnis der dsRNA-Regulation in *A. fumigatus* und im Wirt während pilzlicher Koinkubationen. Beim Pilz konnte ich das Verständnis der RNAi-Biologie von *A. fumigatus* verbessern und einen aufschlussreichen Phänotyp der dsRNA-abhängigen Wachstumshemmung aufdecken. Darüber hinaus konnte ich eine vielversprechende Form der transkriptionellen Reprogrammierung in Immunzellen des Wirts nach *R. oryzae*-Stimulation aufklären, bei der eine dysregulierte alternative Spleißung zu höheren Levels von dsRNA-Hyper-Editing führt. In beiden Fällen hat die Untersuchung der RNA-Prozessierung zu einer spannenden neuen Sicht auf Wirt-Pathogen-Interaktionen beigetragen und unser Verständnis humanpathogener Pilze erweitert.

1 Introduction

1.1 Overview of life-threatening fungal pathogens

Viral and bacterial pathogens tend to attract more attention as sources of infection, in part because of their potential to cause pandemics. In the recent years, however, this has shifted, with increasing awareness and concern regarding the threat fungi pose to humans. Pathogenic fungi cause over one billion human infections annually, resulting in ~3.8 million deaths per year (1). Recognizing the urgent need of understanding pathogenic fungi, the World Health Organization (WHO) took a significant step in 2022 by publishing a list of fungal priority pathogens to underscore the importance of advancing research on fungal infections to guide research, development, and public health action. The list highlights major fungal pathogens, such as *Cryptococcus neoformans*, *Aspergillus fumigatus*, *Candida albicans*, *Candida auris*, *R. delemar var oryzae* (formally *R. arrhizus*) and other *Mucorales* spp., *Fusarium* spp., *Histoplasma* spp., and *Nakaseomyces glabrata*, among others (2). These fungi are responsible for a wide range of diseases, from superficial skin infections to life-threatening systemic illnesses. They can cause life-threatening diseases, particularly in patients with compromised immune systems. Particularly susceptible to invasive fungal diseases (IFDs) are patients who undergo immunotherapy, organ transplantation, or who are otherwise immunocompromised (3,4). Additionally, beyond the predominant fungal pathogens, environmental changes like global warming over the past several decades are altering the ecological niches of many saprophytic and environmental fungi, giving them more opportunities for infections to occur (5,6).

Humans are consistently exposed to fungal spores and yeast cells in their everyday environments. One example is cutaneous mucormycosis caused by *Mucorales* spp. The fungal spores enter through broken skin and germinate, which is usually after trauma, burns, or surgical wounds (7). ~70% of mucormycosis is caused by *R. delemar var oryzae* in the world (8). *A. fumigatus*, a ubiquitous mold, produces airborne spores (conidia) that can be easily inhaled into the lungs, where they can cause respiratory infections. *C. neoformans*, another major fungal pathogen, can be inhaled in its yeast state, leading to pulmonary or even systemic infections, including meningitis (9). Though various parts of the body can be affected (e.g., the mouth, throat, sinuses, lungs, or internal organs such as the liver and brain), the skin and nails are the most common sites for fungal infections.

Various therapeutic approaches can be used to address pathogenic fungal invasion. Notably, the available antifungal agents are currently limited to three main classes, each targeting specific fungal components: 1) azoles (such as fluconazole, itraconazole, and voriconazole), which inhibit ergosterol; 2) echinocandins (including caspofungin, micafungin, and anidulafungin), which inhibit 1,3- β -D-glucan

synthase; 3) polyenes (such as amphotericin B), which interact with fungal membrane sterols (10,11). Additionally, flucytosine is often used in combination with polyenes to enhance efficiency. There are limitations for these antifungals to overcome, including diverse side effects, drug development limitations, and increasing drug resistance. Thus, this urges us to develop novel antifungal therapies but also requires a deeper understanding of the molecular mechanisms underlying host-pathogen interactions.

To have a better view of host-fungal interactions, this thesis focuses on RNA biology as a critical molecular interface. In the following sections, I will first provide a concise overview of host-fungal interactions, with particular emphasis on host recognition and fungal evasion strategies. Then I will focus on dsRNA metabolism in both fungi and the host, including investigation of fungal dsRNA processing to generate small RNAs by RNAi and host dsRNA regulation by dsRNA-binding proteins (dsRBPs).

1.2 Host-fungal interactions

Humans are constantly exposed to fungi in various environments. While the majority of these fungi are harmless or even beneficial, a subset of them can act as opportunistic or primary pathogens under the right conditions. The interactions between host and fungal pathogens are complex and dynamic, as each side strives to gain an advantage during these interactions. Fungi employ a wide range of sophisticated strategies to invade, colonize, and persist within the host. These include mechanisms such as evasion of immune responses, biofilm formation, metabolic adaptation to host environments, production of virulence factors that impair host defense mechanisms (12). In response, the host immune system activates both innate and acquired mechanisms to recognize and eliminate fungal invaders.

1.2.1 Fungal responses during infection

Fungal pathogens are able to mask host recognition, produce virulence factors, and modulate host immune responses in the hostile host environment (13). The fungal cell wall plays a crucial role in interaction with host cells (14). Cell wall components, like polysaccharides (e.g., 1,3- β -D-glucan and chitin), melanin, and others are the first point of contact with immune defenses and serve also as pathogen-associated molecular patterns (PAMPs). PAMPs are molecular structures found in pathogenic microorganisms that can be recognized to be foreign by host immune and antigen-presenting cells (15,16). Usually, PAMPs can be divided into lipids, proteins, carbohydrates, and nucleic acids, and can cause inflammation in mammals (16).

Virulence factors, such as toxins, also actively contribute to host-pathogen interactions. The fungal cytolytic peptide toxin, candidalysin, was the first discovered peptide toxin in a human fungal pathogen and can cause epithelial cell damage to activate immunity (17). A ricin-like toxin, mucoricin, was demonstrated to play a central role in virulence in the *Mucorales* order (including *R. delemar var oryzae*), relying upon its ability to damage human cells in the hyphal state (18). As conidia (asexual spores), these fungi could also invade human cells via binding to cell surface receptors. More specifically, *Mucorales* spp. produce spore coat proteins (e.g., CotH3 and CotH7) that interact with host glucose-regulated protein 78 (GRP78) and integrin $\alpha 3 \beta 1$ of nasal and alveolar epithelial cells, respectively (19). The disease resulting from this invasion is known as mucomycosis (19,20). Three secreted peptides CesT, Colicin, and Ca^{2+} /calmodulin-dependent protein kinases/ligand (CAMK/CAMKL) were reported to target IL-4/IL-13R/STAT-6 to support fungal pathogenesis in *R. delemar var oryzae* (21). A surface heat shock protein of *A. fumigatus* germlings, HscA, was able to reprogram phagosomes into a non-degradative pathway by interacting with host p11 (*S100A10* or light

chain of annexin A2) and associated proteins (22). Fungal-derived toxins are more numerous and better studied in plant fungal pathogens (23–25), but examples like candidalysin encourage further investigation in human pathogenic fungi. As another example, *C. neoformans* and *C. gattii*, which are both listed as WHO priority fungal pathogens, are able to produce polysaccharide capsules that allow them to escape from host phagocytic predators (26).

Additionally, some fungi have been reported to secrete extracellular vesicles (EVs) for host-fungal interactions and virulence (27). EVs are membrane-delimited vesicular compartments containing proteins, lipids, and nucleic acids. They are released by cells into extracellular environment and can be secreted by both host and fungi (28). In a study investigating human-fungal interactions, it was shown that *C. albicans*-released EVs contain proteins related to pathogenesis, cell organization, response to stress, and other functions to mediate the host-fungal interaction (29). EVs from *Paracoccidioides brasiliensis*, a dimorphic fungus causing paracoccidioidomycosis, were shown to contain virulence factors and induce higher expression of host immune receptors, such as Toll-like receptors 4 (TLR4) and dectin-1 (30). While examples have shown that pathogenic fungi manipulate host cells with virulence factors, EVs, and other factors, host cells simultaneously react rapidly to minimize the dangers of infection.

1.2.2 Host recognition strategies during fungal infections

The human immune system, which can be divided into the innate (natural) and acquired (adaptive) immune responses, is an organization of cells and molecules with specialized roles in responding and defending against microbial infection (31). The innate immune system serves as the first line of defense and relies on detection of PAMPs by pattern recognition receptors (PRRs) on immune cells, such as macrophages, monocytes, and neutrophils. Prevalent PRRs in mammals include toll-like receptors (TLRs), RIG-I-like receptors (RLRs), Nod-like receptors (NLRs), mannose-binding lectin, and intracellular DNA sensors such as cGAS (32,33). PRRs differ in their expression across cells of different types. In general, PRRs recognize PAMPs and elicit pro-inflammatory signals and initiate adaptive immune responses. The first host recognition of fungal infection emerged from a *Drosophila-A. fumigatus* infection model. This study identified the importance of toll signaling pathway in controlling expression of antifungal-related genes in *Drosophila* adults and found that mutations in this pathway could dramatically reduce the survival of *Drosophila* after infection (34). Subsequent *in vivo* studies demonstrated that MyD88, a TLR adaptor protein, contributes to multiple antifungal mechanisms. For example, it is required for host defense against *C. neoformans* through the induction of increased TNF- α , IL-12, and IFN- γ (35,36). Dectin-1, as a β -glucan receptor, was first shown in an *in vivo* study to

preferentially recognize *A. fumigatus* swollen conidia and germlings, resulting in the induction of a proinflammatory response (37).

After recognizing the invading pathogens, the innate immune system triggers phagocytosis, which is a process of phagocytic cells digest particles (e.g. pathogens) (9). This mechanism is essential for innate immunity, as it ensures pathogen elimination and contributes to cellular homeostasis. In addition, by processing and presenting pathogen-derived antigens to lymphocytes, phagocytosis links innate defense with the activation of adaptive immune responses (9).

Over time, the adaptive immune response also becomes active, providing a slower but lasting defense mechanism against fungi through three core components: 1) dendritic cells and subtypes that are activated by fungi; 2) the differentiation of T-cells and B-cells guided by fungal molecular patterns, and their corresponding receptors; and 3) the protective mechanisms that reduce fungal evasion while limiting harm to surrounding tissues (38). Further adaptive immune mechanisms are described in (38) as explanation here would go beyond the scope of this thesis.

1.3 dsRNA metabolism in fungi

RNA is a highly structured molecule, which contributes largely to its functions. Transfer RNA (tRNA) for example, is typically folded as a cloverleaf 2D structure and an L-shape 3D structure to carry amino acids and translate the triplet code (39). Double-stranded RNA (dsRNA) forms from two complementary RNA strands, and such structure is commonly found in a variety of RNAs. Unlike the well-known B-form of DNA, long dsRNA is predominantly found in the A-form, with a wider minor groove and narrower major groove (**Figure 1**), which facilitates nucleotide-independent binding by dsRNA binding proteins (dsRBPs) in the minor groove (40). Though certain dsRBPs have structural and sequence-specific binding preferences, most dsRBPs bind to dsRNA with an ~ 11 bp-long interaction interface in a sequence-independent manner (41,42).

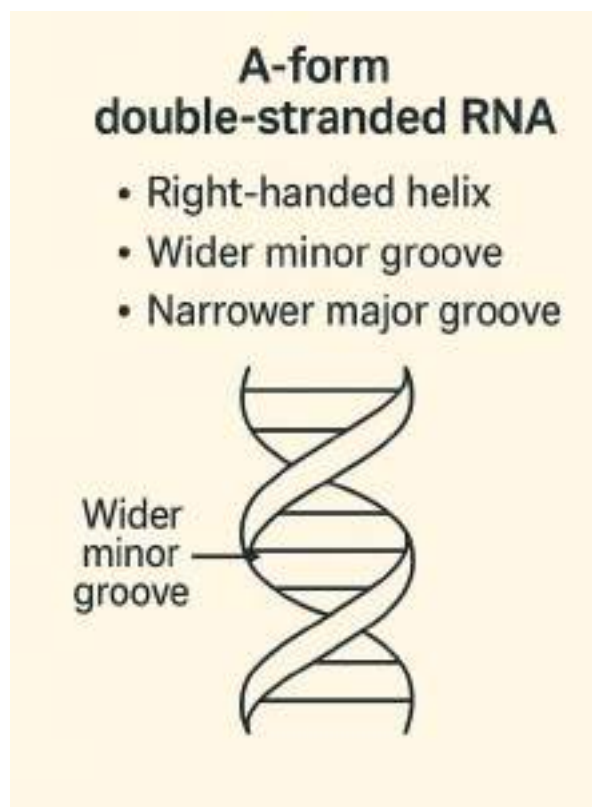


Figure 1: A-form dsRNA

(A) Structure of A-form dsRNA. Image was adapted from (40) and generated from ChatGPT5 (<https://openai.com/>).

1.3.1 dsRNA binding proteins in fungi

dsRBPs are a family of proteins recognizing and interacting with dsRNA. The dsRNA-dsRBP interaction is involved in numerous cellular processes. So far, more than 30 dsRBPs have been identified with a diverse range of important functions in humans and other organisms. For example, RNA editors like

mammalian adenosine deaminase acting on RNA (ADAR) and apolipoprotein B mRNA editing enzyme, catalytic polypeptide (APOBEC) families perform RNA modification by deaminating specific nucleotides (43,44). dsRNA sensors bind dsRNA as part of the innate immune defense (40), RNA-stabilizing proteins play a major role in post-transcriptional control of RNA in splicing, polyadenylation, stabilization, and localization (45), and RNases like Drosha and Dicer execute RNAi in a variety of organisms (46,47). Drosha and Dicer belong to Ribonuclease III (RNase III) enzyme family known to bind and cleave dsRNA for further RNA processing and/or degradation (46). Typically, RNase III enzymes contain two key domains: a catalytic endonuclease domain (RIIID) that is responsible for cleaving dsRNA substrates (48), and the dsRNA binding domain (dsRBD) that enables the enzyme to bind to dsRNA (49). Additionally, RNase III is also involved in the processing of ribosomal RNA (rRNA), small nuclear RNA (snRNA), and small nucleolar RNA (snoRNA).

The representative RNase III superfamily members are bacterial RNase III, yeast Rnt1p, human Drosha, and human Dicer (50–52). Dicer-like proteins (DCLs) and Rnt1 orthologs have been identified in fungi. Rnt1p is the only described RNase III enzyme in *S. cerevisiae*, and contains an uncharacterized N-terminal domain, an RIIID, and a dsRBD (48). Rnt1p recognizes a specific stem-loop structure via a sequence-specific motif at the C-terminus and its special $\alpha\beta\beta\alpha\alpha$ topology (50). Functionally speaking, Rnt1p is 1) essential for the processing of pre-rRNA, 2) found alongside transcriptionally active rRNA gene repeats, and 3) required for the 5' end processing of many snoRNAs (53,54). DCLs have been proven to be involved in vegetative growth, conidiation, pathogenicity, among others (55,56). More importantly, DCLs are one of the key elements in the canonical RNAi pathway, a largely conserved eukaryotic gene-silencing mechanism.

1.3.2 RNAi in fungal biology

Canonical RNAi is a biological process, which initiates with dsRNA cleavage by Dicer or DCLs, resulting in sequence-specific suppression of gene expression. It consists of three core components: Dicer, Argonaute (AGO), and RNA-dependent RNA polymerase (RdRP) (57). Comparative genomic studies have revealed some diversity in *Dicer*-encoding genes counts across eukaryotic species (55,58,59). In mammals, it is widely thought that the RNase III Dicer binds dsRNA and cleaves it into a miRNA or siRNA duplex of ~ 22-nt with a 2-nt overhang at the 3'-end (60). The RNA duplex is then loaded into AGO proteins by Dicer and partner proteins, thus forming the RNA-induced silencing complex (RISC) (61). The strand of the duplex that later functions to guide the complex to a target (guide strand) remains, while the other strand (passenger strand) is discarded and degraded (denoted miRNA*). The selection of the appropriate strand by AGO depends on the preferred binding orientation of the duplex.

This orientation is influenced by two factors: 1) a 5'-terminus that is suitable for loading into the AGO, which favors a 5' terminal pU or pA (62,63); 2) the strand with the least stable 5'-terminal pairing (64–66). RISC is then guided to the target RNA that is complementary in sequence to the interfering RNA. Interference occurs after the RISC complex binds to the target, which results in RNA silencing. In addition, there is a third canonical component, RdRP, that is required for short-interfering RNA (siRNA)-induced silencing to generate dsRNA from ssRNA or amplify sRNA signals in some organisms (**Figure 2**) (67). If sufficient dsRNA is generated through other mechanisms, RdRPs may not be essential for RNA silencing pathway (68). That being said, not all fungi encode active RdRPs.

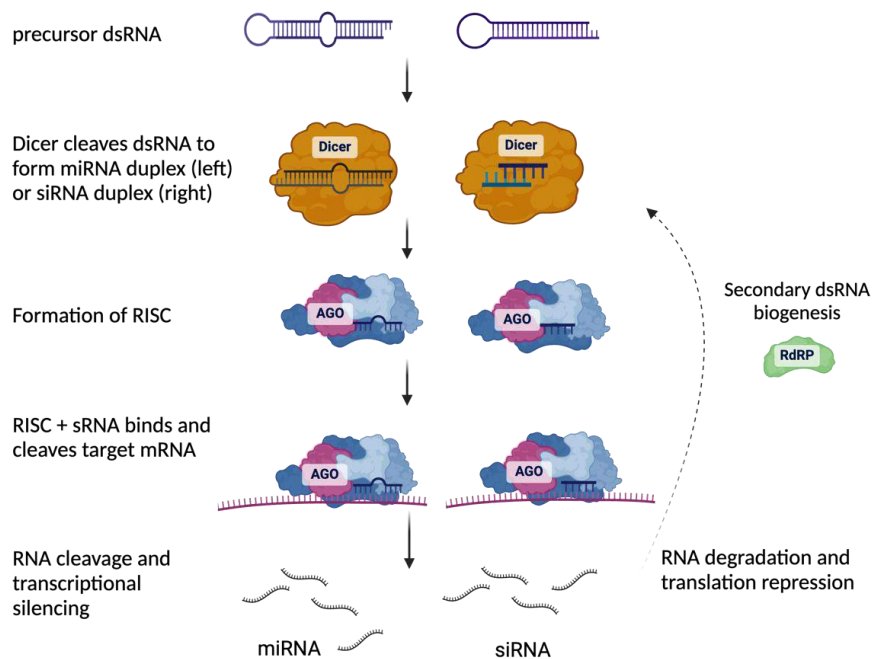


Figure 2: Schema of the canonical RNAi pathway

Dicer proteins recognize and cleave precursor dsRNAs into small RNA duplexes. Shortly after, the sRNA duplex is loaded into AGO proteins by Dicer and partner proteins, thus forming the RNA-induced silencing complex (RISC). The passenger strand is discarded and the guide strand then directs the active RISC to the target mRNA. miRNAs show partial complementarity to the target mRNA and lead to translational repression. siRNAs typically match and bind to the target more completely and result in transcriptional silencing. RdRPs can use sRNA (both miRNA and siRNA) to replicate dsRNA, which can later be recognized and cleaved by Dicer. Image was adapted from (60) and generated using BioRender (BioRender.com).

1.3.3 Small RNA biogenesis in fungi

RNAi produces three main categories of sRNA in mammals: short interfering RNA (siRNA), microRNA (miRNA), and piwi-interacting RNA (piRNA). While siRNA and microRNA-like RNA (miRNA) have been

detected in some fungi, piRNA seem specific to animal cells (69). Due to lack of Drosha in fungi, miRNA differ from miRNA in origin and biogenesis. In terms of biogenesis, siRNA and miRNA are both produced upon Dicer recognition and cleavage of precursor dsRNA. However, there are some general differences between these two classes of sRNAs. First, in terms of parent RNA, siRNA can be synthesized from both endogenous transcripts of repetitive sequences (e.g. transposable elements) and exogenous dsRNA (e.g. viral RNA) (70–72). Whereas miRNAs are products of endogenous single-stranded RNA (ssRNA)-precursor transcripts that form a hairpin structure (73–75) and are less understood in fungi (76). Secondly, from a functional point of view, siRNAs usually bind completely to their targets and trigger AGO-mediated RNA cleavage and transcriptional silencing for genomic defense. miRNAs, on the other hand, don't always fully bind to their target regions, which is usually located in the 3'UTR of mRNAs, and can cause translational repression or mRNA degradation (60,77).

In addition to siRNA and miRNA, qiRNAs (QDE-2-interacting small RNAs) are a class of small RNAs in *N. crassa* (78). They are produced in response to DNA damage during quelling. QDE-3 is a RecQ helicase that recognizes aberrant chromatin structure from highly repetitive sequences (79). QDE-1 associates with these structures and generates aberrant RNA (aRNA) and dsRNA, which are then processed by DCL-1 and DCL-2 into 20-21 nt qiRNA and loaded onto RISC (80,81). The biogenesis of qiRNA in *N. crassa* plays a role in maintaining genome stability by inhibiting protein translation under stress conditions (78). In general, standard siRNA and qiRNA-mediated RNAi display some similarities and differences. First, the two pathways share similar components for silencing (QDE-1, QDE-3, and the Dicers) and the same initial signal (DNA damage). Secondly, qiRNAs (~ 21 nt) are on average shorter than standard siRNAs. Thirdly, quelling is triggered by exogenous factors (e.g. viruses), whereas qiRNAs originate from endogenous factors (repetitive rDNA) (78).

tRNA-derived RNAs (tDRs) have been recognized for their role in regulating gene expression at multiple levels. The parental transfer RNA, well-characterized as the key molecule in facilitating decoding of an mRNA into a protein, has recently been shown to be processed further to produce sRNAs across the kingdoms of life. High-throughput RNA sequencing methods, including AlkB-facilitated RNA methylation sequencing (ARM-seq) (82), multiplex small RNA sequencing (MSR-seq) (83), and others, have led to the identification of tDR species and their associated modifications in numerous systems. tDRs were initially thought to be products of tRNA degradation but were later found to be generated from ribonuclease cleavage at specific positions (84). **Figure 3** illustrates the biogenesis process and classification of tDRs. Briefly, Dicer, Angiogenin (ANG), RNase L, and Rny1p are able to cleavage mature tRNA and generate 5 subtypes of tDRs: 5'-half, 3'-half, 5'-tRF, 3'-tRF, and i-tRF (85). tRNA-halves and i-tRF are typically 30 – 50 nt, whereas tRNA fragments are shorter and, in some cases, even can function

like miRNA when bound to AGO (86). Additionally, 2'-3'- cyclic phosphates at the 3' ends and hydroxyl groups at 5' ends in tRNA-halves and i-tRF also differentiate them from the tRNA fragments (87).

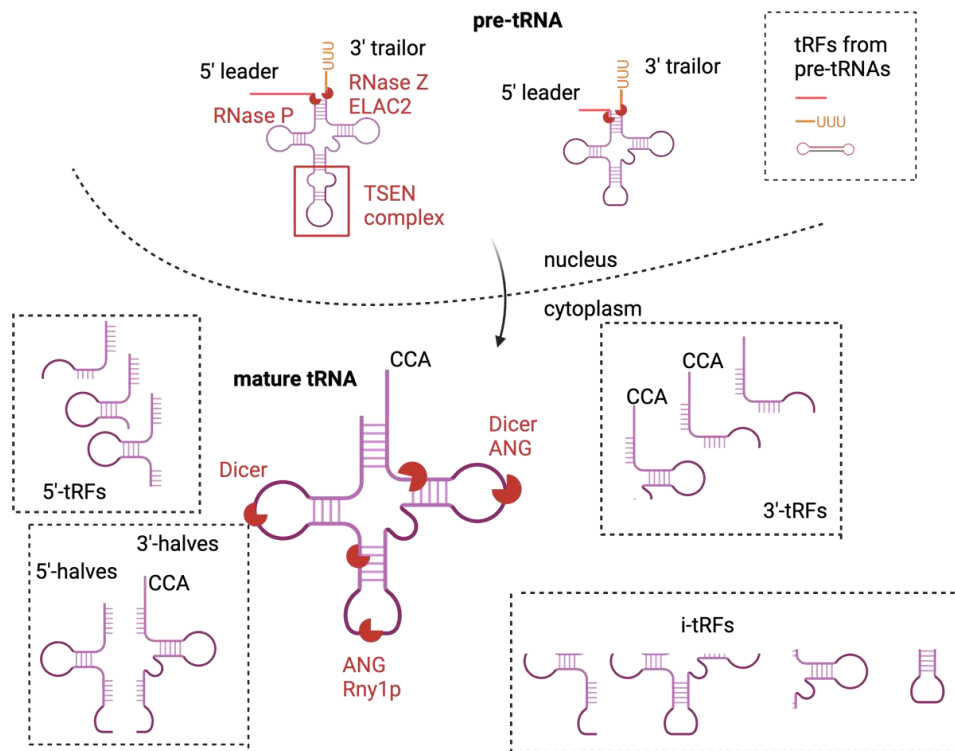


Figure 3: Biogenesis of tDRs from intronless and intron-containing pre-tRNA in fungi

In the nucleus, ribonuclease P cleaves the 5' end of the intronless pre-tRNAs and ribonuclease Z / ELAC2 cleaves the 3' end, whereas pre-tRNAs with introns use the TSEN complex to splice out introns. After intron splicing (when necessary), tRNAs have a "CCA" added at the 3' end. Later, tRNAs are exported to cytoplasm after further processing (e.g. some modifications). In the cytoplasm, Dicer, ANG, RNase L, and Rny1p can cleavage mature tRNA at specific positions and generate 5 subtypes of tDRs: 5'-half, 3'-half, 5'-tRF, 3'-tRF, and i-tRF. Image was adapted from (88) and generated using BioRender (BioRender.com).

1.3.4 Function and regulation of RNAi in genome defense and pathogenesis

RNAi is widespread in the fungal kingdom. Over time, it has evolved to acquire novel functions and is commonly involved in genome defense, heterochromatin formation, and gene regulation (57). RNAi is considered crucial in genome defense against mycoviruses and selfish genetic elements like transposons (89,90). The role of RNAi-mediated gene silencing in antiviral defense was first examined in chestnut blight fungus *Cryphonectria parasitica*. No visible phenotypic change was observed after disruption of the Dicer-like genes (*Dcl-1* and *Dcl-2*). However, a $\Delta Dcl-2$ mutant strain was observed to exhibit severe symptoms and a high level of viral RNA upon infection with mycovirus, whether or not a hypoviral RNA silencing suppressor was present (91), suggesting that DCL-2, but not DCL-1, plays a

role in protecting the fungus from viral infection. RNAi-related heterochromatin formation was previously identified in the fission yeast *Schizosaccharomyces pombe*. It was known that *S. pombe* consisted of three chromosomes, each of which contained large amount of repeats, especially at the centromeres, which were assembled into heterochromatin (92). dsRNA derived from these repeats guided formation and maintenance of heterochromatin via the RNAi machinery (93). sRNAs derived from coding regions can also contribute to gene regulation, as for *M. circinelloides* exonic regions discovered to regulate gene expression by leading to degradation of their corresponding mRNAs (94).

While RNAi is generally regarded as a defense mechanism against viruses, there are notable exceptions, particularly in cases involving beneficial viruses. Viruses are thought to be harmful to their hosts, but in *S. cerevisiae*, this is not always the case. The yeast *S. cerevisiae* can harbor a dsRNA killer virus without harm (95). Killer-positive strains can inhibit the growth of neighboring killer-negative strains and other species, providing a competitive advantage in resource-limited environments (96). This competitive advantage conflicts with a functional RNAi pathway, suggesting an evolutionary trade-off that favored the persistence of RNAi-deficient fungi (97).

RNAi also plays a role in drug resistance and pathogenesis. *C. neoformans*, a yeast that causes ~ 15% of AIDS-related deaths, was found to lose RNAi to become resistant to antifungal drugs (FK506 and rapamycin) through hypermutation (89), which is a process where a specific DNA region exhibits an increased mutation rate (98,99). Loss of RNAi components, e.g., *ZNF3* in clinical isolates, can also result in accumulation of transposable elements like Cnl1 in subtelomeric regions of chromosomes to confer resistance to antifungal drugs (89). Additionally, *M. circinelloides* was discovered to develop transient resistance to FK506, rapamycin, and 5-fluoroorotic acid (5-FOA) through epimutation (100). This is an RNAi-dependent mechanism that can temporarily 'turn off' drug target genes or related pathways, therefore allowing fungi to survive in the presence of antifungal agents.

Lastly, RNAi-dependent sRNA biogenesis is conserved in both humans and fungi, suggesting that sRNAs may act across species and kingdoms to control gene expression. EVs, as mentioned previously, have been shown to carry sRNAs for many purposes, including cell-to-cell communication and immune signaling (28). The export of RNA through EVs was reported in several human fungal pathogens, including *C. neoformans*, *C. albicans*, *H. capsulatum*, *P. brasiliensis*, and *Malassezia sympodialis* (101–103). They may play a role for various biological processes, including cell communication and pathogenesis. Such cross-kingdom communication via extracellular sRNAs is not limited to animal systems, but also in plant-fungal interactions (104,105). The plant fungal pathogen *Botrytis cinerea* delivered sRNAs to plant cells via EVs, hijacking plant AGO1 to suppress host immunity genes (56,106).

In summary, although RNAi is a conserved mechanism, it shows significant variation across fungi. Understanding the role of RNAi in *A. fumigatus* may accelerate potential RNA-based therapeutic options against this high-risk human fungal pathogen. In the next section, I will discuss host responses during host-fungal interactions, focusing on dsRNA-mediated immunity.

1.4 dsRNA metabolism in the host

1.4.1 Human dsRNA sensors and interferon responses

Viruses often contain a dsRNA genome, which is important for the host to recognize and quickly respond to the viral dsRNA. In humans, dsRBPs that recognize dsRNA and activate immune responses are often referred to as dsRNA sensors (41). Most dsRNA sensors contain a helicase domain for dsRNA recognition and a caspase activation and recruitment domain (CARD) for activating downstream signaling pathways through stimulation of the interferon (IFN) response and inhibition of growth (107). IFNs are a group of cytokines that can be divided into three classes (Type I, Type II, and Type III). Functionally speaking, IFNs may be antiviral, antiproliferative, and/or immunomodulatory (108).

The interaction of dsRNA sensors with dsRNA is thought to be a mediator of the IFN system. RIG-I-like receptors (RLRs) and Toll-like receptor 3 (TLR3) are well-characterized dsRNA sensors for triggering IFN responses, while protein kinase R (PKR) and oligoadenylate synthases (OASs) are responsible for growth inhibition through two different manners (41). PKR inhibits growth by disrupting protein synthesis, while OAS induces growth inhibition by triggering RNA degradation (41). The two most representative dsRNA RLR sensors are Retinoic acid inducible gene I (RIG-I) and melanoma differentiation-associated gene 5 (MDA5). Both of these bind to dsRNA and activate the adaptor protein MAVS, which then triggers IFN signaling (109). RIG-I and MDA5 recognize dsRNA through different features. RIG-I recognizes 5' di- or triphosphate-modified blunt dsRNA, which is rarely found in endogenous host transcripts. Thus, this allows RIG-I to distinguish “self” and “non-self” dsRNA (110). In contrast, MDA5 preferentially binds long dsRNA, but cannot discriminate between endogenous and exogenous discrimination (111). This being said, MDA5 can be activated by both pathogen-derived and host-derived dsRNA under specific conditions. However, humans have evolved a mechanism to prevent this inappropriate activation: ADAR proteins (mentioned in **1.3.1**) catalyze the deamination of adenosine to inosine (termed A-to-I RNA editing) in endogenous dsRNA. This editing leads to structural changes, thereby preventing MDA5 binding and activation (112).

1.4.2 Human ADAR proteins and A-to-I RNA editing

The ADAR enzyme family likely evolved from a conserved adenosine deaminase acting on tRNA (ADAT) ancestor, after the addition of a dsRNA binding domain (113). While the ADAT family is present in all eukaryotes, ADAR exists only in some metazoans. So far, no study has reported ADAR being found in choanoflagellates, fungi, or plants (114–116), though some fungi have confirmed ADAR-independent RNA editing (117,118). In humans, there are three ADAR proteins: ADAR1, ADAR2, and ADAR3 encoded

by *ADAR1*, *ADARB1*, and *ADARB2*, respectively. Interestingly, *ADAR1* has two isoforms: *ADAR1-p110* and *ADAR1-p150* driven by different promoters (119,120). *ADAR1-p150* is known to be IFN inducible, but even the “constitutive” isoform was reported to be induced by IFN to some extent (41,121). *ADAR1-p110* protein is mainly responsible for nuclear A-to-I RNA editing (122), while *ADAR1-p150* plays the main role for suppression of dsRNA sensing by MDA5 and PKR in the cytoplasm (123,124). Notably, the two isoforms can both shuttle to the nucleus (125). **Figure 4** shows a detailed feature map of each ADAR protein and the representative *ADAR1*-associated IFN response in humans. *ADAR1* mainly deaminates *A/u* elements, which are a class of SINE retrotransposons, with ~ 300 nt length that comprise 11% of the human genome (126). *ADAR2* mostly contributes to editing of exonic regions and is essential in deaminating the *GRIA2* transcript from CAG to CIG in mice, resulting in a Gln to Arg change (127). Different from *ADAR1* and *ADAR2*, *ADAR3* is predominantly expressed in the brain, structurally has no catalytic activity, and can bind to single-stranded RNA (128). ADAR proteins have been widely studied because of their importance in the immune system. Compared to other RNA modifications, A-to-I RNA editing can be directly detected by RNA sequencing because inosine (I) can be read as a guanosine (G) by the transcriptional and translational machinery (129,130). A-to-I RNA editing has been detected in mammalian pre-mRNA, mature mRNA, and lncRNA in response to different types of pathogens (131–133). Additionally, heavily edited regions rarely found in coding sequences (termed hyper-editing) are also detected across the metazoa using RNA-seq (134,135).

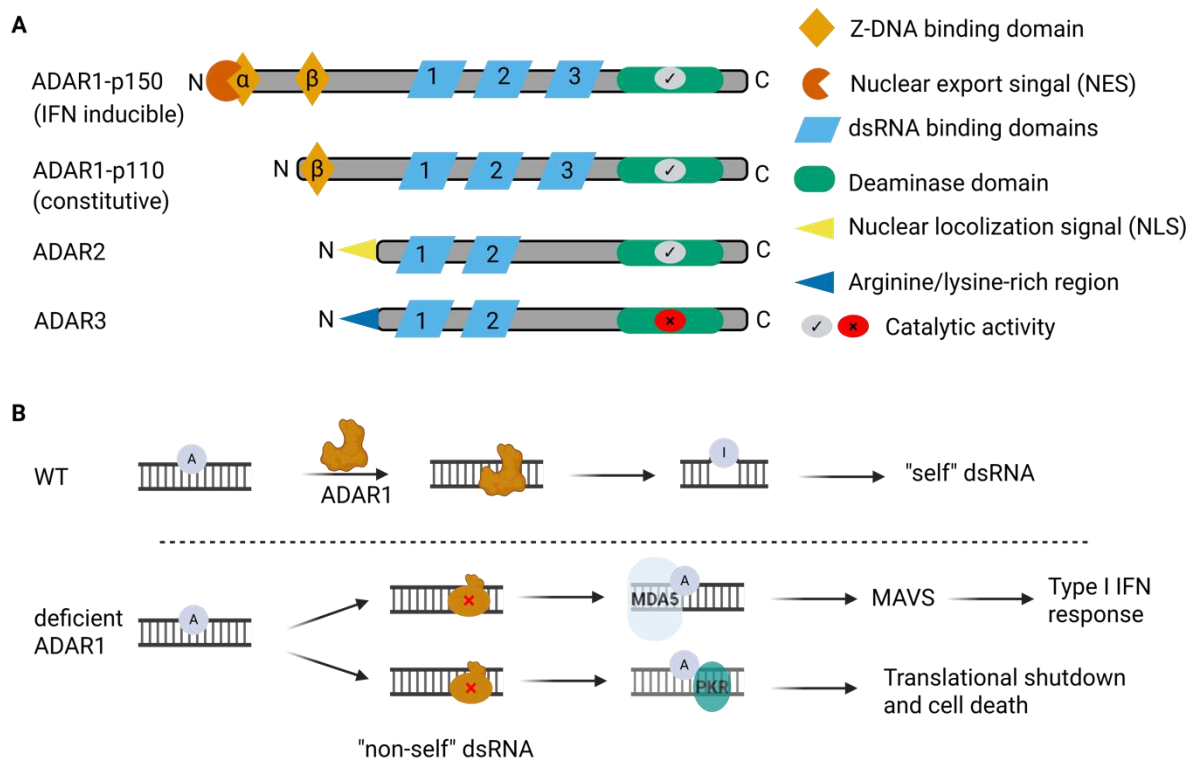


Figure 4: Human ADAR proteins and their roles in the IFN response

(A) Structural domains of human ADAR1-p150, ADAR1-p110, ADAR2, and ADAR3. **(B)** Human ADAR1 prevents “self” dsRNA from activating dsRNA sensors (MDA5 and PKR), which allows IFN response and efficient translation. Image was adapted from (136) and (137), respectively and generated using BioRender (BioRender.com).

A-to-I RNA editing has been studied within the host in response to viral infections, intracellular bacterial infection, and fungal infection (138–140). For viral infections, it’s been reported that the host ADAR1 response to viral infections can both promote or inhibit the outcome depending on the virus, cell type, and stage of infection (141,142). However, it remains less well-studied in the context of fungal infections. One study suggests that *C. albicans* infection can lead to the up-regulation of A-to-I RNA editing (139), but the consequence and mechanism of this editing remain uncharacterized.

1.4.3 ADAR-associated RNA splicing

A-to-I RNA editing in exonic/intronic regions can result in the new synthesis of proteins that are not directly encoded in the genome (termed recoding), a mechanism that is usually driven by ADAR2. For example, it is well-known that ADAR2-mediated A-to-I RNA editing in glutamate-gated ion channels is essential and widespread, with five of the glutamate receptors (GluR-B, GluR-C, GluR-D, GluR-5, GluR-6) known to rely on recoding events (143–145). On the other hand, A-to-I RNA editing could also affect splicing, stability, and other factors through editing in regions like UTRs and introns (146). As two

important post-transcriptional processes, A-to-I RNA editing and RNA splicing share similar time and location (147,148). A handful studies suggest ADARs can affect pre-mRNA splicing in both editing-dependent and editing-independent manners. For example, ADAR1 was reported to edit an intronic splicing silencer, which later led to the recruitment of a splicing factor SRSF7 to suppress exon inclusion (149). ADAR2 was demonstrated to bind to intronic dsRNA regions and inhibit exon inclusion by preventing the spliceosome from recognizing the splice site, a mechanism reliant on ADAR2 binding affinity but not editing (149). In addition to ADAR-mediated RNA splicing, the reverse also appears to be true that RNA splicing can affect A-to-I RNA editing. A study in mice showed global changes in editing patterns in the absence of splicing factors NOVA1 and NOVA2 (150), indicating that splicing factors could influence RNA editing as well. Additionally, it has been shown that slower transcription rates could lead to higher levels of intronic editing (151). These results suggest an intricate interplay of RNA editing and splicing in cells.

2 Aim of study

The WHO report on fungal priority pathogens urges the acceleration of work to diagnose and treat human fungal pathogens (2). In order to develop better diagnostics and therapeutics, we first require a better molecular understanding of these organisms. In this thesis, I focus on RNA regulatory pathways important to both the host and the fungus, with a particular focus on the role of dsRNA processing.

In pathogenic fungi, RNAi was reported to function in gene silencing, growth, defense, and many other responses. In *A. fumigatus*, RNAi was known to functionally suppress complementary mRNA when overexpressing an inverted-repeat transgene, indicating an active RNAi system (152). I hypothesized that the **dsRNA-dependent RNAi machinery contributes to regulation of gene expression in *A. fumigatus***. To test this hypothesis, we aimed to define the molecular function of the RNAi mechanism in *A. fumigatus* using bioinformatic and experimental approaches.

Pathogenic fungi have also been reported to trigger host dsRNA sensors (153,154), but the dynamics of the dsRNA landscape upon fungal infection remains largely uncharted. One study reported elevated A-to-I RNA editing activity in *C. albicans*-infected cells (139), but how RNA editing changes in immune defense in response to different fungal pathogens remains unclear. It is also unclear how hyper-editing (representative of long dsRNA regions) changes in response to fungal infections. I hypothesized that **host-fungal coinubation drives altered A-to-I editing patterns and host post-transcriptional regulation, particularly at the level of dsRNA and dsRBPs**. This hypothesis was tested with three aims, designed to elucidate potential functions of host A-to-I RNA editing in response to fungal coinubations. 1) We aimed to characterize the role of ADAR-dependent A-to-I RNA editing levels in human PBMCs confronted with pathogenic fungi using publicly available datasets. 2) As RNA editing is linked to RNA splicing, we next aimed to investigate differences in host alternative splicing upon fungal stimulations. 3) Furthermore, we aimed to characterize the molecular link between these two post-transcriptional regulators in contributing to the immune response.

Collectively, this thesis improves our understanding of dsRNA metabolism during fungal stimulations by investigating factors both on the host side involved in dsRNA regulation in response to fungal challenge and on the fungal side that contribute to dsRNA processing.

3 Overview of manuscripts

Invasive fungal infections have aroused more and more attention, with 6.5 M infections and 3.8 M deaths annually (1). Additionally, fungi can cause secondary infections as complications of the SARS-CoV-2 infection in immunocompromised patients (155). These COVID-19-associated pathogenic fungi include *R. oryzae*, *A. fumigatus*, and others. Unraveling the molecular intricacies of host-fungal interaction at the RNA level may ultimately inform development of effective RNA-based strategies to treat infection.

During the course of my doctoral research, I first collaborated with fellow researchers from multiple labs and countries to elucidate functions of RNAi components in *A. fumigatus* through several molecular assays, such as RT-qPCR, Southern blot, and mRNA- and sRNA-seq with related computational analyses (detailed in **Manuscript A**). I built upon this research and further explored the canonical RNAi pathway and sRNA biogenesis through bioinformatic analyses and by overexpressing an inverted-repeat transgene. This unveiled evidence of a putative dsRNA-dependent growth arrest (as detailed in **Manuscript B**). Next, I co-authored a literature review of RNA modifications across host-pathogen interactions, a poorly understood regulatory feature of the transcriptome (described in **Manuscript C**). Finally, I investigated host dsRNA processes in human peripheral blood mononuclear cells (PBMCs) relevant to the remodeling of the transcriptome after exposure to three diverse fungal pathogens. By investigating global A-to-I RNA editing levels and AS events from publicly available data, I demonstrated a unique transcriptomic response to *R. oryzae* not induced by *A. fumigatus* or *C. albicans* (elaborated in **Manuscript D**).

4 Manuscript

4.1 Manuscript A

Disruption of the *Aspergillus fumigatus* RNA interference machinery alters the conidial transcriptome

Abdulrahman A. Kelani, Alexander Bruch, Flora Riveccio, Corissa Visser, Thomas Krüger, Danielle Weaver, **Xiaoqing Pan**, Sascha Schäuble, Gianni Panagiotou, Olaf Kniemeyer, Michael J. Bromley, Paul Bowyer, Amelia E. Barber, Axel A. Brakhage, and Matthew G. Blango.

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The candidate is

☐ First author, ☐ Co-first author, ☐ Corresponding author, ☒ Co-author.

Summary:

The RNA interference (RNAi) machinery is an important post-transcriptional regulator, with diverse functions in fungal biology. To enhance our understanding of RNAi mechanisms in *A. fumigatus*, RNAi components were knocked-out. By overexpressing two inverted-repeat transgenes complementary to two genes in *A. fumigatus*, it was determined that DclB and PpdA are essential for RNA silencing in the fungus. mRNA-seq analysis revealed that dicer-like enzymes (DclA/B) and RNA-dependent RNA polymerases (RrpA/B) also contribute to the regulation of conidial ribosome biogenesis genes, although the mechanism remains unclear. Our investigation defined the functional components of hairpin-mediated RNA silencing in *A. fumigatus* and linked the RNAi machinery to regulation of the transcriptome of conidia.

Contribution to the manuscript:

Xiaoqing Pan helped with RNA isolation, and sRNA-seq analysis. Additionally, she contributed in writing the protocol part of the manuscript.

Authors' contributions (in %) to the given categories of the publication

Author	Conceptual	Data analysis	Experimental	Writing the manuscript	Provision of material
Abdulrahman A. Kelani	35%	40%	50%	30%	0%
Alexander Bruch	10%	10%	15%	15%	0%
Flora Riviuccio	0%	0%	7%	5%	0%
Corissa Visser	0%	0%	7%	3%	0%
Thomas Krüger	0%	10%	10%	0%	0%
Danielle Weaver	0%	5%	0%	0%	0%
Xiaoqing Pan	0%	5%	6%	2%	0%
Sascha Schäuble	0%	10%	0%	0%	0%
Amelia E. Barber	0%	5%	5%	0%	0%
Axel A. Brakhage	10%	0%	0%	5%	40%
Matthew G. Blango	40%	15%	0%	35%	60%
Others	5%	0%	0%	5%	0%
Total:	100%	100%	100%	100%	100%

Signature candidate

Signature supervisor (member of the Faculty)

Disruption of the *Aspergillus fumigatus* RNA interference machinery alters the conidial transcriptome

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ABSTRACT

The RNA interference (RNAi) pathway has evolved numerous functionalities in eukaryotes, with many on display in Kingdom Fungi. RNAi can regulate gene expression, facilitate drug resistance, or even be altogether lost to improve growth potential in some fungal pathogens. In the WHO fungal priority pathogen, *Aspergillus fumigatus*, the RNAi system is known to be intact and functional. To extend our limited understanding of *A. fumigatus* RNAi, we first investigated the genetic variation in RNAi-associated genes in a collection of 217 environmental and 83 clinical genomes, where we found that RNAi components are conserved even in clinical strains. Using endogenously expressed inverted-repeat transgenes complementary to a conditionally essential gene (*pabA*) or a nonessential gene (*pksP*), we determined that a subset of the RNAi componentry is active in inverted-repeat transgene silencing in conidia and mycelium. Analysis of mRNA-seq data from RNAi double-knockout strains linked the *A. fumigatus* dicer-like enzymes (DclA/B) and RNA-dependent RNA polymerases (RrpA/B) to regulation of conidial ribosome biogenesis genes; however, surprisingly few endogenous small RNAs were identified in conidia that could explain this broad change. Although RNAi was not clearly linked to growth or stress response defects in the RNAi knockouts, serial passaging of RNAi knockout strains for six generations resulted in lineages with diminished spore production over time, indicating that loss of RNAi can exert a fitness cost on the fungus. Cumulatively, *A. fumigatus* RNAi appears to play an active role in defense against double-stranded RNA species alongside a previously unappreciated housekeeping function in regulation of conidial ribosomal biogenesis genes.

Keywords: RNA interference; filamentous fungi; Argonaute; Dicer-like; fungal pathogen

INTRODUCTION

RNA interference (RNAi) is a conserved molecular mechanism that was likely intact in the last common ancestor of modern eukaryotes (Shabalina and Koonin 2008; Torri et al. 2022). RNAi uses small RNA (sRNA) molecules to suppress gene expression either transcriptionally by hetero-

chromatin recruitment or post-transcriptionally through sequence-specific messenger RNA (mRNA) degradation or translational interference (Moazed 2009; Dang et al. 2011; Torres-Martinez and Ruiz-Vazquez 2017). The first reported example of RNAi in fungi came from the model fungus *Neurospora crassa* as a process termed quelling, where introduction of repetitive transgenes induced

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post-transcriptional gene silencing (Romano and Macino 1992). Since then, several RNAi and small RNA biogenesis pathways have been characterized in fungi that are largely classified as Dicer-dependent (canonical) RNAi or Dicer-independent (noncanonical) RNAi (Ketting 2011; Ruiz-Vazquez et al. 2015; Torres-Martinez and Ruiz-Vazquez 2017).

Dicer is a type IV RNase III enzyme responsible for dsRNA cleavage and biogenesis of a variety of sRNA classes in different organisms (Nicolas et al. 2015; Torres-Martinez and Ruiz-Vazquez 2017). In fungi, the orthologous RNase III domain-containing enzymes are typically referred to as dicer-like (DCL) proteins due to subtle differences in domain structure. The generated sRNAs are then loaded into an Argonaute protein, the main component of the RNA-induced silencing complex that binds and processes sRNAs to guide selective destruction, translational repression, or transcriptional suppression of target mRNAs (Agrawal et al. 2003; Torres-Martinez and Ruiz-Vazquez 2017). In some instances, RNA-dependent RNA polymerases are required for RNAi pathways to amplify the silencing activity by generating additional dsRNA substrates using the single-stranded sRNAs as primers (Torres-Martinez and Ruiz-Vazquez 2017). While the core RNAi proteins are mostly conserved across eukaryotes, some gene losses and duplications have occurred, which further diversified the RNAi pathways and their functional specialization (Hutvagner and Simard 2008; Mukherjee et al. 2013; Torres-Martinez and Ruiz-Vazquez 2017). Loss of RNAi was observed to fuel hypervirulence in *Cryptococcus* due to higher transposon activity (Gusa et al. 2020), and killer virus-infected *Saccharomyces cerevisiae* strains are able to kill and therefore outcompete noninfected strains in the absence of RNAi (Bostian et al. 1980). The RNAi machinery have also been implicated in other functions, ranging from ribosome biogenesis to maintenance of cellular quiescence (Bemstein et al. 2012; Roche et al. 2016), further highlighting the importance of RNAi across the fungal kingdom.

In this study, we improve our understanding of RNAi in the human fungal pathogen, *Aspergillus fumigatus*. *A. fumigatus* is a ubiquitous, filamentous saprophyte and among the leading fungal killers (Latge 1999; Beauvais et al. 2013). It encodes a functional RNAi system consisting of two dicer-like proteins (DclA and DclB), two argonautes (PpdA and PpdB), and two RNA-dependent RNA Polymerases (RrpA and RrpB), together with several other, annotated peripheral factors (Fig. 1A; Mouyna et al. 2004; Hammond and Keller 2005; Hammond et al. 2008b). Here, we show that the *A. fumigatus* RNAi system is conserved among both clinical strains and environmental isolates. We then delineate the RNAi components necessary for inverted-repeat transgene (IRT) silencing and link the RNAi machinery to expression of conidial ribosome biogenesis genes and long-term fitness. Finally, we provide

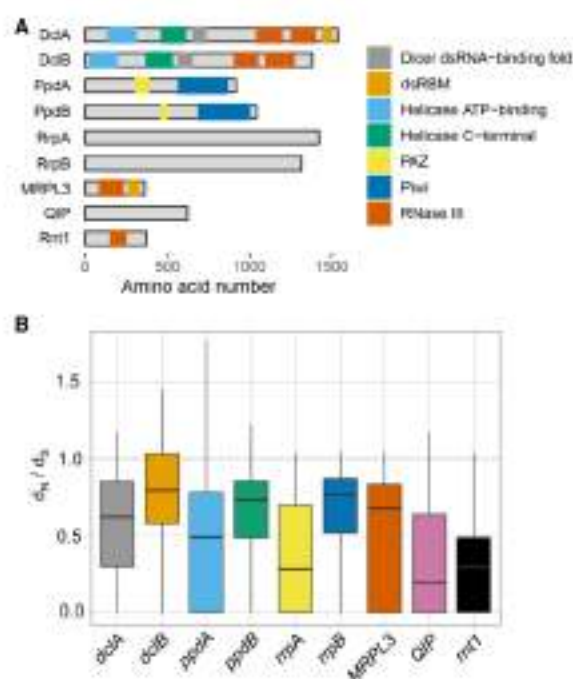


FIGURE 1. RNAi is conserved across 300 sequenced *A. fumigatus* strains. (A) Structural domains of *A. fumigatus* RNAi proteins were analyzed using UniProt (www.uniprot.org). DclA and DclB both contain a DEXH-helicase followed by a helicase-C, a Dicer-dsRNA binding fold that mediates binding with other proteins (Qin et al. 2010), a P-element-induced wimpy testis (PIWI)-Argonaute-Zwille (PAZ) domain for binding the 3' end of the dsRNA, and two RNase III domains necessary for cleavage of bound dsRNA (Schauer et al. 2002; Mukherjee et al. 2013; Song and Rossi 2017). Both argonaute orthologs (PpdA/B) have the characteristic PAZ domain and a PIWI domain present in all eukaryotic argonautes (Jinek and Doudna 2009; Ender and Meister 2010). (B) Calculation of the ratio of nonsynonymous to synonymous base changes (d_N/d_S) for each of the core RNAi components and peripheral factors using 300 genomes from Barber et al. (2021). Ratios under one indicate that each gene is under neutral-to-purifying selection.

a brief characterization of the endogenous small RNA pool of *A. fumigatus*, which appears to be almost entirely separable from the observed differential expression and protein abundance in the knockout strains.

RESULTS

RNAi pathway genes are conserved in environmental and clinical isolates of *A. fumigatus*

The RNAi pathway has diversified extensively since its emergence around the time of the last eukaryotic common ancestor (Torri et al. 2022). Functions range from gene regulation to genome defense, but in some specific instances the machinery has been lost to provide a growth

advantage (Billmyre et al. 2013). To determine whether the RNAi machinery is stably maintained in *A. fumigatus*, we explored the core RNAi components for genetic variation across 300 *A. fumigatus* genomes collected from diverse clinical and environmental niches (Barber et al. 2020, 2021). All the RNAi pathway genes were present as a single copy in the genomes, indicating a high degree of evolutionary conservation. While many strains contained genetic variation, such as missense substitutions or alterations to the 3'UTR (Supplemental Fig. S1; Supplemental Data Set S1), high impact variants like frameshifts or premature stops were rare and only observed in 1%–6% of samples ($n=4$ for *dclA*, $n=1$ for *rpA*, $n=19$ for *rpB*). Of note, the *rpA* and *rpB* genes are thought to be dispensable for silencing in the related organism *Aspergillus nidulans* (Hammond and Keller 2005), whereas orthologs of the *dclA* gene contribute to the process of meiotic silencing of unpaired DNA during the sexual cycle in other organisms (Alexander et al. 2008). Calculation of the ratio of non-synonymous to synonymous base changes (d_N/d_S) indicated that all members of the pathway are under purifying selection, often with one paralogous copy of the gene under a stronger degree of selection (Fig. 1B). Overall, we found a strong degree of genomic conservation among the *A. fumigatus* RNAi system, which together with the evidence of purifying selection suggests that the pathway likely plays a conserved, beneficial function across *A. fumigatus* strains and that loss does not strongly facilitate virulence in humans as has been hypothesized for other fungal pathogens (D'Souza et al. 2011; Billmyre et al. 2013).

Engineered loss of the RNAi machinery provides no growth advantage in stress or drug challenge

Genomic analysis suggested that the RNAi machinery is not frequently lost under clinical or environmental conditions, but we wanted to explicitly test the hypothesis that RNAi is dispensable for stress response and drug resistance. To do this, we created a series of knockouts in the core RNAi machinery, including single knockouts of *dclA*, *dclB*, *ppdA*, *ppdB*, *rpA*, and *rpB* (Supplemental Fig. S2). In addition, we made a series of pairwise knockouts in *dclA/B*, *ppdA/B*, and *rpA/B*. None of the knockouts showed growth defects on minimal or rich media (Fig. 2A). Next, we examined the function of RNAi components in response to various stress conditions, including the cell wall stressors congo red, caffeine, and caspofungin; cell membrane stress agent sodium dodecyl sulfate (SDS); and oxidative stressor H_2O_2 (Supplemental Fig. S3). We also tested for sensitivity to azole drugs like voriconazole and tebuconazole, as well as thiabendazole (TBZ), which is a microtubule inhibitor previously used to link RNAi components to chromosome segregation in fission yeast (Hall et al. 2003). In each case, the RNAi knockout strains

showed no obvious phenotypic difference in comparison with the wild type.

DclB and PpdA are necessary for inverted-repeat transgene silencing

The lack of phenotypic differences upon deletion of RNAi components led us to question the functionality and redundancy of the individual RNAi proteins in *A. fumigatus* and determine which core RNAi factors of *A. fumigatus* are required for silencing. It is known that exogenous delivery of RNA directly into the fungal cytoplasm is challenging due to a sequestration of negatively charged RNA by the cationic cell wall (Jochl et al. 2009). We first confirmed this finding in our own system using confocal microscopy of an exogenously delivered, 500-bp, fluorescein-labeled in vitro-transcribed dsRNA after incubation for 16 h at 37°C (Supplemental Fig. S4A). The dsRNA clearly accumulated in the cell wall of the fungus costained with calcofluor white as expected.

Next, we turned to heterologous expression of two inverted-repeat transgenes from the *A. fumigatus* chromosome to target a nonessential gene and a conditionally essential gene (Supplemental Fig. S4B). We reengineered a previously described system (Mouyna et al. 2004) using the pSilent-1 gene silencing vector (Nakayashiki et al. 2005) to produce a wild-type strain harboring a 500-bp, Tet-inducible *pkpP* or *pabA* inverted-repeat transgene on the *A. fumigatus* chromosome at the *pyrG* locus. The non-essential *pkpP* gene is required for dihydroxynaphthalene (DHN)-melanin biosynthesis, and silencing of *pkpP*, results in an easy phenotypic readout via the production of pigmentless, white conidia (Langfelder et al. 1998). The second inverted-repeat transgene was complementary to the *A. fumigatus pabA* gene, which encodes a para-aminobenzoic acid synthetase (PabA). PabA catalyzes the biosynthesis of para-aminobenzoic acid (PABA), a precursor for folate biosynthesis in the fungus. Previous work has shown that deletion of *pabA* in *A. fumigatus* and *A. nidulans* results in loss of virulence and severe growth reduction, respectively, lending credence to *pabA* as a potential drug target (Brown et al. 2000). In fact, *A. fumigatus pabA* is conditionally essential, and only grows with appropriate supplementation (e.g., folate or p-aminobenzoic acid).

We first tested the *pkpP* inverted-repeat transgene in the wild-type strain, where we observed white conidia after induction with doxycycline on AMM plates, indicative of successful post-transcriptional silencing of the *pkpP* gene (Fig. 2B). Ultimately, we created two versions of these plasmids with different resistance cassettes so that we could also incorporate the construct onto the chromosome in each of our single and double knockouts. While similar silencing was observed in most transformed knockout strains, $\Delta dclB$ and $\Delta ppdA$ remained able to produce melanized conidia, suggesting that both DclB and PpdA are essential

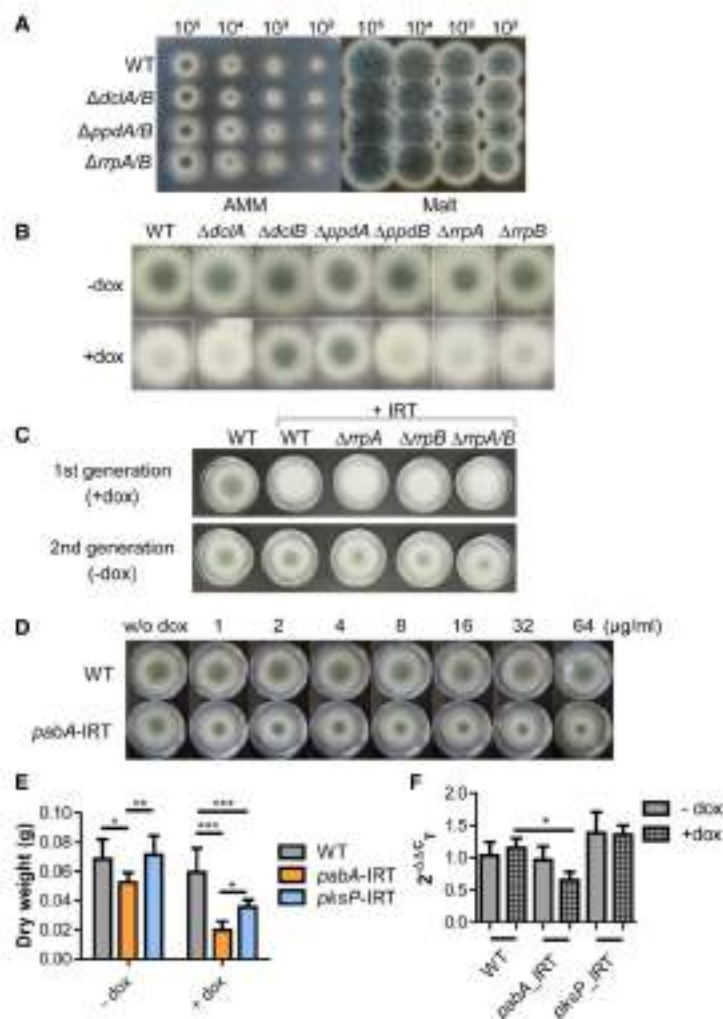


FIGURE 2. DclB and PpdA are required for IRT-mediated silencing in *A. fumigatus*. (A) Droplet growth assays for wild-type (WT), $\Delta dclA/B$, $\Delta ppdA/B$, and $\Delta rpaA/B$ strains grown on AMM or Malt agar plates for 48 h. Image shown is representative of three independent experiments. (B) Wild-type and RNAi knockout strains transformed with IRT constructs were spot inoculated on AMM with and without inducer (10 μ g/ml doxycycline [dox]) for 2 d. Image shown is representative of three independent experiments. (C) Wild-type and IRT-transformed strains were spot inoculated on AMM agar plates with inducer (10 μ g/ml doxycycline) for 2 d. Spores were collected and the experiment was repeated without IRT induction. (D) The wild-type (CEA17 $\DeltaakuB^{52100}$) strain was transformed with a plasmid (pSilent-pabA) expressing a dox-inducible IRT targeting *pabA*, a para-aminobenzoic acid synthetase-encoding gene. Wild-type and *pabA*-IRT-expressing strains were spot inoculated on AMM agar plates supplemented with different concentrations of dox. Images were taken after 2 d of incubation at 37°C. (E) Wild-type and *pabA*-IRT-expressing strains, as well as a control *pksP*-IRT-expressing strains were grown with and without dox induction for 24 h. Mycelia were collected and dried for 3 d at 60°C before measuring weight. Plot displays data from four biological replicates, with each replicate having three technical repeats. Significance was determined by two-way ANOVA with Bonferroni-corrected post hoc *t*-test comparing all columns. (F) Validation of silencing of *pabA* gene after induction of *pabA*-IRT using qRT-PCR. Plot shows standard error of the mean of three biological replicates performed in technical triplicates using the $2^{-\Delta\Delta C_t}$ method (Livak and Schmittgen 2001). Significance was determined by one-way ANOVA followed by Bonferroni corrected post hoc *t*-test comparing all columns and assuming Gaussian distributions.

for inverted-repeat transgene silencing and cannot be rescued by their respective paralogs, DclA and PpdB.

As mentioned, the RdRPs contribute to the RNAi pathway in some organisms by transcribing additional dsRNA substrates and amplifying the silencing response (Dang et al. 2011). To eliminate the possibility that the overexpression of our inverted-repeat transgene negated the necessity of the amplification role of RdRPs in *A. fumigatus*, we also passaged the induced RdRP strains onto fresh plates lacking inducer. This experiment allowed us to determine whether the silencing persisted after the induction of the inverted-repeat transgene was removed. In the wild-type, $\Delta rpaA$, and $\Delta rpaB$ strains silencing was quickly lost (Fig. 2C), indicating that these proteins likely do not contribute to amplification of silencing in *A. fumigatus* and in agreement with results observed previously in *A. nidulans* (Hammond and Keller 2005). A curious observation is that the loss of silencing occurs to a lower extent in the *rpa* knockouts than in wild type, perhaps suggesting an additional layer of complexity surrounding the RdRP proteins in *A. fumigatus*.

Using the *in vitro*-transcribed dsRNA with complementarity to the conditionally essential gene *pabA*, we observed no cessation of growth in our imaging experiments (Supplemental Fig. S4A), consistent with numerous previous studies showing that only minimal silencing can be achieved by the addition of exogenous RNA to *Aspergillus* species (Khatri and Rajam 2007; Abdel-Hadi et al. 2011; Kallada et al. 2013; Mousavi et al. 2015; Nami et al. 2017). To test whether effective silencing of *pabA* could lead to growth repression, we again used a heterologously expressed dsRNA targeting the *pabA* gene from the chromosome of *A. fumigatus* as described above (Fig. 2D). We grew both the *pabA*-IRT-transformed strain and the wild-type strain with and without doxycycline induction on AMM agar plates and

liquid cultures. In the absence of doxycycline, the *pabA*-IRT led to minor but significant growth inhibition, likely due to leaky expression (Fig. 2D,E). Upon induction with doxycycline, growth of the *pabA*-IRT-transformed strain was reduced on both AMM agar plates and liquid cultures when compared with the wild-type strain (Fig. 2D,E). Growth was more clearly inhibited in liquid culture than on solid media, likely due to more homogenous delivery of the doxycycline inducer. As a control, we also included the *pksP*-IRT strain in our liquid culture experiments and found that this construct also had a growth-inhibiting effect, albeit significantly less than the *pabA*-IRT strain (Fig. 2E). As a control, we confirmed by qRT-PCR that the *pabA*-IRT strain specifically targets *pabA*, resulting in a modest 43% decrease in message (Fig. 2F). Collectively, these results indicated that *A. fumigatus* RNAi can be leveraged to silence both a nonessential gene (*pksP*) and a conditionally essential gene (*pabA*) in a *DclB*- and *PpdA*-dependent manner.

mRNA-seq analysis suggests distinct contributions to conidial gene expression

The inverted-repeat transgene experiments suggested that only select components of the *A. fumigatus* RNAi machinery are required for transgene silencing. We were therefore interested to elucidate any additional roles of RNAi in *A. fumigatus* biology. To this end, we performed mRNA-seq on poly(A)-enriched RNA isolated from conidia and mycelium grown in liquid culture for 24 or 48 h (Supplemental Fig. S5), as RNAi is known to serve a role in asexual reproduction in other filamentous ascomycetes (Gaffar et al. 2019). We determined the differentially expressed genes (Supplemental Data Set S2) and found that each of the RNAi knockouts showed significant differences compared with wild type in the conidia samples but very few changes in expression in mycelium using this stringent cutoff (Fig. 3A). These findings were consistent with the lack of phenotypes observed in stress and growth assays. We performed Gene Ontology (GO)-slim analysis to improve our understanding of these changes in conidia and were surprised to see a clear link between the *dcl* and *rrp* genes with ribosome biogenesis and the nucleolus, where 85 genes associated with ribosome biogenesis were significantly down-regulated in the $\Delta dclA/B$ strain and 24 genes were down-regulated in the $\Delta rrpA/B$ strain compared with wild type (Fig. 3B; Supplemental Fig. S6; Supplemental Data Set S3). The argonaute proteins appeared to contribute to the conidial transcriptome differently, as we observed decreased expression of genes associated with transmembrane transport and the regulation of transcription, alongside up-regulation of genes associated with mRNA and tRNA metabolic processes.

To expand on our mRNA-seq results, we performed LC-MS/MS-based proteomics on conidia and 48-h-old mycelium for each of the double knockouts and the wild type.

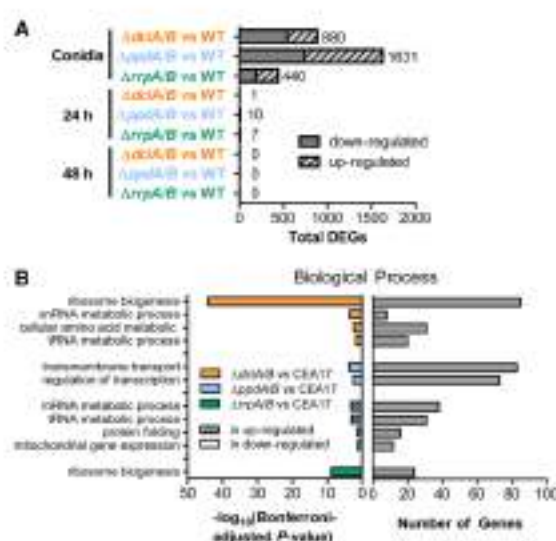


FIGURE 3. Deletion of RNAi components results in altered conidial transcriptome. (A) Genes that were identified as significantly differentially expressed in mRNA-seq. Differences were considered significant if $q \leq 0.05$ and $|\log_2(MRN)| \leq 1$ by all four tools. The number next to the bar indicates the total number of differentially expressed genes (DEGs). (B) GO-Slim analysis (using FungiDB.org) for Biological Process generated using differentially expressed genes from conidia as input gene sets. Up- and down-regulated differentially expressed genes (left panel) as well as the number of differentially expressed genes (right panel) are indicated.

Among the most abundant proteins detected in conidia were several histones (HhfA, Htb1), a Cu/Zn superoxide dismutase (SodC), and elongation factor 1- α (AFUA_1G06390) (Supplemental Data Set S4). At 48 h, we observed a high abundance of alcohol dehydrogenase (AFUA_7G01010), the same Cu/Zn superoxide dismutase (SodC) as in conidia, and several ubiquitin isoforms (AFUA_1G04040 and AFUA_3G11260); results that are consistent with the metabolic changes associated with carbon and iron starvation (Supplemental Data Set S5; Oberegger et al. 2000). The data set from conidia identified roughly 5810 proteins, likely making this one of the most thorough descriptions of the proteome of *A. fumigatus* conidia to date. For the 48-h mycelium, we identified 2380 proteins. Interestingly, from the top 200 most-abundant proteins from each data set, we only observed 26 in common, reiterating the massive changes that are undergone during the *A. fumigatus* asexual life cycle.

We next determined the differentially abundant proteins between knockout and wild type using a two-tailed Welch's t-test with Benjamini-Hochberg correction to control for multiple comparisons, which is known to be a stringent statistical test for proteomics analysis (Pascovici et al. 2016). The use of the Benjamini-Hochberg correction resulted

in only three significant changes (adjusted *P*-value of ≤ 0.05), all in the conidia condition (Supplemental Data Set S4). Two proteins had lower abundance in the $\Delta dclA/B$ strain (AFUA_4G10100, AFUA_4G06250) and one protein (AFUA_7G08590) showed slightly increased abundance in the $\Delta ppdA/B$ strain compared with wild type (Supplemental Data Set S4). The changes in AFUA_4G06250, a predicted ribosome biogenesis protein orthologous to Nop4, and AFUA_7G08590, a protein with a predicted BTB/POZ domain, were consistent with the change observed in our mRNA-seq analysis.

In many cases, controlling for multiple comparisons eliminates true hits from proteomics analyses, particularly in the context of low power associated with the standard three biological replicates (Pascovici et al. 2016). To more comprehensively analyze our data sets, we also reformed the statistical analysis using only a two-tailed Welch's *t*-test, which revealed more widespread alterations to protein abundance in both conidia and 48-h-old mycelium as expected (Fig. 4; Supplemental Fig. S7; Supplemental Data Sets S4, S5). In conidia, we observed 164 proteins with at least twofold increase in abundance over wild-type in the $\Delta dclA/B$ strain and 113 proteins that significantly decreased in abundance. The $\Delta ppdA/B$ and $\Delta rrpA/B$ strains showed fewer significantly altered proteins, with 49 proteins increased and 49 decreased in abundance compared with wild type for the $\Delta ppdA/B$ strain and 37 proteins increased and 70 decreased for $\Delta rrpA/B$ (Supplemental Data Set S4). Interestingly, we observed very little agreement between the mRNA-seq and proteomics analysis, implying that the conidial transcriptome differs substantially from the conidial proteome. In 48-h-old mycelium, we again observed significantly different protein abundances using a two-tailed Welch's *t*-test. In particular, we found 35 proteins to be significantly more abundant (greater than twofold change) in the $\Delta dclA/B$ strain than wild type and 19 with decreased abundance, 61 proteins increased and 34 decreased in the $\Delta ppdA/B$ strain compared with wild type, and 86 proteins increased and 36 decreased in abundance in $\Delta rrpA/B$ (Supplemental Data Set S5). Despite numerous cases of differential abundance, no clear pattern of regulation emerged from this analysis, potentially consistent with the idea that RNAi is pleiotropic in its contribution to gene regulation. Collectively, these results suggest that the RNAi machinery in *A. fumigatus* influences fungal gene expression in multiple ways, with a particularly strong effect on genes associated with ribosome biogenesis in conidia.

RNAi genes exhibit growth stage-specific expression

The significant changes observed in conidia and mycelium led us to assess the expression of the different core and accessory RNAi components more closely in conidia versus mycelium for the wild-type and double-knockout strains

using the data collected in the mRNA-seq experiment. The results clearly showed growth stage-specific expression for much of the RNAi machinery (Fig. 5). The *dclA* gene was more highly expressed in conidia than mycelium, whereas *dclB* increased with the age of the mycelium in a pattern similar to the mitochondrial ribosomal protein L3 (MRPL3) gene—known to encode an additional RNase III activity. The *mt1* gene, which encodes a nuclear RNase III enzyme that could potentially compensate for the *dcl*-like RNase III activity in some instances, showed higher normalized expression in conidia than mycelium like *dclA*. The *ppdB* gene was expressed in all stages, whereas *ppdA* was essentially absent in conidia, consistent with a predicted role in post-transcriptional gene silencing that is likely unnecessary in relatively transcriptionally dormant conidia. Instead, we observed high expression of the exonuclease *QIP* in conidia, and not in mycelium, possibly suggesting *QIP* contributes to asexual as well as sexual reproduction as in other systems (Xiao et al. 2010).

Intriguingly, expression of many of the RNAi genes was significantly altered in the knockout strains as determined by two-way ANOVA with a Bonferroni posttest, with a few particularly notable examples. First, in the absence of the *dcl* or *ppd* genes, *mt1* levels were significantly increased in conidia (Fig. 5), perhaps suggesting some level of compensation by this alternative RNase III enzyme. The expression of *QIP* and *rrpA* increased in the *dcl* double knockout in line with a known interaction between *QIP* and *rrpA*, whereas expression of both decreased in the *ppd* double knockout. Finally, MRPL3 was greatly increased in conidia of the $\Delta ppdA/B$ strain, potentially connecting the argonaute proteins to regulation of translation in mitochondria and consistent with the observation of an increase in GO-slim genes associated with mitochondrial gene expression.

RNAi contributes to fungal fitness over time

Next, to determine whether the differential expression and protein abundance observed in the mRNA-seq and proteomics experiments might influence the long-term fitness of *A. fumigatus* RNAi knockouts, we performed serial passaging of the $\Delta dclA/B$, $\Delta ppdA/B$, and $\Delta rrpA/B$ double knockouts and a wild-type control strain for six generations on AMM agar plates as separate lineages. Spores were spotted in the center of the plate and allowed to grow at 37°C for 7 d with minimal disturbance to provide numerous rounds of starvation and sporulation to recapitulate the asexual lifecycle of *A. fumigatus*. One sublineage of each of the RNAi knockout strains $\Delta dclA/B$ and $\Delta ppdA/B$ displayed subtle phenotypes that included white sectoring and altered production of conidia (Supplemental Fig. S8). Previous reports have indicated that these white colonies typically result from mutation of the *pkcP* gene, which encodes a polyketide synthase enzyme essential for the initial step in the biosynthesis of DHN melanin (Langfelder et al.

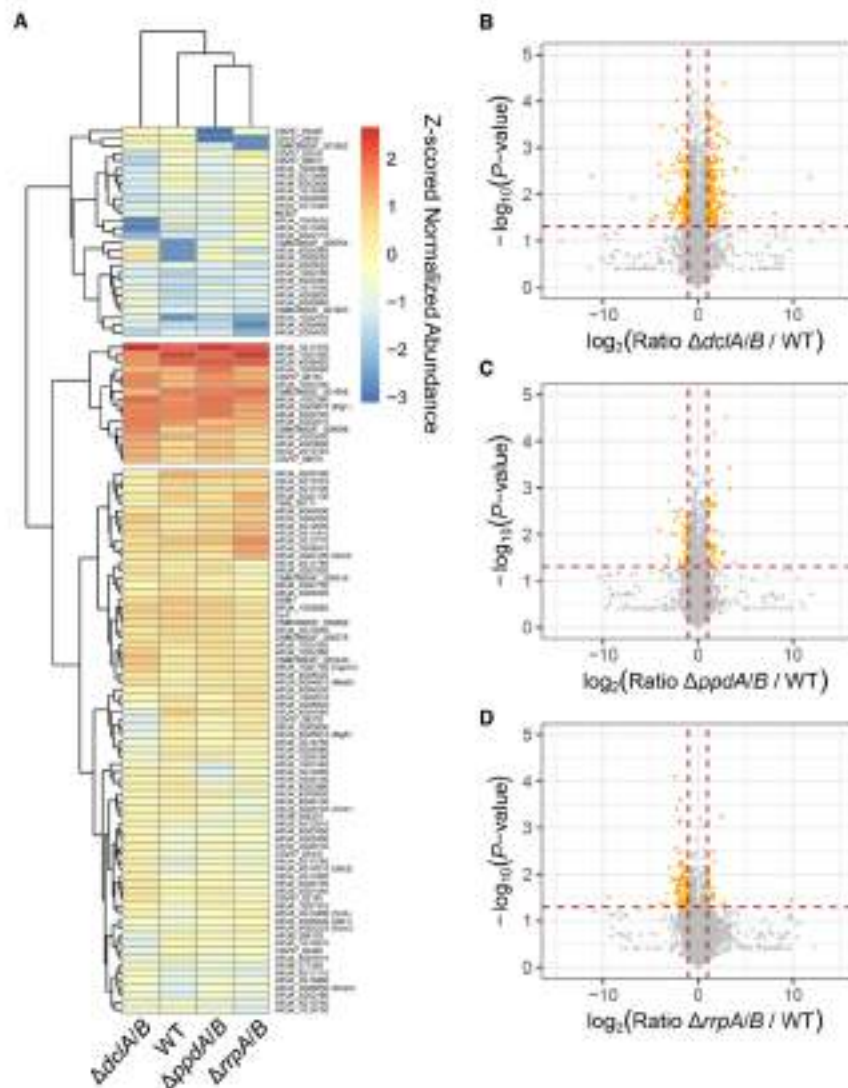


FIGURE 4. Conidia exhibit an altered proteome different from the transcriptome. (A) Clustered heat map depicting Z-scored normalized average abundance data from conidia of double knockouts and wild type for cases where proteins exhibited at least a $\log_2(\text{FC}) = 2$ and a P-value of ≤ 0.05 in one of the conditions. Normalized abundances were determined using area under the curve from label-free quantification. (B–D) Volcano plots showing expression ratio for double knockout compared with wild type for $\Delta dclA/B$ (B), $\Delta ppdA/B$ (C), and $\Delta rrpA/B$ (D). The y-axis shows the $-\log_{10}$ transformed P-value, where higher values are more significant. Orange circles show proteins that are at least $\log_2(\text{FC}) = 1$ and with a P-value of ≤ 0.05 .

1998; Gibbons et al. 2022). Conidia collected from the white sectors in our experiment remained white upon transfer to new plates and were confirmed by PCR to hold the appropriate resistance cassette of the knockout. We then counted spore concentrations after a final passaging and observed a decrease in spore abundance from several lineages of the $\Delta dclA/B$ and $\Delta ppdA/B$ double knockouts (Fig. 6), suggesting that the $\Delta dclA/B$

and $\Delta ppdA/B$ knockouts can become less reproductive over time.

***A. fumigatus* produces a limited pool of high-abundance endogenous small RNAs**

Finally, we assessed the endogenous pool of small RNAs generated in *A. fumigatus* CEA10 wild type to uncover

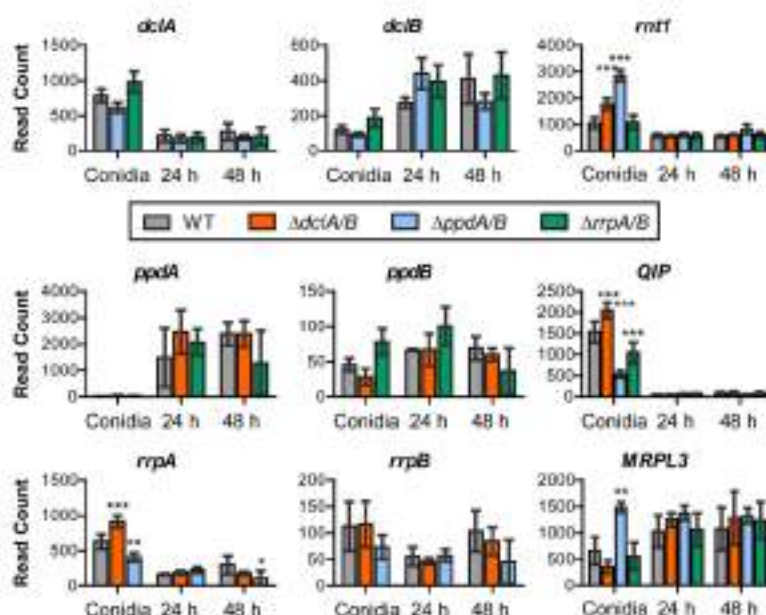


FIGURE 5. The RNAi genes are differentially expressed in conidia and mycelium. Median ratio normalization (MRN) expression indicated as read count of individual genes in conidia, 24-h and 48-h mycelium using the mRNA-seq experiment described in Figure 3. Significance was determined by two-way ANOVA with Bonferroni-corrected post hoc t-test, resulting in some minor differences from the data analyzed in Figure 3. (*) $P \leq 0.05$, (**) $P \leq 0.01$, (***) $P \leq 0.001$.

potential small RNAs with complementarity to the observed differentially expressed genes and proteins. We performed small RNA sequencing using RNA again isolated from resting conidia and mycelium grown for 24 or 48 h in liquid AMM culture. The RNA was processed with RNA 5' pyrophosphohydrolase (RppH) to remove any 5' triphosphates before library preparation to facilitate small RNA discovery. The resulting library concentrations were quite low for resting conidia and mycelium grown for 24 h, so we did not normalize the library input before sequencing, and instead sequenced all the available libraries to maximize sRNA discovery. Not surprisingly, we found the largest number of reads from the 48-h mycelium samples where we also observed the largest amount of small RNA in preparative analyses (Supplemental Table S1). We predicted small RNAs using Manatee, a web-based algorithm for identification and quantification of sRNAs (Handzlik et al. 2020). Three-hundred-twenty-seven sRNAs were initially predicted by the program for the three time points (conidia, 24-h mycelium, and 48-h mycelium) (Supplemental Data Set S6). Forty-two small RNA regions were identified in the conidia data set, 77 in the 24-h data, and 208 from the 48-h data. Removal of duplicated regions found in multiple samples resulted in 297 unique identifications, with 14 found in two conditions and eight found at all three time points (Supplemental Data Set S6). After applying a cutoff to remove transcripts with fewer than 200 total reads,

we were left with a total of 23 high-abundance sRNA candidates (Table 1). These small RNA loci range from 19 to 25 nt in length, although the exact 5' and 3' nucleotides were not experimentally defined. The limited number of high-abundance candidates is consistent with previous work on *A. fumigatus* endogenous small RNAs (Ozkan et al. 2017).

The most abundant small RNAs in conidia were antisense to AFUB_098420 (AFUA_4G04590), an uncharacterized gene predicted to be associated with ubiquinone biosynthesis, and AFUB_068210 (AFUA_4G11180), a predicted AAA family ATPase. As expected, we observed a higher abundance of small RNAs at 48 h, but these small RNAs also had limited complementarity to proteins with altered abundance in 48-h-old mycelium or conidia samples. Certainly, a more extensive survey of the small RNAs produced by *A. fumigatus* using the different RNAi knockout strains generated here will be re-

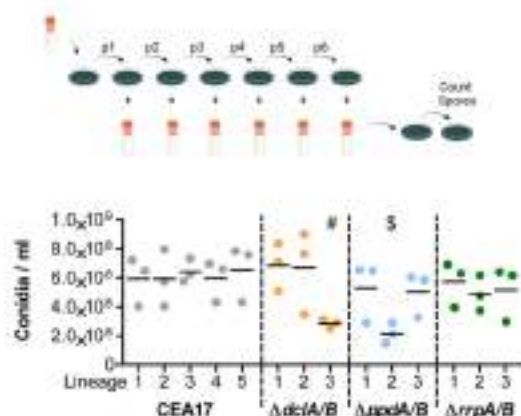


FIGURE 6. Deletion of RNAi components affects long-term fitness. (Top) Wild-type and double-knockout strains were grown as more than three lineages and passaged for six generations on AMM agar plates at 37°C for 7 d. Model created with Biorender.com. The evolved strains were then grown for an additional week and the spores were counted using a CASY counter as three biological replicates. The plot shows the values from each biological replicate of each lineage and the mean value of the biological replicates (black line). Significance was determined by one-way ANOVA with Bonferroni-corrected post hoc t-test comparing all columns. (#) $P \leq 0.05$, (\$) $P \leq 0.01$ for all five comparisons with wild-type lineages.

TABLE 1. *A. fumigatus* small RNAs identified by Manatee with over 200 reads

Name	Cluster length	Total count	Detected in	Antisense to	Genomic sequence
AAU_sRNA_1	21	8361	Conidia, 48	AFUB_098420	GTAAGATCGGAAGAGCACTA
AAU_sRNA_2	19	2657	24	AFUB_037210	GGTCCCCGAGCGGGGAG
AAU_sRNA_3	19	2096	48	AFUB_089310	CGCCGTGGCGGGCAGAT
AAU_sRNA_4	19	1344	24		TCAGATCAACAGATAG
AAU_sRNA_5	19	1015	Conidia, 24, 48	AFUB_068210	TCGATGTCGGCTCTCAA
AAU_sRNA_6	19	841	48		TGCTCCCGCCGACGTCTG
AAU_sRNA_7	20	593	24		TCCACCGCAGGCTCCACGT
AAU_sRNA_8	19	571	Conidia, 24		ATCTGGTTGAATCTGCA
AAU_sRNA_9	24	517	Conidia, 24, 48	AFUB_030860	ATGACGTGCGCGGGTCTGAGC
AAU_sRNA_10	20	506	48		GAATTTGAAAGCTGGCCAA
AAU_sRNA_11	22	466	Conidia, 24, 48	AFUB_021700	CTGGGTCTGGCTGGCCGGTTC
AAU_sRNA_12	19	446	Conidia, 24, 48		CCCCGTGACAACTGGA
AAU_sRNA_13	23	445	Conidia, 24, 48		CTGACATGGTTAGTCGATCAC
AAU_sRNA_14	19	404	48		GGCGTCCCGAGGACAGCC
AAU_sRNA_15	23	390	48	AFUB_057330	GCGGAACCCCGCCGACGCCAC
AAU_sRNA_16	23	376	48	AFUB_044170	ATCAGGGATCGGGCGGTTTCA
AAU_sRNA_17	21	313	48		TGAACGACCCCGACTATCTT
AAU_sRNA_18	24	235	24, 48	AFUB_068490	TAGATCCCGCGGGCGTCCGAT
AAU_sRNA_19	19	220	48	AFUB_076450	CTCCGGGGCGGTTCCGAC
AAU_sRNA_20	23	218	48		CCAAACACCAACATCCAAAAAC
AAU_sRNA_21	20	214	48		ACTCCGCTACCCACATTA
AAU_sRNA_22	19	209	48		TTTGGCGCGCGGTCGACG
AAU_sRNA_23	21	206	48		GCGCAGCCTGCCTGGACTGG

quired to ultimately tease apart the complex regulation across the asexual lifecycle.

DISCUSSION

The *A. fumigatus* RNAi pathway is functional in post-transcriptional gene silencing and involved in defense against mycoviruses (Mouyna et al. 2004; Ozkan et al. 2017). While work done in *A. nidulans* laid the groundwork for understanding further RNAi in *A. fumigatus*, distinct differences between the organisms, including truncations of the *A. nidulans* proteins (Hammond and Keller 2005; Hammond et al. 2008a,b), suggested that additional experimentation could be beneficial. We were able to show that the truncations of some core RNAi components in *A. nidulans* are not commonly repeated in *A. fumigatus*, as 300 sequenced genomes of *A. fumigatus* showed only rare truncation events (e.g., frameshifts) and relatively minor genetic variation. This relatively high level of conservation coupled with the observation that modest silencing can result in growth inhibition (Fig. 2F) suggests that we may even be able to exploit this machinery for therapeutic purposes if we can better facilitate RNA delivery in the future, as is being pursued with agricultural pests (Cai et al. 2019; Bruch et al. 2021).

Using a previously established inverted-repeat transgene approach (Mouyna et al. 2004), we created an inducible dsRNA against the *pksP* gene of *A. fumigatus*, which is required for production of DHN-melanin and the gray-green color of conidia. With this approach, we were able to determine that DclB and PpdA are required for inverted-repeat transgene silencing in *A. fumigatus*, whereas the other core components are dispensable for this form of RNA silencing. The finding that DclB and PpdA are essential for silencing matches previous work in *N. crassa*, where DCL-2 and QDE-2, orthologs of DclB and PpdA, were shown to be important for quelling (Dang et al. 2011). The result that *A. fumigatus* *ppdA* is necessary for silencing also matches results from *A. nidulans* with ortholog *RsdA* (Hammond and Keller 2005). Unlike *N. crassa*, DclA was not able to compensate in a Δ dclB deletion strain of *A. fumigatus* in our *pksP*-inverted-repeat transgene silencing experiments, suggesting discrete functions for DclA, perhaps during the poorly understood *A. fumigatus* sexual cycle. QDE-1 is important for silencing in *N. crassa*, but lost in *A. nidulans* (Hammond and Keller 2005), whereas the ortholog encoded by *mpA* is conserved, but dispensable for IRT silencing, in *A. fumigatus* (Fig. 2B,C). This result appears to be consistent with the described loss of the activity in *A. nidulans*.

We showed that a 500-bp double-stranded RNA with complementarity to the *pabA* mRNA is capable of

silencing *pabA* expression to limit fungal growth when expressed from the chromosome. Surprisingly, we also observed that the *pksP-IRT* led to growth inhibition when heterologously expressed. We suspect that the *pksP-IRT* sequence either results in production of small RNAs that have off-target effects or that the overproduction of a double-stranded RNA is mistaken for a mycoviral infection, resulting in altered fungal behavior. It is possible that overexpression of double-stranded RNA, which may also mimic a mycoviral infection, sequesters RNase III proteins from their normal functions in ribosome biogenesis and limits fungal growth. It is worth noting that in the literature, heterologous overexpression of a much less structured human mRNA, RBP7, in *A. fumigatus* did not detrimentally alter growth (Shopova et al. 2020), suggesting that the observed growth inhibition is not due solely to resource depletion, e.g., nucleotide pool depletion. A closer inspection of the small RNAs produced from the inverted-repeat transgene constructs will be required to fully understand this observation and detail the interplay between the different functionalities of RNAi in *A. fumigatus*.

The RNAi pathway in fungi has been linked to diverse regulation, including in sexual differentiation, development, stress response, and growth (Ruiz-Vazquez et al. 2015; Nicolas and Garre 2016; Torres-Martinez and Ruiz-Vazquez 2017). Despite all these important functions in other organisms, *A. fumigatus* appears to be relatively robust in tolerating deletion of the core RNAi machinery. Minimal functional differences were observed in a wide array of stress assays, although we did observe a decrease in fitness in our serial passaging experiment for some lineages of the $\Delta dclA/B$ and $\Delta pdsA/B$ strains, implying that loss of RNAi can have consequences. RNAi pathways also frequently play an important role in maintaining genome integrity, as has been described in distantly related fungi *Cryptococcus neoformans* (Janbon et al. 2010; Wang et al. 2012), *Mucor circineoides* (Nicolas et al. 2010; Calo et al. 2017), *Fusarium graminearum* (Chen et al. 2015), *Magnaporthe oryzae* (Kadotani et al. 2003; Murata et al. 2007), *Schizosaccharomyces pombe* (Roche et al. 2016), and *Trichoderma atroviride* (Carreras-Villasenor et al. 2013). Investigation of fungal RNAi has proven a function in preserving the integrity of the host genome against invasive nucleic acids including transgenes, transposons, and invading viruses during both the vegetative and sexual cycles (Nuss 2011; Torres-Martinez and Ruiz-Vazquez 2017). Future studies into the RNAi components that contribute to genome protection are now possible using the tools we created here.

Additional work will be needed to determine the mechanism of regulation imparted by the dicer-like proteins in conidia, but some clear differences from previous reports should be noted. For example, *A. fumigatus* has a conserved, seemingly intact Rnt1 enzyme and the action of the dicer-like proteins seems unlikely to match that

observed in *C. albicans*, where the only active RNase III enzyme performs the function of both dicer and Rnt1 in processing of rRNA (Bernstein et al. 2012). RNAi has also been linked to maintenance of quiescence in *S. pombe*, but again differences exist between the two organisms. *A. fumigatus* RNAi knockouts showed no defects in growth in our assays, and no change was observed when the RNAi knockouts were stressed with TBZ. The effects in conidia could also be due to regulation of heterochromatin formation as has been shown in other organisms (Martenssen and Moazed 2015), but at least preliminary studies in *A. nidulans* suggest that RNAi is uncoupled from regulation of heterochromatin in *Aspergillus* (Hammond and Keller 2005).

In the proteomics analyses, we observed only minor differences using a strict statistical analysis, whereas a more relaxed analysis led to disparate changes. As RNAi is known to play pleiotropic roles in gene regulation, we can envisage that *A. fumigatus* may use RNAi to control gene expression in a variety of different ways. This pleiotropic regulation is likely evident in the limited pool of endogenous small RNAs in *A. fumigatus*. The majority of reads from small RNA-seq were derived from ribosomal RNAs and tRNAs, as was observed previously (Jochl et al. 2009), with only 23 high-abundance small RNA loci. These matched poorly to observed changes in protein abundance in the knockouts, potentially suggesting alternative biogenesis pathways or regulation in the conidiophore structure. A more in-depth analysis using the inverted-repeat transgene and deletion strains produced here will be required to understand the pool of small RNAs in *A. fumigatus*.

In conclusion, we have shown that *A. fumigatus* RNAi appears to play an active role in post-transcriptional gene silencing that runs parallel to a previously unappreciated housekeeping function in regulation of conidial ribosomal biogenesis genes. The strains created and data sets collected here will likely be useful in further delineating the functions of this complex, evolutionarily diverse RNA regulatory pathway.

MATERIALS AND METHODS

Domain mapping and genomic analysis

Domain structures were created using UniProt annotations (<https://www.uniprot.org>; Q4WSH7, Q4WY64, Q4WEW6, Q4WVR4, Q4WUR8, Q4WC92, Q4WXX3, Q4WA22, and Q4WVE3) and created with the drawProteins (Brennan 2018) package using R version 4.1.0 (2021-05-18).

The genomic data analyzed in this study were generated in Barber et al. (2021) and are publicly available as raw reads and assembled genomes under PRJNA697844 and PRJNA595552. Isolates originated from environmental and clinical sources and displayed a global distribution. Presence/absence variation in the genomes was quantified using OrthoFinder using peptide

sequences from 300 genomes. Given that no cases of gene loss or amplification were detected among the genome assemblies, analysis of sequence variation was performed by aligning raw sequencing reads to the *Af293* reference genome (FungiDB release 39). PCR duplicates were marked using MarkDuplicate from Picard v.2.18.25. Variant calling to detect SNVs and short indels was performed using HaplotypeCaller and high-quality variants identified using the following filtering criteria: SNP: $QD < 2.0$ | $MQ < 40.0$ | $FS > 60.0$ | $MQRankSum < -12.5$ | $ReadPosRankSum < -8.0$; indel: $QD < 2.0$ | $FS > 200.0$ | $ReadPosRankSum < -20.0$. Variant function was predicted using SnpEff v.4.3.

Strains and culture conditions

Aspergillus fumigatus strains were grown on *Aspergillus* minimal medium (AMM) agar plates (70 mM NaNO_3 , 11.2 mM KH_2PO_4 , 7 mM KCl, 2 mM MgSO_4 , 1% [w/v] glucose and 1 $\mu\text{L/mL}$ trace element solution at pH 6.5). The trace element solution was composed of 1 g of $\text{FeSO}_4 \cdot 7 \text{H}_2\text{O}$, 8.8 g of $\text{ZnSO}_4 \cdot 7 \text{H}_2\text{O}$, 0.4 g of $\text{CuSO}_4 \cdot 5 \text{H}_2\text{O}$, 0.15 g of $\text{MnSO}_4 \cdot \text{H}_2\text{O}$, 0.1 g of $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10 \text{H}_2\text{O}$, 0.05 g of $(\text{NH}_4)_6\text{Mo}_7\text{O}_{24} \cdot 4 \text{H}_2\text{O}$, and ultrafiltered water to 1000 mL. Spores were harvested after 5 d (or 7 d for the evolution experiment) with 10 mL of sterile distilled water using a T-shaped inoculation spreader, and spore suspensions were filtered through a 30- μm cell strainer (MACS; Miltenyi Biotec GmbH) to exclude mycelium. *A. fumigatus* transformants were screened on AMM agar plates supplemented with 150 $\mu\text{g/mL}$ hygromycin, 0.1 $\mu\text{g/mL}$ pyrimethamine, or 160 $\mu\text{g/mL}$ phleomycin for selection. For in vitro transcription and confocal laser scanning microscopy experiments, *A. fumigatus* AfS150 was grown on malt agar (Sigma-Aldrich) plates supplemented with agar to a final concentration of 2% (w/v) for 5 d. *Escherichia coli* strains were grown at 37°C on LB broth or agar supplemented with 100 $\mu\text{g/mL}$ carbenicillin when necessary.

Genetic manipulation strategies

Knockout constructs were created using several strategies; namely, Fusion PCR or the Gibson cloning method (Szewczyk and Krappmann 2010). To make constructs by Fusion PCR, an antibiotic selection gene and ~1 kb of DNA flanking regions (5' and 3') of the target gene were amplified. PCR fragments were purified using the GeneJET PCR purification kit (Thermo Fisher Scientific) and concentrations were measured by NanoDrop 2000 spectrophotometer. Next, equimolar amounts of the three fragments, two flanking regions, and selection gene, were combined using Fusion PCR with the 2 $\times$ Phusion Flash High Fidelity Master Mix and 10 pmol of each forward and reverse primer for the 5' flank and 3' flank, respectively. Fusion PCR products were purified and validated by agarose gel electrophoresis. For Gibson assembly, a selection gene and ~1 kb flanking regions of the target gene were amplified by PCR using primers that create overhangs (Supplemental Table S2). In addition, pUC18 vector backbone was also amplified from plasmid "pUCHph Volker." DNA fragments were assembled and the resulting plasmid was used to transform chemically competent *E. coli* following the NEBuilder HiFi DNA Assembly protocol (New England Biolabs). Knockouts of *A. fumigatus* were generated by homologous recombination following the transformation of protoplasts (Weidner et al. 1998).

Growth and viability assays

The growth capacity of different *A. fumigatus* strains in AMM media was investigated by determining the dry weight from shaking cultures. 1×10^7 spores were inoculated in 50 mL of AMM and incubated at 37°C and 200 rpm for 24 h. Mycelia were harvested using Miracloth (Merck KGaA), washed with deionized water, and dried at 60°C for 3 d. The susceptibility of different *A. fumigatus* strains to stress conditions was investigated by spotting 5 μL of spore suspension with concentrations ranging from 10^2 to $10^8/\text{mL}$ on AMM agar plates supplemented with different stress-inducing agents for 2 d. All plates were incubated at 37°C.

Inverted-repeat transgene silencing assays

To silence *A. fumigatus* *pksP*, sense and antisense *pksP* PCR products (500 bp each) were amplified with primer pairs $\alpha\text{ARK0031}$ & $\alpha\text{ARK0032}$ and $\alpha\text{ARK0033}$ & $\alpha\text{ARK0034}$, respectively. These PCR products were then cloned into the two multicloning sites in the p-Silent1 plasmid vector, which already has a hygromycin resistant cassette. The *trpC* promoter upstream of the inverted-repeat region originally present from the p-Silent1 vector was replaced by a tetracycline-inducible promoter (TetON). The resulting plasmid was then digested with restriction enzymes *SacI* and *FspI* to excise the IRT-expressing region and hygromycin (hph) resistant cassette, which was then cloned into a pUC18 vector backbone flanked by 5' and 3' regions (~1 kb) of a nonfunctional *pyrG* locus in the CEA17 $\Delta\text{akuB}^{\text{KUSO}}$ (A1163, wild type) strain to generate pSilent-pksP1. Because some of the RNAi knockout strains already have a hygromycin resistant cassette, we constructed pSilent-pksP2 by replacing the selection marker with a phleomycin (ble) resistant gene. Both plasmids were linearized by *BstI* restriction enzyme before transformation of strains (Supplemental Table S3). To generate a para-aminobenzoic acid synthetase (*pabA*) silencing construct, *pabA*-IRT sequence plus a 3' *trpC* terminator was synthesized. This synthesized DNA was then cloned using Gibson assembly into a PCR product amplified from pSilent-pksP2 with primer pair $\alpha\text{ARK0096}$ & $\alpha\text{ARK0038}$. The latter PCR product consist of the pUC18 backbone, *pyrG* flanks, phleomycin resistant cassette, and TetON promoter to drive the *pabA*-IRT expression. The plasmid was linearized by *PstI* restriction enzyme before transformation of CEA17 $\Delta\text{akuB}^{\text{KUSO}}$ wild-type strain (da Silva Ferreira et al. 2006).

Southern blot

Target gene deletion was confirmed by Southern blot, as described previously (Southern 2006). Digested genomic DNA samples were separated in a 1% (w/v) agarose gel. Separated DNA fragments on gel were transferred onto a nylon membrane (pore size: 0.45 μm , Carl Roth GmbH). The DNA was crosslinked to the membrane by UV irradiation followed by overnight hybridization with a DIG-11-dUTP-labeled DNA probe at 65°C. The membrane was washed twice for 5 min at 65°C (1 $\times$ SSC, 0.1 SDS), followed by two washes with wash buffer II (0.5 $\times$ SSC, 0.1 SDS) for 20 min at 65°C. The membrane was next rinsed with maleic acid wash buffer, incubated with blocking solution for 30 min to prevent unspecific antibody binding, followed by another incubation with 20 mL of blocking solution plus 0.5 μL of

anti-digoxigenin-AP, Fab fragments (Roche) for 30 min. The unbound antibodies were removed by washing twice with 50 mL of maleic acid wash buffer for 15 min. Finally, the membrane was equilibrated with detection buffer, developed with several drops of the chemiluminescent substrate CDP-Star, ready-to-use (Roche) and detected with a chemiluminescence imaging Fusion FX SPECTRA instrument (Vibber Loumat).

In vitro transcription and confocal laser scanning microscopy

For in vitro dsRNA transcription, genomic DNA was isolated from mycelium of *A. fumigatus* CEA17Δ*akuB*^{K180} grown overnight in liquid cultures and used as a template for the amplification of 500 bp of the *pabA* sequence by PCR. The PCR reaction mix consisted of 200–300 ng of template DNA, 10 pmol of each primer, and 25 µL of 2× Phusion Flash high-fidelity PCR master mix (Thermo Fisher Scientific). Both the forward and the reverse primer were designed to contain a T7 promoter at their 5' end for subsequent dsRNA transcription (Supplemental Table S2; T7_ *pabA*_500_1_for/T7_ *pabA*_A500_1_rev). PCR products were loaded onto a 1% agarose gel to check for the correct size and purified with the GeneJET PCR purification kit (Thermo Fisher Scientific). The products were then used as a template for in vitro dsRNA transcription using the MEGAscript RNAi kit (Invitrogen) according to the manufacturer's protocol with slight modifications. Briefly, the transcription reaction was assembled as specified with the addition of 0.5 µL fluorescein-labeled-UTP solution (Sigma-Aldrich). The reaction was incubated for 4 h at 37°C and the RNA products were annealed by heating the reaction for 5 min to 75°C and subsequently left to cool at room temperature. Digestion of DNA and single-stranded RNA and the purification of dsRNA was performed as described in the protocol with the exception that the purified dsRNA was eluted in two steps using a total of 100 µL of 95°C ultrapure H₂O. Quantity and quality of the dsRNA was assessed by fluorometric analysis using a Qubit 4 Fluorometer (Thermo Fisher Scientific). dsRNA was stored in the dark at –80°C until usage.

For fluorescence microscopy, 1×10^4 freshly collected *A. fumigatus* AF5150 conidia [Lothar et al. 2014] were seeded into each well of an eight-well chamber slide (µ-Slide 8 Well, ibidi) in a total of 100 µL of RPMI medium (Sigma-Aldrich), treated with 2 µg of labeled dsRNA, and incubated for 16 h at 37°C and 5% CO₂ (v/v). The following day, old media was removed from the samples and fresh media containing calcofluor white was added to the fungus. After 30 min of incubation at room temperature in the dark, hyphae were carefully washed with sterile phosphate-buffered saline (PBS). An amount of 100 µL of RPMI was added to each chamber and samples were observed by confocal laser scanning microscopy with a Zeiss LSM 780 microscope (Carl Zeiss). Images were processed using the ZEN blue edition software (version 3.4, Carl Zeiss).

RNA isolation

Fungal mycelium was collected from liquid culture using Miracloth (Millipore) and disrupted in liquid nitrogen using a pre-cooled mortar and pestle. Roughly 0.5 g of homogenized mycelium was transferred into a 2-mL Eppendorf tube. An amount of

800 µL TRIzol was added to the ground mycelia and vortexed vigorously. Tubes were frozen briefly for 5 sec in liquid nitrogen and allowed to thaw on ice. An amount of 160 µL of chloroform was added to the thawed mixture, vortexed, and centrifuged for 5 min at 4°C at full speed. The aqueous upper phase was transferred to a fresh 2-mL tube without disturbing the interphase. RNA extraction from aqueous phase was done with one volume of phenol/chloroform/isoamyl alcohol (25:24:1, v/v). Brief vortexing preceded centrifugation for 5 min at 4°C. Extraction was repeated until no more interphase was observable, and this was followed by another extraction with 400 µL of chloroform. RNA was precipitated using 400 µL of isopropanol for 20 min, followed by pelleting by centrifugation for 20 min at 4°C. The pellet was washed with 700 µL of 70% ethanol and air dried for 5 min at 37°C prior to resuspension in RNase-free water. The RNA isolation was followed by a DNase treatment using 2 units of TURBO DNase (Thermo Fisher) per 10 µg of RNA for 30 min at 37°C in 100 µL of total volume. Total RNA was collected using the RNA Clean and Concentrator-25 kit (Zymo Research) according to the manufacturer's instructions.

Quantitative RT-PCR (qRT-PCR)

RNA was isolated from cultures of CEA17Δ*akuB*^{K180}, *pabA*-IRT, and *pksP*-IRT with and without doxycycline induction after 24 h. Complementary DNA (cDNA) was synthesized from RNA according to the iScript cDNA synthesis kit protocol (Bio-Rad Laboratories, Inc.), and the concentration was determined using a Qubit Flex Fluorometer (Thermo Fisher Scientific). A 20-µL reaction volume per well was prepared in a 96-well plate. The PCR reaction mix consisted of 10 ng of cDNA, 4 pmol of each primer pair, 1 µL of 20× Eva Green (Biotium), and 10 µL of 2× MyTaq HS Mix (Biolane). Specific primers (qRT_ *coxV*_for / qRT_ *coxV*_rev) for *A. fumigatus* cytochrome C oxidase subunit V (AFUA_5G10560) were used as internal controls to normalize the expression of target genes. The cDNAs from four biological samples were used for analysis, and all the reactions were run in triplicate. The reaction was carried out using the QuantStudio 3 Real-Time PCR system (Thermo Fisher Scientific) with the following thermal cycling profile: 20 sec at 95°C, followed by 40 cycles of amplification (5 sec at 95°C, 34 sec at 60°C). Relative expression was calculated according to the 2^{–ΔΔC_T} method [Livak and Schmittgen 2001].

mRNA seq and analysis

Total RNA was isolated from three biological replicates of CEA17Δ*akuB*^{K180}, Δ*ppdA/B*, Δ*dcuA/B*, and Δ*ppaA/B* knockout strains from conidia, 24-, and 48-h-old mycelium. RNA was isolated from mycelia as described above. For *A. fumigatus* conidia, the spores were first homogenized in a bead beater (0.5-mm beads; FastPrep-24, MP Biomedicals) in TRIzol before proceeding with the first chloroform extraction step as described in the RNA isolation method above. The RNA isolation was followed by a DNase treatment using two units of TURBO DNase (Thermo Fisher) per 10 µg of RNA for 30 min at 37°C in 100-µL total volume. Total RNA was then collected using the RNA Clean and Concentrator-25 kit (Zymo Research) according to the manufacturer's instructions. Total RNA collected were then sent for sequencing. Poly(A) selection and directional sequencing of

mRNA was performed by Novogene using paired-end 150-bp read sequencing on a NovaSeq 6000.

Preprocessing of raw reads including quality control and gene abundance estimation was done with the GEO2RNAseq pipeline (v0.9.12) [Seelbinder et al. 2019] in R (version 3.5.1). Quality analysis was done with FastQC (v0.11.8) before and after trimming. Read-quality trimming was done with Trimmomatic (v0.36). Reads were rRNA-filtered using SortMeRNA (v2.1) with a single rRNA database combining all rRNA databases shipped with SortMeRNA. Reference annotation was created by extracting and combining exon features from corresponding annotation files. Reads were mapped against the reference genome of *A. fumigatus* (Af293, ASM265v1) using HISAT2 (v2.1.0, paired-end mode). Gene abundance estimation was done with featureCounts (v1.28.0) in paired-end mode with default parameters. MultiQC version 1.7 was finally used to summarize and assess the quality of the output of FastQC, Trimmomatic, HISAT, featureCounts, and SAMtools. The count matrix with gene abundance data without and with median-of-ratios normalization (MRN) [Anders and Huber 2010] were extracted. Raw files are accessible under Gene Expression Omnibus accession number GSE223618.

Differential gene expression was analyzed using GEO2RNAseq. Pairwise tests were performed using four statistical tools (DESeq v1.30.0, DESeq2 v1.18.1, limma voom v3.34.6, and edgeR v3.20.7) to report *P*-values and multiple testing corrected *P*-values using the false-discovery rate method $q = \text{FDR}(p)$ for each tool. Gene expression differences were considered significant if they were reported significant by all four tools ($q \leq 0.05$ and $|\log_2(\text{MRN})| \geq 1$). GO-Slim analysis was performed on genes identified as significantly differentially expressed by all four methods described above using <https://fungi.db.org> [Amos et al. 2022] with a *P*-value cutoff of 0.05 and limiting the search to only GO-Slim terms.

Proteome analysis

Spores were harvested from 5-d-old AMM agar plates incubated at 37°C. Spores were pelleted in 2-mL screw cap tubes and a 300- μ L equivalent of 0.5-mm beads were added. Bead beating was done in lysis buffer [1% (w/v) SDS, 150 mM NaCl, 100 mM TEAB (triethyl ammonium bicarbonate), with the addition of each one tablet cOmplete ULTRA protease inhibitor cocktail and PhosSTOP [both Roche] per 10 mL of lysis buffer], three times for 20 sec with 2-min rest on ice in between. Harvested mycelium from 48-h cultures was disrupted by mortar and pestle with liquid nitrogen. Cell debris was then homogenized in lysis buffer. All samples were then incubated at 37°C in a water bath sonicator for 30 min with 0.5 μ L of Benzonase nuclease (250 U/ μ L). Proteins were separated from insolubilized debris by centrifugation at 18,000g for 15 min. Each 100 μ g of total protein per sample was diluted with 100 mM TEAB to gain a final volume of 100 μ L. Subsequently, cysteine thiols were reduced and carbamidomethylated for 30 min at 70°C by addition of 2 μ L each of 500 mM TCEP [tris(2-carboxyethyl)phosphine] and 625 mM 2-chloroacetamide (CAA). The samples were further cleaned up by methanol-chloroform-water precipitation using the protocol of Wessel and Flugge [1984]. Protein precipitates were resolubilized in 5% trifluoroethanol of aqueous 100 mM TEAB and digested for 18 h with Trypsin+LysC (Promega) at a ratio of 25:1 protein:protease. Samples were evaporated in a vacuum concen-

trator (Eppendorf). Tryptic peptides were resolubilized in 30 μ L of 0.05% TFA and 2% ACN in H₂O filtered through 10-kDa MWCO PES membrane spin filters (VWR) at 14,000g for 15 min and transferred to HPLC vials.

LC-MS/MS analysis and protein database search

An improvement in technology was implemented between the two LC-MS/MS proteomics experiments described in this manuscript. Each method is described in detail below for clarity. Method 1 (M1) is related to 48-h mycelial proteomics (three biological replicates) and method 2 (M2) to conidial proteomics (four biological replicates). Each biological replicate was measured in triplicate (three analytical replicates). LC-MS/MS analysis was performed on an Ultimate 3000 nano RSLC system connected to either a QExactive Plus (M1) or an Orbitrap Exploris 480 mass spectrometer (both Thermo Fisher Scientific) equipped with a FAIMS ion mobility interface (M2). Peptide trapping for 5 min on an Acclaim Pep Map 100 column (2 cm $\times$ 75 μ m, 3 μ m) at 5 μ L/min was followed by separation on an analytical Acclaim Pep Map RSLC nano column (50 cm $\times$ 75 μ m, 2 μ m). Mobile phase gradient elution of eluent A (0.1% [w/v] formic acid in water) mixed with eluent B (0.1% [w/v] formic acid in 90/10 acetonitrile/water) was performed using the following gradients. M1: 0–5 min at 4% B, 30 min at 7% B, 60 min at 10% B, 100 min at 15% B, 140 min at 25% B, 180 min at 45% B, 200 min at 65% B, 210–215 min at 96% B, 215.1–240 min at 4% B. M2: 0 min at 4% B, 20 min at 6% B, 45 min at 10% B, 75 min at 16% B, 105 min at 25% B, 135 min at 45% B, 150 min at 65% B, 160–165 min at 96% B, 165.1–180 min at 4% B. Positively charged ions were generated at spray voltage of 2.2 kV using a stainless steel emitter attached to the Nanospray Flex Ion Source (Thermo Fisher Scientific). The quadrupole/orbitrap instrument was operated in Full MS/data-dependent MS2 mode to trigger fragmentation of the top 15 precursor ions (M1). For M2 the number of data-dependent scans was related to the cycle time of 1.5 sec (M2). Precursor ions were monitored at m/z 300–1500 (M1) or m/z 300–1200 (M2) at a resolution of 140,000 (M1) or 120,000 (M2) FWHM (full width at half maximum) using a maximum injection time (T_{max}) of 120 msec (M1) or 50 msec (M2) and an AGC (automatic gain control) target of 3×10^5 (M1) and a normalized AGC target of 300% (M2), respectively. Precursor ions with a charge state of $z = 2–5$ were filtered at an isolation width of m/z 1.6 (M1) or m/z 4.0 (M2) for further fragmentation at 30% (M1) or 28% (M2) HCD collision energy. MS2 ions were scanned at 17,500 FWHM (T_{max} = 120 msec, AGC = 2×10^5) for M1 and 15,000 FWHM (T_{max} = 40 msec, AGC = 200%) for M2. Three different compensation voltages were applied (–48 V, –63 V, –78 V) for the FAIMS interface (M2).

Tandem mass spectra were searched against the FungiDB database (https://fungi.db.org/common/downloads/Current_Release/AfumigatusAf293/fasta/data/FungiDB-S7_AfumigatusAf293_AnnotatedProteins.fasta) of *Aspergillus fumigatus* Af293 (July 9, 2021) for M1 and against the UniProt pan proteome databases (December 21, 2022) of *Neosartorya fumigata*/*Aspergillus fumigatus* (https://ftp.uniprot.org/pub/databases/uniprot/current_release/knowledgebase/pan_proteomes/UP000002530.fasta.gz) for M2 using Thermo Proteome Discoverer (PD) 2.4 (M1) or 3.0 (M2). The following database search engines and respective

scoring thresholds (in parentheses) have been applied for the different methods. M1: Mascot 2.4.1 (>30), Sequest HT (>4), MS Amanda 2.0 (>300), and MS Fragger 3.2 (>8). M2: Mascot 2.8 (>30), Comet (>3), MS Amanda 2.0 (>300), Sequest HT with and without INFERYS Rescoring (>3), and CHIMERY5 (>2). Two missed cleavages were allowed for the tryptic digestion. The precursor mass tolerance was set to 10 ppm and the fragment mass tolerance was set to 0.02 kDa. Modifications were defined as dynamic Met oxidation, phosphorylation of Ser, Thr, and Tyr (M2), protein N-terminal acetylation with and without Met loss as well as static Cys carbamidomethylation. A strict false discovery rate (FDR) of <1% (peptide and protein level) and aforementioned search engine score threshold values were required for positive protein hits. The Percolator node and a reverse decoy database was used for q-value validation of spectral matches. Only rank 1 proteins and peptides of the top scored proteins were counted. Label-free protein quantification was based on the Minora algorithm using the precursor abundance based on intensity and a signal to noise ratio >5. Normalization was performed by using the total peptide amount method. Imputation of missing values was applied by using abundance values of 75% of the lowest abundance identified. Differential protein abundance was defined as a fold change of >2, P-value/ABS(log₂) ≤ 0.05 and at least identified in two of three (M1) or three of four replicates (M2) of the sample group with the highest abundance (Supplemental Data Sets S4, S5). Heat maps were created in R v. 4.2.2 using pheatmap v. 1.0.12 by first log₂-transforming normalized abundance data and then calculating Z-scores for proteins that exhibited at least a log₂(FC) = 2 and a P-value ≤ 0.05 in one of the conditions.

Library preparation for small RNA-sequencing and bioinformatic analysis

Total RNA was isolated from three separate biological experiments using *A. fumigatus* CEA10 conidia (5 d on AMM agar plates at 37°C) and mycelium after 24 and 48 h of growth in AMM liquid culture at 37°C using the Universal RNA Kit (Roboklon) with slight modifications. Namely, conidia and mycelia at different time points were harvested and resuspended in RL buffer followed by bead beating three times for 30 sec each using 0.5-mm glass beads (Carl Roth). The total RNA was then isolated according to the yeast protocol provided by the manufacturer (Roboklon) using an on-column DNase treatment (Lucigen Baseline-ZERO DNase). The RNA isolation was followed by an additional DNase treatment using four units of TURBO DNase (Thermo Fisher) for 30 min at 37°C in 100-μL total volume. The small RNAs were then collected using the RNA Clean and Concentrator-25 kit (Zymo Research) according to the manufacturer's instructions for small RNA enrichment.

Since the 5' end of *A. fumigatus* small RNAs are not well-defined, the isolated RNA was treated with 1 μL (5 units) of RppH enzyme (New England Biolabs) in NEB buffer 2 (50 mM NaCl, 10 mM Tris-HCl, 10 mM MgCl₂, 1 mM DTT at pH 7.9) for 1 h at 37°C to facilitate removal of a pyrophosphate from the 5' end of triphosphorylated RNA. The reaction was ended with EDTA for 5 min at 65°C followed by cleanup using the RNA Clean and Concentrator-25 kit (Zymo Research) as described above. RNA was quantified using the Qubit 4 Fluorometer (Thermo Fisher). Small RNA libraries were then prepared from 10 ng of small RNA using the NEBNext Multiplex Small RNA library preparation

set for Illumina (New England Biolabs #E73005) according to the manufacturer's recommendation using NEBNext Multiplex Oligos for Illumina (Index Primer Set 1, #E73355) (Supplemental Table S1). Size selection was performed according to the manufacturer using AMPure XP beads (Beckman Coulter). Libraries were validated for the correct size by polyacrylamide gel electrophoresis. The barcoded libraries were pooled without further adjustment, followed by 50-cycle single-end sequencing using the Illumina HiSeq 4000 platform (Eurofins Genomics). In this case, the sRNA data produced are not strictly suitable for comparison of read abundance between time points but were deemed appropriate for sRNA discovery.

Adaptors were trimmed from sRNA sequencing data using BBDuk from BBTools v38.22 (<https://sourceforge.net/projects/bbmap/>) and quality profiles assessed using FastQC. Candidate sRNAs were identified from merged replicate data using Manatee v1.2 (Handzik et al. 2020) along with the *A. fumigatus* A1163 genome (ASM15014v1) sequence and annotated transfer RNAs as a reference. Sequence hits between 19 and 25 nt in length and with more than 200 counts were retained for further analysis. Hits were visually inspected using sRNA- and mRNA-seq data and the complete *A. fumigatus* gene annotation viewed in Integrative Genomics Viewer (IGV) to validate their existence and identify antisense overlap with gene transcription (Thorvaldsdottir et al. 2013). FASTA sequences for the final candidate sRNAs were extracted using Gffread from GFF utilities (Pertea and Pertea 2020).

Data availability

The mass spectrometry proteomics data were deposited in the ProteomeXchange Consortium via the PRIDE partner repository (Perez-Riverol et al. 2022) with the data set identifiers PXD033984 (48-h mycelium) and PXD039739 (conidia). Small RNA and mRNA sequencing data sets can be found on GEO under identifiers GSE223619 and GSE223618, respectively.

Statistical analysis

The proteomics data were analyzed as described. For the remaining analyses, P-values were determined by ANOVA or two-tailed Student's t-tests where appropriate using Prism 5.0.0 software (GraphPad Software). Differences between the groups were considered significant at a P-value of <0.05. Throughout the article, significance is denoted as follows: P ≤ 0.05 (*), P ≤ 0.01 (**), P ≤ 0.001 (***), and nonsignificant (ns).

SUPPLEMENTAL MATERIAL

Supplemental material is available for this article.

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REFERENCES

- Abdel-Hadi AM, Caley DP, Carter DR, Magan N. 2011. Control of aflatoxin production of *Aspergillus flavus* and *Aspergillus parasiticus* using RNA silencing technology by targeting *aflD* (*nor-1*) gene. *Toxins (Basel)* **3**: 647–659. doi:10.3390/toxins3060647
- Agrawal N, Dasari PV, Mohammed A, Malhotra P, Bhatnagar RK, Mukherjee SK. 2003. RNA interference: biology, mechanism, and applications. *Microbiol Mol Biol Rev* **67**: 657–685. doi:10.1128/MMBR.67.4.657-685.2003
- Alexander WG, Raju NB, Xiao H, Hammond TM, Perdue TD, Metzberg RL, Pukkila PJ, Shiu PK. 2008. DCL-1 colocalizes with other components of the MSUD machinery and is required for silencing. *Fungal Genet Biol* **45**: 719–727. doi:10.1016/j.fgb.2007.10.006
- Amos B, Aureococcus C, Barba M, Barreto A, Basenko EY, Bazant W, Behar R, Blevins AS, Bohme U, Brestelli J, et al. 2022. VEPATHDB: the eukaryotic pathogen, vector and host bioinformatics resource center. *Nucleic Acids Res* **50**: D898–D911. doi:10.1093/nar/glab929
- Anders S, Huber W. 2010. Differential expression analysis for sequence count data. *Genome Biol* **11**: R106. doi:10.1186/gb-2010-11-10-r106
- Barber AE, Riedel J, Sae-Ong T, Kang K, Bräbets W, Panagiotou G, Deising HB, Kurai O. 2020. Effects of agricultural fungicide use on *Aspergillus fumigatus* abundance, antifungal susceptibility, and population structure. *MBio* **11**: e02213–20. doi:10.1128/mBio.02213-20
- Barber AE, Sae-Ong T, Kang K, Seelbinder B, Li J, Walther G, Panagiotou G, Kurai O. 2021. *Aspergillus fumigatus* pan-genome analysis identifies genetic variants associated with human infection. *Nat Microbiol* **6**: 1526–1536. doi:10.1038/s41564-021-00993-x
- Beauvais A, Bozza S, Kniemeyer O, Formosa C, Balloy V, Henry C, Roberson RW, Daigle E, Chignard M, Brakhage AA, et al. 2013. Deletion of the α -(1,3)-glucan synthase genes induces a restructuring of the conidia cell wall responsible for the avirulence of *Aspergillus fumigatus*. *PLoS Pathog* **9**: e1003716. doi:10.1371/journal.ppat.1003716
- Bernstein DA, Vyas VK, Weinberg DE, Drinnenberg IA, Bartel DP, Fink GR. 2012. *Candida albicans* Dicer (*CaDcr1*) is required for efficient ribosomal and spliceosomal RNA maturation. *Proc Natl Acad Sci* **109**: 523–528. doi:10.1073/pnas.1118559109
- Billmyre RB, Calo S, Feretaki M, Wang X, Heitman J. 2013. RNAi function, diversity, and loss in the fungal kingdom. *Chromosome Res* **21**: 561–572. doi:10.1007/s10577-013-9388-2
- Bostian KA, Hopper JE, Rogers DT, Tipper DJ. 1980. Translational analysis of the killer-associated virus-like particle dsRNA genome of *S. cerevisiae*: M dsRNA encodes toxin. *Cell* **19**: 403–414. doi:10.1016/0092-8674(80)90514-0
- Brennan P. 2018. drawProteins: a Bioconductor/R package for reproducible and programmatic generation of protein schematics. *FT000Res* **7**: 1105. doi:10.12688/ft000research.14541.1
- Brown JS, Aufaure-Brown A, Brown J, Jennings JM, Axt H Jr, Holden DW. 2000. Signature-tagged and directed mutagenesis identify PABA synthetase as essential for *Aspergillus fumigatus* pathogenicity. *Mol Microbiol* **36**: 1371–1380. doi:10.1046/j.1365-2958.2000.01953.x
- Bruch A, Ketani AA, Blango MG. 2021. RNA-based therapeutics to treat human fungal infections. *Trends Microbiol* **30**: 411–420. doi:10.1016/j.tim.2021.09.007
- Cai Q, He B, Weiberg A, Buck AH, Jin H. 2019. Small RNAs and extracellular vesicles: new mechanisms of cross-species communication and innovative tools for disease control. *PLoS Pathog* **15**: e1008090. doi:10.1371/journal.ppat.1008090
- Calo S, Nicolas FE, Lee SC, Vila A, Cervantes M, Torres-Martinez S, Ruiz-Vazquez RM, Cardenas ME, Heitman J. 2017. A non-canonical RNA degradation pathway suppresses RNAi-dependent epimutations in the human fungal pathogen *Mucor circinelloides*. *PLoS Genet* **13**: e1006686. doi:10.1371/journal.pgen.1006686
- Careras-Villaseñor N, Esquivel-Naranjo EU, Villalobos-Escobedo JM, Abreu-Goodger C, Herrera-Estralla A. 2013. The RNAi machinery regulates growth and development in the filamentous fungus *Trichoderma atroviride*. *Mol Microbiol* **89**: 96–112. doi:10.1111/mmi.12261
- Chen Y, Gao Q, Huang M, Liu Y, Liu Z, Liu X, Ma Z. 2015. Characterization of RNA silencing components in the plant pathogenic fungus *Fusarium graminearum*. *Sci Rep* **5**: 12500. doi:10.1038/srep12500
- Dang Y, Yang Q, Xue Z, Liu Y. 2011. RNA interference in fungi: pathways, functions, and applications. *Eukaryot Cell* **10**: 1148–1155. doi:10.1128/EC.05109-11
- da Silva Ferreira ME, Kress MR, Savoldi M, Goldman MH, Hartl A, Heinekamp T, Brakhage AA, Goldman GH. 2006. The *akuB* (*KU80*) mutant deficient for nonhomologous end joining is a powerful tool for analyzing pathogenicity in *Aspergillus fumigatus*. *Eukaryot Cell* **5**: 207–211. doi:10.1128/EC.5.1.207-211.2006
- D'Souza CA, Kronstad JW, Taylor G, Warren R, Yuen M, Hu G, Jung WH, Sham A, Kidd SE, Tangen K, et al. 2011. Genome variation in *Cryptococcus gattii*, an emerging pathogen of immunocompetent hosts. *MBio* **2**: e00342-10.
- Ender C, Meister G. 2010. Argonaute proteins at a glance. *J Cell Sci* **123**: 1819–1823. doi:10.1242/jcs.055210
- Gaffar FY, Imeni J, Karlovsky P, Koch A, Kogel KH. 2019. Different components of the RNA interference machinery are required for conidiation, ascospore germination, virulence, decaryinol production, and fungal inhibition by exogenous double-stranded RNA in the head blight pathogen *Fusarium graminearum*. *Front Microbiol* **10**: 1662. doi:10.3389/fmicb.2019.01662
- Gibbons JG, D'Avino P, Zhao S, Cox GW, Rinker DC, Fortwendel JR, Latge J-P. 2022. Comparative genomics reveals a single nucleotide deletion in *pkxP* that results in white-spore phenotype in natural variants of *Aspergillus fumigatus*. *Front Fungal Biol* **3**. doi:10.3389/ffunb.2022.897954
- Gusa A, Williams JD, Cho JE, Averette AF, Sun S, Shouse EM, Heitman J, Alspaugh JA, Jinks-Robertson S. 2020. Transposon mobilization in the human fungal pathogen *Cryptococcus* is mutagenic during infection and promotes drug resistance in vitro. *Proc Natl Acad Sci* **117**: 9973–9980. doi:10.1073/pnas.2001451117
- Hall IM, Noma K, Grewal SI. 2003. RNA interference machinery regulates chromosome dynamics during mitosis and meiosis in fission yeast. *Proc Natl Acad Sci* **100**: 193–198. doi:10.1073/pnas.232688099
- Hammond TM, Keller NP. 2005. RNA silencing in *Aspergillus nidulans* is independent of RNA-dependent RNA polymerases. *Genetics* **169**: 607–617. doi:10.1534/genetics.104.035964
- Hammond TM, Andrews MD, Roossinck MJ, Keller NP. 2008a. *Aspergillus mycoviruses* are targets and suppressors of RNA silencing. *Eukaryot Cell* **7**: 350–357. doi:10.1128/EC.00356-07

- Hammond TM, Bok JW, Andrewski MD, Reyes-Dominguez Y, Sczarczewski C, Keller NP. 2008b. RNA silencing gene truncation in the filamentous fungus *Aspergillus nidulans*. *Eukaryot Cell* **7**: 339–349. doi:10.1128/EC.00355-07
- Handzik JE, Tastsoglou S, Vlachos IS, Hatzigeorgiou AG. 2020. Manatee: detection and quantification of small non-coding RNAs from next-generation sequencing data. *Sci Rep* **10**: 705. doi:10.1038/s41598-020-57495-9
- Hutvagner G, Simard MJ. 2008. Argonaute proteins: key players in RNA silencing. *Nat Rev Mol Cell Biol* **9**: 22–32. doi:10.1038/nrm2321
- Janbon G, Maeng S, Yeng DH, Ko YJ, Jung KW, Moyrand F, Floyd A, Heitman J, Bahn YS. 2010. Characterizing the role of RNA silencing components in *Cryptococcus neoformans*. *Fungal Genet Biol* **47**: 1070–1080. doi:10.1016/j.fgb.2010.10.005
- Jinek M, Doudna JA. 2009. A three-dimensional view of the molecular machinery of RNA interference. *Nature* **457**: 405–412. doi:10.1038/nature07755
- Joch C, Loh E, Floner A, Haas H, Huttenhofer A. 2009. Development-dependent scavenging of nucleic acids in the filamentous fungus *Aspergillus fumigatus*. *RNA Biol* **6**: 179–186. doi:10.4161/ma.6.2.7717
- Kadotani N, Nakayashiki H, Tosa Y, Mayama S. 2003. RNA silencing in the phytopathogenic fungus *Magnaporthe oryzae*. *Mol Plant Microbe Interact* **16**: 769–776. doi:10.1094/MPMI.2003.16.9.769
- Kalida N, Naorem A, Manchikarla RV. 2013. Targeting fungal genes by diced siRNAs: a rapid tool to decipher gene function in *Aspergillus nidulans*. *PLoS ONE* **8**: e75443. doi:10.1371/journal.pone.0075443
- Ketting RF. 2011. The many faces of RNAi. *Dev Cell* **20**: 148–161. doi:10.1016/j.devcel.2011.01.012
- Khatir M, Rajam MV. 2007. Targeting polyamines of *Aspergillus nidulans* by siRNA specific to fungal ornithine decarboxylase gene. *Med Mycol* **45**: 211–220. doi:10.1080/13693780601158779
- Langfelder K, Jahn B, Gehring H, Schmidt A, Wanner G, Brakhage AA. 1998. Identification of a polyketide synthase gene (*pkdP*) of *Aspergillus fumigatus* involved in conidial pigment biosynthesis and virulence. *Med Microbiol Immunol* **187**: 79–89. doi:10.1007/s004300050077
- Latge JP. 1999. *Aspergillus fumigatus* and aspergillosis. *Clin Microbiol Rev* **12**: 310–350. doi:10.1128/CMR.12.2.310
- Livak KJ, Schmittgen TD. 2001. Analysis of relative gene expression data using real-time quantitative PCR and the 2^{-ΔΔC_T} method. *Methods* **25**: 402–408. doi:10.1006/meth.2001.1262
- Lothar J, Breitschopf T, Krappmann S, Merton CO, Bouzani M, Kurzi O, Gunzer M, Hasenberg M, Einsele H, Loeffler J. 2014. Human dendritic cell subsets display distinct interactions with the pathogenic mould *Aspergillus fumigatus*. *Int J Med Microbiol* **304**: 1160–1168. doi:10.1016/j.ijmm.2014.08.009
- Martenssen R, Moazed D. 2015. RNAi and heterochromatin assembly. *Cold Spring Harb Perspect Biol* **7**: a019323. doi:10.1101/cshperspect.a019323
- Moazed D. 2009. Small RNAs in transcriptional gene silencing and genome defence. *Nature* **457**: 413–420. doi:10.1038/nature07756
- Mousavi B, Hedayati MT, Teimoori-Toolabi L, Guillot J, Alizadeh A, Badali H. 2015. *cyp51A* gene silencing using RNA interference in azole-resistant *Aspergillus fumigatus*. *Mycoses* **58**: 699–706. doi:10.1111/myc.12417
- Mouyna L, Henry C, Doering TL, Latge JP. 2004. Gene silencing with RNA interference in the human pathogenic fungus *Aspergillus fumigatus*. *FEMS Microbiol Lett* **237**: 317–324. doi:10.1111/j.1574-6968.2004.tb09713.x
- Mukherjee K, Campos H, Kolaczowski B. 2013. Evolution of animal and plant dicers: early parallel duplications and recurrent adaptation of antiviral RNA binding in plants. *Mol Biol Evol* **30**: 627–641. doi:10.1093/molbev/mss263
- Murata T, Kadotani N, Yamaguchi M, Tosa Y, Mayama S, Nakayashiki H. 2007. siRNA-dependent and -independent post-transcriptional cosuppression of the LTR-retrotransposon MAGGY in the phytopathogenic fungus *Magnaporthe oryzae*. *Nucleic Acids Res* **35**: 5987–5994. doi:10.1093/nar/gkm646
- Nakayashiki H, Hanada S, Nguyen BQ, Kadotani N, Tosa Y, Mayama S. 2005. RNA silencing as a tool for exploring gene function in ascomycete fungi. *Fungal Genet Biol* **42**: 275–283. doi:10.1016/j.fgb.2005.01.002
- Nami S, Baradaran B, Mansoori B, Kordbacheh P, Rezaei S, Falahati M, Mohamed Khosroshahi L, Safara M, Zaini F. 2017. The utilization of RNA silencing technology to mitigate the voriconazole resistance of *Aspergillus flavus*; lipofectamine-based delivery. *Adv Pharm Bull* **7**: 53–59. doi:10.15171/apb.2017.007
- Nicolas FE, Garra V. 2016. RNA interference in fungi: retention and loss. *Microbiol Spectr* **4**. doi:10.1128/microbiolspec.FUNK-0008-2016
- Nicolas FE, Moxon S, de Haro JP, Calo S, Grigoriev IV, Torres-Martinez S, Moulton V, Ruiz-Vazquez RM, Dalmay T. 2010. Endogenous short RNAs generated by Dicer 2 and RNA-dependent RNA polymerase 1 regulate mRNAs in the basal fungus *Mucor circinellus*. *Nucleic Acids Res* **38**: 5535–5541. doi:10.1093/nar/gkq301
- Nicolas FE, Vila A, Moxon S, Cascales MD, Torres-Martinez S, Ruiz-Vazquez RM, Garra V. 2015. The RNAi machinery controls distinct responses to environmental signals in the basal fungus *Mucor circinellus*. *BMC Genomics* **16**: 237. doi:10.1186/s12864-015-1443-2
- Nuss DL. 2011. Mycoviruses, RNA silencing, and viral RNA recombination. *Adv Virus Res* **80**: 25–48. doi:10.1016/B978-0-12-385987-7.00002-6
- Oberegger H, Zadra I, Schoeser M, Haas H. 2000. Iron starvation leads to increased expression of Cu/Zn-superoxide dismutase in *Aspergillus*. *FEBS Lett* **485**: 113–116. doi:10.1016/S0014-5793(00)02206-7
- Ozkan S, Mohorianu I, Xu P, Dalmay T, Coutts RHA. 2017. Profile and functional analysis of small RNAs derived from *Aspergillus fumigatus* infected with double-stranded RNA mycoviruses. *BMC Genomics* **18**: 416. doi:10.1186/s12864-017-3773-8
- Pascovici D, Handler DC, Wu JX, Haynes PA. 2016. Multiple testing corrections in quantitative proteomics: a useful but blunt tool. *Proteomics* **16**: 2448–2453. doi:10.1002/pmic.201600044
- Perez-Riverol Y, Bai J, Bandla C, Garcia-Seisdedos D, Hewapathirana S, Kamathirathan S, Kundu DJ, Prakash A, Fiercks-Zipper A, Eisenacher M, et al. 2022. The PRIDE database resources in 2022: a hub for mass spectrometry-based proteomics evidences. *Nucleic Acids Res* **50**: D543–D552. doi:10.1093/nar/gkab1038
- Pertea G, Pertea M. 2020. GFF utilities: GFFRead and GFFCompare. *F1000Res* **9**: 304. doi:10.12688/f1000research.23297.1
- Qin H, Chen F, Huan X, Machida S, Song J, Yuan YA. 2010. Structure of the *Arabidopsis thaliana* DCL4/DUF283 domain reveals a non-canonical double-stranded RNA-binding fold for protein-protein interaction. *RNA* **16**: 474–481. doi:10.1261/ma.1965310
- Roche B, Arcangeli B, Martienssen RA. 2016. RNA interference is essential for cellular quiescence. *Science* **354**: aah5651. doi:10.1126/science.aah5651
- Romano N, Macino G. 1992. Quelling: transient inactivation of gene expression in *Neurospora crassa* by transformation with homologous sequences. *Mol Microbiol* **6**: 3343–3353. doi:10.1111/j.1365-2958.1992.tb02202.x
- Ruiz-Vazquez RM, Nicolas FE, Torres-Martinez S, Garra V. 2015. Distinct RNAi pathways in the regulation of physiology and

- development in the fungus *Mucor circinelloides*. *Adv Genet* **91**: 55–102. doi:10.1016/bs.adgen.2015.07.002
- Schauer SE, Jacobsen SE, Meinke DW, Ray A. 2002. DICER-LIKE1: blind men and elephants in *Arabidopsis* development. *Trends Plant Sci* **7**: 487–491. doi:10.1016/S1360-1385(02)02355-5
- Seebinder B, Wolf T, Priebe S, McNamara S, Geilber S, Guthke R, Linda J. 2019. GEO2RNAseq: an easy-to-use R pipeline for complete preprocessing of RNA-seq data. *bioRxiv* doi:10.1101/771063
- Shabalina SA, Koonin EV. 2008. Origins and evolution of eukaryotic RNA interference. *Trends Ecol Evol* **23**: 578–587. doi:10.1016/j.tree.2008.06.005
- Shopova IA, Belyaev I, Dasari P, Jahreis S, Stroe MC, Cseresnyes Z, Zimmermann AK, Medyukhina A, Svensson CM, Kruger T, et al. 2020. Human neutrophils produce antifungal extracellular vesicles against *Aspergillus fumigatus*. *MBio* **11**: e00596–20. doi:10.1128/mBio.00596-20
- Song MS, Rossi JJ. 2017. Molecular mechanisms of Dicer: endonuclease and enzymatic activity. *Biochem J* **474**: 1603–1618. doi:10.1042/BCJ20160759
- Southern E. 2006. Southern blotting. *Nat Protoc* **1**: 518–525. doi:10.1038/ingrot.2006.73
- Szewczyk E, Krappmann S. 2010. Conserved regulators of mating are essential for *Aspergillus fumigatus* cleistothecium formation. *Eukaryot Cell* **9**: 774–783. doi:10.1128/EC.00375-09
- Thorvaldsdottir H, Robinson JT, Mesirov JP. 2013. Integrative Genomics Viewer (IGV): high-performance genomics data visualization and exploration. *Brief Bioinform* **14**: 178–192. doi:10.1093/bib/bbs017
- Torres-Martinez S, Ruiz-Vazquez RM. 2017. The RNAi universe in fungi: a varied landscape of small RNAs and biological functions. *Annu Rev Microbiol* **71**: 371–391. doi:10.1146/annurev-micro-090816-093352
- Torri A, Jaeger J, Pradeu T, Saleh MC. 2022. The origin of RNA interference: adaptive or neutral evolution? *PLoS Biol* **20**: e3001715. doi:10.1371/journal.pbio.3001715
- Wang X, Wang P, Sun S, Darwiche S, Idnurm A, Heitman J. 2012. Transgene induced co-suppression during vegetative growth in *Cryptococcus neoformans*. *PLoS Genet* **8**: e1002885. doi:10.1371/journal.pgen.1002885
- Weidner G, d'Enfert C, Koch A, Mol PC, Brakhage AA. 1998. Development of a homologous transformation system for the human pathogenic fungus *Aspergillus fumigatus* based on the *pyrG* gene encoding orotidine 5'-monophosphate decarboxylase. *Curr Genet* **33**: 378–385. doi:10.1007/s002940050350
- Wessel D, Flugge UI. 1984. A method for the quantitative recovery of protein in dilute solution in the presence of detergents and lipids. *Anal Biochem* **138**: 141–143. doi:10.1016/0003-2697(84)90782-6
- Xiao H, Alexander WG, Hammond TM, Boone EC, Perdue TD, Pukkila PJ, Shiu PK. 2010. QIP, a protein that converts duplex siRNA into single strands, is required for meiotic silencing by unpaired DNA. *Genetics* **186**: 119–126. doi:10.1534/genetics.110.116273

4.2 Manuscript B

An improved description of the small RNA landscape of the human fungal pathogen

***Aspergillus fumigatus*.**

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The candidate is

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Summary:

Aspergillus fumigatus is a dangerous human fungal pathogen that produces limited small RNAs (sRNA) under standard conditions. To have a better understanding of the sRNA biogenesis, we overexpressed two inverted-repeat transgenes (IRT) to form dsRNA structures, which revealed dicer-like-dependent sRNAs enriched for a 5'U and roughly 20 nt in length as well as a dsRNA-dependent fungal growth inhibition phenotype. A more thorough investigation of wild-type strains also revealed dicer-like-independent tRNA-derived RNAs. Collectively, this work improves our understanding of RNA processing in *A. fumigatus* and opens many questions for future investigations.

Contribution to the manuscript:

Xiaoqing Pan took a lead role in designing and executing the experiments and data analysis. She participated in interpreting and drawing scientific conclusion and the writing.

Authors' contributions (in %) to the given categories of the publication

Author	Conceptual	Data analysis	Experimental	Writing the manuscript	Provision of material
Xiaoqing Pan	30%	60%	20%	30%	0%
Abdulrahman A. Kelani	30%	10%	60%	30%	0%
Lukas Schrettenbrunner	0%	0%	10%	0%	0%
Swatika Prabakar	0%	0%	5%	0%	0%
Bhawana Israni	0%	0%	5%	0%	0%
Matthew G. Blango	40%	30%	0%	40%	100%
Total:	100%	100%	100%	100%	100%

Signature candidate

Signature supervisor (member of the Faculty)



RESEARCH ARTICLE

An Improved Description of the Small RNA Landscape of the Human Fungal Pathogen *Aspergillus fumigatus*

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ABSTRACT

Aspergillus fumigatus is a ubiquitous filamentous fungus and dangerous human pathogen that produces a limited pool of small RNAs under standard laboratory conditions. To better understand the rules of small RNA production in *A. fumigatus*, we induced canonical RNA interference (RNAi) via overexpression of two separate inverted-repeat transgenes. We observed production of predominantly 20-nt, 5' uridine-containing small RNAs from the 3' end of each transgene nearest to the loop region and dependent on dicer-like B RNase III enzyme. Using this refined knowledge, we assessed small RNA biogenesis by sRNA-seq in three double knockout strains of the RNAi pathway, namely the dicer-like proteins (*ΔdclA/B*), argonautes (*ΔagoA/B*), and RNA-dependent RNA polymerases (*ΔrpaA/B*). In each case, we found limited evidence for production of 5' U-containing small RNAs reliant on the RNAi machinery under standard laboratory conditions. We did observe 5' U-containing small RNAs amongst the abundant tRNA-derived RNAs (tDRs); however, biogenesis of tDRs was predominantly Dicer-like independent. To more accurately define the complex tDR repertoire, we employed a cutting-edge tDR-sequencing approach that improved tRNA-half detection and revealed qualitative morphotype-specific changes in the small RNA fraction of conidia relative to mycelium. Finally, leveraging the limited sRNA repertoire of *A. fumigatus*, we tested the consequences of inverted-repeat transgene overexpression in the *ΔdclA/B* double knockout, which revealed growth inhibition even in the absence of double-stranded RNA (dsRNA) processing and small RNA production. We hypothesize that the RNAi substrate-limited landscape of *A. fumigatus* facilitates sensitivity to increases in dsRNA, offering an intriguing system for future studies of dsRNA metabolism.

1 | Introduction

The canonical RNA interference (RNAi) pathway processes double-stranded RNA (dsRNA) through cleavage to small RNAs (sRNA) that are then used as guide molecules to facilitate silencing of target transcripts (Wilson and Doudna 2013). The dsRNA precursor molecule can be produced endogenously via transcription of self-genes or exogenously, for example, from a virus or repetitive transgene. The dsRNA A-form helix

is characterized by a narrowed major groove compared to B-form DNA, meaning that the molecule itself can be a substrate for multiple dsRNA binding proteins (dsRBPs) independent of the nucleotide sequence (Cottrell et al. 2024). The result is competition for dsRNA substrates by dsRBPs that bind, cleave, or edit the dsRNA molecule through interactions with the sugar-phosphate backbone (Cottrell et al. 2024). In humans for example, this manifests as a balance between processing (RNA cleavage or modification) and activation of inflammatory

Xiaoqing Pan and Abdulrahman A. Kefani contributed equally to this manuscript and are listed in reverse alphabetical order.

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cascades by dsRBP sensors like MDA5 (Liddicott et al. 2015), whereas in microbes like the bacteria *Escherichia coli*, dsRNA is thought to be rapidly degraded or processed to smaller RNAs (Lybecker et al. 2014). In many cases, how fungi process dsRNA remains poorly understood.

The sRNA landscape of fungi (like many eukaryotes) consists of a heterogeneous group of RNA molecules, ranging from fragments of housekeeping RNA like tRNA and rRNA to small interfering RNAs (siRNA) and microRNA-like RNAs (miRNA) produced from both perfectly duplexed dsRNA and dsRNA structures with bulges and/or mismatches (Torres-Martinez and Ruiz-Vazquez 2017). Fungal sRNAs are produced from these precursors via diverse biogenesis pathways, often including members of the canonical RNAi machinery such as the RNase III enzyme Dicer (or Dicer-like), Argonaute, and/or RNA-dependent RNA polymerases (RdRPs), together with peripheral factors (Lee et al. 2016). Some fungi have lost RNAi to gain an advantage in a particular niche (Billmyre et al. 2015), but many, including the ubiquitous human pathogen *Aspergillus fumigatus*, harbor an active RNAi pathway (Mouysa et al. 2004) capable of producing small RNAs in response to mycovirus infection (Orkan et al. 2017). Interestingly, *A. fumigatus* appears to produce a limited pool of endogenous sRNAs under standard growth conditions, consisting in large part of tRNA-derived RNAs (tDRs; Figure S1) despite encoding two Dicer-like proteins, two Argonautes, and two RdRPs (Table S1; Kelani et al. 2023; Ozkan et al. 2017; Jochl et al. 2008).

The tDRs are an emerging class of heterogeneous 16- to 50-nucleotide (nt) sRNAs that include the 5' and 3' tRNA halves (tRHs; ~31 to 40 nt), tRNA fragments (tRF) of 17 to 25 nt in length processed from the 5' (tRF-5) and 3' ends (tRF-3, tRF-3 CCA), internal tRFs (i-tRF), and other miscellaneous fragments (Tycowski and Grzywniak 2022) (Figure S1). The generation of tDRs is multifactorial and dynamic across development, relying on enzymes like the RNase A ribonuclease angiogenin, RNase T2-type enzymes like Rny1p in yeast, or even Dicer in some species (Donovan et al. 2017; Ivanov et al. 2011; Jochl et al. 2008; Thompson et al. 2008; Luhtala and Parker 2010; Loweland et al. 2024; Babiarz et al. 2008; Cole et al. 2009; Haissecker et al. 2010; Kuscu et al. 2018; Halber et al. 2008; Thompson and Parker 2009). In *A. fumigatus*, tRHs accumulate after germination from conidia to hyphae independently of mature tRNA levels; however, during early conidiogenesis there is a rapid increase in tRHs and a decrease in mature tRNA levels (Jochl et al. 2008). In that study, it was posited that the decrease in the mature tRNA pool in conidia may contribute to the metabolic dormancy of spores as active translation is shut down. The tRF-3 RNAs are known to interact with the Argonaute-containing RNA-induced silencing complex (RISC) to guide silencing of target mRNA based on 3' UTR seed sequence complementarity (Maute et al. 2013; Thompson and Parker 2009) and to facilitate ribosomal RNA processing in association with Argonaute (Couvillion et al. 2010). Despite the existence of foundational work in this area, the exact small tRNA-derived RNAs produced in conidia versus hyphae remain unknown.

Here, we provide an improved global characterization of the sRNA landscape and dsRNA metabolism of the World Health

Organization critical priority pathogen *A. fumigatus* (Fisher and Denning 2023) using sRNA-seq and tDR-seq approaches. Although the endogenous repertoire of sRNAs is limited, we show that heterologous overexpression of inverted-repeat transgenes results in production of canonical 20-nt 5' U-containing small RNAs and limits fungal growth even in the absence of Dicer-like proteins, suggesting that excess dsRNA may itself serve as a check on growth in *A. fumigatus*.

2 | Results

2.1 | The *A. fumigatus* RNAi Machinery Processes Inverted-Repeat Transgene dsRNA to 20-nt, 5' U-Containing sRNAs

We previously showed that overexpression of long dsRNA in *A. fumigatus* as an inverted-repeat transgene (IRT) with complementarity to a nonessential gene (the polyketide synthase *pkpP*) results in target gene silencing and surprising growth inhibition in liquid culture (Kelani et al. 2023). To refine our understanding of dsRNA processing in *A. fumigatus*, we employed sRNA-seq of *pkpP* IRT-expressing wild-type and dicer-like double knockout strains ($\Delta dclA/B$ -IRT; Kelani et al. 2023) after induced expression of the transgene in liquid culture. We grew the *pkpP* IRT strains for 16 h plus 8 h of induction to ensure sufficient biomass prior to collection of RNA and sRNA-seq analysis (Figure S2A,B). This allowed us to assess the ability of *A. fumigatus* to produce sRNAs from the transgene transcript. Untreated wild-type fungus was included as a control for standardization with previous experiments. sRNAs were mapped to the *pkpP* IRT sequence, and we observed an accumulation of 20-nt RNAs with a 5' uridine produced from the transgene in the wild-type strain (Figure 1A-C), revealing that the *A. fumigatus* RNAi machinery can function in a manner consistent with the fungal literature (Regmi et al. 2013; Mi et al. 2008). These peaks of sRNA were absent in the wild-type and $\Delta dclA/B$ -IRT strains but did remain when we performed the same experiment in $\Delta dclA/B$ and $\Delta rpsA/B$ *pkpP* IRT-expressing strains (Figure S2C-E). Closer inspection of the mapped reads revealed that the largest accumulation originated from the 3' half of the transcript (antisense) primarily as approximately two 20-nt RNAs (Figure 1C, Figure S2D,E). Prediction of target binding sites using these two relatively abundant sRNA sequences revealed no additional high-quality targets beyond the known complementarity to the *pkpP* gene, potentially suggesting that the toxicity observed by the expression of the inverted-repeat transgene in our previous study is not facilitated through small RNA off-target effects (Kelani et al. 2023).

sRNA-seq of *pkpP* IRT-expressing single knockouts of *dclA* and *dclB* revealed that *DclB*, but not *DclA*, is required for production of the observed 5' U-containing sRNAs (Figure 1D). Interestingly, even in this second, separate IRT-sequencing experiment, the pattern of cleavage of the hairpin RNA was nearly identical to that observed in Figure 1C providing corroboration of *DclB* specificity preferences. As an additional control, we confirmed the activity of *DclB* through complementation of the $\Delta dclB$ -IRT strain with a functional copy of *dclB* driven by its native promoter (Figure S3). To confirm that the length of 20 nt and a 5' U were general features of *DclB*

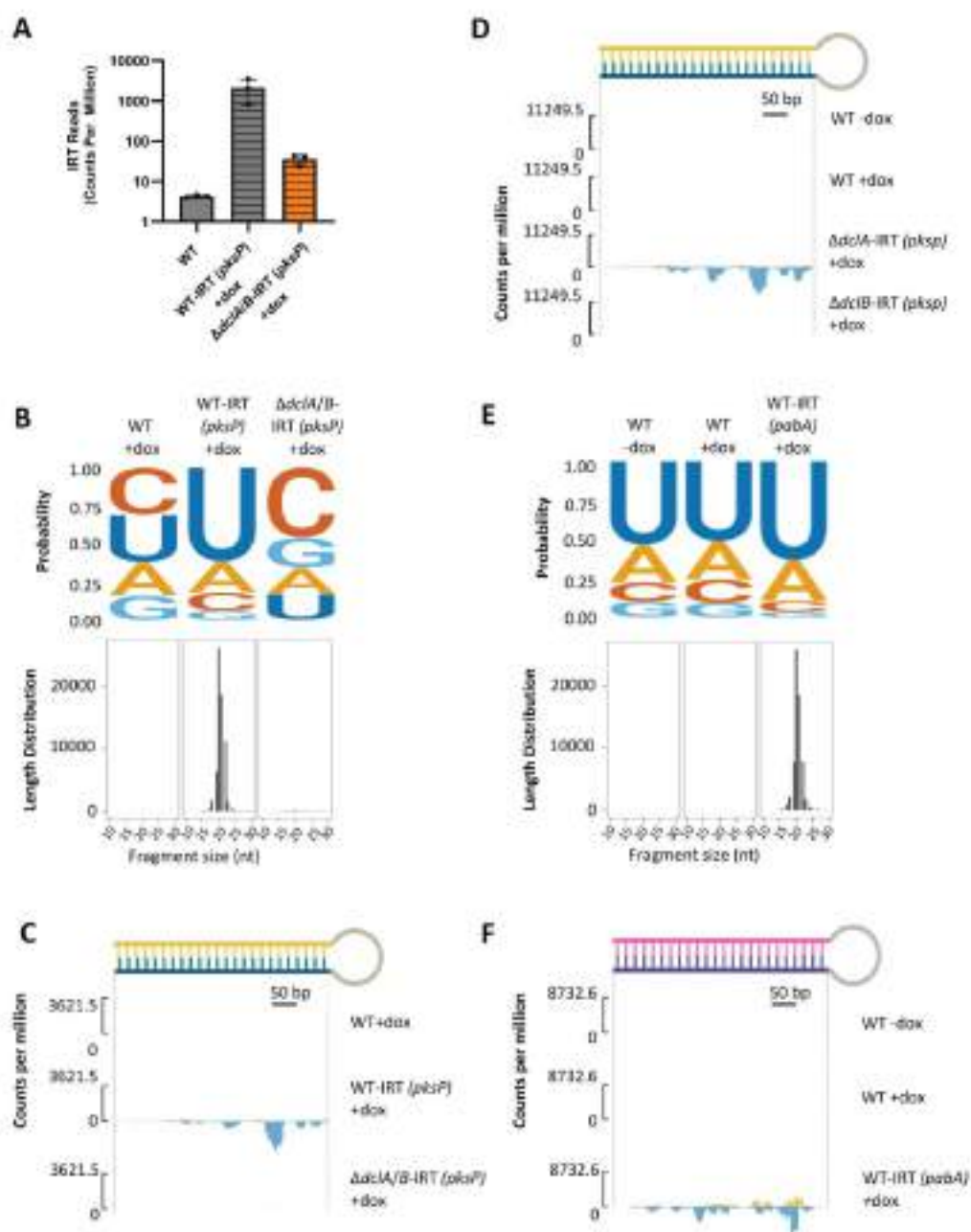


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FIGURE 1 | Inverted-repeat transgene expression induces 20-nt, 5' U-containing sRNA production. (A) sRNA reads mapped to *pkdP* inverted-repeat transgene (IRT) sequence per million mapped reads for each of the denoted strains. Data is accumulated from three biological replicate sRNA-seq experiments. (B) The distribution of the 5' nucleotide for the respective strains (top) and histogram showing the average sRNA read length for the sRNAs that mapped to the *pkdP* IRT (bottom). (C) Graph showing read coverage of *pkdP* inverted-repeat transgene region with reads from the 5' end of the inverted repeat shown as positive strand and from the 3' end shown as negative strand. (D) Graph showing read coverage of *pkdP* inverted-repeat transgene region in the $\Delta dclA$ and $\Delta dclB$ knockout strains. (E) The distribution of the 5' nucleotide for the respective strains (top) and histogram showing the average sRNA read length for the sRNAs that mapped to the *pkdA* IRT (bottom). (F) Graph showing read coverage of *pkdA* inverted-repeat transgene region in the respective strains. Graphs in (C, D, F) were created with the SpicK package (Kurianthach and William Harbour 2009) with reads from the 5' end of the inverted repeat shown as positive strand and from the 3' end shown as negative strand.

processing of hairpin RNAs, we performed additional sequencing of a *pkdA* IRT-expressing wild-type strain that was created previously (Kefani et al. 2023), and again observed 20-nt, 5' U-containing sRNAs primarily produced from the 3' end of the transcript (Figure 1E,F). Additional biochemical experiments in vitro with purified enzymes and defined substrates will be required to fully elucidate the mechanism of DclB cleavage, but these results already suggest some DclB preferences in *A. fumigatus*.

2.2 | The Composition of *A. fumigatus* Endogenous sRNAs Is Morphotype Specific

After improving our understanding of small RNA biogenesis, we next employed previously described knockout strains in the canonical RNA interference machinery ($\Delta dclA/B$, $\Delta pdsA/B$, and $\Delta rpsA/B$) created in the wild-type CEA17 Δ oku80 background to simultaneously profile expression of small noncoding RNA (ncRNA; this study) and mRNA (Kefani et al. 2023) from the same biological samples under normal growth conditions (conidia, 24-h mycelium, and 48-h mycelium; Figure 2A; Figure S4A; Data S1). We mapped the expression of distinct small ncRNA classes, including sRNA, snRNA, and tDRs (Figure 2B). In conidia, many reads mapped to rRNA, whereas the proportion of tDRs and other small ncRNAs increased in the mycelial samples, consistent with an expected, more diverse transcriptome in metabolically active mycelium. Across all the sequencing samples, rRNA made up the majority of the mapped reads, followed by short reads mapping to tRNA genes. An investigation of the 19–25 nt small RNAs derived from rRNA revealed no 5' nucleotide sequence bias or change after deletion of RNAi components (Figure S4B).

We next aimed to identify sRNAs that might be masked by the high abundance of rRNA reads. Removal of reads matching rRNA and restriction of read length to 18 to 30 nt led to identification of similar numbers of sRNAs from each of the double knockout strains compared to wild type (Figure S4C,D). In conidia, a clear population of 18 to 21 nucleotides was identified, whereas in 24- and 48-h mycelium, we generally observed two or more peaks centered at ~19 nt and/or ~28 nt, often indicative of tRNA-derived RNAs. We then used Manatee (Handzlik et al. 2020) to identify and predict sRNAs, including microRNA-like RNAs (miRNA; Data S2). We found no sequence bias in the 19–25 nt small RNA fraction from any condition (Figure 2C). We observed few low abundance candidate sRNAs in conidia, and a large number of slightly higher abundance sRNAs from 24- and 48-h mycelium. Interestingly, deletion of *dclA/B* did not change

the overall number of candidate sRNAs identified (1524 compared to 1615 in wild type), whereas loss of *pdsA/B* and *rpsA/B* decreased the number of sRNAs identified by Manatee (873 and 1016 respectively), although most of the candidates identified were not of particularly high abundance (Data S2). Intersection of the Manatee-identified sRNAs using UpSetR showed limited overlap between the different strains (Figure S5A), with differences in abundance driven by a few seemingly random tDRs (Figure S5B; Data S2). These results are consistent with studies showing a limited pool of sRNAs meeting the criteria of miRNA in *A. fumigatus* (Ozkan et al. 2017; Kefani et al. 2023; Jochl et al. 2008), and instead more RNAs derived from structural RNAs (e.g., tRNAs and rRNA).

2.3 | tDR Biogenesis Is Primarily Independent of the RNAi Machinery in *A. fumigatus*

The identification of abundant tDRs in our sRNA-seq dataset coupled with previous reports of tRH production in *A. fumigatus* during starvation (Jochl et al. 2008) led us to assess the tDR fraction in more detail. In agreement, we observed an increase in the relative abundance of nuclear-encoded 5'-tRHs from conidia to older mycelium; however, we also detected a large abundance of tRFs, likely generated by cleavage of either mature or precursor tRNA (e.g., by RNase T2). For nuclear-encoded tRFs, we noticed that the relative abundance of tRF-5 in the small RNA fraction increases from conidia to 24- and 48-h-old mycelium where nutrients are more limited, while the tRF-3 fraction decreased across those samples (Figure 2D). However, within mitochondrial tDRs, reads mapped mostly to tRF-3, 5'-tRHs, and tRF-5 (Figure 2D).

Next, we created heat maps of normalized abundance for the tDRs of each tRNA in the wild type fungus (Figure S6, Data S3) using the unika software package as previously described (Isakova et al. 2020). We observed large differences in the relative abundance of tDRs in each morphotype, suggesting differential stability of different fragments or potentially poor ligation of select tDRs during sequencing library preparation. We saw slight, statistically significant differences in ratios of tDR abundance for some of the RNAi knockout strains (Figure S7, Data S3), especially $\Delta pdsA/B$, as compared to the wild type. In conidia, $\Delta pdsA/B$ appeared to produce fewer tRF-5 derivatives from many tRNAs. Surprisingly, we saw minimal differences in the production of tDRs in the $\Delta dclA/B$ strain. These results suggest that the RNAi machinery, particularly the Dicer-like proteins, do not play a major role in tRNA fragment production in *A. fumigatus* under standard growth conditions.

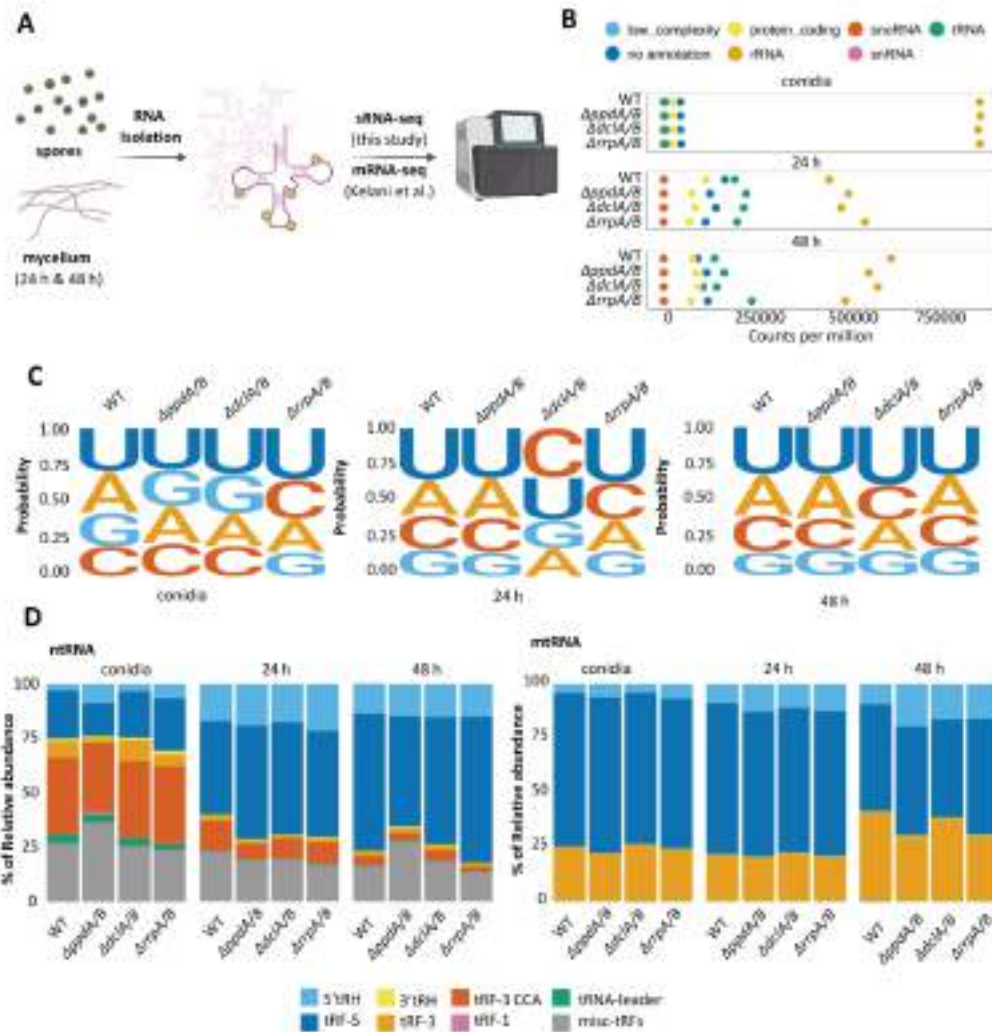


FIGURE 2 | Small RNA abundance in *A. fawcettii* conidia and mycelium. (A) Strains and conditions profiled in this study. The wild-type (CEA17 Δ ada18^{trp}) and three knockout strains ($\Delta ppa4/\Delta$, $\Delta dcl4/\Delta$, $\Delta rps4/\Delta$) were collected ($n=3$ biological replicates per condition). RNA samples were collected from fungal conidia and hyphae (24 and 48 h) and Illumina sequenced. Created in BioRender. Blango, M. (2023) <https://BioRender.com/r3sp643>. (B) Coverage of small ncRNA types within the profiled samples. Small ncRNAs were annotated with low complexity, no annotation, protein coding, rRNA, snRNA, sRNA, and tRNA (including nuclear tRNA and mitochondrial tRNA). (C) The distribution of the 5' nucleotide for the respective strains from sRNA-seq data across conidia, 24- and 48-h mycelium. (D) Comparison of tDR relative abundance across samples, and between wild-type and RNAi double knockouts for nuclear-encoded tRNA. $n=3$ biological replicates per condition.

2.4 | tDR-Seq Reveals Altered tDR Patterns Between Fungal Morphotypes

The abundance of tDRs in our sRNA-seq, together with the known difficulties of sequencing tRNAs using standard methodologies (e.g., poor adapter ligation and decreased reverse transcription efficiency), led us to perform a tRF & tRNA-seq (here

denoted tDR-seq) approach, where RNA ends are repaired and common methylation modifications are removed to facilitate ligation and readthrough, respectively (Figure 3A; Data S4). We focused on the wild-type fungal morphotypes for simplicity as only minor differences were observed between the RNAi knockouts and wild type in the previous analysis. Isolation of RNA again confirmed that the proportion of small RNAs increases

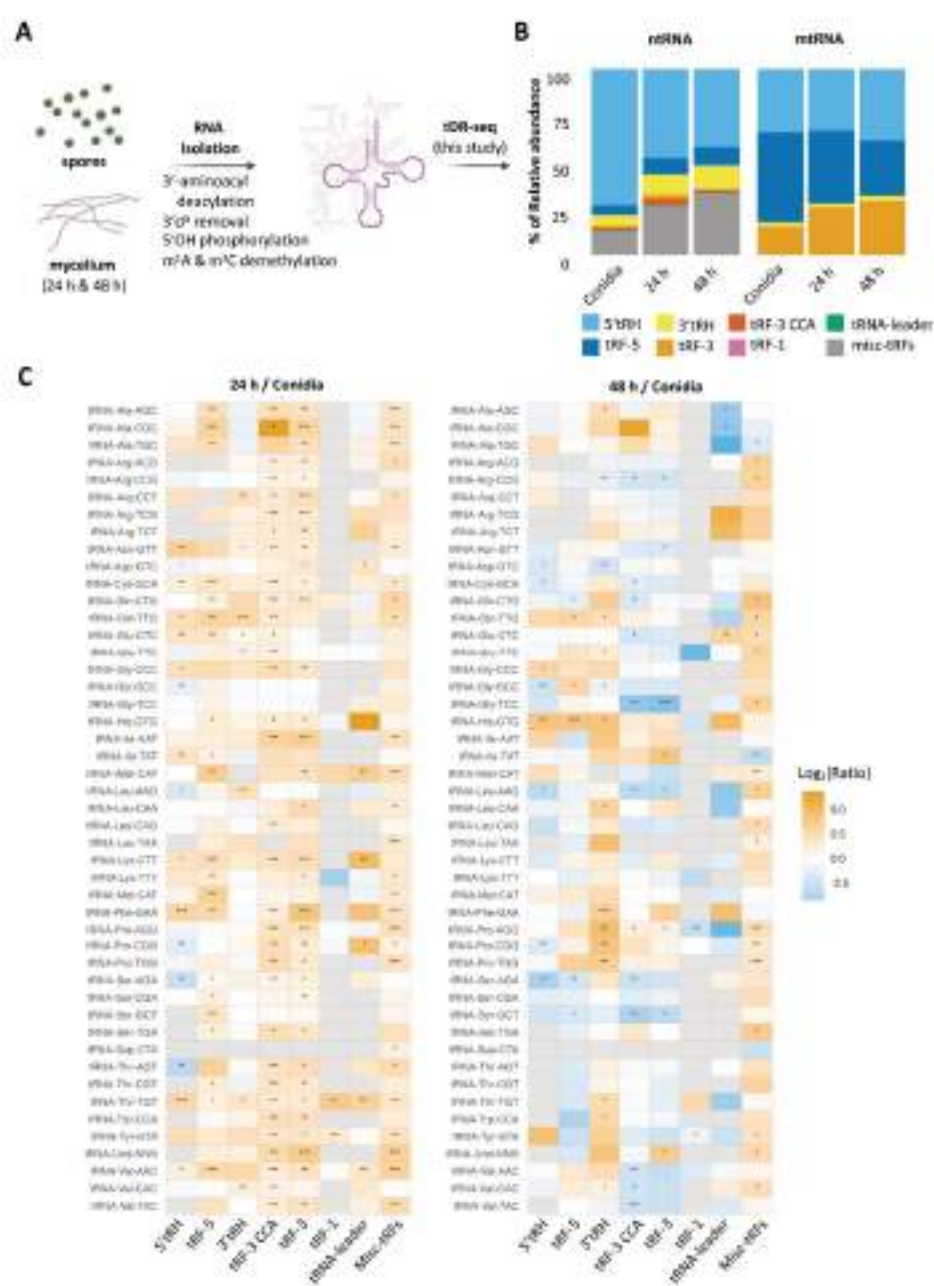


FIGURE 3 | Legend on next page.

at 48-h compared to 24-h mycelium and conidia (Figure S8A). After unitas analysis of tDR-seq, a larger relative proportion of reads mapped to tRNA fragments (tRFs) than in the sRNA-seq

however, a closer look at the distribution of tDRs indicated a higher proportion of tRFs compared to the sRNA-seq analysis for the nuclear-encoded tRNAs (Figure S8B; Data S5). As

FIGURE 3 | Dynamics of tDR expression in *A. fumigatus* conidia and mycelium revealed by tDR-seq. (A) RNA samples ($n = 3$) were collected from wild-type *A. fumigatus* conidia and hyphae. Samples were pre-treated to remove specific modifications and resolve tDR ends prior to library preparation. RNA molecules of up to 50 nt were selected for tDR-seq using Illumina sequencing. Created in BioRender: Blango, M. (2025) <https://biorender.com/e74f017>. (B) Relative abundance of different tDRs across samples for tDRs derived from nuclear- (nRNA) and mitochondria-encoded (mtRNA) tRNAs. (C) Heatmap showing ratio of abundance of nuclear-encoded tDR species and their tRNA of origin (with reads collapsed into a single barcode) for mycelium and conidia timepoints from tDR-seq dataset. Plots show ratio of counts per million mapped reads based on fractionated counts produced with unitary $\times 1.74$ (Gibert et al., 2017) for mycelium (24-h or 48-h) versus conidia. *Denotes significant differences as defined in the methods; $n = 3$ biological replicates per condition.

compared to the tDR distribution from sRNA-seq (Figure 2D), tDR-seq revealed more 5' and 3' tRHs relative to other tDRs, consistent with an approach that fixes RNA ends prior to library preparation (Figure 3B). Interestingly, the relative abundance of 5' tRHs with nuclear origin was much higher in the conidia (~75%) after the tDR-seq approach (Figure 3B). Further analysis revealed that a majority (~73%) of tDRs identified in the conidia came from a particular Asp(GTC) nuclear-encoded tRNA, with the 5' tRH being the most abundant fragment (Data S5). The length of the tRNA fragments was more consistent with tRHs for the nuclear-encoded tRNAs, but the mitochondria-encoded tDRs had a wider range, with a relative enrichment of smaller fragments in hyphae (Figure S8C), consistent with the literature (Isakova et al., 2020).

A heat map of tDR abundances for each tRNA again revealed substantial differences in abundance and fragment distribution between tRNAs with different or even the same isocodons (Figure S9, Data S5). Even though many of the detected tRHs derive from Asp(GTC), other tRNAs like His(GTG) and Pro(CGG) also produce a relatively high relative abundance of tRHs. Many of the nuclear-encoded tRNAs appeared to have stable tRF-3 CCA fragments, with the previously identified Tyr(GTA)-tRF-3 CCA being among the most abundant fragments (Figure S9; Kelani et al., 2023). The counts per million reads of tDRs generally increased in 24-h mycelium relative to conidia, as demonstrated by tDR ratios between the morphotypes (Figure 3C). Collectively, these results corroborate that the sRNA-seq data that the *A. fumigatus* small transcriptome consists mostly of sRNAs derived from structural RNAs (e.g., rRNAs and tRNAs) under standard growth, with different relative abundances in the small RNA fraction depending on the morphotype.

2.5 | dsRNA Overexpression Limits *A. fumigatus* Growth Independent of Dicer-Like Proteins

Our observation that the small RNAs produced from the *pksP* IRT had no predicted off-target sites aside from the target *pksP* mRNA led us to hypothesize that high levels of dsRNA may itself be detrimental to *A. fumigatus*, perhaps through competition for substrates with endogenous pathways. In fact, *A. fumigatus* has 10 annotated dsRBPs (FungiDB.org; Alvarez-Jarreta et al., 2024), with 5 being putative RNase III enzymes (InterPro). These contain two Dicer-like proteins (DclA, AFUB_099350; DclB, AFUB_100200; Table S1), a mitochondrial RNase III enzyme that is an ortholog of MRPL3 from *N. crassa* (AFUB_012260; Lee et al., 2010), the nucleolar RNase III enzyme Rnt1 (AFUB_083000), and a small RNase III domain-containing protein of unknown function found

only in *A. fumigatus* and a few close relatives (AFUB_054390) (Figure S10A). The five other dsRBPs are less well-studied, but in several cases have roles in rRNA assembly or function (Ribosomal Protein S5, AFUB_088030; 37S Ribosomal Protein S5, AFUB_059110; RAD52 ortholog, AFUB_064040; AFUB_031820; and AFUB_067420). Two of these dsRBPs are predicted to reside in the mitochondria (MRPL3; 37S Ribosomal Protein S5), with the others likely found in the cytoplasm and/or nucleus. Expression profiling from previously published mRNA-seq data on conidia, 24-h mycelium, and 48-h mycelium revealed stage-specific expression of many of these genes often influenced by expression of the core RNAi components (Kelani et al., 2023 and Figure S10B), suggesting that specific subsets of proteins might compete for dsRNA binding at different growth stages.

We tested the contribution of sRNA production to the observed growth inhibitory effect of the *pksP* IRT in liquid culture. As expected, induction of the inverted-repeat transgene in the wild-type strain limited fungal biomass as before (Kelani et al., 2023), but surprisingly, this phenotype was also similar in the $\Delta dclA/B$ strain after induction, albeit less pronounced (Figure 4A). Low expression of dsRNA via leaky TetON induction in the knock-out strain resulted in a mild decrease in growth even in the uninduced RNAi double knockout strain, suggesting that dsRNA may itself be growth inhibitory. Although the untreated culture of the $\Delta dclA/B$ strain was significantly decreased compared to the control strains, there remained a difference between the induced and uninduced condition in the $\Delta dclA/B$ *pksP* IRT strain. As a control, we grew each of these strains in the absence of the inverted-repeat transgene, with no major differences in growth observed for the $\Delta dclA/B$ double knockout strain (Figure S10C). Overexpression of the *pksP* IRT was also inhibitory in the single *dclA* and *dclB* deletion strains, again with a stronger inhibition upon induction with doxycycline compared to the wild type (Figure 4B, Figure S10D), indicating that the observed effect was not due to the pairwise deletion of the dicer-like proteins. Future study will be required to tease apart the complete regulation of dsRNA in the sRNA-limited environment of *A. fumigatus* under laboratory growth conditions.

3 | Discussion

Small ncRNAs are key regulators of gene expression and stress response in many organisms, including fungi. Here, we provide an improved description of the sRNA landscape of *A. fumigatus* by leveraging sRNA-seq and tDR-seq. The pool of *A. fumigatus* tDRs appears to be heterogeneous, dynamic, and primarily independent of the canonical RNAi machinery under standard

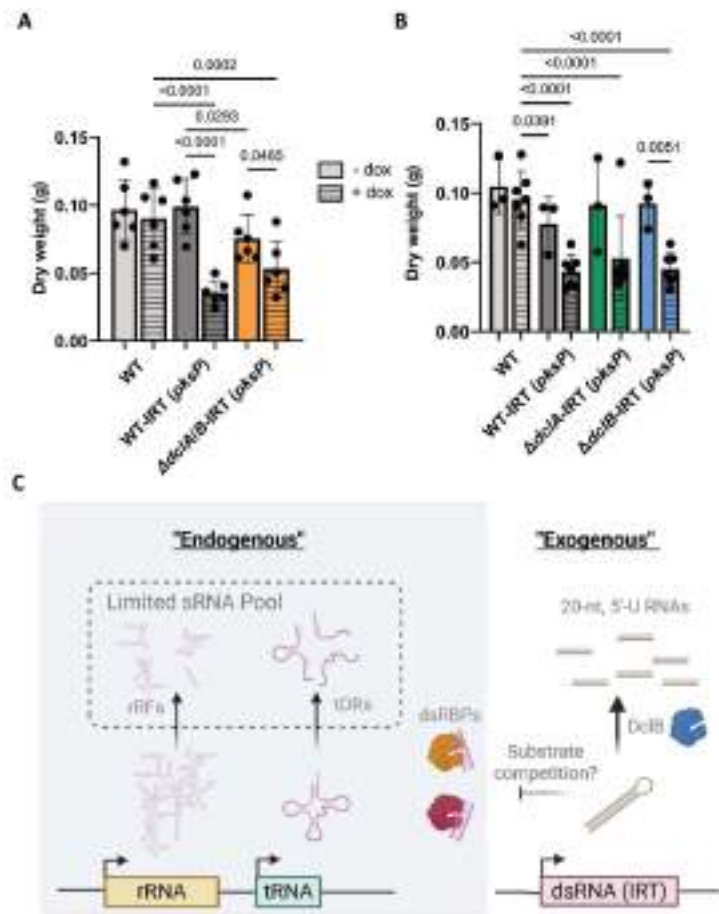


FIGURE 4 | Excess double-stranded RNA induces growth arrest. (A) Measurement of fungal dry weight (biomass) after 24-h growth in liquid culture in the presence or absence of doxycycline as an inducer of *pksP*IRT in the RNAi knockout strains or compared to a wild-type strain. Relevant comparisons were performed with two-way ANOVA and Sidak's multiple comparisons posttest from six biological replicates representing the average of two technical replicates each. (B) Measurement of fungal dry weight (biomass) after 24-h growth in liquid culture in the presence or absence of doxycycline as an inducer of *pksP*IRT in the *dcl* single-knockouts or wild-type strain and a wild-type strain lacking the IRT. Relevant comparisons were performed with two-way ANOVA and Sidak's multiple comparisons posttest from at least three biological replicates representing the average of two technical replicates each. (C) Working model of small RNA landscape in *A. fumigatus*. High levels of exogenous dsRNA can be induced to produce 20-nt 5'-U-containing RNAs and sequester dsRBPs from normal functions to limit growth. Created in BioRender (<https://BioRender.com/r33p645>). Blango, M. (2026) <https://BioRender.com/r33p645>.

growth conditions. We took advantage of the limited endogenous sRNA landscape to assess the features of the sRNAs produced by *A. fumigatus* upon induction of an inverted-repeat transgene. We were not surprised to observe an accumulation of 5'-U-containing 20-nt sRNAs derived from the *pksP*IRT transgene, but it was notable that these sRNAs were primarily generated from two consecutive loci in the 3' half of the 500-bp dsRNA. We noted that sRNAs were produced in the wild-type and *ΔdclA* IRT strains, but not the *ΔdclB* or *ΔdclA/B* strains, corroborating our previous finding that DclB is required for

small RNA generation from the IRT in *A. fumigatus*. A second round of sequencing of a previously created *pksP* IRT expressing strain again showed an enrichment of 5'-U-containing 20-nt sRNAs from primarily the 3' end of the transcript, suggesting a conserved feature of hairpin cleavage by DclB.

We performed sRNA-seq to evaluate the dynamics of small ncRNAs during two distinct morphological phases of *A. fumigatus* growth and identified expected classes of small ncRNAs, with reads derived from rRNA and tRNA exhibiting the highest

abundance in conidia and mycelium. Although a potential sign of RNA degradation, our previous experiences with *A. fumigatus* in combination with our quality control steps and the literature (Jochl et al. 2008), suggest that rRNA- and tRNA-derived RNAs do accumulate over the lifecycle, particularly at later growth timepoints (e.g., 48-h mycelium). It remains possible that artifacts could be introduced by the RNA isolation method (Kim et al. 2012), a feature which should be more systematically explored in the future. Interestingly, the abundance of tDRs observed in our sRNA-seq and tDR-seq did not appear to correlate with our previous measurement of the overall tRNA abundances from mycelium under the same growth conditions using Nano-tRNAseq, where for example, Ser(GCT) was found to be the most abundant tRNA isocodexon (Bruch et al. 2025). This supports the hypothesis that tDR biogenesis is a regulated process in *A. fumigatus*, and not solely a product of RNA degradation (Jochl et al. 2008).

Some previous studies have indicated that RNAi components like Dicer play a role in tDR biogenesis (Babiarz et al. 2008; Cole et al. 2009; Haussecker et al. 2010), but our results suggest that *A. fumigatus* Dicer-like proteins play a minor role under standard growth conditions. As compared to wild type, we did observe a slight decrease in the production of some tDRs in the conidia of $\Delta pdcA/B$ and an increase at later growth stages (i.e., 48-h-old hyphae). Argonaute proteins are known to stabilize tDRs in some systems (Kuscu et al. 2018), and it remains possible that the total pool of mature tRNAs susceptible to cleavage is lower in the $\Delta pdcA/B$ strain, based on the downregulation of genes associated with transcription observed previously (Kelani et al. 2023). It is worth noting that we did not observe loss of sRNAs in the $\Delta pdcA/B$ *pkdP* IRT sRNA-seq experiment (Figure S2D), suggesting that the argonautes may not play a major role in sRNA stabilization in *A. fumigatus*. Further study will be required to assign the enzymes responsible for fragment generation.

To better assess the tDR repertoire of *A. fumigatus*, we performed tDR-seq (Zhang et al. 2011), which revealed an enriched pool of tDRs compared to sRNA-seq consistent with the literature (Jochl et al. 2008). The tDR-seq dataset showed an abundance of tRNA reads, a smaller fraction of tDRs, and other RNAs consistent with previous work (Gustafsson et al. 2025). We were not surprised to observe an increase in tDR detection using tDR-seq compared to sRNA-seq (Figure S8), but it was somewhat surprising how strong this enrichment was for conidia compared to the mycelium conditions. It will be interesting in the future to more accurately compare the end status of RNAs from each of these conditions, as this result may suggest interesting biochemistry in the conidia, perhaps for longer term protection of the RNAs.

We next assessed the ability of overexpression of the inverted-repeat transgene to induce growth arrest in liquid culture as previously observed in (Kelani et al. 2023), but now in the absence of dicer-like protein components. We were surprised to see that growth was limited, even when the RNAi system was disrupted by deletion of the dicer-like proteins. The growth disruption in the $\Delta dclA/B$ and $\Delta dclB$ strains incapable of producing sRNAs (Figure 4A,B) suggested that the growth disruption is independent of sRNA production and that off-target effects of

the sRNAs are not the major driver of growth arrest. Instead, we hypothesized that the dsRNA itself may serve as a signal for growth arrest. In this case, high levels of dsRNA may compete with dsRBPs for endogenous substrates and impede normal biology, akin to an actively replicating retrotransposible element (Figure 4C). Such a mechanism could offer a natural brake on growth in response to abnormally high levels of dsRNA and give an advantage to unaffected cells, but this remains speculative. Additional work using complementation with catalytically inactive RNAi machinery will be required to fully understand the mechanism(s) of action. Of note, previous work showed that overexpression of a single-stranded mRNA in similar growth conditions does not lead to growth inhibition (Shopova et al. 2020), providing additional support for a potential inhibitory role of dsRNA itself.

In conclusion, our study (1) reveals DclB-dependent production of canonical 20-nt, 5' U-containing small RNAs by *A. fumigatus* upon dsRNA induction; (2) further defines the dynamics of the limited pool of endogenous small RNAs consisting largely of sRNA and tRNA-derived RNAs using sRNA- and tDR-seq; and (3) provides an intriguing hypothesis for a link between dsRNA metabolism and growth arrest in *A. fumigatus* independent of sRNA production that should be investigated further in the future.

4 | Experimental Procedures

4.1 | Strains, Culture Conditions, and Growth Assays

Throughout this study, *A. fumigatus* strains (Table S2) were cultivated on *Aspergillus* minimal medium (AMM) agar plates and liquid culture as described previously (Kelani et al. 2023). It is important to note that the strain termed wild type here lacks nonhomologous end joining, which may have unintended impacts on the biology observed. For measuring the dry weight, *A. fumigatus* strains were cultivated on AMM-agar plates for 5 days, then 1×10^6 spores were added to 30-mL AMM liquid culture. Strains with inverted repeat elements were supplemented with 10 μ g/mL of doxycycline. Flasks with AMM liquid culture were shaken at 200 rpm at 37°C in the dark for 24 h. Mycelium was collected through Miracloth (Millipore) and dried in the oven at 65°C for 7 days prior to measurement of the dry weight. For the sRNA-seq of inverted-repeat transgene strains, 1×10^6 freshly harvested spores were inoculated into 50 mL AMM and incubated at 37°C and 200 rpm. After 16 h, 10 μ g/mL of doxycycline was added to the denoted inverted-repeat transgene strains, followed by 8 h of additional growth (total of 24 h).

The DclB complementation plasmid was created using primers defined in Table S3 (plasmid map in Figure S3). Fungal protoplasts were created of the $\Delta dclB$ + IRT (*pkdP*) strain and transformed with linearized plasmid according to the protocol described in (Kelani et al. 2023). Colonies were selected for integration of the pyrithiamine resistance gene (*ptr4*) in the safe haven 2 (SH2) region; Pham et al. 2020). After three restreaks, DNA was collected and southern blots performed according to (Kelani et al. 2023), using restriction enzymes *NruI* and *KpnI*.

and flanking sequence as probe. Positive clones were confirmed in growth assays after ~40h by assessing for the presence of silencing after complementation of *dcB* (white colonies indicate functional silencing of *pkpA* gene).

4.2 | RNA Isolation for IRT-sRNA-Seq

For IRT-sRNA-seq, RNA was isolated as in (Kelani et al. 2023) with several improvements. Collected mycelium was first transferred to a 2-mL screw-cap tube filled 1/3 full with 0.5 mm soda lime glass beads (BioSpec). One milliliter of TRIzol (Thermo-Fisher Scientific, Dreieich, Germany) was added to efficiently lyse the cells using the FastPrep-24 Homogenizer by subjecting the cells to three rounds of 30s, 4.0 m/s homogenization (1 min break between repetitions) and 5 min of incubation at room temperature. Two hundred microliters of chloroform was added to each sample, vigorously mixed and incubated for 3 min at room temperature before centrifuging the samples at 12,000xg for 15 min at 4°C. The upper aqueous phase containing the RNA was carefully transferred to a clean RNase-free tube and an equal volume of pure ice-cold ethanol was added and mixed thoroughly. Purification of RNA was carried out by using the RNA Clean and Concentrator kit (Zymo Research GmbH, Freiburg, Germany) following the manufacturer's instructions. To eliminate DNase contamination, total RNA was treated with DNase I, RNase-free (Thermo-Fisher Scientific, Darmstadt, Germany) at 1U per µg of RNA. Following a 1-h incubation at 37°C, the reaction was terminated by adding EDTA (5 mM final concentration) at 65°C for 10 min and total RNA was collected using the RNA Clean and Concentrator-25 kit (Zymo Research) following the manufacturer's instructions. DNase-free RNA concentrations were measured with the Qubit RNA BR Assay kit (Thermo-Fisher Scientific, Darmstadt, Germany). Purified RNA was quantified using the Qubit RNA BR Assay kit (Thermo-Fisher Scientific, Darmstadt, Germany) on the Qubit Flex Fluorometer.

4.3 | RNA Isolation for sRNA- and tDR-Seq

RNA isolation was performed as described previously (Kelani et al. 2023). Briefly, fungal mycelium was harvested from liquid culture using Miracloth (Millipore) and then disrupted in liquid nitrogen with a precooled mortar and pestle. Approximately 0.5 g of homogenized mycelium was transferred to a 2-mL Eppendorf tube. To this, 800 µL of TRIzol was added and vigorously vortexed. The tubes were briefly frozen in liquid nitrogen and then allowed to thaw on ice. Following thawing, 160 µL of chloroform was added to the mixture, vortexed, and centrifuged for 5 min at 4°C at maximum speed. The resulting aqueous upper phase was carefully transferred to a fresh 2-mL tube without disturbing the interphase. RNA extraction from the aqueous phase involved the addition of 1 volume of phenol:chloroform:isoamyl alcohol (25:24:1, v/v), followed by brief vortexing and centrifugation for 5 min at 4°C. This extraction process was repeated until no interphase was visible, and an additional extraction with 400 µL chloroform was performed. RNA was precipitated by adding 400 µL isopropanol for 20 min, followed by centrifugation for 20 min at 4°C. The resulting pellet was washed with 700 µL

70% ethanol and air-dried at 37°C for 5 min before being resuspended in RNase-free water. To remove any remaining DNA, a DNase treatment was carried out using 2 units of TURBO DNase (Thermo-Fisher) per 10 µg RNA for 30 min at 37°C in a total volume of 100 µL. The resulting DNase-free RNA was then processed as described above.

4.4 | Library Preparation and Sequencing

sRNA-seq libraries were prepared by Novogene after RNA quality assessment (Figures S2A-C and S4A) using the TruSeq Small RNA Library Preparation Kit according to the manufacturer's protocol, then sequenced using the NovaSeq 6000 instrument in a single-read, 50-bp based mode at 10 million reads of sequencing depth. For tDR-seq performed by CD Genomics (tRF&tRNA-seq), the following treatments for modifications were performed: (1) 3'-aminoacyl (charged) deacylation to 3'-OH for 3' adaptor ligation, (2) 5'-cP removal to 5'-OH for 3' adaptor ligation, (3) 5'-OH (hydroxyl group) phosphorylation to 5'-P for 5' adaptor ligation, and (4) m¹A and m³C demethylation for efficient reverse transcription. Finally, RNA libraries were sequenced using an Illumina HiSeq 2500 instrument in a 50-bp single-read mode at 10 million reads of sequencing depth.

4.5 | Data Analysis

TrimGalore version 0.6.8 (<https://github.com/FelixKruenger/TrimGalore>) was used for quality checks and removing adapters of sequencing reads. SortMeRNA version 4.3.7 (Kopylova et al. 2012) was used to remove potential rRNA reads, with the rRNA database generated from Rfam (<https://rfam.org/>) and silva (<https://www.arb-silva.de/>). In silico removal of rRNA reads did not alter analysis results and was subsequently omitted for downstream analyses. Sequencing data were analyzed with unitas v. 1.7.0 (Gebert et al. 2017) using default parameters and *A. fumigatus* A1163 ASM15014v1 as the reference genome. *A. fumigatus* nuclear tRNAs were referenced from tRNAdb (Thornlow et al. 2020), mitochondrial tRNAs were scanned from the mitochondrial genome (Joardar et al. 2012) using the 27 tRNAs predicted by tRNAscan-SE (Chan et al. 2021). All analyses were performed using "R" version 4.3.1 (2023-06-16) and Rstudio (2022-02-02), an integrated development environment (IDE) for R or Prism 10.4.0 software (GraphPad Software). Figures were made with the ggplot2 package (version 3.4.2) in R. Graphs for read coverage were created using the Spark package (Kurtenbach and Harbour 2019). P-values were determined by ANOVA or two-tailed Student's *t*-tests where appropriate. For the tRNA fragment heatmaps, pairwise comparisons were performed using linear mixed-effects models (LMMs) with sample as a fixed effect and replicate as a random effect. When the LMMs were singular or failed to converge, fixed effects linear models were employed instead. We computed post hoc pairwise contrasts using the emmeans package (Lenth 2025), and controlled for multiple comparisons using the Benjamini-Hochberg method. In all cases, differences between the groups were considered significant at a *p*-value of <0.05. **p*-value of <0.05, **<0.01, ***<0.001.

Author Contributions

Xiaoqing Pan: conceptualization, experimental work, data analysis and interpretation, writing – original draft, writing – review and editing. **Abdulrahman A. Kolani:** conceptualization, experimental work, data analysis and interpretation, writing – original draft, writing – review and editing. **Lukas Schrottenbrunner:** experimental work. **Swatika Prabhakar:** experimental work. **Bhawana Israni:** experimental work. **Matthew G. Blanga:** conceptualization, funding acquisition, data analysis and interpretation, writing – original draft, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The sRNA-seq data are available under NCBI Gene Expression Omnibus (GEO) identifier GSE267454. These data were collected simultaneously with previously published mRNA-seq data (Kolani et al. 2023) available using identifier GSE229618. The tDR-seq data are available under identifier GSE267453. The sRNA-seq for the inverted-repeat transgene overexpression is available using GSE286229 and GSE300894.

References

- Alvarez-Jarrett, J., B. Amos, C. Asencio-Cabeza, et al. 2024. "VETuPathDB: The Eukaryotic Pathogen, Vector and Host Bioinformatics Resource Center in 2023." *Nucleic Acids Research* 52: D806–D816.
- Babiarz, J. E., J. G. Bojby, Y. Wang, D. P. Bartel, and R. Bellocq. 2008. "Mouse ES Cells Express Endogenous siRNAs, siRNAs, and Other Microprocessor-Independent, Dicer-Dependent Small RNAs." *Genes & Development* 22: 2773–2785.
- Bilimiyev, R. B., S. Caki, M. Peretzaki, X. Wang, and J. Heitman. 2013. "RNAi Function, Diversity, and Loss in the Fungal Kingdom." *Chromosomes Research* 21: 561–572.
- Bruch, A., V. Lazarova, M. Berg, et al. 2023. "tRNA Hypomodification Facilitates 5-Fluorocytosine Resistance via Cross-Pathway Control System Activation in *Aspergillus fumigatus*." *Nucleic Acids Research* 51: gkac205.
- Chan, F. F., B. Y. Lin, A. J. Mak, and T. M. Lowe. 2021. "tRNAscan-SE 2.0: Improved Detection and Functional Classification of Transfer RNA Genes." *Nucleic Acids Research* 49: 9077–9096.
- Cole, C., A. Sobala, C. Lu, et al. 2009. "Filtering of Deep Sequencing Data Reveals the Existence of Abundant Dicer-Dependent Small RNAs Derived From tRNAs." *RNA* 15: 2147–2160.
- Cornell, K. A., R. J. Andrews, and B. L. Bass. 2024. "The Competitive Landscape of the dsRNA World." *Molecular Cell* 84: 107–119.

- Couvillier, M. T., R. Sachidanandam, and K. Collins. 2010. "A Growth-Essential Tetrahymena P160 Protein Carries tRNA Fragment Cargo." *Genes & Development* 24: 2762–2767.
- Dorovan, J., S. Rath, D. Kotli-Mandrikov, and A. Korniykh. 2017. "Rapid RNase L-Driven Arrest of Protein Synthesis in the dsRNA Response Without Degradation of Translation Machinery." *RNA* 23: 1600–1671.
- Fisher, M. C., and D. W. Bessing. 2023. "The WHO Fungal Priority Pathogens List as a Game-Changer." *Nature Reviews Microbiology* 21: 211–212.
- Gehert, D., C. Hessel, and D. Rosenkranz. 2017. "tRNAs: The Universal Tool for Annotation of Small RNAs." *BMC Genomics* 18: 644.
- Gentofen, H. T., L. Porgason, C. Galen, et al. 2023. "Deep Sequencing of Yeast and Mouse tRNAs and tRNA Fragments Using OTTH." *eLife* 14: e77604.
- Heiser, H. J., F. V. Kargl, G. J. Hannon, and M. A. Elliot. 2008. "Developmentally Regulated Cleavage of tRNAs in the Bacterium *Streptococcus meliloti*." *Nucleic Acids Research* 36: 732–741.
- Handzik, J. E., S. Tsatsoglou, I. S. Vlachos, and A. G. Hatzigeorgiou. 2020. "Manual Detection and Quantification of Small Non-Coding RNAs From Next-Generation Sequencing Data." *Scientific Reports* 10: 705.
- Hauswirth, D., Y. Huang, A. Liu, P. Parameswaran, A. Z. Fire, and M. A. Kay. 2010. "Human tRNA-Derived Small RNAs in the Global Regulation of RNA Silencing." *RNA* 16: 673–685.
- Isakova, A., T. Fehrmann, A. Keller, and S. R. Quake. 2020. "A Mouse Tissue Atlas of Small Noncoding RNA." *Proceedings of the National Academy of Sciences of the United States of America* 117: 25634–25643.
- Ivanov, P., M. M. Elmar, J. Villen, S. P. Gygi, and P. Anderson. 2011. "Angiostatin-Induced tRNA Fragments Inhibit Translation Initiation." *Molecular Cell* 43: 613–623.
- Jasdar, V., N. F. Abrams, J. Hoisler, et al. 2012. "Sequencing of Mitochondrial Genomes of Nine *Aspergillus* and *Penicillium* Species Identifies Mobile Introns and Accessory Genes as Main Sources of Genome Size Variability." *BMC Genomics* 13: 698.
- Jeckl, C., M. Rederhoff, J. Hertel, et al. 2008. "Small ncRNA Transcriptome Analysis From *Aspergillus fumigatus* Suggests a Novel Mechanism for Regulation of Protein Synthesis." *Nucleic Acids Research* 36: 2677–2689.
- Kolani, A. A., A. Bruch, F. Rivicci, et al. 2023. "Disruption of the *Aspergillus fumigatus* RNA Interference Machinery Alters the Conidial Transcriptome." *RNA* 29: 1055–1060.
- Kim, Y. K., J. Yeo, B. Kim, M. Ha, and V. N. Kim. 2012. "Short Structured RNAs With Low GC Content Are Selectively Lost During Extraction From a Small Number of Cells." *Molecular Cell* 46: 893–895.
- Kopylova, E., L. Noe, and H. Touny. 2012. "SortMeRNA: Fast and Accurate Filtering of Ribosomal RNAs in Metatranscriptomic Data." *Bioinformatics* 28: 3211–3217.
- Kurtenbach, S., and J. W. Harbauer. 2019. "SpurK: A Publication-Quality NOS Visualization Tool." *bioRxiv*, 843524.
- Kwack, C., P. Kumar, M. Kiran, Z. Su, A. Malik, and A. Datta. 2018. "tRNA Fragments (tRFs) Guide Ago to Regulate Gene Expression Post-Transcriptionally in a Dicer-Independent Manner." *RNA* 24: 1093–1105.
- Lee, H. C., L. Li, W. Gu, et al. 2010. "Diverse Pathways Generate microRNA-Like RNAs and Dicer-Independent Small Interfering RNAs in Fungi." *Molecular Cell* 38: 803–814.
- Lenth, R., and J. Piskowski. 2023. "emmeans: Estimated Marginal Means, aka Least-Squares Means." R package version 2.0.1.

- Laddicot, B. J., R. Piskol, A. M. Chalk, et al. 2015. "tRNA Editing by ADAR1 Prevents MDA5 Sensing of Endogenous dsRNA as Nonself." *Science* 349: 1115–1120.
- Loveland, A. B., C. S. Koh, R. Ganesan, A. Jacobson, and A. A. Kozlov. 2024. "Structural Mechanism of Angiogenin Activation by the Ribosome." *Nature* 630: 769–776.
- Lukhina, N., and R. Parker. 2010. "T2 Parity Ribonuclease: Ancient Enzymes With Diverse Roles." *Trends in Biochemical Sciences* 35: 253–259.
- Lybecker, M., B. Zimmermann, I. Bleski, N. Tskitubava, and R. Schneider. 2014. "The Double-Stranded Transcriptome of *Escherichia coli*." *Proceedings of the National Academy of Sciences of the United States of America* 111: 3134–3138.
- Maue, R. L., C. Schneider, P. Samarin, et al. 2013. "tRNA-Derived microRNA Modulates Proliferation and the DNA Damage Response and Is Down-Regulated in B Cell Lymphoma." *Proceedings of the National Academy of Sciences of the United States of America* 110: 1404–1409.
- Mi, S., T. Cai, Y. Hu, et al. 2008. "Sorting of Small RNAs Into Arachidonic Acid-Sensitive Complexes Is Directed by the 5' Terminal Nucleotide." *Cell* 133: 116–127.
- Mouyria, L., C. Henry, T. L. Doering, and J. P. Latge. 2004. "Gene Silencing With RNA Interference in the Human Pathogenic Fungus *Aspergillus fumigatus*." *FEMS Microbiology Letters* 237: 317–324.
- Oskan, S., I. Mohorovic, P. Xu, T. Dalmay, and B. H. A. Coutts. 2007. "Profile and Functional Analysis of Small RNAs Derived From *Aspergillus fumigatus* Infected With Double-Stranded RNA Mycoviruses." *BMC Genomics* 18: 436.
- Pham, T., X. Xie, and X. Lin. 2020. "An Intergenic 'Safe Haven' Region in *Aspergillus fumigatus*." *Molecular Mycology* 58: 1178–1186.
- Ragot, R., T. E. Newman, Y. Khetry, L. G. Kamparis, and M. C. Derhyshid. 2023. "Genome-Wide Identification of Sclerotinia sclerotiorum Small RNAs and Their Endogenous Targets." *BMC Genomics* 24: 582.
- Shayeva, J. A., L. Belyaev, P. Duart, et al. 2020. "Human Neutrophils Produce Antifungal Extracellular Vesicles Against *Aspergillus fumigatus*." *RNA* 11: e00596–20.
- Thompson, D. M., C. Lu, B. J. Green, and R. Parker. 2008. "tRNA Charge Is a Conserved Response to Oxidative Stress in Eukaryotes." *RNA* 14: 2055–2103.
- Thompson, D. M., and R. Parker. 2006. "The RNase Rnp1p Cleaves tRNAs and Promotes Cell Death During Oxidative Stress in *Saccharomyces cerevisiae*." *Journal of Cell Biology* 185: 43–50.
- Thorellow, B. P., J. Armstrong, A. D. Holmes, J. M. Howard, R. B. Corbett-Delg, and T. M. Lewis. 2021. "Predicting Transfer RNA Gene Activity From Sequence and Genomic Context." *Genome Research* 31: 93–94.
- Torres-Martinez, S., and R. M. Ruiz-Vergara. 2017. "The tRNA Universe in Fungi: A Varied Landscape of Small RNAs and Biological Functions." *Annual Review of Microbiology* 71: 371–391.
- Tycenewska, A., and K. Grzywa. 2022. "tRNA-Derived Fragments as New Players in Regulatory Processes in Yeast." *Yeast* 40: 283–289.
- Wilson, B. C., and J. A. Doudna. 2013. "Molecular Mechanisms of RNA Interference." *Annual Review of Biophysics* 42: 217–239.
- Zhang, J., L. Li, L. Luo, et al. 2021. "Screening and Potential Role of tRFs and tRNAs Derived From tRNAs in the Carcinogenesis and Development of Lung Adenocarcinoma." *Oncology Letters* 22: 508.
- pdf, **Data S1:** mm70042-sup-0001-DataS1.xlsx, **Data S2:** mm70042-sup-0003-DataS2.xlsx, **Data S3:** mm70042-sup-0004-DataS3.xlsx, **Data S4:** mm70042-sup-0005-DataS4.xlsx, **Data S5:** mm70042-sup-0006-DataS5.xlsx.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** mm70042-sup-0001-AppendixS1.

4.3 Manuscript C

Past, Present, and Future of RNA Modifications in Infectious Disease Research

Xiaoqing Pan*, Alexander Bruch*, Matthew G. Blango

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Manuscript published in *ACS Infectious Diseases Journal* 10(12): 4017-4029; Dec 2024.

The candidate is

☐ First author, ☒ Co-first author, ☐ Corresponding author, ☐ Co-author.

Summary:

RNA modifications serve as an important post-transcriptional regulator for gene regulation, profoundly shaping RNA functionality. This review article underscores the urgency of exploring pathogen-specific RNA modification patterns, which vary across infections and may reveal novel mechanisms of immune evasion or host adaptation. Additionally, it identifies key challenges, such as technical limitations in mapping modifications and interpreting their context-dependent roles, to address these barriers.

Contribution to the manuscript:

Xiaoqing Pan co-contributed in writing the review section that address the current understanding of RNA modifications for infectious disease.

Authors' contributions (in %) to the given categories of the publication

Author	Conceptual	Data analysis	Experimental	Writing the manuscript	Provision of material
Xiaoqing Pan	30%	0%	0%	30%	0%
Alexander Bruch	30%	0%	0%	30%	0%
Matthew G. Blango	40%	0%	0%	40%	100%
Total:	100%	0%	0%	100%	100%

Signature candidate

Signature supervisor (member of the Faculty)

Past, Present, and Future of RNA Modifications in Infectious Disease Research

Xiaoqing Pan,^{*} Alexander Bruch,[#] and Matthew G. Blango^{*}

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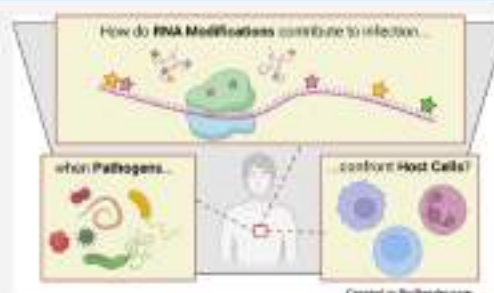
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ABSTRACT: In early 2024, the National Academies of Sciences, Engineering, and Medicine (NASEM) released a roadmap for the future of research into mapping ribonucleic acid (RNA) modifications, which underscored the importance of better defining these diverse chemical changes to the RNA macromolecule. As nearly all mature RNA molecules harbor some form of modification, we must understand RNA modifications to fully appreciate the functionality of RNA. The NASEM report calls for massive mobilization of resources and investment akin to the transformative Human Genome Project of the early 1990s. Like the Human Genome Project, a concerted effort in improving our ability to assess every single modification on every single RNA molecule in an organism will change the way we approach biological questions, accelerate technological advance, and improve our understanding of the molecular world. Consequently, we are also at the start of a revolution in defining the impact of RNA modifications in the context of host–microbe and even microbe–microbe interactions. In this perspective, we briefly introduce RNA modifications to the infection biologist, highlight key aspects of the NASEM report and exciting examples of RNA modifications contributing to host and pathogen biology, and finally postulate where infectious disease research may benefit from this exciting new endeavor in globally mapping RNA modifications.

KEYWORDS: RNA modification, RNA editing, m⁶A, fungi, bacteria, virus



NASEM REPORT AS A FRAMEWORK FOR THE ADVANCEMENT OF RNA MODIFICATIONS RESEARCH

The new report from the National Academies of Sciences, Engineering, and Medicine (NASEM) entitled “Charting a future for sequencing RNA and its modifications: A new era for biology and medicine” issues a blueprint for the future of RNA and RNA modification research.¹ The report, coordinated and assembled by a diverse list of international leaders, first provides a historical overview of RNA modification research and its vast impact on our understanding of biology and disease. The authors then go to great lengths to explain the cutting-edge of technologies for mapping RNA modifications and where we are likely to see rapid advances in the near and long-term. In defining the required drivers of acceleration for RNA modifications research, the authors identify the importance of not only technological advances, but also improved chemical standards and computational solutions, e.g., databases, in facilitating this revolution in our understanding of RNA. In addition to the need for advances in technology and reagents, including novel modalities for modification detection, improved biochemical standards, and overhauled or novel

computational approaches, the report finds that we will also require buildup of infrastructure, training of a skilled workforce, and increased public awareness in parallel. The authors end with a vision for the future of RNA modification research, including a series of succinct recommendations for the community and society at large moving forward. The document is expansive and inspirational in its scope for the future of RNA research, covering everything from the basic biology and biochemistry of RNA modifications to a concrete plan for coordination and allocation of resources and effort. It will certainly serve as a tool for education, a guide for policy makers, and a roadmap for future RNA researchers. Collectively, the NASEM report is a call to arms, encouraging further investment in RNA and RNA modification research

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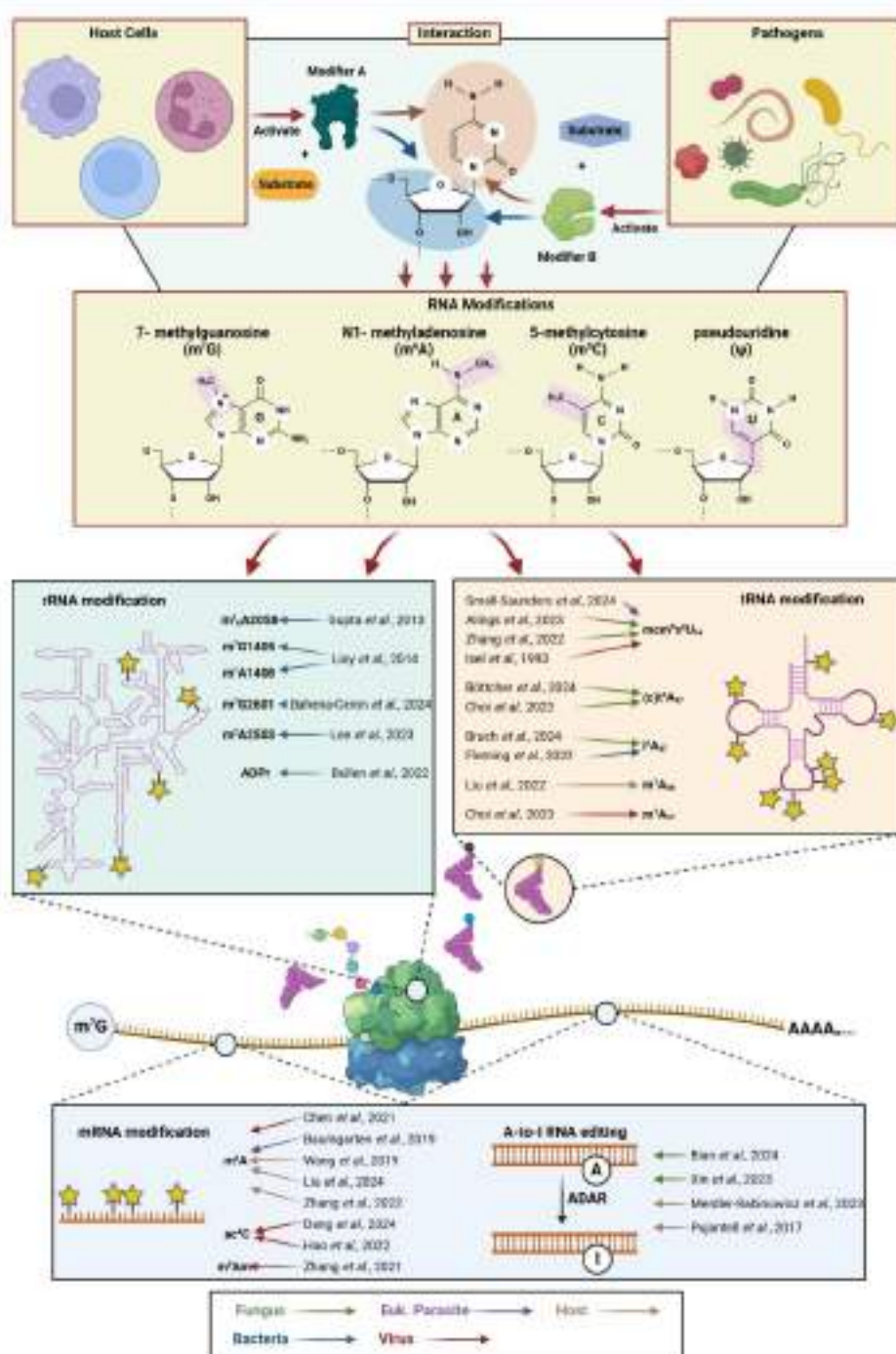


Figure 1. RNA modifications mediate microbial pathogenesis and host response. Despite an abundance of research into RNA modifications and microbial pathogenesis, we still know far too little about the importance of RNA modifications for the outcome of the host–pathogen interactions. The chemical structures of several well-described RNA modifications are highlighted (top).⁸ Several examples are given indicating common themes in RNA modification regulation as it relates to microbial pathogenesis and the host response to infection by viruses, bacteria, eukaryotic parasites, and fungi (bottom). Created in BioRender. [Blangu, M. (2024) BioRender.com/552i122].

and global cooperation in improving our understanding of these crucial chemical changes.

Throughout the report, the impact of RNA modifications on disease and infection is touched upon, but as expected from such an expansive report, a full dissection of the field is

impossible. Here, we aim to add to the importance of this seminal report with additional insights into where the field of RNA modification research intersects that of infection biology. We provide a more elaborate description of key references and posit just a few ways that the success of the proposed path from the NASEM report could change the way we consider infectious diseases. We encourage the readers of this perspective to also visit the NASEM report to better appreciate the entire landscape of RNA modifications research, through the lens of many of the very researchers that made seminal discoveries to establish the field.

■ WHAT IS AN RNA MODIFICATION AND HOW DO WE MAP THEM?

RNA is decorated with over 170 known chemical modifications that influence RNA function, stability, structure, and interaction with other macromolecules including nucleic acids and proteins. These chemical modifications range from small alterations like methylations and acetylations to bulky modifications like those of glutamyl-queuosine (GluQ) or 5-carboxymethylaminomethyl-2-geranylthiouridine (cmnm⁵ges²U).² Sometimes these modifications appear to be permanent additions to the RNA, whereas in other cases, modifications like N⁶-methyladenosine (m⁶A) or 4-thiouridine (s⁴U) are reversible and highly regulated.^{3,4} Importantly, a single RNA molecule can harbor multiple modifications, resulting in numerous structures and functionalities. The implication here is that one gene (DNA sequence) can encode for myriad variations of mature RNA molecules based on these chemical decorations, the collection of which is termed the epitranscriptome. It is now appreciated that there is not one epitranscriptome as there is one genome, but instead a collection of epitranscriptomes depending on the cell type, environmental growth conditions, lifecycle stage, etc.¹ tRNAs harbor the largest number of modifications in both prokaryotes and eukaryotes with roughly 14 per molecule on average;⁵ however, rRNA also require abundant modification for proper function. In eukaryotes, modifications are also regularly observed on mRNA and noncoding RNAs, whereas these modifications appear to be less common on the nonstructural prokaryotic RNAs.⁶ We definitely have a long way to go to completely understand all of the RNA modifications within even the simplest of organisms, but the NASEM report encourages the first steps toward defining a single epitranscriptome in full and the long-term goal of being able to even assess more complex and dynamic epitranscriptomes.

RNA modifications can currently be incompletely mapped using a variety of techniques, including mass spectrometry, next-generation RNA-sequencing technologies, and the emerging direct RNA sequencing approaches like the Oxford Nanopore Technologies (ONT) sequencing platform, among many others (Figure 1).⁶ Each of these techniques has strengths and weaknesses, but none are presently able to sequence all RNA modifications on a single RNA to determine the full epitranscriptome, a point firmly made by the NASEM report.¹ The developing ONT sequencing approaches harbor a lot of promise for a future of mapping full epitranscriptomes, but as with any field in the early days, experts are still learning of the biases in the technology and improving de novo modification identification. Single-cell RNA-seq (scRNA-seq) is another approach that is pushing the boundaries of sensitivity and with great potential to improve our ability to assess all RNA modifications on all RNA of one individual cell.

Approaches like these are currently being applied widely to the mapping of RNA modifications in the context of diverse human pathogens, including viruses, bacteria, fungi, and even archaea.^{5,7,8} In the remainder of this perspective, we will introduce some of the more compelling examples of the importance of RNA modifications in host–pathogenesis from the last few decades (Figure 1) and posit where the future of RNA modifications research may take us in combating infectious disease (Figure 2).

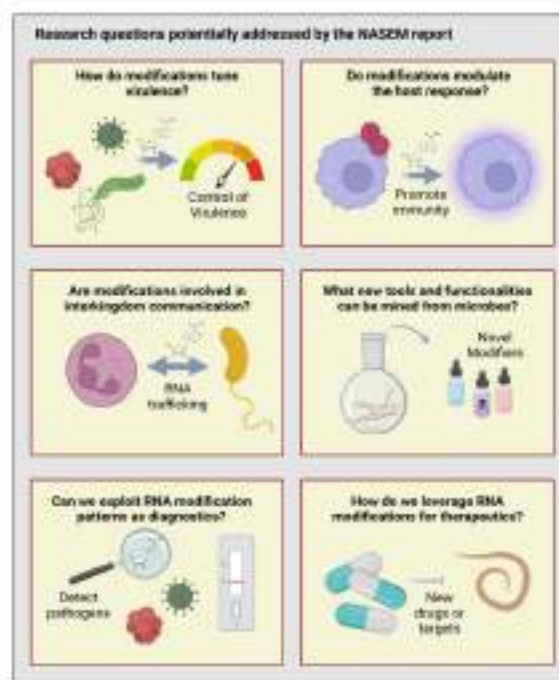


Figure 2. The future of RNA modifications in infectious disease research. A schematic showing where the study of RNA modifications may take us in terms of understanding pathogens and infectious disease, including RNA therapeutics. Created in BioRender. [Blanco, M. (2024) BioRender.com/n17b280].

■ MICROBIAL GENE REGULATION IS MODULATED BY RNA MODIFICATIONS

Microbes have long been a source of pivotal research in RNA modifications, for example the study of *Saccharomyces cerevisiae* led to the first report of an RNA modification, pseudouridine, in 1957.¹⁰ Foundational work followed in abundance in many organisms, including bacteria like *Salmonella enterica*, both a human pathogen and laboratory workhorse.¹¹ Studies of RNA modifications in *Salmonella* led to identification of a link between RNA modification and *his*¹² and *leu*¹³ operon regulation and even supported the idea of flexibility within the genetic code.¹⁴ The field has expanded rapidly since then in many directions, and despite its origins in microbes, myriad intriguing questions remain to be answered surrounding the influence of modifications on pathogenesis, drug resistance, and stress response. The literature already supports the hypothesis that RNA modifications play essential roles in each of these processes, and many more (Table 1). In this article, we will only scratch the surface and demonstrate the

Table 1. Influence of RNA Modifications on Pathogen Stress Resistance and Virulence: Selected Studies of Modifications in Human Pathogens^a

modification	RNA	role(s) in pathogens	pathogen	ref
Bacterial				
m ² A	tRNA	Virulence and stress response controlled by modification of select tRNAs at A ₃₇	<i>E. coli</i> (UPEC)	24
cmcm ⁵ U	tRNA	Pathogenicity controlled by modification of select tRNAs at U ₃₄	<i>P. aeruginosa</i>	28
m ² A	rRNA, tRNA	Reactive oxygen sensing and control of virulence	<i>E. faecalis</i>	35
Dimethylation	rRNA	Dimethylation of 235 A ₂₃₅ promotes altered translation and resistance to erythromycin	<i>S. aureus</i>	30
m ⁷ G	rRNA	Confer aminoglycoside resistance	<i>E. coli</i>	95
m ⁷ G	rRNA	Methylation of the 235 at G ₂₃₅ modulates virulence, growth, and biofilm formation	<i>S. aureus</i>	31
s ² U	tRNA	Control of intracellular growth	<i>M. tuberculosis</i>	26
Quinosine	tRNA	LPS production	<i>E. coli</i> (ST131)	27
ADP-ribose	tRNA, rRNA	Bacterial RlsP2 functions as RNA-modifying effector protein to disrupt host structured noncoding RNAs	<i>P. aeruginosa</i>	90
Multiple	rRNA, tRNA	Numerous modification enzymes control resistance and susceptibility to a broad range of drugs	<i>V. cholerae</i>	29
Fungal				
i ⁶ A	tRNA	Stress response and drug resistance influenced by modification of A ₃₇	<i>A. fumigatus</i>	21
cmcm ⁵ U	tRNA	In vivo virulence via modification of select tRNAs at U ₃₄	<i>S. cerevisiae</i>	17
i ⁶ A	tRNA	Influences adhesion and invasion	<i>C. albicans</i>	18
i ⁶ A	tRNA	Regulation of virulence factor production	<i>C. neoformans</i>	96
A-to-I	mRNA	A-to-I editing contributes to the sexual cycle	<i>F. graminearum</i>	97, 98
Multiple	tRNA	tRNA modifications are stable in response to ionizing radiation	<i>C. neoformans</i>	99
Viral				
cmcm ⁵ U	tRNA	Modulate viral reverse transcription	HIV-1	100
m ² A	Viral RNA	Maintains viral RNA stability and translation	HIV-1	101
m ² A	Viral RNA	ALKBH5 coordinates viral and cellular response to low oxygen	Hepatitis B Virus	102
m ² A	Viral sRNA	Modification of viral-encoded small RNA promotes replication	Parvovirus	103
m ² A	Viral RNA	Modification promotes virus evasion of innate immune sensing	HIV-1	104
m ² A	tRNA	Virus elicits modification removal to promote tRNA fragment production	RSV	105
2'-O-Me	Viral RNA	Internal modifications of viral genome promote escape	HIV-1	106
m ² C	Viral RNA	Necessary for Aly/REF export factor recognition to promote mRNA export and translation	Hepatitis B Virus	107
Parasites				
cmcm ⁵ U	tRNA	Regulation of s ² U impacts artemisinin resistance	<i>P. falciparum</i>	108
m ² A	mRNA	Altered m ² A levels over asexual growth cycle	<i>P. falciparum</i>	18
Quinosine	tRNA	Promotes oxidative stress response and represses virulence	<i>E. histolytica</i>	39

^aAbbreviations: m²A₃₇, 2-methylthio-*N*⁶-isopentenyladenosine 37; i⁶A₃₇, *N*⁶-isopentenyladenosine 37; m²A, *N*⁶-methyladenosine; cmcm⁵U₃₄, 5-methoxycarbonyl-methyl-2-thiouridine 34; cmcm⁵U₃₄, 5-carboxymethylaminomethyluridine 34; m²A, *N*⁶-methyladenosine; m⁷G, *N*⁷-methylguanosine; s²U, 2-thiouridine; i⁶A, *N*⁶-threonylcarbamoyladenosine; A-to-I, adenosine to inosine; m²A, *N*⁶-methyladenosine; 2'-O-Me, 2'-O-methylation

amazing potential of RNA modifications in regulation of gene expression. We focus primarily on human pathogens, drawing also from studies performed using cell culture systems and mouse models in later sections. First, we will look at the pathogens themselves, as much work has already been done to understand how microbes control stress response and virulence pathways via RNA modifications.

■ tRNA MODIFICATIONS IN PATHOGENS CONTROL STRESS RESPONSE AND VIRULENCE

A lot of RNA modification research of pathogens to date has occurred in the context of tRNA modifications and their regulation of translational capacity in comparison to model bacteria (e.g., *Escherichia coli*) and fungi (e.g., *Saccharomyces cerevisiae*). Although powerhouses of genetics, laboratory strains of these organisms are often not particularly robust in terms of stress response.¹⁵ It is therefore not surprising that further investigation of core, conserved RNA modification enzymes in nonmodel organisms has already revealed previously unappreciated functionalities.

An important concept in the study of tRNA modifications is that of the modification tunable transcripts (MoTTs), which are transcripts with specific/biased codon usage whose translation can be modulated by specific tRNA modification levels.¹⁶ Alteration of tRNA modifications can thus lead to fine-tuning of the proteome via adjustment of the levels of tRNA modification enzymes or even the precursor metabolites required for modification. MoTTs provide the organism with an additional layer of regulatory control that coupled with the central importance of tRNA in gene expression gives prominence to modifications of the tRNA. This regulation manifests in many ways. For example, in a clinical isolate of the fungus *S. cerevisiae*, the Ncs2 enzyme required for 2-thiolation of tRNA was shown to harbor a single point mutation (Ncs2^K) that facilitated growth at higher temperature and increased stress response and virulence compared to laboratory yeast strains.¹⁷ The authors determined that the point mutation led to increased 2-thiolation in the pathogenic strain at 37 °C, linking RNA modification and optimal translation firmly to virulence capacity in this case. A similar phenotype was observed in the opportunistic pathogenic yeast and frequent commensal *Candida albicans* for the Ncs2 ortholog; however,

these functions are not always conserved. Another tRNA modification enzyme, Hmal, facilitated two distinguishable phenotypes through its threonylcarbamoyladenosine (t^6A) dehydratase activity in the closely related *Candida* species, *C. albicans* and *C. dubliniensis*.¹³ This suggests that RNA modifications can be co-opted and evolved to fine-tune regulatory networks differently in each organism, hinting that we are likely to find a lot of new biology by broadening our search for RNA modification mechanisms.

In *Aspergillus fumigatus*, a filamentous fungal pathogen capable of causing infections ranging from allergy to deadly invasive aspergillosis, depletion of a particular tRNA modification via deletion of the catalytic subunit of the elongator complex resulted in numerous growth and stress phenotypes. Elongator is a conserved protein complex responsible for the 5-methoxycarbonylmethyl-2-thiouridine (mcm^5s^2U) modification of tRNA wobble uridine (U_{34}) in a subset of tRNA isoacceptors. Modification of this position is typically necessary for efficient mRNA decoding and proper translation of proteins. In *A. fumigatus* the phenotypes associated with loss of this modification could be rescued by deletion of a transcription factor, CpcA, which is a yeast Gcn4 ortholog known to serve as an important sensor of altered translation efficiency caused by hypomodified tRNAs.¹⁹ This result linking the modification to control of metabolism differs to some degree from the conventional knowledge in yeast (and some higher eukaryotes), where elongator deletions result in phenotypes that cannot be rescued by Gcn4 deletion, e.g., proteotoxic stress and altered translation of mRNAs with higher ratios of target codon abundance, akin to MoTTs introduced above.²⁰ The differences observed here suggest that even highly conserved pathways of modification can be leveraged for alternative roles in different organisms.

The concept of alternative functionalities between organisms continues with the isopentenyltransferase, ModS, of *A. fumigatus* that modifies A_{37} of select tRNAs. Deletion of *modS* results in increased resistance to the antifungal drug flucytosine, opposite to phenotypes observed in model fungi.²¹ Orthologs of ModS in bacteria, e.g., MiaA, are also linked to drug sensitivity²² and contribute to *vir* gene expression in the plant pathogen *Agrobacterium tumefaciens*,²³ *lex* operon expression in *Salmonella*,¹³ and stress response and virulence of extraintestinal pathogenic *E. coli* (ExPEC),²⁴ adding a layer of importance atop a variety of phenotypes observed upon deletion in K12 laboratory *E. coli* strains.²⁵ From these examples, it should not be surprising then that many tRNA modifications contribute to optimal virulence and stress response. For example, RNA modifications are linked to intracellular survival within the host for the devastating human pathogenic bacteria *Mycobacterium tuberculosis*, where MnmA-dependent tRNA uridine sulfation (s^2U) was required for optimal intracellular growth.²⁶ In *E. coli* ST131, the predominant ExPEC lineage worldwide, queuosine modification of tRNAs by QueF was tied to lipopolysaccharide production,²⁷ and the GidA protein of the opportunistic bacterial pathogen *Pseudomonas aeruginosa* was shown to introduce a carboxymethylaminomethyl modification in selected tRNAs that modulates a switch between a pathogenic and general growth state.²⁸ This switch was mediated through control of translation of virulence regulators by facilitating readthrough of rare codons requiring modification of wobble base 34 in select tRNAs. Such small changes leading to altered translational profiles highlights the rapidity and breadth of

impact of even a single tRNA modification in the response of microbes to their environment.

Applying similar principles, a recent study of the role of tRNA and tRNA modifications on drug resistance in *Vibrio cholerae* intriguingly uncovered unanticipated roles for many RNA modification enzymes in control of resistance or susceptibility to aminoglycosides, fluoroquinolones, β -lactams, chloramphenicol, and trimethoprim.²⁹ A better understanding of the mechanisms of drug resistance for the causative agent of cholera has obvious potential to influence the way we treat *V. cholerae* infections in the future. Collectively, these studies indicate an importance for tRNA modifications generally, but also demonstrate the potential of tRNA modifications of particular tRNA isoacceptors to serve as regulatory nodes capable of responding to stressful situations. Despite these important phenotypes, only a handful of tRNA modifications have been studied in depth in human pathogens, making this field ripe for the harvest of new regulatory mechanisms and impactful biology.

■ rRNA MODIFICATIONS PROVIDE REGULATORY FLEXIBILITY TO HIGHLY CONSERVED MOLECULES

Ribosomal RNA, like tRNA, is also heavily modified with important biological implications. Multiple studies have now documented the role of rRNA modifications in controlling diverse cellular responses relevant to virulence and stress resistance. For example, the erythromycin resistance methyltransferases (ERMs) of the Gram-positive bacterial pathogen *Staphylococcus aureus* promote resistance to the antibiotic erythromycin via dimethylation of a nucleotide in the large ribosomal subunit. Modification of the rRNA results in altered translation and a varied proteome.³⁰ Interestingly, m^6A_{1684} modified ribosomes of *S. aureus* are outcompeted by unmodified ribosomes, but during antibiotic treatment with macrolide, lincosamide, and streptogramin B antibiotics this phenotype is reversed, and the modified ribosomes give the bacteria an advantage.³¹ In fact, rRNA modification is a common way to modulate susceptibility to antibiotics, as shown repeatedly with the Cfr radical SAM enzyme that methylates A_{1583} (m^8A_{1583}) of *E. coli* 23S rRNA.³² Modifications of rRNA also impact virulence related phenotypes. Modification by RlmQ of *S. aureus* m^7G_{2801} of the 23S rRNA impacts growth, virulence, and biofilm formation via modulation of the tRNA accommodation channel,³³ the *rsmJ* and *rsmH* genes of *S. aureus* were determined to be virulence genes responsible for 2'-O- and N⁶-methylations (m^4Cm_{1412}) of 16S rRNA,³⁴ and the *Enterococcus faecalis*-modifying enzyme RlmN was recently shown to selectively alter modification upon direct sensing of reactive oxygen species (ROS).³⁵ There, ROS, or sublethal doses of antibiotics capable of inducing ROS, led to decreases in N³-methyladenosine (m^3A) for both the 23S rRNA and tRNAs, with implications for an environmentally responsive, dynamic RNA modification system. Similarly, approaches using direct nanopore sequencing have confirmed the modification patterns of rRNA to be complex and dynamic under stress just like those of tRNA,³⁶ spotlighting again the importance of mapping the epitranscriptome under many different conditions as proposed in the NASEM report.¹

Table 2. Influence of RNA Modifications on Host Response to Infection: Selected Studies of RNA Modifications Involved in the Host Response to Infection^a

modification	RNA	role(s) in host	host	ref
m ⁶ A	tRNA	Improve MYC translation to promote T cell expansion	Mouse	109
m ⁶ A	mRNA	hnRNP2B1 facilitates modification and trafficking of CGAS, IFI16, and STING mRNAs	Human/Mouse	110
m ⁶ A	mRNA	Modifications are dynamically regulated across infection	Mouse	53, 54
m ⁶ A	mRNA	Negative regulator of immunity in CMV infection	Human/Mouse	65
m ⁶ A	mRNA	Promote granulopoiesis and neutrophil mobilization	Human/Mouse	59
m ⁶ A	mRNA	Demethylation results in antiviral transcripts trapped in nucleus	Mouse	58
m ⁶ A	mRNA	Modification of α -ketoglutarate dehydrogenase (OGDH) mRNA reduces stability and protein expression to limit viral replication	Human/Mouse	57
m ⁶ A	mRNA	Inhibition of hepatitis B virus protein expression	Human	111
m ⁶ Am	mRNA	HIV viral protein R (Vpr) interacts with PCIF1 methyltransferase to facilitate ubiquitination and degradation preventing m ⁶ A adjacent to m ⁷ G cap	Human	112
NAD	mRNA/ miRNA	miRNA and snoRNA lost NAD ⁺ cap when infected with HIV-1	Human	113
m ⁵ C	mRNA	m ⁵ C of IFR3 mRNA negatively regulates IFN I responses during viral infections	Human	114
m ⁵ C	lncRNA	Depletion of NSUN1 (m ⁵ C methyltransferase) leads to increased interferon I response and viral suppression	Human/Mouse	115
A-to-I	dsRNA	A-to-I editing of viral RNA restricts infection	Human/Mouse	63–65
A-to-I	mRNA	A-to-I editing varied from tissues and cell types during a variety of infections	Human/Mouse	70–73, 116
A-to-I	dsRNA	ADAR1 regulates immune functions	Human	117
A-to-I	Viral dsRNA	ADAR promotes viral evolution	Human	67, 68
ac ⁴ C	mRNA	Control of alphavirus and enterovirus 71	Human/African Green Monkey	79, 80
Ψ	mRNA, ncRNA	Pseudouridine is added to host and HIV-1 viral transcripts	Human	88
C-to-U	mRNA	C-to-U editing by APOBEC3A restricts viral infection	Human	75, 76
Glycosylation	mRNA	Control of neutrophil recruitment	Mouse	118
Multiple	Total RNA	<i>Toxoplasma gondii</i> infection changes modification patterns in mouse spleen and liver	Mouse	119

^aAbbreviations: m⁶A, N⁶-methyladenosine; m⁶Am, N⁶,2'-O-dimethyladenosine; NAD, nicotinamide adenine dinucleotide; m⁵C, 5-methylcytosine; A-to-I, adenosine to inosine; ac⁴C, N⁴-acetylcytosine; Ψ, pseudouridine; C-to-U, cytosine to uracil

■ ENVIRONMENTAL AND PHYSIOLOGICAL ADAPTATIONS OF MICROBES ARE FUELED BY RNA MODIFICATIONS

A variety of RNA modifications have been studied in the context of parasitic protozoan biology,⁵⁷ with many of the same themes emerging as occur in the fungi or even higher eukaryotes. For example, the asexual life cycle of the unicellular protozoan parasite *Plasmodium falciparum* relies on dynamic amounts of m⁶A methylation to fine-tune gene expression via modulation of RNA stability and translational efficiency across the lifecycle.⁵⁸ Another example linking environment to pathogenesis comes from the anaerobic parasitic amoebozoan *Entamoeba histolytica*, where the hypermodified nucleobase queuine modulates oxidative stress response and serves as a virulence attenuator.⁵⁹ Queuine, which must be taken up from the environment by eukaryotes, is incorporated into tRNA by a tRNA-guanine transglycosylase (EhTGT) in *E. histolytica*, resulting in a queuosine-modified ribonucleoside in a small subset of tRNAs. An abundance of queuine represses virulence by downregulating expression of virulence-associated genes. This case highlights the importance of the environment and nutritional availability/supplementation on modification status and the conclusions we draw about modification patterns in controlled laboratory settings. Surely, much more can be learned from studying modifications in unconventional, albeit challenging systems like the protozoan parasites.

It is well-documented that RNA modifications play essential roles in viral pathogenesis.^{60–62} They contribute to viral RNA stability⁴³ as well as mRNA capping to limit detection by host

pattern recognition receptors, e.g., by encoding their own 2'-O-methyltransferase (2'-O-MTase) for cap addition.^{44,45} Modifications are also often required for full viral functionality, as observed with cytidine methylation of the pregenomic RNA required for a proper hepatitis B virus life cycle.⁴⁶ Similarly, m⁶A modification of the human immunodeficiency virus (HIV) viral RNA mediates increased stability through binding of the host m⁶A reader protein YTHDF2.⁴⁷ In this case, the virus relies on modification to exploit a host protein to stabilize its genome. In another variation of this theme, it was recently postulated that the human apolipoprotein B mRNA-editing enzyme, catalytic polypeptide-like (APOBEC) family proteins contribute to SARS-CoV-2 evolution directly in the host by introducing cytosine-to-uracil (C-to-U) transitions in the viral genome.⁴⁸ The overall mechanism is unclear, but the observed overrepresentation of C-to-U transitions suggests that host RNA modification pathways are likely being co-opted to facilitate viral pathogenesis and evolution at least in some cases. The role of RNA modifications in the host response, particularly regarding control of viral replication, is an expansive topic on its own, which will be addressed in more detail in the following sections.

■ REGULATION OF mRNA METHYLATION IN HIGHER EUKARYOTES CONTROLS HOST RESPONSE

The host response to infection is complex, relying on both broad general responses and specific actions against individual pathogens; all of which must be coordinated and, in many

cases, recalcitrant to attack by the pathogen itself. The mechanistic role of RNA modifications in immunity remains poorly explored, but it is already obvious that modifications play a significant role in regulating the immune response to a wide variety of pathogens by influencing RNA structure and stability or interaction with RNA binding proteins (Table 2).^{49,50}

One of the most abundant modifications in higher eukaryotes is methylation of the N⁶ position of adenosine, denoted N⁶-methyladenosine (m⁶A).⁵¹ This modification contributes expansively to gene regulation, including during immunity.^{51,52} The "writer", "reader", and "eraser" proteins of m⁶A metabolism that add, interact with, or remove m⁶A use this modification to coordinate multiple aspects of the immune response, including immune cell differentiation, proliferation, activation, and even polarization, among others.⁵² The coordinated rewriting of m⁶A during infection has been linked to viral repression in the case of the murine pathogen vesicular stomatitis virus (VSV) and binding of reader proteins to m⁶A-modified transcripts is linked to control of antiviral defense against murine cytomegalovirus (CMV)⁵³ reminiscent of the example described above for HIV.⁴⁷ Fungal pathogens are capable of influencing host m⁶A RNA modification patterns during infection as well. For example, m⁶A levels appear to increase throughout the course of infection in the eye during *Fusarium solani* infection.⁵⁴ So far, the implications of large-scale rewriting of m⁶A modifications in host defense are mostly descriptive, but as techniques and sensitivities improve, we will likely gain a better view of the complexities of modifications during infection. This is just one area where the advances gained in line with the NASEM report will have concrete effects on the ability to better interrogate RNA modifications during infection.

The m⁶A modification can also serve as a negative regulator of immunity, as occurs during human CMV infection, where m⁶A modification is relied upon to repress the interferon response and maintain cellular homeostasis.⁵⁵ A similar outcome is observed for the m⁶A reader YTHDF3, which regulates translation of the important transcriptional regulator FOXO3 independently of its m⁶A activity to selectively inhibit interferon (IFN)-stimulated gene expression.⁵⁶ In this case, the RNA modification machinery has been repurposed for an additional modification-independent regulation, underscoring the dangers of assigning functions for modifications based only on purported protein activities. The immune response is a highly complex regulatory network that requires numerous inputs and adjustments for proper function. Although limited, the examples described here hint that this network is commonly fine-tuned or even drastically altered by the presence, absence, or dynamics of RNA modifications during infection. In the future, higher resolution mapping of RNA modifications during the host response should allow us to better appreciate the influence of modification dynamics on host response.

As with any complex biological system, multiple pathways can lead to the same outcome. For example, a host may add an RNA modification like m⁶A directly using a writer enzyme or alternatively control the removal of RNA modifications by eraser enzymes to influence their response and maintain a modification, as occurs in the mouse host during vesicular stomatitis virus (VSV) infection. Here, the murine host inhibits the activity of the RNA m⁶A demethylase ALKBH5 to limit viral replication by restricting production of itaconate through

destabilization of the α -ketoglutarate dehydrogenase (OGDH) mRNA.⁵⁷ In parallel, ALKBH5 itself can be recruited by the DEAD-box (DDX) RNA helicase DDX46 to demethylate m⁶A-modified antiviral transcripts and trap them in the nucleus limiting innate immunity during VSV infection.⁵⁸ Interestingly, during bacterial infection ALKBH5 performs a different function and promotes granulopoiesis and neutrophil mobilization during cecal ligation-induced polymicrobial sepsis.⁵⁹ Collectively these studies reveal the dynamic networks of "writer", "reader", and "eraser" proteins required during different infection situations and show that varied strategies of regulation are used against different microbial challenges.

■ HOST ORGANISMS USE RNA MODIFICATIONS TO MODULATE THE RESPONSE TO PATHOGENS

Another system that is leveraged for a variety of cellular functions relies on Adenosine-to-Inosine (A-to-I) editing, which occurs widely in pre-mRNA, mature mRNA, and ncRNA. In mammals, this modification is broadly mediated by adenosine deaminase acting on RNA (ADAR) and more specifically on tRNA by adenosine deaminase acting on tRNA (ADAT) proteins. Most organisms harbor orthologs of the ADATs, important for installing inosine in the wobble position of some tRNAs, whereas only higher eukaryotes appear to have canonical ADAR proteins. The ADAR proteins have been linked to gene regulation of immunity in multiple systems,⁶⁰ but the most extensive studies relevant to this perspective come from mouse and human studies.^{61,62} The specificity of ADAR proteins to bind and (hyper)edit long stretches of perfectly duplexed double-stranded RNA (dsRNA) leads to the obvious hypothesis that ADAR could modify viral dsRNA to restrict infection by scrambling the genome of RNA viruses; however, the story is much more complex. While in some cases direct modification of viral RNA does in fact seem inhibitory (e.g., measles virus),^{63–65} ADAR activity can sometimes facilitate infection⁶⁶ or even promote viral evolution.^{67,68} These results suggest that the host response to viral infection can be both promoted and inhibited by the activity of ADAR proteins depending on the virus, cell type, and stage of infection.⁶⁹ RNA editing patterns within the host also change in a pathogen-dependent manner, as observed now in numerous studies of viral infection^{70–72} but also intracellular bacterial infections.⁷³ All these interactions must be considered in the context of the normal housekeeping functions of ADAR in marking host dsRNA as "self" to limit inflammation by dsRBP sensor proteins like MDA-5,⁷⁴ complicating further the already heterogeneous RNA modification landscape during infection.

Finally, several other RNA modifications have been investigated to some detail in the host response to pathogens. For example, human APOBEC3A mediates C-to-U RNA editing as discussed above and functions like ADAR editing in regard to restriction of viral pathogens and coordination of the host response to infection.^{75,76} APOBEC3A is a member of a larger family of enzymes, many of which contain RNA editing activity. Certainly, more investigation of the contributions of these enzymes will reveal novel biology relevant to infection. Hints already imply that a complex interplay exists from studies of SARS-CoV-2.⁷⁷ 5-methylcytosine (m⁵C), another modification of cytosine, has been linked to negative regulation of the interferon response, where it was shown that depletion of the NSUN2 m⁵C methyltransferase enhanced the interferon response and viral suppression.⁷⁸ These results are consistent

with additional work on H1N1 influenza A virus, where m⁵C was observed to be deposited on lncRNAs following infection of human A549 epithelial cells.⁶⁰ Cytidine is also modified by N-acetyltransferases (ac⁴C) and again linked to control of viral infection. N-acetyltransferase 10 was shown to regulate replication of alphavirus⁷⁹ and enterovirus 71,⁸⁰ two viruses capable of causing severe central nervous system destruction. Improved mapping of modifications like ac⁴C and m⁵C will likely reveal additional mechanisms associated with viral pathogenesis and link this modification to other host-microbe interactions in the future.

■ TECHNOLOGICAL ADVANCES HAVE IMPROVED OUR ASSESSMENT OF MODIFICATIONS DURING INFECTION

Technology has advanced rapidly around RNA modifications, and efforts have been made to map a variety of modifications during the host response as discussed throughout this perspective. Advances in RNA-seq technology have facilitated the large-scale identification of specific modifications in a variety of pathogens (Table 1), albeit with moderate levels of noise.^{64,81} Chemical- and antibody-assisted methods have provided orthogonal evidence and removed some bias from the initial direct sequencing approaches,⁸ and the latest generation of nanopore direct RNA sequencing has further refined our understanding.^{21,82,83} Emerging combinations of chemical ligation approaches^{84,85} with sequencing technology will likely prove better still, but are unlikely to facilitate the identification of all modifications on a single RNA in the near-term. The story is similar for mass spectrometry analysis, with new more sensitive devices constantly reaching the market and improved analysis techniques,^{86,87} we now have an unprecedented and frankly amazing view of many modifications. Our incomplete set of chemical standards remains a limitation, and in nearly all cases, many exciting questions remain. The mapping of RNA modifications on each and every RNA in a cell will facilitate the identification of novel biological mechanisms of regulation and improve our understanding of the host response to microbial challenge but will also force us to ask more questions and evolve our thinking around RNA modifications and infection. For example, a recent study mapping pseudouridine on ncRNA and mRNA during HIV-1 infection observed that the enzyme required for pseudouridylation of mRNA in humans remains surprisingly poorly understood, highlighting the fact that we still have only a rudimentary knowledge of many of the mechanisms behind modifications, despite all these technological advances.⁴⁸ In the next section, we will explore a few more unanswered questions that could be answered by efforts stemming from the NASEM report.

■ OUTSTANDING QUESTIONS: WHAT DO WE STAND TO GAIN FROM MAPPING ALL RNA MODIFICATIONS?

As seen in the previous sections, RNA modifications are clearly playing important, impactful roles in regulation of host pathogen interactions, but the elucidation of all RNA modifications within a particular epitranscriptome is likely to be further transformative in numerous ways. We expect that such an RNA modification moonshot will result in 1) an improved mechanistic understanding of pathogenesis at the molecular level and reveal novel players in infection biology, 2)

facilitate new and more accessible technology at lower cost, and 3) promote realization of new therapeutics and treatment strategies (Figure 2). As highlighted in the NASEM report, such strategic investments in science are not only a boon to research but also provide a quantifiable return on investment to the benefit of society.

The first benefit of defining RNA modifications in their entirety will be a more mechanistic understanding of host-pathogenesis. The interaction of host and pathogen is frequently described as an evolutionary molecular arms race to gain the advantage over the opposing organism. The role of RNA modifications in shaping these interactions and facilitating flexibility beyond that encoded in the genome remains poorly investigated, but certainly a compelling question. Examples mentioned here do support the ability of RNA modifications to facilitate viral evolution,^{67,68} but how these modifications may prime the host for more rapid evolution remains poorly understood.⁸⁸ Relatedly, there is likely much to learn about the evolution of RNA modification distribution across single RNA molecules in hosts and pathogens, which could potentially be revealed by the biological weaknesses exploited by each opposing partner during interaction. Another area where intensified interest in RNA modification biology may uncover new regulation is surrounding pathogen effectors, which are best described in bacteria and plant fungal/oomycete pathogens as secreted proteins involved in modulating or manipulating host responses. Currently, limited examples exist of RNA modification enzymes capable of serving as effectors to impart a cross-kingdom RNA modification (e.g., bacterial or fungal enzymes secreted to modify human RNA),⁹⁰ but it seems likely that many more instances will be discovered with time. We are particularly intrigued by examples from human fungal pathogens, where research typically lags that of viral and bacterial pathogens. Advances in *de novo* mapping of RNA modifications will likely prove pivotal in defining these rare modifications introduced by pathogens and that may only appear in conjunction with known, well-characterized modifications or at low frequency in dying cells during infection. This is an exciting area that could benefit tremendously from improved modification mapping sensitivities.

The NASEM report mentions on several occasions the importance of mining new enzymatic activities from microbes to advance the technology required to map RNA modifications. The elucidation of novel microbial factors that can be leveraged for RNA modification research will in turn reveal new information about the microbes themselves. These advancements in technology in parallel with improved computing capacity will certainly supercharge research efforts into RNA modifications during infection. Currently, investigating RNA modifications during the host pathogen interaction is challenging due to limited biological materials and complicated samples due to nucleic acid contributions from multiple partners. Increased sensitivity and versatility of methods will hopefully improve the chances of success in this area. Challenges will remain. How can we assign modifications to the host when we do not know the complete modification repertoire of the pathogen? We will also need solutions to mapping modifications in complex disease states and polymicrobial communities, where all the microbial players may not even be yet known.

The goal of studying pathogens and host-pathogenesis is to inform better strategies for treatment of these invaders in the

clinic. RNA modification research has already proven to be valuable in this regard by demonstrating an importance in influencing drug resistance, modulating pathogenesis, and facilitating new biotechnology, e.g., the mRNA vaccines that ultimately limited further SARS-CoV-2 spread. More generally, the study of RNA in the context of RNA-based therapeutics is already changing the way we consider treatment of infectious disease,^{91,92} with new vaccines in the pipeline and RNA-based drugs to fight viral infections in development. The NASEM report provides a roadmap, but how this plan will be used and adapted by individual subfields will be an additional challenge, particularly when ensuring that the technological gains are shared openly and widely in a reproducible manner.

The list of outstanding questions in infectious disease research that may be answerable with a newfound capability to map full epitranscriptomes is enormous, but we definitely had a few personal favorites in brainstorming for this perspective. First, we are excited to learn how RNA modification patterns influence RNA trafficking, particularly in the context of RNA secretion in association with extracellular vesicles. The primary advances in this area may not initially come from the study of host pathogenesis,^{93,94} but it seems likely that understanding how chemical changes to RNA influences their intracellular, intercellular, or even interkingdom trafficking will be of major importance in therapeutically delivering RNA to specific niches in the future. We are also excited to see studies of the full epitranscriptome from individual cell types during infection. Just as mRNA isoforms differ widely between cell types, we expect that RNA modification patterns will follow suit, particularly in response to different pathogens. Coalescing all this information into usable models and tractable databases will be a huge challenge for the future, likely requiring clever computational solutions as mentioned by the NASEM report. Finally, how will mapping of RNA modifications change diagnostics? It seems possible that in the future, rapid mapping of cellular RNA modifications from a host blood or urine sample could be used together with deep learning to identify all sorts of maladies, including genetic disorders, cancers, or even pathogens typically hidden below detection limits.

CONCLUSIONS

The future of RNA modifications research holds amazing potential to understand the molecular world. As the human genome project completely changed our research landscape, so too will a major investment in unraveling the mysteries of RNA modifications. As in many cases, studies of host–pathogenesis are likely to mostly follow major breakthroughs from more tractable systems, but as in the past, investigation of host–pathogen interactions will also reveal features of the molecular world that may be concealed in these same tractable model systems: likely we will see entirely new functions for RNA and RNA modifications emerge from inspection of pathogenic microbes. Major investment in strategic initiatives tends to simultaneously improve accessibility to previously considered “specialist” assays and make more commonplace technically complex equipment, while also bolstering availability and lowering prices of rare reagents. If the NASEM report is followed even partially toward completion, we anticipate a future alight with our understanding of the role of RNA in microbial pathogenesis and host immunity.

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Notes

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REFERENCES

- (1) Charting a Future for Sequencing RNA and Its Modifications: A New Era for Biology and Medicine; National Academies Press: Washington, D.C., 2024; DOI: 10.17126/27165.
- (2) Suzuki, T. The expanding world of tRNA modifications and their disease relevance. *Nat. Rev. Mol. Cell Biol.* **2021**, *22* (6), 375–392.
- (3) Fu, Y.; Dominissini, D.; Rachavi, G.; He, C. Gene expression regulation mediated through reversible m⁶A RNA methylation. *Nat. Rev. Genet.* **2014**, *15* (5), 293–306.
- (4) Jamontas, R.; Laurynėnas, A.; Povilaitytė, D.; Meškys, R.; Aućyraitė, A. RudS: Bacterial Desulfidase Responsible for tRNA 4-thiouridine de-modification. *Nucleic Acids Res.* **2024**, *52* (17), 10543–10562.
- (5) Höfer, K.; Jäschke, A. Epitranscriptomics: RNA modifications in bacteria and archaea. *Microbiol. Spectr.* **2018**, *6* (3), DOI: 10.1128/microbiolspec.RWR-0015-2017.
- (6) Zhang, Y.; Lu, L.; Li, X. Detection technologies for RNA modifications. *Exp. Mol. Med.* **2022**, *54* (10), 1601–1616.
- (7) Albeke, J. S.; Verstraten, R.; Depledge, D. P. Nanopore-based detection of viral RNA modifications. *mBio* **2022**, *13* (3), e03702–21.
- (8) Teichert, I. Fungal RNA editing: who, when, and why? *Appl. Microbiol. Biotechnol.* **2020**, *104* (13), 5689–5695.
- (9) Qiu, L.; Jing, Q.; Li, Y.; Han, J. RNA modification: mechanisms and therapeutic targets. *Mol. Biomed.* **2023**, *4* (1), 25.
- (10) Davis, F. F.; Allen, F. W. Ribonucleic acids from yeast which contain a fifth nucleotide. *J. Biol. Chem.* **1957**, *227* (2), 907.
- (11) Koh, C. S.; Sarin, L. P. Transfer RNA modification and infection – implications for pathogenicity and host responses.

<https://doi.org/10.1021/acsinfecdis.4c00598>
ACS Infect. Dis. XXXX, XXX, XXX–XXX

Biochim. Biophys. Acta BBA - Gene Regul. Mech. **2018**, *1861* (4), 419–432.

(12) Singer, C. E.; Smith, G. R.; Cortese, R.; Ames, B. N. Mutant tRNA^{His} ineffective in repression and lacking two pseudouridine modifications. *Nature. New Biol.* **1972**, *238* (81), 72–74.

(13) Blum, P. H. Reduced *lex* operon expression in a *miaA* mutant of *Salmonella typhimurium*. *J. Bacteriol.* **1988**, *170* (11), 5125–5133.

(14) Bossi, L.; Roth, J. R. The influence of codon content on genetic code translation. *Nature* **1980**, *286* (5769), 123–127.

(15) Denamur, E.; Clermont, O.; Bonacorsi, S.; Giedon, D. The population genetics of pathogenic *Escherichia coli*. *Nat. Rev. Microbiol.* **2021**, *19* (1), 37–54.

(16) Dedon, P. C.; Begley, T. J. A system of RNA modifications and biased codon use controls cellular stress response at the level of translation. *Chem. Res. Toxicol.* **2014**, *27* (3), 330–337.

(17) Allings, F.; Schumann, K.; Eggers, C.; Böttcher, B.; Sokolowski, M.; Shvetsova, E.; Sharma, P.; Roth, J.; Rashit, L.; Glatt, S.; Brünke, S.; Leidel, S. A. Ncs2⁺ mediates *in vivo* virulence of pathogenic yeast through sulphur modification of cytoplasmic transfer RNA. *Nucleic Acids Res.* **2023**, *51* (15), 8133–8149.

(18) Böttcher, B.; Kienast, S. D.; Leufken, J.; Eggers, C.; Sharma, P.; Leufken, C. M.; Morgner, B.; Drexler, H. C. A.; Schulz, D.; Allert, S.; Jacobsen, I. D.; Vylkova, S.; Leidel, S. A.; Brünke, S. A highly conserved tRNA modification contributes to *C. albicans* filamentation and virulence. *Microbiol. Spectr.* **2024**, *12* (5), No. e0425522.

(19) Zhang, Y.; Wang, Y.; Fan, J.; Zhu, G.; Lu, L. *Aspergillus fumigatus* Elongator complex subunit 3 affects hyphal growth, adhesion and virulence through wobble uridine tRNA modification. *PLoS Pathog.* **2022**, *18* (11), No. e1010976.

(20) Klassen, R.; Bruch, A.; Schaffrath, R. Induction of protein aggregation and starvation response by tRNA modification defects. *Curr. Genet.* **2020**, *66* (6), 1053–1057.

(21) Bruch, A.; Lazarova, V.; Berg, M.; Krueger, T.; Schaeuble, S.; Kelan, A. A.; Mertens, B.; Lehenberger, P.; Kniemeyer, O.; Kaiser, S.; Panagiotou, G.; Gsaller, F.; Hlang, M. G. tRNA hypomodification facilitates 5-fluorocytosine resistance via cross-pathway control system activation in *Aspergillus fumigatus*. *bioRxiv*, October 16, 2024, 578369. DOI: 10.1101/2024.02.19.578369.

(22) Taylor, D. E.; Trisler, C. A.; Trescher, G.; Bekkering, M. Host mutations (*miaA* and *rplA*) reduce tetracycline resistance mediated by Tet(O) and Tet(M). *Antimicrob. Agents Chemother.* **1998**, *42* (1), 59–64.

(23) Gray, J.; Wang, J.; Gebm, S. B. Mutation of the *miaA* gene of *Agrobacterium tumefaciens* results in reduced *vir* gene expression. *J. Bacteriol.* **1992**, *174* (4), 1086–1098.

(24) Fleming, B. A.; Blango, M. G.; Romek, A. A.; Kincannon, W. M.; Tran, A.; Lewis, A. J.; Russell, C. W.; Zhou, Q.; Baird, L. M.; Barber, A. E.; Brannon, J. R.; Beebout, C. J.; Bandarian, V.; Hadjifrangiskou, M.; Howard, M. T.; Mulvey, M. A. A tRNA modifying enzyme as a tunable regulatory nexus for bacterial stress responses and virulence. *Nucleic Acids Res.* **2022**, *50*, 7570.

(25) Aubee, J. L.; Williams, K.; Adigun, A.; Oluwaya, O.; Nurse, J.; Thompson, K. M. Post-transcriptional regulation of the *miaA* prenyl transferase by CsrA and the small RNA CsrB in *E. coli*. *bioRxiv*, February 28, 2024, 582573. DOI: 10.1101/2024.02.28.582573.

(26) Tomasi, F. G.; Kimura, S.; Rubin, E. J.; Wukow, M. K. A tRNA modification in *Mycobacterium tuberculosis* facilitates optimal intracellular growth. *eLife* **2023**, *12*, RP87146. DOI: 10.7554/eLife.87146.2.

(27) Diaz-Ruilo, J.; González-Pastor, J. E. tRNA queuine modification is involved in biofilm formation and virulence in bacteria. *Nucleic Acids Res.* **2023**, *51* (18), 9821–9837.

(28) Krueger, J.; Preusse, M.; Oswaldo Gomez, N.; Frommeyer, Y. N.; Doberenz, S.; Lorenz, A.; Kordes, A.; Grobe, S.; Mücken, M.; Depledge, D. P.; Svensson, S. L.; Weiss, S.; Karver, V.; Pich, A.; Sharma, C. M.; Ignatova, Z.; Häussler, S. tRNA epitranscriptome determines pathogenicity of the opportunistic pathogen *Pseudomonas aeruginosa*. *Proc. Natl. Acad. Sci. U. S. A.* **2024**, *121* (11), No. e2312874121.

(29) Babosan, A.; Fruchard, L.; Krin, E.; Carvalho, A.; Mazel, D.; Baharoglu, Z. Nonessential tRNA and rRNA modifications impact the bacterial response to sub-MIC antibiotic stress. *microLife* **2022**, *3*, No. uqac019.

(30) Gupta, P.; Sothilvelam, S.; Vázquez-Laslop, N.; Mankin, A. S. Deregulation of translation due to post-transcriptional modification of rRNA explains why *Emm* genes are inducible. *Nat. Commun.* **2013**, *4* (1), 1984.

(31) Shields, K. E.; Ranara, D.; Tan, Y.; Zhang, D.; Yap, M.-N. F. Epitranscriptional m⁶A modification of rRNA negatively impacts translation and host colonization in *Staphylococcus aureus*. *PLOS Pathog.* **2024**, *20* (1), No. e1011968.

(32) Tsai, K.; Stojković, V.; Noda-Garcia, L.; Young, I. D.; Myasnikov, A. G.; Kleinman, J.; Palla, A.; Flour, S. N.; Frost, A.; Fraser, J. S.; Tawfik, D. S.; Fujimori, D. G. Directed Evolution of the tRNA Methylating Enzyme Cfr Reveals Molecular Basis of Antibiotic Resistance. *eLife* **2022**, *11*, No. e70017.

(33) Bahena-Ceron, R.; Teixeira, C.; Ponce, J. R. J.; Wolff, P.; Connon, F.; François, P.; Klaholz, B. P.; Vandenesch, F.; Romby, P.; Moreau, K.; Marzi, S. RlmQ: A Newly discovered tRNA modification enzyme bridging RNA modification and virulence traits in *Staphylococcus aureus*. *RNA N. Y. N* **2024**, *30* (3), 200–212.

(34) Kyuma, T.; Kimura, S.; Hanada, Y.; Suzuki, T.; Sekimizu, K.; Kaito, C. Ribosomal RNA methyltransferases contribute to *Staphylococcus aureus* virulence. *FEBS J.* **2015**, *282* (13), 2570–2584.

(35) Lee, W. L.; Sinha, A.; Lam, L. N.; Loo, H. L.; Liang, J.; Ho, P.; Chu, L.; Chan, C. S. C.; Begley, T.; Kline, K. A.; Dedon, P. An RNA modification enzyme directly senses reactive oxygen species for translational regulation in *Enterococcus faecalis*. *Nat. Commun.* **2023**, *14* (1), 4093.

(36) Fleming, A. M.; Bonanietti, P.; Xiao, S.; Bandarian, V.; Burrows, C. J. Direct nanopore sequencing for the 17 RNA modification types in 36 locations in the *E. coli* ribosome enables monitoring of stress-dependent changes. *ACS Chem. Biol.* **2023**, *18* (10), 2211–2223.

(37) Guo, G.; Lin, Y.; Zhu, X.; Ding, F.; Xue, X.; Zhang, Q. Emerging roles of the epitranscriptome in parasitic protozoan biology and pathogenesis. *Trends Parasitol.* **2024**, *40* (3), 214–229.

(38) Baumgarten, S.; Bryant, J. M.; Sinha, A.; Reyser, T.; Preiser, P. R.; Dedon, P. C.; Scherf, A. Transcriptome-wide dynamics of extensive m⁶A mRNA methylation during *Plasmodium falciparum* blood stage development. *Nat. Microbiol.* **2019**, *4* (12), 2246–2259.

(39) Nagaraja, S.; Cai, M. W.; Sun, J.; Varet, H.; Sarid, L.; Trebbick-Geffen, M.; Shalov, Y.; Maramba, M.; Legendre, R.; Coppée, J.-Y.; Begley, T. J.; Dedon, P. C.; Gourmatis, S.; Guillen, N.; Saito-Nakano, Y.; Shimokawa, C.; Hisaeda, H.; Ankit, S. Queuine is a nutritional regulator of *Enterococcus histolytica* response to oxidative stress and a virulence attenuator. *mBio* **2021**, *12* (2). DOI: 10.1128/mBio.03549-20.

(40) Potuznik, J. F.; Cahová, H. It's the little things (in viral RNA). *mBio* **2020**, *11* (5). DOI: 10.1128/mBio.02131-20.

(41) Gokhale, N. S.; Horner, S. M. RNA modifications go viral. *PLoS Pathog.* **2017**, *13* (3), No. e1006188.

(42) Rajendren, S.; Kartjohel, J. The impact of RNA modifications on the biology of DNA virus infection. *Eur. J. Cell Biol.* **2022**, *101* (3), No. 151239.

(43) Henry, B. A.; Kanarek, J. P.; Kotter, A.; Helm, M.; Lee, N. 5-methylcytosine modification of an Epstein-Barr Virus noncoding RNA decreases its stability. *RNA N. Y. N* **2020**, *26* (8), 1038–1048.

(44) Wang, X.; Chen, C.; Sun, H.; Mao, K.; Yao, J.; Zhang, W.; Zhan, M.; Li, H.-B.; Zhang, Z.; Zhu, S.; Lu, L. m⁶A mRNA modification potentiates Th17 functions to inflame autoimmunity. *Sci. China Life Sci.* **2023**, *66* (11), 2543–2552.

(45) Chen, Y.; Guo, D. Molecular mechanisms of coronavirus RNA capping and methylation. *Virol. Sin.* **2016**, *31* (1), 3–11.

(46) Su, P.-Y.; Chang, C.-H.; Yen, S.-C. B.; Wu, H.-Y.; Tung, W.-J.; Hu, Y.-P.; Chen, Y.-Y. L.; Lin, M.-H.; Shih, C.; Chen, P.-J.; Tsai, K. Epitranscriptomic cytidine methylation of the Hepatitis B viral RNA is

essential for viral reverse transcription and particle production. *Proc. Natl. Acad. Sci. U. S. A.* **2024**, *121* (24), No. e2400378121.

(47) Tsai, K.; Bogerd, H. P.; Kennedy, E. M.; Emery, A.; Swannstrom, R.; Cullen, B. R. Epitranscriptomic addition of m⁶A regulates HIV-1 RNA stability and alternative splicing. *Genes Dev.* **2021**, *35* (13–14), 992–1004.

(48) Simmonds, P.; Ansari, M. A. Extensive C→U Transition biases in the genomes of a wide range of mammalian RNA viruses: potential associations with transcriptional mutations, damage- or host-mediated editing of viral RNA. *PLOS Pathog.* **2021**, *17* (6), No. e1009596.

(49) Cui, L.; Ma, R.; Cai, J.; Guo, C.; Chen, Z.; Yao, L.; Wang, Y.; Fan, R.; Wang, X.; Shi, Y. RNA modifications: importance in immune cell biology and related diseases. *Signal Transduct. Target. Ther.* **2022**, *7* (1), 334.

(50) Han, D.; Xu, M. M. RNA modification in the immune system. *Annu. Rev. Immunol.* **2023**, *41* (1), 73–98.

(51) Jiang, X.; Liu, B.; Nie, Z.; Duan, L.; Xiong, Q.; Jin, Z.; Yang, C.; Chen, Y. The role of m⁶A modification in the biological functions and diseases. *Signal Transduct. Target. Ther.* **2021**, *6* (1), 1–16.

(52) Wang, Y.; Li, L.; Li, J.; Zhao, B.; Huang, G.; Li, X.; Xie, Z.; Zhou, Z. The emerging role of m⁶A modification in regulating the immune system and autoimmune diseases. *Front. Cell Dev. Biol.* **2021**, *9*, 755691.

(53) Ma, S.; Yan, J.; Bai, T.; Zhang, J.; Chen, Z.; Wang, L.-S.; Sun, J. C.; Chen, J.; Caligiuri, M. A.; Yu, J. The RNA m⁶A reader YTHDF2 controls NK cell antitumor and antiviral immunity. *J. Exp. Med.* **2021**, *218* (8), No. e20210279.

(54) Tang, H.; Huang, L.; Hu, J. Inhibition of the m⁶A methyltransferase METTL3 attenuates the inflammatory response in *Fusarium solani*-induced keratitis via the NF- κ B signaling pathway. *Invest. Ophthalmol. Vis. Sci.* **2022**, *63* (11), 2.

(55) Winkler, R.; Gills, E.; Lasman, L.; Safra, M.; Geula, S.; Soyris, C.; Nachshon, A.; Tai-Schmiedel, J.; Friedman, N.; Le-Trilling, V. T. K.; Trilling, M.; Mandelboim, M.; Hanna, J. H.; Schwartz, S.; Stern-Ginossar, N. m⁶A modification controls the innate immune response to infection by targeting type I interferons. *Nat. Immunol.* **2019**, *20* (2), 173–182.

(56) Zhang, Y.; Wang, X.; Zhang, X.; Wang, J.; Ma, Y.; Zhang, L.; Cao, X. RNA-binding protein YTHDF3 suppresses interferon-dependent antiviral responses by promoting FOXO3 translation. *Proc. Natl. Acad. Sci. U. S. A.* **2019**, *116* (3), 976–981.

(57) Liu, Y.; You, Y.; Lu, Z.; Yang, J.; Li, P.; Liu, L.; Xu, H.; Niu, Y.; Cao, X. N⁶-methyladenosine RNA modification-mediated cellular metabolism rewiring inhibits viral replication. *Science* **2019**, *365* (6458), 1171–1176.

(58) Zheng, Q.; Hou, J.; Zhou, Y.; Li, Z.; Cao, X. The RNA helicase DDX46 inhibits innate immunity by entrapping m⁶A-demethylated antiviral transcripts in the nucleus. *Nat. Immunol.* **2017**, *18* (10), 1094–1103.

(59) Liu, Y.; Song, R.; Lu, Z.; Zhao, L.; Zhan, X.; Li, Y.; Cao, X. The RNA m⁶A demethylase ALKBH5 drives emergency granulopoiesis and neutrophil mobilization by upregulating G-CSF expression. *Cell. Mol. Immunol.* **2024**, *21* (1), 6–18.

(60) Dhakal, A.; Salim, C.; Skelly, M.; Amichan, Y.; Lamm, A. T.; Hundley, H. A. ADARs regulate cuticle collagen expression and promote survival to pathogen infection. *BMC Biol.* **2024**, *22*, 37.

(61) Qian, J.; Sedmik, J.; Vukić, D.; Khan, A.; Keegan, L. P.; O'Connell, M. A. ADAR RNA modifications, the epitranscriptome and innate immunity. *Trends Biochem. Sci.* **2021**, *46* (9), 758–771.

(62) Jarmoskaite, I.; Li, J. B. Multifaceted roles of RNA editing enzyme ADAR1 in innate immunity. *RNA* **2024**, *30* (5), 500–511.

(63) Ward, S. V.; George, C. X.; Welch, M. J.; Lion, L.-Y.; Hahn, B.; Lewicki, H.; de la Torre, J. C.; Samuel, C. E.; Oldstone, M. B. RNA editing enzyme adenosine deaminase is a restriction factor for controlling measles virus replication that also is required for embryogenesis. *Proc. Natl. Acad. Sci. U. S. A.* **2011**, *108* (1), 331–336.

(64) Pfäler, C. K.; Dunhuve, R. C.; Nersisyan, S.; Brodsky, L.; Cattaneo, R. Extensive editing of cellular and viral double-stranded RNA structures accounts for innate immunity suppression and the

proviral activity of ADAR1p150. *PLOS Biol.* **2018**, *16* (11), No. e2006577.

(65) Suspène, R.; Petit, V.; Puyraimond-Zemmour, D.; Aynaud, M.-M.; Henry, M.; Guétard, D.; Rusniok, C.; Wain-Hobson, S.; Vartanian, J.-P. Double-stranded RNA adenosine deaminase ADAR1-induced hypermutated genomes among inactivated seasonal influenza and live attenuated measles virus vaccines. *J. Virol.* **2011**, *85* (5), 2458–2462.

(66) Clerius, G.; Gilman, J.-F.; Daher, A.; Bonnet, M.; Meurs, E. F.; Gagnon, A. ADAR1 interacts with PKR during human immunodeficiency virus infection of lymphocytes and contributes to viral replication. *J. Virol.* **2009**, *83* (19), 10119–10124.

(67) Ringlander, J.; Pingul, J.; Kann, H.; Prakash, K.; Rydell, G.; Andersson, M.; Martner, A.; Lindh, M.; Horal, P.; Hellstrand, K.; Kann, M. Impact of ADAR-induced editing of minor viral RNA populations on replication and transmission of SARS-CoV-2. *Proc. Natl. Acad. Sci. U. S. A.* **2022**, *119* (6), No. e2112663119.

(68) Piontkivska, H.; Frederick, M.; Miyamoto, M. M.; Wayne, M. L. RNA editing by the host ADAR system affects the molecular evolution of the Zika virus. *Ecol. Evol.* **2017**, *7* (12), 4475–4485.

(69) Piontkivska, H.; Wales-McGrath, B.; Miyamoto, M.; Wayne, M. L. ADAR editing in viruses: an evolutionary force to reckon with. *Genome Biol. Evol.* **2021**, *13* (11), No. eab240.

(70) Mendler-Rabinowicz, R.; Gordik, D.; Park, J.; Meydan, C.; Fox, J.; Karmon, M.; Roth, H. S.; Cohen-Fukheim, R.; Shohat-Ofir, G.; Eisenberg, E.; Rupp, E.; Mason, C. E.; Levano, E. Y. Elevated A-to-I RNA editing in COVID-19 infected individuals. *NAR Genomics Bioinform.* **2023**, *3* (4), No. lqad092.

(71) Wales-McGrath, B.; Mercer, H.; Piontkivska, H. Changes in ADAR RNA editing patterns in CMV and ZIKV congenital infections. *BMC Genomics* **2023**, *24* (1), 685.

(72) Crooks, P. S.; Tossberg, J. T.; Poetter, K. P.; Aune, T. M. Reduced A-to-I editing of endogenous alu RNAs in lung after SARS-CoV-2 infection. *Curr. Res. Immunol.* **2021**, *2*, 52–59.

(73) Wei, Z.-Y.; Wang, Z.-X.; Li, J.-H.; Wen, Y.-S.; Gao, D.; Xia, S.-Y.; Li, Y.-N.; Pan, X.-B.; Liu, Y.-S.; Jin, Y.-Y.; Chen, J.-H. Host A-to-I RNA editing signatures in intracellular bacterial and single-strand RNA viral infections. *Front. Immunol.* **2023**, *14*, No. 1121096.

(74) Liddcoat, B. J.; Piskol, R.; Chalk, A. M.; Ramaswami, G.; Higuchi, M.; Hartner, J. C.; Li, J. B.; Seeburg, P. H.; Walkley, C. R. RNA editing by ADAR1 prevents MDAs sensing of endogenous dsRNA as nonself. *Science* **2015**, *349* (6252), 1115–1120.

(75) Alqassim, E. Y.; Sharma, S.; Khan, A. N. M. N. H.; Emmons, T. R.; Cortes Gomez, E.; Alahmari, A.; Singel, K. L.; Mark, J.; Davidson, B. A.; Robert McGray, A. J.; Liu, Q.; Licht, B. D.; Moysich, K. B.; Wang, J.; Odunsi, K.; Segal, B. H.; Baysal, B. E. RNA editing enzyme APOBEC3A promotes pro-inflammatory M1 macrophage polarization. *Commun. Biol.* **2021**, *4*, 102.

(76) Di Giorgio, S.; Martignani, F.; Torcia, M. G.; Mattiuz, G.; Conticello, S. G. Evidence for host-dependent RNA editing in the transcriptome of SARS-CoV-2. *Sci. Adv.* **2020**, *6* (25), No. eabb5813.

(77) Kim, K.; Calabrese, P.; Wang, S.; Qin, C.; Rao, Y.; Feng, P.; Chen, X. S. The roles of APOBEC-mediated RNA editing in SARS-CoV-2 mutations, replication and fitness. *Sci. Rep.* **2022**, *12* (1), No. 14972.

(78) Feng, C.; Xin, K.; Du, Y.; Zou, J.; Xing, X.; Xia, Q.; Zhang, Y.; Zhang, R.; Huang, W.; Wang, Q.; Jiang, C.; Wang, X.; Kang, Z.; Xu, J.-R.; Liu, H. Unveiling the A-to-I mRNA editing machinery and its regulation and evolution in fungi. *Nat. Commun.* **2024**, *15* (1), 3934.

(79) Dang, Y.; Li, J.; Li, Y.; Wang, Y.; Zhao, Y.; Zhao, N.; Li, W.; Zhang, H.; Ye, C.; Ma, H.; Zhang, L.; Liu, H.; Dong, Y.; Yao, M.; Lei, Y.; Xu, Z.; Zhang, F.; Ye, W. N-acetyltransferase 10 regulates alphavirus replication via N4-acetylcytidine (Ac4C) modification of the lymphocyte antigen six family member E (LY6E) mRNA. *J. Virol.* **2024**, *98* (1), No. e01350-23.

(80) Hao, H.; Liu, W.; Miao, Y.; Ma, L.; Yu, B.; Liu, L.; Yang, C.; Zhang, K.; Chen, Z.; Yang, J.; Zheng, Z.; Zhang, B.; Deng, F.; Gong, P.; Yuan, J.; Hu, Z.; Guan, W. N4-acetylcytidine regulates the

replication and pathogenicity of enterovirus 71. *Nucleic Acids Res.* **2022**, *50* (16), 9339–9354.

(81) Kong, Y.; Mead, E. A.; Fang, G. Navigating the pitfalls of mapping DNA and RNA modifications. *Nat. Rev. Genet.* **2023**, *24* (6), 363–381.

(82) Lucas, M. C.; Pryszcz, L. P.; Medina, R.; Milenkovic, I.; Camacho, N.; Marchand, V.; Motorin, Y.; Rihac de Poaplana, L.; Novoa, E. M. Quantitative analysis of tRNA abundance and modifications by nanopore RNA sequencing. *Nat. Biotechnol.* **2024**, *42*, 72.

(83) Piątkowski, J.; Koźlak, K.; Golk, P. Mitochondrial transcriptome of *Candida albicans* in flagrant — direct RNA sequencing reveals a new layer of information. *BMC Genomics* **2024**, *25* (1), 869.

(84) George, J. T.; Srivatsan, S. G. Bioorthogonal Chemistry-based RNA labeling technologies: evolution and current state. *Chem. Commun. Camb. Engl.* **2020**, *56* (82), 12307–12318.

(85) Liu, H.; Wang, Y.; Zhou, X. Labeling and sequencing nucleic acid modifications using bio-orthogonal tools. *RSC Chem. Biol.* **2022**, *3* (8), 994–1007.

(86) Lauman, R.; Garcia, B. A. Unraveling the RNA modification code with mass spectrometry. *Mol. Omics* **2020**, *16* (4), 305–315.

(87) Ammann, G.; Berg, M.; Dahwig, J. F.; Kaiser, S. M. Pitfalls in RNA modification quantification using nucleoside mass spectrometry. *Acc. Chem. Res.* **2023**, *56* (22), 3121–3131.

(88) Campos, C. M.; Tsai, K.; Courtney, D. G.; Bogerd, H. P.; Holley, C. L.; Cullen, B. R. Mapping of pseudouridine residues on cellular and viral transcripts using a novel antibody-based technique. *RNA* **2021**, *27* (11), 1400.

(89) Linnat, A.; Melamed, D. Evolutionary honing in and mutational replacement: how long-term directed mutational responses to specific environmental pressures are possible. *Theory Biosci.* **2023**, *142* (2), 87–105.

(90) Bullen, N. P.; Sychantha, D.; Thang, S. S.; Culver, P. H.; Rudez, M.; Ahmad, S.; Shah, V. S.; Filloux, A.; Pehina, G.; Whitney, J. C. An ADP-ribosyltransferase toxin kills bacterial cells by modifying structured non-coding RNAs. *Mol. Cell* **2022**, *82* (18), 3484–3498.e11.

(91) Bruch, A.; Kelani, A. A.; Blango, M. G. RNA-based therapeutics to treat human fungal infections. *Trends Microbiol.* **2022**, *30*, 411.

(92) Zhu, Y.; Zhu, L.; Wang, X.; Jin, H. RNA-based therapeutics: an overview and prospectus. *Cell Death Dis.* **2022**, *13* (7), 1–15.

(93) Pan, H.-Y.; Yu, Y.; Cao, T.; Liu, Y.; Zhou, Y.-L.; Zhang, X.-X. Systematic profiling of exosomal small RNA epigenetic modifications by high-performance liquid chromatography–mass spectrometry. *Anal. Chem.* **2021**, *93* (45), 14907–14911.

(94) Kaur, S.; Saldana, A. C.; Elkahloun, A. G.; Peterson, J. D.; Arakelyan, A.; Singh, S. P.; Jenkins, L. M.; Kuo, B.; Reginald, B.; Jordan, D. G.; Tran, A. D.; Wu, W.; Zimmerberg, J.; Margolis, L.; Roberts, D. D. CD47 interactions with exportin-1 limit the targeting of m⁶G-modified RNAs to extracellular vesicles. *J. Cell Commun. Signal.* **2022**, *16* (3), 397–419.

(95) Liroy, V. S.; Goussard, S.; Guerin, V.; Yoon, E.-J.; Courvalin, P.; Galimand, M.; Grillot-Courvalin, C. Aminoglycoside resistance 16S rRNA methyltransferases block endogenous methylation, affect translation efficiency and fitness of the host. *RNA* **2014**, *20* (3), 382–391.

(96) Choi, Y.; Hyeon, H.; Lee, K.; Bahn, Y.-S. Ssa5 catalyzing universal t^rA tRNA modification is responsible for multifaceted functions of the KEOPS complex in *Cryptococcus neoformans*. *mSphere* **2024**, *9* (1), No. e00557-23.

(97) Bian, Z.; Wang, Z.; Wang, D.; Xu, J.-R. Sexual stage-specific A-to-I mRNA editing is mediated by tRNA-editing enzymes in fungi. *Proc. Natl. Acad. Sci. U. S. A.* **2024**, *121* (12), No. e2319235121.

(98) Xin, K.; Zhang, Y.; Fan, L.; Qi, Z.; Feng, C.; Wang, Q.; Jiang, C.; Xu, J.-R.; Liu, H. Experimental evidence for the functional importance and adaptive advantage of A-to-I RNA editing in fungi. *Proc. Natl. Acad. Sci. U. S. A.* **2023**, *120* (12), No. e2219029120.

(99) Kelley, M.; Paulines, M. J.; Yoshida, G.; Myers, R.; Jora, M.; Levoy, J. P.; Addepalli, B.; Benoit, J. B.; Limbach, P. A. Ionizing

radiation and chemical oxidant exposure impacts on *Cryptococcus neoformans* transfer RNAs. *PLoS One* **2022**, *17* (3), No. e0266239.

(100) Isel, C.; Marquet, R.; Keith, G.; Ehresmann, C.; Ehresmann, B. Modified nucleotides of tRNA(3Lys) modulate primer/template loop-loop interaction in the initiation complex of HIV-1 reverse transcription. *J. Biol. Chem.* **1993**, *268* (34), 25269–25272.

(101) Baek, A.; Lee, G.-E.; Golconda, S.; Rayhan, A.; Manganiaris, A. A.; Chen, S.; Tiranuru, N.; Yu, H.; Kim, S.; Kimmel, C.; Zablocki, O.; Sullivan, M. B.; Addepalli, B.; Wu, L.; Kim, S. Single-molecule epitranscriptomic analysis of full-length HIV-1 RNAs reveals functional roles of site-specific m⁶As. *Nat. Microbiol.* **2024**, *9* (5), 1340–1355.

(102) Trukuda, S.; Harris, J. M.; Magri, A.; Balfe, P.; Siddiqui, A.; Wing, P. A. C.; McKeating, J. A. The N6-methyladenosine demethylase ALKBH5 regulates the hypoxic HBV transcriptome. *PLOS Pathog.* **2024**, *20* (1), No. e1011917.

(103) Ning, K.; Zhao, J.; Feng, Z.; Park, S. Y.; McFarlin, S.; Cheng, F.; Yan, Z.; Wang, J.; Qiu, J. N6-methyladenosine modification of a parvovirus-encoded small noncoding RNA facilitates viral DNA replication through recruiting Y-family DNA polymerases. *Proc. Natl. Acad. Sci. U. S. A.* **2024**, *121* (25), No. e2320782121.

(104) Chen, S.; Kumar, S.; Espada, C. E.; Tiranuru, N.; Cahill, M. P.; Hu, L.; Hu, C.; Wu, L. N6-methyladenosine modification of HIV-1 RNA suppresses Type-I interferon induction in differentiated monocyte cells and primary macrophages. *PLoS Pathog.* **2021**, *17* (3), No. e1009421.

(105) Choi, E.-J.; Wu, W.; Zhang, K.; Yuan, X.; Deng, J.; Ismail, D.; Back, D. L.; Thumassery, K. S.; Garofalo, R. P.; Zhang, S.; Bao, X. Parent tRNA modification status determines the induction of functional tRNA-derived RNA by respiratory syncytial virus infection. *Viruses* **2023**, *15* (1), 57.

(106) El Kazzi, P.; Rabah, N.; Chamestin, C.; Poulain, L.; Ferron, F.; Debat, F.; Canard, B.; Missé, D.; Contard, B.; Niole, S.; Decroly, E. Internal RNA 2'-O-methylation in the HIV-1 genome counteracts ISG20 nuclease-mediated antiviral effect. *Nucleic Acids Res.* **2023**, *51* (6), 2501–2515.

(107) Ding, S.; Liu, H.; Liu, L.; Ma, L.; Chen, Z.; Zhu, M.; Liu, L.; Zhang, X.; Hao, H.; Zuo, L.; Yang, J.; Wu, X.; Zhou, P.; Huang, F.; Zhu, F.; Guan, W. Epigenetic addition of m⁶C to HBV transcripts promotes viral replication and evasion of innate antiviral responses. *Cell Death Dis.* **2024**, *15* (1), 1–15.

(108) Small-Saunders, J. L.; Sinha, A.; Blasham, T. S.; Hagena, L. M.; Sun, G.; Preiser, P. R.; Dedon, P. C.; Fidock, D. A. tRNA modification reprogramming contributes to artemisinin resistance in *Plasmodium falciparum*. *Nat. Microbiol.* **2024**, *9* (6), 1483–1498.

(109) Liu, Y.; Zhou, J.; Li, X.; Zhang, X.; Shi, J.; Wang, X.; Li, H.; Miao, S.; Chen, H.; He, X.; Dong, L.; Lee, G. B.; Zheng, J.; Liu, R.-J.; Su, B.; Ye, Y.; Flavell, R. A.; Yi, C.; Wu, Y.; Li, H.-B. tRNA-m⁶A modification promotes T cell expansion via efficient MYC protein synthesis. *Nat. Immunol.* **2022**, *23* (10), 1433–1444.

(110) Wang, L.; Wen, M.; Cao, X. Nuclear hnRNP2B1 initiates and amplifies the innate immune response to DNA viruses. *Science* **2019**, *365* (6454), No. eaav0758.

(111) Kim, G.-W.; Siddiqui, A. Hepatitis B Virus X protein expression is tightly regulated by N6-methyladenosine modification of its mRNA. *J. Virol.* **2022**, *96* (4), No. e01655-21.

(112) Zhang, Q.; Kang, Y.; Wang, S.; Gonzalez, G. M.; Li, W.; Hui, H.; Wang, Y.; Rana, T. M. HIV reprograms host m⁶A RNA methylome by viral vpr protein-mediated degradation of PCIF1. *Nat. Commun.* **2021**, *12* (1), 3543.

(113) Benoni, B.; Potužnik, J. F.; Škrba, A.; Benoni, R.; Trylčova, J.; Tulpa, M.; Spustová, K.; Grab, K.; Mititelu, M.-B.; Pačes, J.; Weber, J.; Stanek, D.; Kowalska, J.; Bednarcova, L.; Keckesova, Z.; Vopaleňský, P.; Gaharova, L.; Čabova, H. HIV-1 infection reduces NAD capping of host cell snRNA and snoRNA. *ACS Chem. Biol.* **2024**, *19* (6), 1243–1249.

(114) Jiang, S.; Hu, J.; Bai, Y.; Hao, R.; Liu, L.; Chen, H. Transcriptome-wide 5-methylcytosine modification profiling of long

non-coding RNAs in A549 cells infected with H1N1 influenza A virus. *BMC Genomics* **2023**, *24* (1), 316.

(115) Zhang, Y.; Zhang, L.-S.; Dai, Q.; Chen, P.; Lu, M.; Kairis, E. L.; Murugiah, V.; Xu, J.; Shukla, R. K.; Liang, X.; Zou, Z.; Coemet-Boyaka, E.; Qiu, J.; Peeples, M. E.; Sharma, A.; He, C.; Li, J. 5-methylcytosine (m^5C) RNA modification controls the innate immune response to virus infection by regulating type I interferons. *Proc. Natl. Acad. Sci. U. S. A.* **2022**, *119* (42), No. e2123338119.

(116) Jin, Y.-Y.; Liang, Y.-P.; Huang, W.-H.; Guo, L.; Cheng, L.-L.; Ran, T.-T.; Yao, J.-P.; Zhu, L.; Chen, J.-H. Ocular A-to-I RNA editing signatures associated with SARS-CoV-2 infection. *BMC Genomics* **2024**, *25* (1), 431.

(117) Pujantell, M.; Rivera-Muñoz, E.; Badia, R.; Castellvi, M.; Garcia-Vidal, E.; Sirera, G.; Puig, T.; Ramirez, C.; Clotet, B.; Esté, J. A.; Ballana, E. RNA editing by ADAR1 regulates innate and antiviral immune functions in primary macrophages. *Sci. Rep.* **2017**, *7* (1), No. 13339.

(118) Zhang, N.; Tang, W.; Torres, L.; Wang, X.; Ajaj, Y.; Zhu, L.; Luan, Y.; Zhou, H.; Wang, Y.; Zhang, D.; Karbatov, V.; Khan, S. A.; Kumar, P.; Hadaigü, A.; Wu, D.; Lu, J. Cell surface RNAs control neutrophil recruitment. *Cell* **2024**, *187* (4), 846–860.e17.

(119) Zhang, X.-X.; Sun, Y.-Z.; Wang, W.; Gao, Y.; Wei, X.-Y.; Sun, H.-C.; Wang, C.-R.; Ni, H.-B.; Yang, X.; Elsheikha, H. M.; Guo, H.-P. Altered landscape of total RNA, tRNA and snRNA modifications in the liver and spleen of mice infected by *Toxoplasma gondii*. *PLoS Negl. Trop. Dis.* **2024**, *18* (6), No. e0012281.

4.4 Manuscript D

Human RNA editing patterns reveal disrupted mRNA splicing upon *Rhizopus oryzae* stimulation

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Summary:

A-to-I RNA editing is a post-transcriptional modification of RNA that is important in immunity, yet its role during host-fungal interaction remains poorly understood. By analyzing a publicly available RNA-seq dataset of human PBMCs coincubated with three fungi (*A. fumigatus*, *C. albicans*, and *R. oryzae var delemar*), we observed an increase of A-to-I hyper-editing patterns in response to *R. oryzae*. Interestingly, *R. oryzae* stimulation also induced a large number of significant alternative splicing events, whereas *A. fumigatus* and *C. albicans* triggered no such response. The remainder of the manuscript focuses on several interesting consequences of this altered splicing effect in potential modulation of the immune response. Our investigation extended the role of host A-to-I RNA editing and RNA splicing in response to fungal stimulations and revealed an intriguing link between *R. oryzae* challenge and altered RNA splicing.

Contribution to the manuscript:

Xiaoqing Pan took a lead role in designing and executing the project. She participated in data analysis, data visualization, and played a key role in writing the manuscript.

Authors' contributions (in %) to the given categories of the publication

Author	Conceptual	Data analysis	Experimental	Writing the manuscript	Provision of material
Xiaoqing Pan	30%	90%	70%	40%	0%
Michael Richter	20%	0%	30%	30%	0%
Delf-Magnus Kummerfeld	5%	10%	0%	0%	0%
Alexander Bruch	5%	0%	0%	0%	0%
Katarzyna Winek	5%	0%	0%	0%	10%
Matthew G. Blango	35%	0%	0%	30%	90%
Total:	100%	100%	100%	100%	100%

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Human RNA editing patterns reveal disrupted mRNA splicing upon *Rhizopus oryzae* stimulation

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Abstract

Fungal pathogens trigger well-defined transcriptomic responses, yet our understanding of the epitranscriptome during infection remains incomplete. To partially address this knowledge gap, we focused on host adenosine deaminases acting on RNA (ADAR)-mediated adenosine-to-inosine (A-to-I) RNA editing. Leveraging publicly available RNA-seq data from human peripheral blood mononuclear cells (PBMCs) and small airway epithelial cells (HSAECs), we observed a significant increase in A-to-I hyper-editing in response to the human fungal pathogen *Rhizopus oryzae*, particularly in intronic regions. Such a signature was notably absent from the response to two other dangerous fungal pathogens, *Aspergillus fumigatus* and *Candida albicans*. We observed *R. oryzae* to trigger a specific dysregulation of host splicing genes and an increased frequency of exon inclusion/skipping and intron retention associated with splicing factors YBX1/SRSF2. *R. oryzae*-driven alternative splicing led to mature transcripts with lower folding free energy and higher levels of hyper-editing, indicative of more double-stranded RNA structure. Hyper-edited RNAs are known to inhibit the interferon response, consistent with a reported downregulation of pattern recognition receptors and limited type-I interferon signaling during *R. oryzae* infection. Exon skipping events observed in immune genes like the Nod-like receptor NLRC5, led to transcripts with a predicted premature stop codon and likely further impair immune signaling. Taken together, these results suggest that *R. oryzae* disrupts host RNA splicing by inducing YBX1/SRSF2, resulting in increased levels of hyper-editing and altered antifungal immunity.

Introduction

Pathogenic fungi infect over one billion and kill ~3.75 million people per year ¹. In 2022, the World Health Organization (WHO) created a fungal priority pathogens list highlighting the need for more intense investigation of significant human pathogens like *Aspergillus fumigatus*, *Candida* spp. (including *Candida albicans*, *Candida auris*, and others), *Mucorales* spp. (including *Rhizopus* spp., *Mucor* spp., and *Lichtheimia* spp.), and many others ². These fungi infect diverse human niches, including skin, lungs, and other tissues and cause life-threatening diseases mainly in immunocompromised patients. *A. fumigatus* is an environmental mold and the primary causative agent of invasive aspergillosis via asexual spores (called conidia) that are easily inhaled into the alveoli of the lung ³. *C. albicans* is a dimorphic yeast that is commonly found as a commensal member of the gut microbiota in humans. However, in the right conditions, *C. albicans* is also an opportunistic pathogen of the mucosa (mouth, throat, gut, vagina, and skin) in a disease known as candidiasis. Similar to *A. fumigatus*, *Mucorales* spp. are molds capable of producing spores that predominantly affect the lungs, sinuses, skin, and other soft tissues, with *Rhizopus* species accounting for more than 70% of mucormycoses worldwide ⁴, largely in immunocompromised or

diabetic patients⁵. Despite the existence and continued development of effective antifungals, overall case-fatality rates range from 23% to 80%^{1,6}.

Similar to bacterial and viral infections, a growing number of studies suggest that fungi like *A. fumigatus* and *Candida* species trigger the interferon (IFN) response⁷⁻⁹, crucial in initiating immune defenses. For example, *C. albicans* secretes extracellular vesicles, which induce type I IFN signaling through the cyclic GMP-AMP synthase (cGAS)-stimulator of IFN genes (STING) pathway¹⁰, whereas during vaginal *Candidiasis*, mitochondrial-associated type I IFN signaling is induced⁹. One study of *A. fumigatus*-stimulated Signal Transducer and Activator of Transcription 3 (STAT3)-deficient patients showed a reduction of IFN- γ and IL-17 in those patients compared with healthy controls, indicating STAT3-deficiency leads to lower IFN- γ and IL-17 and susceptibility to aspergillosis¹¹. In human bronchial epithelial cells (HBECS), it was reported that double-stranded RNA (dsRNA) from resting conidia of *A. fumigatus* is recognized by Toll-like receptor-3, resulting in the induction of IFN- β signaling pathway and its associated genes⁷.

Host dsRNA sensors trigger an IFN response upon substrate recognition¹². While some dsRNA sensors distinguish host-derived endogenous dsRNA (self) from pathogen-derived exogenous dsRNA (non-self), most lack this “self” versus “non-self” discrimination. This includes MDA5 (Melanoma Differentiation-Associated gene 5), a key protein in human immune response¹³. As a member of RIG-I (retinoic acid-inducible gene-I-like receptors, RLRs) family, it functions as a pattern recognition receptor (PRR) detecting pathogens and inducing the IFN response¹³. Interestingly, base substitutions of adenosine (A) to inosine (I) in endogenous dsRNA limits activation of MDA5 and subsequent immune responses otherwise elicited by pathogen-derived dsRNA. In metazoans, these substitutions, termed A-to-I RNA editing, are catalyzed by an enzyme family called ADAR (Adenosine Deaminases Acting on RNA)^{14,15}. While hyper-editing refers to a promiscuous form of RNA editing on long dsRNA that is rarely found in coding sequences¹⁶.

Three enzymes constitute the ADAR family in humans: ADAR1, ADAR2, and ADAR3. ADAR1 and ADAR2 are found in many tissues, whereas ADAR3 is tissue-specific and only found in the brain¹⁷. ADAR1 exists in two isoforms: p110 and p150, driven by alternative promoters^{17,18}. p110 exclusively localizes in the nucleus while p150 accumulates in the cytoplasm after induction but also transfers between cytoplasm and nucleus¹⁹. It is typically thought that p150 is responsible for the suppression of dsRNA sensors MDA5 by A-to-I RNA editing and PKR (protein kinase R) by competing for its binding sites on dsRNA²⁰⁻²³. While the role of ADAR in modifying endogenous dsRNA to prevent immune activation has been shown in many studies, recent studies also indicate that ADAR1 modifies pathogen-derived RNA

molecules^{23,24}. Numerous studies of host RNA editing patterns after viral infection have been reported, but the molecular mechanisms in the host upon fungal infections are largely unknown. A study on *C. albicans* infection showed increased A-to-I editing activity of human epithelial cells and endothelial cells upon infection²⁵, but whether this is a unique feature of *C. albicans* or more widely relevant to other fungal infections remains unknown.

A-to-I RNA editing in exonic regions can result in altered amino acid sequence (termed recoding), which is usually driven by ADAR2²⁶. In many cases, A-to-I editing occurs more frequently in UTRs and introns, depending on the organism, tissue, and other conditions, particularly where inverted-repeat regions like *Alu* elements are abundant²⁷. RNA splicing shares similar time and space with A-to-I editing and the two post-transcriptional regulators seem to be linked^{28,29}. For example, pre-mRNA splicing was indicated to be both editing-dependent and editing-independent depending on the study and conditions²⁹⁻³². In addition, A-to-I editing can also be influenced by pre-mRNA splicing. For example, it was reported that the lack of splicing factors NOVA1 and NOVA2 changes global editing levels, hinting that splicing efficiency could regulate RNA editing levels in mice³³. Overall, the mechanisms of RNA editing events that are associated with alternative splicing still remain unclear, especially upon infection.

Here, using publicly available data we analyzed RNA editing in human PBMCs following stimulation with three inactivated opportunistic invasive fungal pathogens (*A. fumigatus*, *C. albicans*, and *R. oryzae* var *delemar*). In contrast to *A. fumigatus* and *C. albicans*, we found that *R. oryzae* resulted in editing of host genes uniquely involved in RNA splicing. Further analysis revealed widespread low-level disruption of alternative splicing, resulting in retained introns and skipped exons, often in immune-relevant genes like *NLRC5*, a PRR not previously linked to *R. oryzae* defense. We hypothesize that *R. oryzae* may actively dysregulate *NLRC5* splicing to suppress innate immune response.

Results

Human PBMCs show distinct RNA editing patterns upon fungal challenge.

The contribution of RNA editing and double-stranded RNA regulation to the host response against fungal infections remains poorly understood. To address this question, REDITools and the *hyper-editing* software packages were used to identify RNA editing and hyper-editing events, respectively, using previously published poly(A)-enriched, unstranded RNA-seq data of human peripheral blood mononuclear cells (PBMCs) confronted with dead *A. fumigatus* (paraformaldehyde-fixed), *C. albicans* (heat-killed), and *R. oryzae* (paraformaldehyde-fixed) (NCBI GEO Identifier: GSE162746; ³⁴) (**Fig. 1A**).

We observed global levels of A-to-I RNA editing to be comparable between infections using REDITools, but hyper-editing indicated that samples coincubated with *R. oryzae* showed almost double the hyper-editing events compared with the control (**Fig. 1B**). Hyper-editing occurs mostly in long dsRNA derived from repeat elements, especially in *Alu* repeats³⁵. We next mapped the hyper-edited sites to different regions and found that *R. oryzae* stimulation heterogeneously increased hyper-editing across the human transcriptome (**Fig. 1C, Fig. S1A**). *C. albicans* coincubation led to a small but significant reduction, while no change was observed from coincubation with *A. fumigatus*. Since ADAR1 is the main A-to-I editor in humans, we investigated the expression of *ADAR1* in relation to the calculated A-to-I RNA editing index in *Alu* elements³⁶. Interestingly, stimulated PBMCs upon *R. oryzae* coincubation showed the lowest *ADAR1* expression but the highest A-to-I RNA editing index in *Alu* elements (**Fig. 1C**), potentially due to more specific, less promiscuous editing at lower ADAR1 concentrations. Principal component analysis (PCA) showed that the host response to *R. oryzae* had the highest inter-individual variation compared with *A. fumigatus* and *C. albicans* stimulation (**Fig. S1B**), which aligned with previous work³⁴. Additionally, 5,657 differentially expressed genes (DEGs) were identified upon *R. oryzae* stimulation compared to 552 upon *A. fumigatus* stimulation and 622 with *C. albicans* challenge (**Supplementary Table 1**).

We subsequently focused on identifying genes specifically edited in fungal-stimulated samples. REDITools revealed 121 edited genes in *A. fumigatus*-treated PBMCs that were not edited in the control, 128 in *C. albicans*-treated PBMCs compared to control, and 172 in *R. oryzae*-treated PBMCs compared to control (**Fig. 1D; Supplementary Table 2; Fig. S1D (analysis from hyper-editing)**). Strikingly, direct comparison of these fungal-specific gene sets showed minimal overlap between stimulation conditions; only seven genes (*IFNAR1*, *CCNL1*, *USP47*, *PURB*, *ZNF789*, *ZNF394*, *FAM177A1*) were common to all three fungal treatments, indicating that A-to-I RNA editing patterns are largely stimulus-specific across different fungi (**Fig. S1C**). GO enrichment analysis revealed several pathways as targets of editing in response to fungal stimulations. Proliferation-related pathways were predicted in each comparison, while unique pathways were also observed in the host responding to *C. albicans* and *R. oryzae*. Genes uniquely edited in response to *C. albicans* were enriched in virus-responsive and innate immunity regulating pathways, consistent with a well-documented induction of type I and type II interferon-related pathway genes in response to *C. albicans*³⁴. Host genes involved in RNA splicing and processing were observed in response to *R. oryzae* stimulation, suggesting a potential interplay between RNA editing and RNA splicing upon fungal challenge.

***R. oryzae* coincubation results in differential host alternative splicing.**

To further investigate if *R. oryzae* coinubation dysregulates host RNA splicing, we next focused on DEGs in the RNA splicing pathway (GO: 0008380). 52 RNA splicing associated genes were differentially expressed in response to *R. oryzae* (45 up-regulated and 7 down-regulated), while only 9 and 3 RNA splicing-associated genes were differentially regulated in response to *A. fumigatus* and *C. albicans*, respectively (**Fig. 2A**). Since many splicing factors were observed to be differentially expressed, we next performed an alternative splicing (AS) analysis using rMATS-turbo³⁷. 12318, 12171, and 19738 total AS events were detected when comparing *A. fumigatus*, *C. albicans*, and *R. oryzae* treatment relative to the unstimulated control, respectively. 72, 83, and 5728 significant AS events (FDR $\leq$ 0.01, $|\Delta\text{PSI}| \geq 0.05$) were detected for *A. fumigatus*, *C. albicans*, or *R. oryzae* coinubation, respectively (**Fig. 2B**). Among the five common AS events (alternative 3' splice sites (A3SS), alternative 5' splice sites (A5SS), mutually exclusive exons (MXE), retained introns (RI), and skipped exons (SE)), SE was the most prolific splicing event type.

To address the caveat that the dataset analyzed was generated using inactivated fungi, we confirmed that live *R. oryzae* similarly dysregulates AS in the host, using a public dataset of human small airway epithelial cells (HSAECs) infected with *R. oryzae* for 0, 3, 6, and 16 h³⁸. We subjected this paired-end Illumina stranded RNA-seq data to with rMATS-turbo AS analysis (**Supplementary Fig. S3A**). We examined AS events at 3, 6, and 16 h of *R. oryzae* infection compared with 0 h infection and observed a total 72874, 76014, and 68559 AS events with 3844, 3665, and 6247 determined to be significant, respectively (**Supplementary Fig. S3B**).

***R. oryzae*-induced intron retention is associated with YBX1 and SRSF2 binding motifs.**

Since A-to-I RNA editing, particularly hyper-editing, is enriched in introns in humans³⁹, we focused first on RI events. In response to *R. oryzae* coinubation, PBMCs exhibited 1,021 examples of intron retention and 131 events where a retained intron was spliced out upon infection, while *A. fumigatus* and *C. albicans* only showed 10 and 17 differential RI events, respectively (**Fig. 3A**). Interestingly, 130 retained introns also contained RNA editing sites. To further explore the role of AS in response to *R. oryzae*, we focused on understanding the molecular mechanisms driving this aberrant splicing. We extracted differentially expressed splicing factors from **Fig. 2A** and performed motif enrichment analysis using rMAPS2 (RNA Map Analysis and Plotting Server 2;⁴⁰). We observed RNA binding motifs (RBMs) for 14 splicing factors, including multiple motifs from YBX1 (Y box binding protein 1) and a single motif from SRSF2 (Serine/Arginine-rich splicing factor 2) (**Fig. 3B**). Moreover, YBX1 showed two RBMs that increased in association with RI events, which were located upstream of the intron retention (**Fig. 3C**).

Altered host RNA splicing is associated with increased A-to-I RNA editing.

To determine if changes in RNA splicing were influencing RNA editing patterns, we next intersected our AS analysis and A-to-I editing, together with DEGs to better understand the impact of these changes on gene expression. We detected a total of 5,656 DEGs, 1,850 DASGs (differentially alternatively spliced genes), and 1,374 AIEGs (A-to-I RNA edited genes) in PBMCs treated with *R. oryzae* (**Fig. 4A**). 116 appeared in all three categories. As expected, GO analysis revealed these genes were mainly associated with RNA splicing (**Fig. 4B**). By intersecting DASGs and AIEGs, ~25% of DASGs were found to associate with editing events in response to *R. oryzae*, while two other tested fungal coinfections did not show such a high percentage (**Fig. 4C**). This is in line with the HSAEC dataset we analyzed with live *R. oryzae* infection, where roughly 20% of AS events overlapped with A-to-I RNA editing at all three infection time points (**Supplementary Fig. S3C**). To validate the association, we next analyzed the folding free energies of predicted RNA secondary structure after AS. We determined the minimum free energy (MFE) structures of spliced-in (longer isoforms) and spliced-out (shorter isoforms) transcripts using RNAfold⁴¹. As expected, spliced-in transcripts showed lower MFE values within AS events like A3SS, A5SS, RI, and SE (**Fig. 4D**), suggesting more stability of transcripts and higher likelihood to form dsRNA. Overall, these results suggest that *R. oryzae* coinfection induces intron retention events and an accumulation of dsRNA, which may then serve as substrates for ADAR1 hyper-editing.

R. oryzae stimulation results in exon skipping in the pattern recognition receptor NLRC5

Exon skipping events were determined to be the most significantly altered in response to *R. oryzae* coinfection (**Fig. 2B**). To assess the consequence of this abundant exon skipping, we determined the directionality of these events and revealed 1014 skipped exons in response to *R. oryzae*, compared with 20 and 30 skipped exons upon *A. fumigatus* and *C. albicans* challenge, respectively (**Fig. 5A**). Among these events, the pattern recognition receptor NOD-like receptor family CARD domain containing 5 (NLRC5), important as a driver of an inflammatory cell death called PANoptosis, was shown to undergo significant alternative splicing in the form of exon 27 skipping, while exhibiting no change in mRNA expression (normalized by TPM) compared with unstimulated samples (**Fig. 5B-D**). Further protein domain prediction identified that exon skipping resulted in the likely loss of a substrate binding site at the C-terminus due inclusion of a premature stop codon (**Fig. 5E**). Although not previously linked directly to defense against *R. oryzae*, the putative disruption of the NLRC5 protein begets the potential hypothesis that *R. oryzae* dysregulates NLRC5 through altered alternative splicing to suppress inflammatory cell death upon infection of PBMCs.

Discussion

The aim of this study was to elucidate common and species-specific RNA processing patterns in the host transcriptome upon challenge with the pathogenic fungi *A. fumigatus*, *C. albicans*, and *R. oryzae*. ADAR1-dependent A-to-I RNA editing is an important mechanism regulating inflammation by preventing excessive activation of dsRNA sensors like MDA5 and PKR²⁰. Previous studies have shown ADAR1 to be interferon inducible and responsive to various infections²¹, though its expression and A-to-I RNA editing activity have not been well-characterized in response to fungal infection. In this study, we re-analyzed mRNA-seq data from³⁴, with a particular focus on A-to-I RNA editing and hyper-editing imparted by the ADAR proteins. A-to-I RNA editing detected using REDITools was slightly decreased during the tested fungal stimulations, but surprisingly A-to-I hyper-editing in PBMCs showed a significant increase in response to *R. oryzae* compared to the unstimulated control. Since hyper-editing typically occurs in dsRNA regions > 40 bp in a promiscuous manner³⁹, the increase in hyper-editing suggested that *R. oryzae* stimulation induced transcription or promoted accessibility of a new population of long dsRNA in the cell. A separate analysis from airway epithelial cells upon live *R. delemar* infection identified increased A-to-I RNA editing events from both REDITools and *hyper-editing* (**Fig. S2**), suggesting that host editing events are dependent on cell type, infection status, and likely other environmental conditions.

GO enrichment analysis of editing-associated genes showed several interesting features upon fungal infection. First, genes edited in response to *C. albicans* coincubation were enriched in pathways involved in antiviral defense regulation of innate immune responses, which was expected given that *C. albicans* can induce host genes of the type I and type II interferon pathway typically associated with antiviral defense⁴². The retinoic acid-inducible gene I (RIG-I) for example, initiates antiviral immunity and was predicted to be edited only after *C. albicans* stimulation; IFN induced genes like *IFI6*, *IFI44*, *IFIT3* were also predicted to follow this pattern. Second, *R. oryzae* stimulation of PBMCs revealed editing of RNA splicing genes, which was not previously reported. A closer look at host differential gene expression after infection revealed an obvious upregulation of genes associated with splicing upon *R. oryzae* coincubation (**Fig. 2A**), with 52 RNA splicing-associated genes (from GO Category 0008380) significantly upregulated when stimulated with *R. oryzae*, compared to only 9 and 3 genes in response to *A. fumigatus* and *C. albicans*, respectively. Collectively, these results confirmed that *R. oryzae* coincubation with formaldehyde-fixed or live spores leads to massive transcriptomic changes in host cells. We suspect that these effects are driven by signaling through the GRP78 chaperone and surface receptor upon activation by the *R. oryzae* CotH proteins⁴³. It is also possible that an effector is released

by *R. oryzae*, as happens with the symbiotic arbuscular mycorrhizal fungus *R. irregularis*, which releases SP7-like effectors to fine tune the plant host transcriptome ⁴⁴. Other examples are also in the literature, for example arginine-rich effector Pst_A23 from wheat pathogenic fungus *Puccinia striiformis* f. sp. *tritici* (*Pst*) can be secreted and directly localized in nuclear regions enriched in splicing factors, thereby interfering with host immunity ⁴⁵. Plant-pathogens like the oomycete *Phytophthora infestans* also secrete effectors that interact with host snRNPs and affect pre-mRNA splice sites ⁴⁶.

To better understand the consequences of splicing dysregulation and refine our model, we further explored both RI and SE events. Though RI might result in more dsRNA, because the main substrates of ADAR1 occur in introns and UTRs ³⁹, we only observed 130 out of 73200 editing events to be located in retained introns, likely due to the challenges of detecting hyper-editing in low read coverage. Nevertheless, by predicting RBMs, we observed YBX1, SRSF2, and their corresponding binding motifs to associate with RI events. YBX1 functions in stabilizing mRNA and regulating transcripts ⁴⁷, and multiple studies have shown YBX1 to affect AS. For example, YBX1 knockdown embryos in mammals were observed to promote SE in Breast Cancer Type 1 (BRCA1), but inhibit SE in Ekaryotic Translation Initiation Factor 3 Subunit 1 (EIF3I), Embryonic Ectoderm Development (EED), and Heterogeneous Nuclear Ribonucleoprotein M (HNRNPM) ⁴⁸. Interestingly, our results also revealed that some HNRNP genes (*HNRNPH1*, *HNRNPH3*, *HNRNPC*, *HNRNPA2B*, *HNRNPD*, and *HNRNPK*) showed significant alternative splicing events (**Table 3**). Thus, we hypothesize that *R. oryzae* stimulation induces splicing factor YBX1 through the CoH-GRP78 axis, which triggers intron retention and further disruption to RNA splicing.

GRP78 has been heavily studied as a receptor in regard to infection by members of the *Mucorales*. Fungal surface proteins CoH2 and CoH3 both bind to GRP78 in nasal and alveolar epithelial cells ⁴, and GRP78 expression was previously shown to be upregulated upon infection in PBMCs as well ³⁴. The interaction of CoH-GRP78 has varied effects, for example enabling invasion of endothelial cells or facilitating hyphal escape of phagocytosis by neutrophils to ultimately cause more cell damage ⁴⁹. Interestingly, *GRP78* mRNA is alternatively spliced to produce several isoforms. One isoform, cell surface GRP78 (CS-GRP78), is produced by retention of intron 1 to trigger alternative translation initiation, and this results in an isoform lacking the endoplasmic reticulum (ER) signal peptide. The full-length isoform is localized to the ER to control ER stress ⁵⁰. Whereas CS-GRP78 promotes cell survival under ER stress, full-length GRP78 regulates ER stress by regulating the activation of ER stress transducers to facilitate activation of the unfolded protein response (UPR) ⁵¹. The UPR was previously shown to be activated in *R. oryzae*-treated PBMCs; however, we did not observe differential alternative

splicing of GRP78, suggesting that CS-GRP78 production may not be an outcome of the altered RNA splicing upon *R. oryzae* treatment.

Overall, through investigation of host A-to-I RNA editing, we revealed a disruption of RNA splicing during *R. oryzae* coinubation, identifying RNA splicing-related genes specifically involved in the host response to *R. oryzae* treatment. We propose that fungal CotH proteins drive activation of YBX1 through the GRP78 axis, resulting in altered alternative splicing, increased hyper-editing, and inhibition of host dsRNA signaling pathways. Future studies will be required to elucidate the full mechanism of action of *R. oryzae*-dependent disruption of RNA splicing during infection.

Materials and Methods

RNA-seq data

The data analyzed in this manuscript was generated by ³⁴, specifically NCBI GEO Identifier: GSE162746 using samples from the 4 h stimulation time point as described above. Data was also reanalyzed from ³⁸, specifically NCBI SRA: SRS21091027 – SRS21091037 and SRS210910858 as described above.

RNA-seq analysis

As mentioned in ³⁴, PBMCs (n = 8 donors) in 6-well plates were stimulated with three fungi separately. For *R. oryzae* stimulation, 6 replicates (A - F) were used for further analysis. Of note, one replicate of *A. fumigatus*-infected PBMCs that was determined to be a partial outlier was kept in the analysis, which may serve as a confounding factor in some downstream analyses. After trimming the adaptors and filtering low quality reads via Trim Galore (version 0.6.8) (<https://github.com/FelixKrueger/TrimGalore>), sequencing reads were mapped to the human genome (GRCh38 v43) using STAR (version 2.7.10a) ⁵². FeatureCounts (version 2.0.3) was used to quantify the read counts per gene based on the annotation using the default parameters. Differential expressed genes were predicted by R package DESeq2 ⁵³ with threshold ($p < 0.05$, FoldChange >2). R package ClusterProfiler ⁵⁴ was used for GO analysis.

RNA editing prediction

To further analyse RNA editing, we aligned clean reads to the human genome with strict parameters (`--outFilterMultimapNmax 1, --alignSJoverhangMin 8, --alignSJDBoverhangMin 1, --outFilterMismatchNoverLmax 0.04, --alignIntronMin 20, --alignIntronMax 1000000, --alignMatesGapMax 1000000`). REDITools ^{55, 56} was then used for predicting RNA editing sites. We applied stringent filtering with a coverage of at least 10 reads in at least 2 replicates and an editing

frequency of $\geq 1\%$. For non-stranded RNA-seq dataset, both A-to-G and T-to-C were considered as A-to-G editing. Hyper-editing pipeline (<https://github.com/ryncuddleston/RNA-hyper-editing>) was used for hyper-editing analysis with default parameters, and RNAEditingIndexer (<https://github.com/a2iEditing/RNAEditingIndexer>) in *Alu* regions to identify the RNA editing index, defined as the percentage of A-to-G mismatches divided by total aligned nucleotides within the repeats. annotatePeaks.pl under HOMER software suite ⁵⁷ was used to analyze editing sites across human genome (hg38).

Alternative splicing prediction

We used clean reads to identify AS events via rMATS-turbo (version 4.3.0) ³⁷ with “--s” option. Events with *P* value < 0.05, FDR < 0.01, and |PSI| > 0.05 were regarded as differential AS events between control and stimulated fungi. The rMATS-turbo output was visualized with rmats2sashimiplot ³⁷. Junction reads (JC) were analyzed for AS event analysis. rMAPS online website ⁴⁰ was performed to predict RNA binding motifs with the raw rMATS-turbo result as the input.

Data analysis

All downstream analyses were performed using R version 4.2.2 (2022-10-31) and its IDE RStudio (2022-02-02). Figures were drawn with the ggplot2 package (version 3.4.2). *P*-values were determined by ANOVA for multiple-group comparisons or two-tailed Student’s *t*-tests for pairwise comparisons where appropriate and differences between the groups with a significance set a *P*-value of <0.05.

Acknowledgement

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Reference

1. Denning, D.W. Global incidence and mortality of severe fungal disease. *Lancet Infect Dis* (2024).
2. Fisher, M.C. & Denning, D.W. The WHO fungal priority pathogens list as a game-changer. *Nat*

- Rev Microbiol*, 1–2 (2023).
3. Latge, J.P. & Chamilos, G. *Aspergillus fumigatus* and Aspergillosis in 2019. *Clin Microbiol Rev* **33** (2019).
 4. Alqarihi, A., Kontoyiannis, D.P. & Ibrahim, A.S. Mucormycosis in 2023: an update on pathogenesis and management. *Front Cell Infect Microbiol* **13**, 1254919 (2023).
 5. Petrlikos, G. *et al.* Epidemiology and clinical manifestations of mucormycosis. *Clin Infect Dis* **54 Suppl 1**, S23–34 (2012).
 6. Spellberg, B., Edwards, J., Jr. & Ibrahim, A. Novel perspectives on mucormycosis: pathophysiology, presentation, and management. *Clin Microbiol Rev* **18**, 556–569 (2005).
 7. Beisswenger, C., Hess, C. & Bals, R. *Aspergillus fumigatus* conidia induce interferon-beta signalling in respiratory epithelial cells. *Eur Respir J* **39**, 411–418 (2012).
 8. Pekmezovic, M., Dietschmann, A. & Gresnigt, M.S. Type I interferons during host-fungus interactions: Is antifungal immunity going viral? *PLoS Pathog* **18**, e1010740 (2022).
 9. Pekmezovic, M. *et al.* *Candida* pathogens induce protective mitochondria-associated type I interferon signalling and a damage-driven response in vaginal epithelial cells. *Nature Microbiology* (2021).
 10. Brown Harding, H. *et al.* *Candida albicans* extracellular vesicles trigger type I IFN signalling via cGAS and STING. *Nature Microbiology* (2024).
 11. Danion, F. *et al.* *Aspergillus fumigatus* Infection in Humans With STAT3-Deficiency Is Associated With Defective Interferon-Gamma and Th17 Responses. *Front Immunol* **11**, 38 (2020).
 12. Wang, Y., Swiecki, M., McCartney, S.A. & Colonna, M. dsRNA sensors and plasmacytoid dendritic cells in host defense and autoimmunity. *Immunol Rev* **243**, 74–90 (2011).
 13. Cottrell, K.A., Andrews, R.J. & Bass, B.L. The competitive landscape of the dsRNA world. *Mol Cell* (2023).
 14. Bass, B.L. & Weintraub, H. An unwinding activity that covalently modifies its double-stranded RNA substrate. *Cell* **55**, 1089–1098 (1988).
 15. Bass, B.L. & Weintraub, H. A developmentally regulated activity that unwinds RNA duplexes. *Cell* **48**, 607–613 (1987).
 16. Porath, H.T., Knisbacher, B.A., Eisenberg, E. & Levanon, E.Y. Massive A-to-I RNA editing is common across the Metazoa and correlates with dsRNA abundance. *Genome Biol* **18**, 185 (2017).
 17. Yanai, M. *et al.* ADAR2 Is Involved in Self and Nonself Recognition of Borna Disease Virus Genomic RNA in the Nucleus. *J Virol* **94** (2020).
 18. Desterro, J.M. *et al.* Dynamic association of RNA-editing enzymes with the nucleolus. *J Cell Sci* **116**, 1805–1818 (2003).
 19. Savva, Y.A., Rieder, L.E. & Reenan, R.A. The ADAR protein family. *Genome Biol* **13**, 252 (2012).
 20. Hu, S.B. *et al.* ADAR1p150 prevents MDA5 and PKR activation via distinct mechanisms to avert fatal autoinflammation. *Mol Cell* (2023).
 21. Liddicoat, B.J. *et al.* RNA editing by ADAR1 prevents MDA5 sensing of endogenous dsRNA as nonself. *Science* **349**, 1115–1120 (2015).
 22. Maurano, M. *et al.* Protein kinase R and the integrated stress response drive immunopathology caused by mutations in the RNA deaminase ADAR1. *Immunity* **54**, 1948–1960 e1945 (2021).
 23. Pfaller, C.K., George, C.X. & Samuel, C.E. Adenosine Deaminases Acting on RNA (ADARs) and Viral Infections. *Annu Rev Virol* **8**, 239–264 (2021).
 24. Nie, Y., Hammond, G.L. & Yang, J.H. Double-stranded RNA deaminase ADAR1 increases host susceptibility to virus infection. *J Virol* **81**, 917–923 (2007).
 25. Huang, Y. *et al.* A survey on cellular RNA editing activity in response to *Candida albicans* infections. *BMC Genomics* **19**, 43 (2018).
 26. Gabay, O. *et al.* Landscape of adenosine-to-inosine RNA recoding across human tissues. *Nat Commun* **13**, 1184 (2022).
 27. Deininger, P. Alu elements: know the SINEs. *Genome Biol* **12**, 236 (2011).

28. Bentley, D.L. Coupling mRNA processing with transcription in time and space. *Nat Rev Genet* **15**, 163–175 (2014).
29. Hsiao, Y.E. *et al.* RNA editing in nascent RNA affects pre-mRNA splicing. *Genome Res* **28**, 812–823 (2018).
30. Tang, S.J. *et al.* Cis- and trans-regulations of pre-mRNA splicing by RNA editing enzymes influence cancer development. *Nat Commun* **11**, 799 (2020).
31. Kapoor, U. *et al.* ADAR-deficiency perturbs the global splicing landscape in mouse tissues. *Genome Res* **30**, 1107–1118 (2020).
32. Blow, M., Futreal, P.A., Wooster, R. & Stratton, M.R. A survey of RNA editing in human brain. *Genome Res* **14**, 2379–2387 (2004).
33. Licht, K. *et al.* A high resolution A-to-I editing map in the mouse identifies editing events controlled by pre-mRNA splicing. *Genome Res* **29**, 1453–1463 (2019).
34. Bruno, M. *et al.* Comparative host transcriptome in response to pathogenic fungi identifies common and species-specific transcriptional antifungal host response pathways. *Computational and Structural Biotechnology Journal* (2020).
35. Porath, H.T., Carmi, S. & Levanon, E.Y. A genome-wide map of hyper-edited RNA reveals numerous new sites. *Nat Commun* **5**, 4726 (2014).
36. Roth, S.H., Levanon, E.Y. & Eisenberg, E. Genome-wide quantification of ADAR adenosine-to-inosine RNA editing activity. *Nat Methods* **16**, 1131–1138 (2019).
37. Wang, Y. *et al.* rMATS-turbo: an efficient and flexible computational tool for alternative splicing analysis of large-scale RNA-seq data. *Nat Protoc* **19**, 1083–1104 (2024).
38. Kavaliauskas, P. *et al.* Multiple roles for hypoxia inducible factor 1-alpha in airway epithelial cells during mucormycosis. *Nat Commun* **15**, 5282 (2024).
39. Blango, M.G. & Bass, B.L. Identification of the long, edited dsRNAome of LPS-stimulated immune cells. *Genome Res* **26**, 852–862 (2016).
40. Hwang, J.Y. *et al.* rMAPS2: an update of the RNA map analysis and plotting server for alternative splicing regulation. *Nucleic Acids Res* **48**, W300–W306 (2020).
41. Lorenz, R. *et al.* ViennaRNA Package 2.0. *Algorithms Mol Biol* **6**, 26 (2011).
42. Smeeckens, S.P. *et al.* Functional genomics identifies type I interferon pathway as central for host defense against *Candida albicans*. *Nat Commun* **4**, 1342 (2013).
43. Gebremariam, T. *et al.* CoH3 mediates fungal invasion of host cells during mucormycosis. *J Clin Invest* **124**, 237–250 (2014).
44. Betz, R. *et al.* Alternative splicing regulation in plants by SP7-like effectors from symbiotic arbuscular mycorrhizal fungi. *Nat Commun* **15**, 7107 (2024).
45. Tang, C. *et al.* A rust fungus effector directly binds plant pre-mRNA splice site to reprogram alternative splicing and suppress host immunity. *Plant Biotechnol J* **20**, 1167–1181 (2022).
46. Huang, J. *et al.* *Phytophthora* Effectors Modulate Genome-wide Alternative Splicing of Host mRNAs to Reprogram Plant Immunity. *Mol Plant* **13**, 1470–1484 (2020).
47. Feng, M. *et al.* YBX1 is required for maintaining myeloid leukemia cell survival by regulating BCL2 stability in an m6A-dependent manner. *Blood* **138**, 71–85 (2021).
48. Deng, M. *et al.* YBX1 mediates alternative splicing and maternal mRNA decay during pre-implantation development. *Cell Biosci* **12**, 12 (2022).
49. Gebremariam, T. *et al.* Anti-CoH3 antibodies protect mice from mucormycosis by prevention of invasion and augmenting opsonophagocytosis. *Sci Adv* **5**, eaaw1327 (2019).
50. Zhang, Y., Liu, R., Ni, M., Gill, P. & Lee, A.S. Cell surface relocalization of the endoplasmic reticulum chaperone and unfolded protein response regulator GRP78/BiP. *J Biol Chem* **285**, 15065–15075 (2010).
51. Lee, A.S. The ER chaperone and signaling regulator GRP78/BiP as a monitor of endoplasmic reticulum stress. *Methods* **35**, 373–381 (2005).
52. Dobin, A. *et al.* STAR: ultrafast universal RNA-seq aligner. *Bioinformatics* **29**, 15–21 (2013).
53. Love, M.I., Huber, W. & Anders, S. Moderated estimation of fold change and dispersion for

- RNA-seq data with DESeq2. *Genome Biol* **15**, 550 (2014).
54. Yu, G., Wang, L.G., Han, Y. & He, Q.Y. clusterProfiler: an R package for comparing biological themes among gene clusters. *OMICS* **16**, 284–287 (2012).
 55. Lo Giudice, C., Tangaro, M.A., Pesole, G. & Picardi, E. Investigating RNA editing in deep transcriptome datasets with REDIttools and REDlportal. *Nat Protoc* **15**, 1098–1131 (2020).
 56. Picardi, E. & Pesole, G. REDIttools: high-throughput RNA editing detection made easy. *Bioinformatics* **29**, 1813–1814 (2013).
 57. Heinz, S. *et al.* Simple combinations of lineage-determining transcription factors prime cis-regulatory elements required for macrophage and B cell identities. *Mol Cell* **38**, 576–589 (2010).

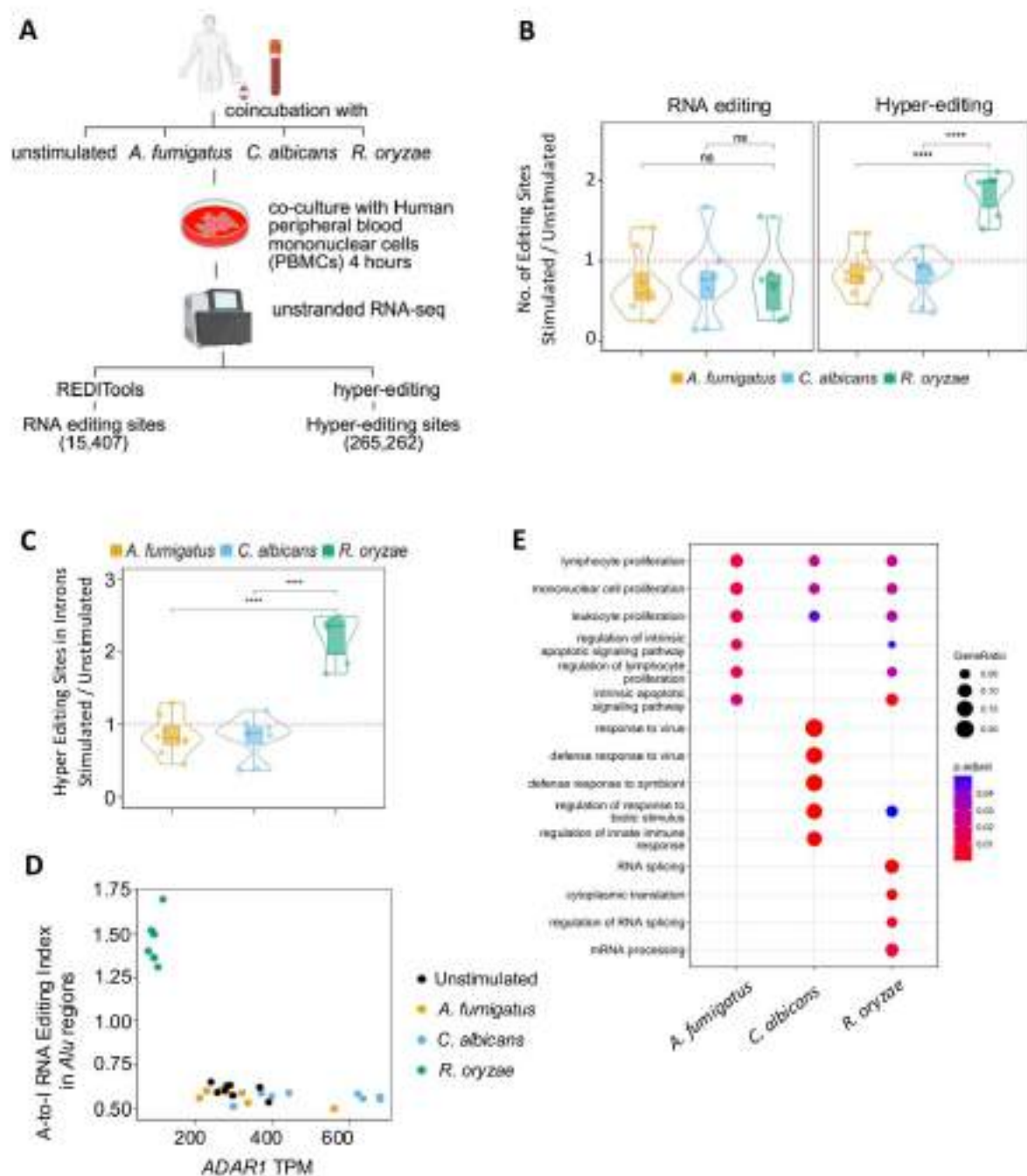
Figure 1.

Figure 1. A-to-I editing patterns in human PBMCs upon three fungal stimulations.

(A) Workflow of PBMCs exposed to three fungal pathogens (heat- or, chemically-killed pathogens). (B) standard RNA editing and hyper-editing editing ratio of PBMCs in response to *A. fumigatus*, *C. albicans*, and *R. oryzae* analyzed by REDITools (RNA editing) and *hyper-editing* (hyper-editing). (C) Ratio of hyper-editing sites in intron regions in response to *A. fumigatus*, *C. albicans*, and *R. oryzae*. (D) The A-to-I RNA editing index in *Alu* regions relative to the normalized expression of *ADAR1* (TPM, normalized by transcripts per million). (E) Biological process gene ontology (GO) term analysis of unique A-to-I edited genes upon fungal coincubations.

Figure 2.

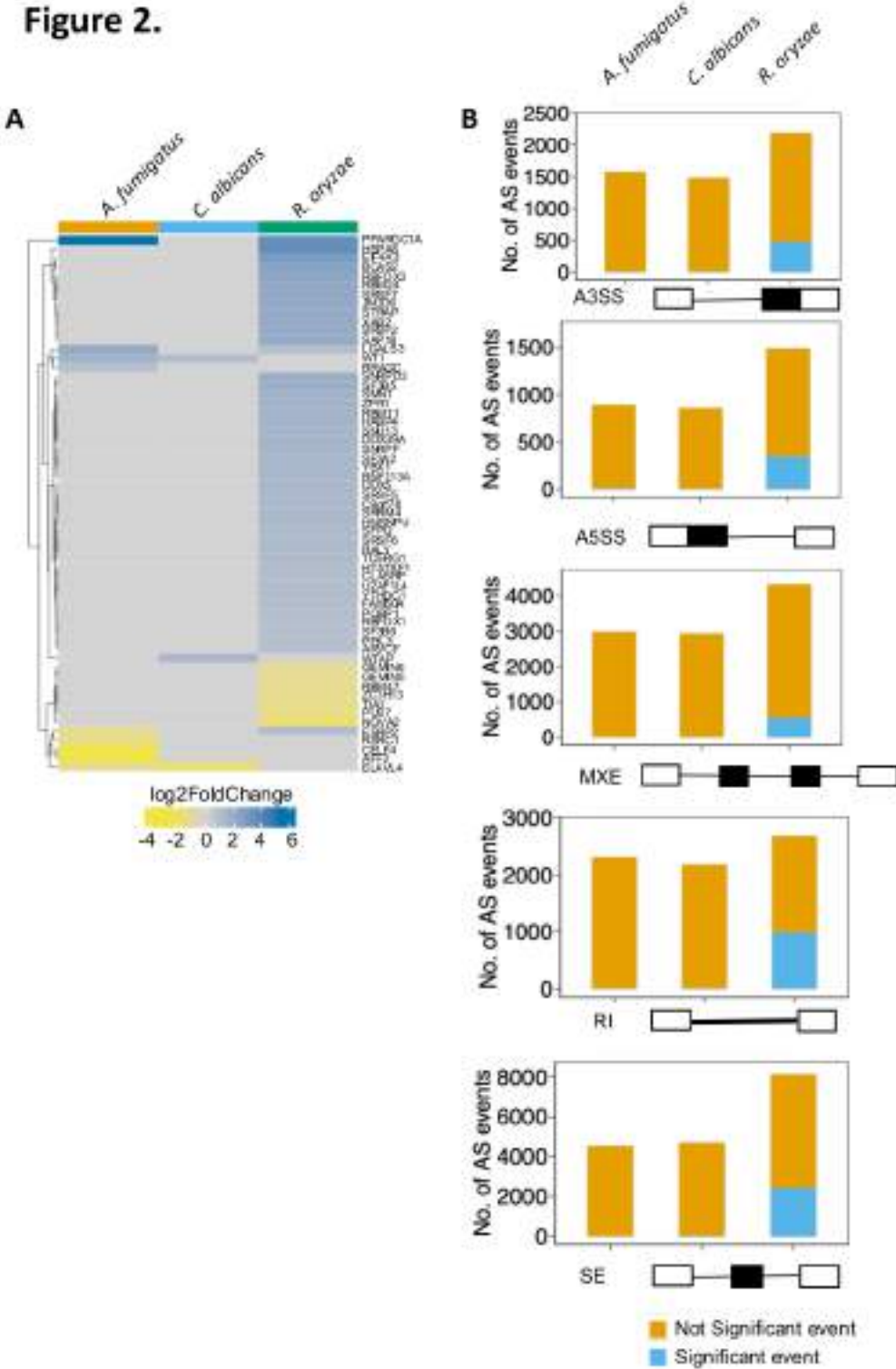


Figure 2. *R. oryzae* dysregulating host splicing factors and RNA splicing.

(A) Heatmap of differentially expressed genes involved in RNA splicing pathway in response to three fungal stimulations. (B) Number of detected AS events across three fungal stimulations, colored by significance ($FDR \leq 0.01$, $|\Delta PSI| \geq 0.05$). Alternative 3' splice sites (A3SS), alternative 5' splice sites (A5SS), mutually exclusive exons (MXE), retained introns (RI), and skipped exons (SE).

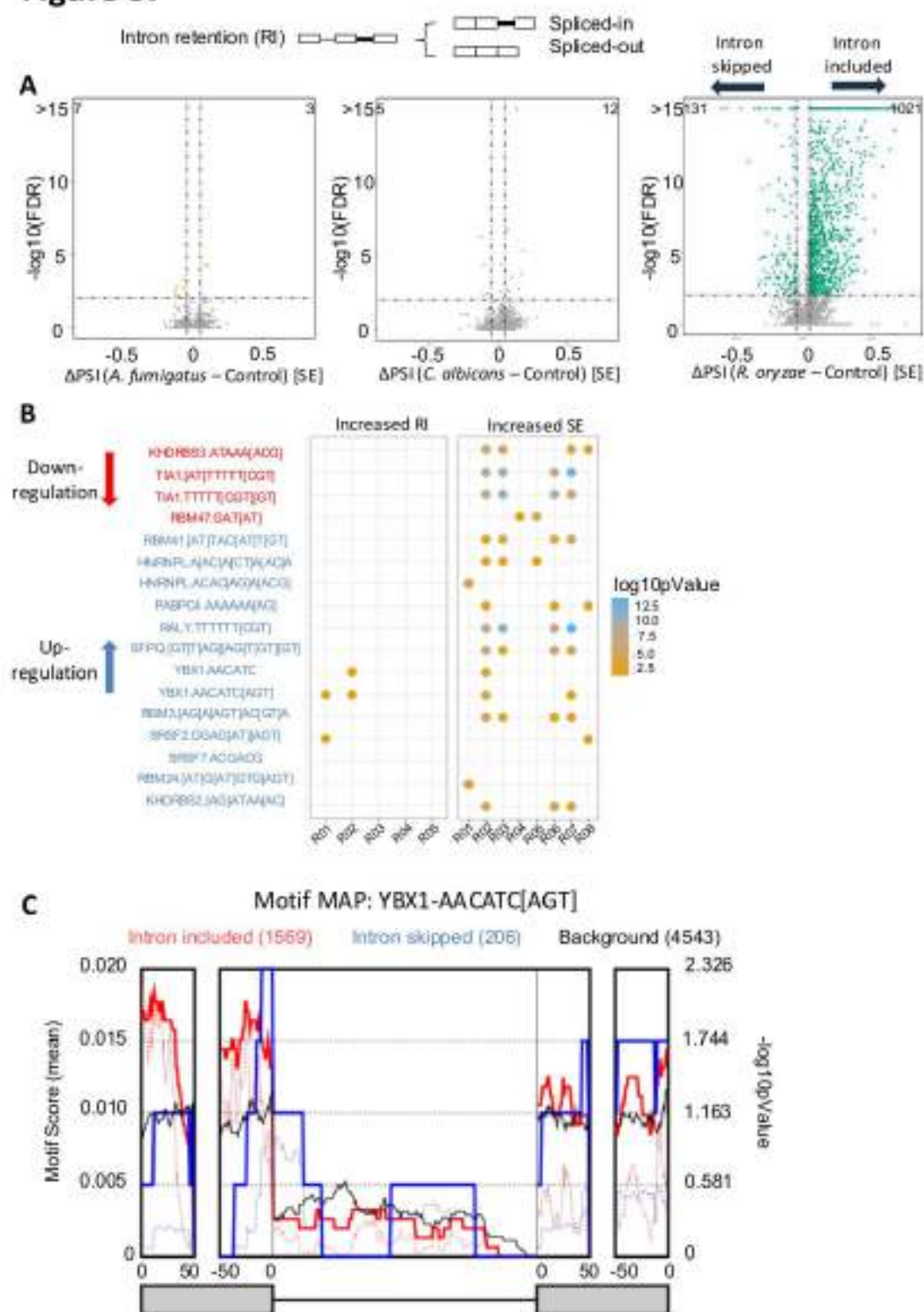
Figure 3.

Figure 3. *R. oryzae*-induced RI events associate with YBX1 and SRSF2 binding motifs.

(A) Volcano plots of significant RI events in different fungal treatments. Each dot represents one AS event. (B) Significantly differentially abundant SFs and associated motifs enriched around AS events. Y-axis describes SFs and their associated binding motifs, colored by up- or down-regulations analyzed from DESeq2. The position (R) of each splicing event relative to the event type is represented in the x-axis, 5' to 3'. Each panel shows a single event type (RI was analyzed from up-regulated events with a *P* value less than 0.05 while SE was analyzed from down-regulated events *P* value less than 0.05). The color indicates the $-\log_{10}$ adjusted *P* value of SFs and associated motifs. (C) Motif analysis of YBX1 for RI event. The middle section of the figure shows the retained intron (black line), 3' end of the preceding exon (left white box), and the 5' end of the downstream exon (right white box). The y-axis on the left represents the motif score of the solid lines, the y-axis on the right indicates the $-\log_{10}$ adjusted *P* value as dotted lines. The black line indicates the background signal, red and blue lines show enrichment of YBX1 binding motifs around introns more (red) or less (blue) utilized by *R. oryzae*.

Figure 4.

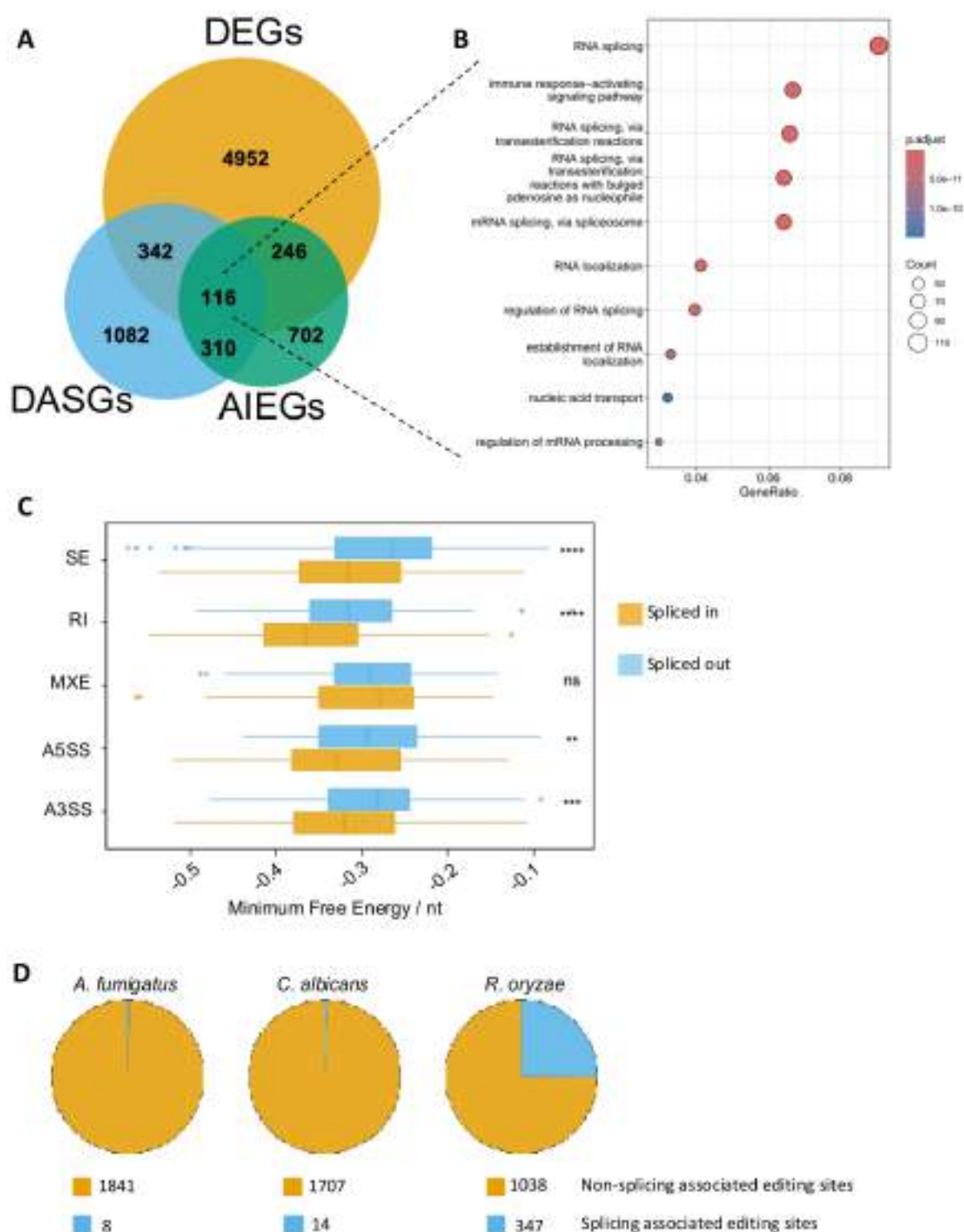


Figure 4. RNA splicing is associated with RNA editing upon *R. oryzae* treatment.

(A) Venn diagram of differential expressed genes (DEGs), differential alternative spliced genes (DASGs), and A-to-I edited genes (AIEGs) from *R. oryzae* stimulated samples. (B) GO enrichment of 116 genes overlapped in all three categories in A. (C) Bar plot of RNA secondary structure stability before and after AS. Minimal free energy is presented in x axis, five tested AS event types are in y axis, colored by spliced in transcripts (yellow) and spliced out transcripts (blue). (D) Pie chart of splicing- (blue) or non-splicing- (red) associated editing events upon three fungal treatments.

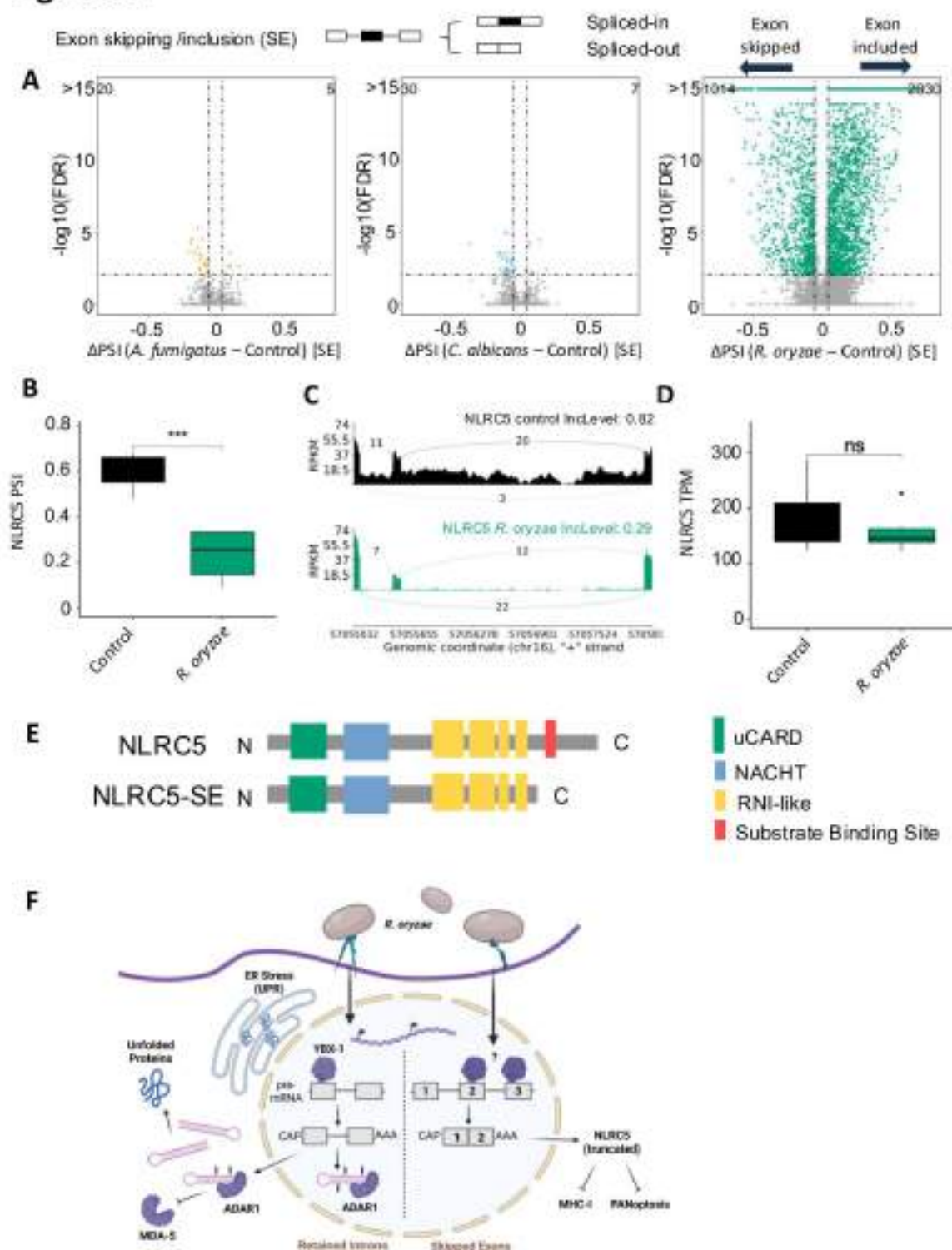
Figure 5.

Figure 5. *R. oryzae* stimulation results in exon skipping in *NLRC5*.

(A) Volcano plots of significant SE events in different fungal treatments. Each dot represents one AS event. (B) The Percent Spliced In (PSI) value of *NLRC5* in unstimulated and *R. oryzae* stimulated samples. (C) Sashimi plot showing differential AS events between the unstimulated (top) and *R. oryzae* stimulated (bottom) samples. (D) Boxplot of *NLRC5* expression normalized by transcript per million (TPM) in unstimulated and *R. oryzae* stimulated samples. (E) Structural domains of *NLRC5* with or without SE event, analyzed using NCBI Conserved Domain Search (<https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>). (F) Working model of PBMC-*R. oryzae* cocubation. Stimulation of *R. oryzae* causes up-regulation of splicing factor YBX1. By binding to its associated motifs, as a result, introns are retained, which form into dsRNA as the substrate for ADAR1 hyper-editing. Increased hyper-editing sites effectively reduce dsRNA sensor MDA5 and may increase unfolded proteins. In parallel, SE events are increased upon *R. oryzae* treatment, led to truncated *NLRC5* transcripts, and likely further inhibit related immune responses like MHC class I presentation and PANoptosis.

Supplementary figure S1.

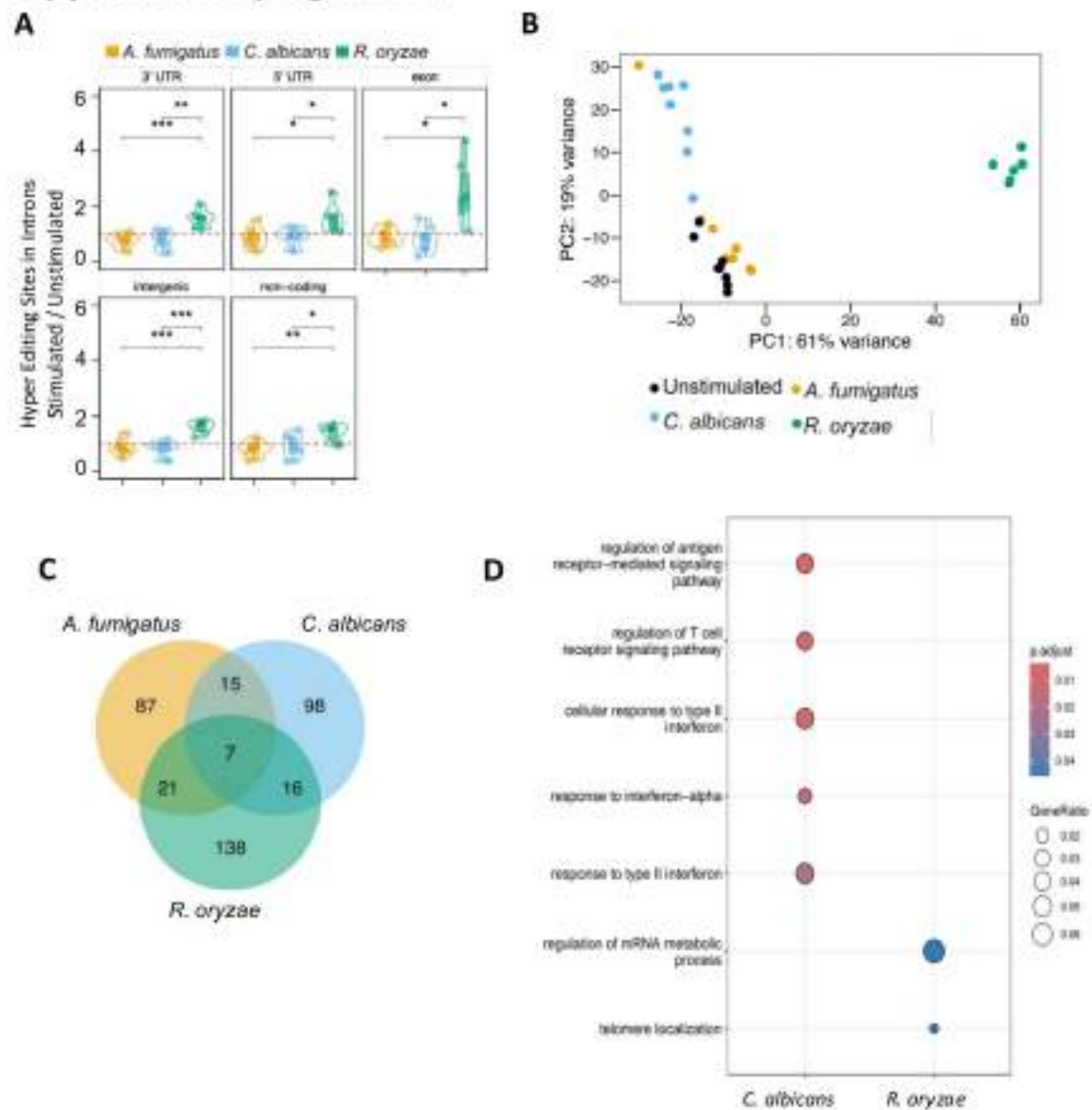


Figure S1

(A) Ratio of hyper-editing sites in human genomes. (B) Principal component analysis of the transcriptional response signature. (C) Venn diagram of genes being uniquely edited from REDITools. (D) GO analysis of genes being uniquely hyper-edited in the biological process.

Supplementary figure S2.

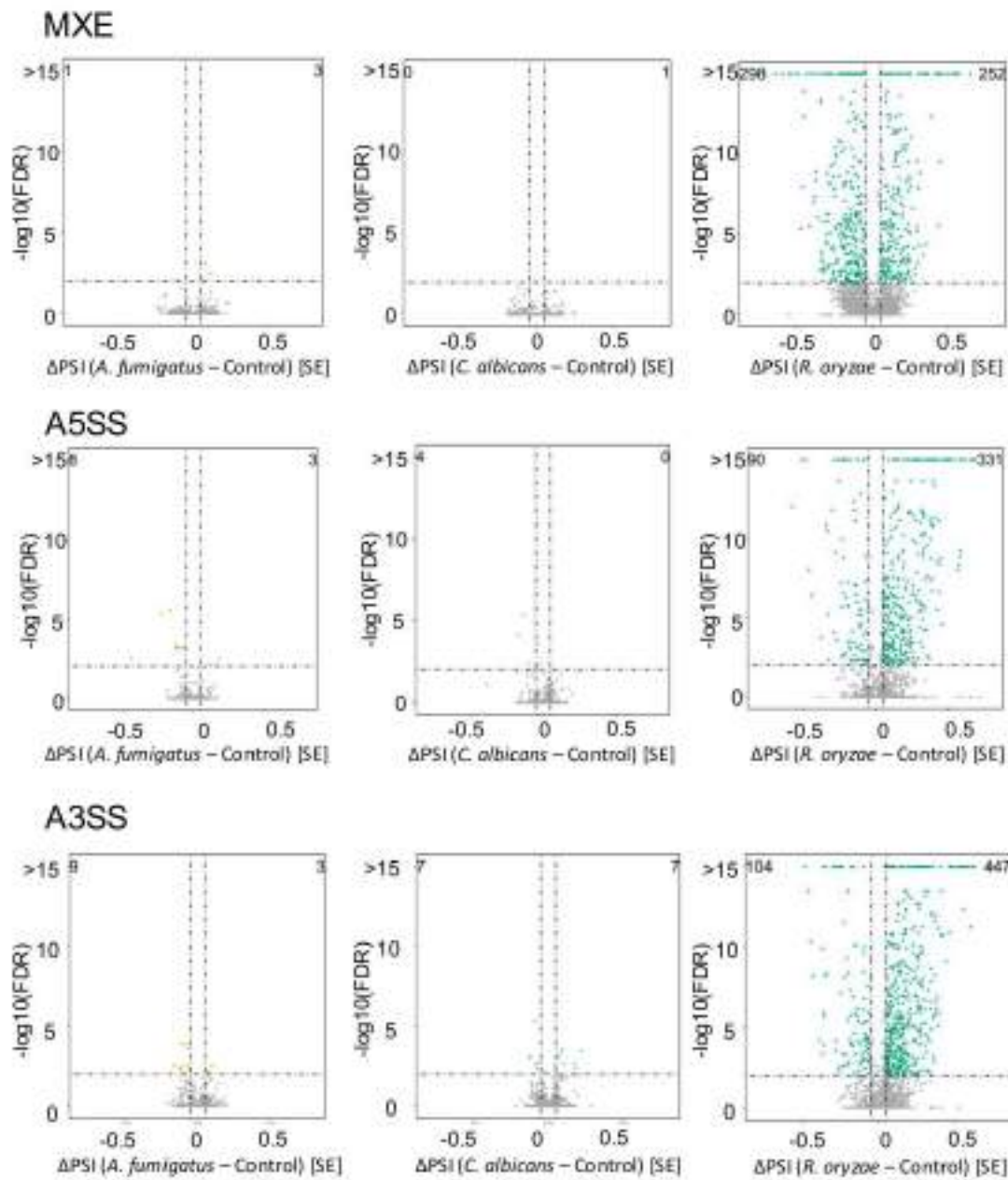


Figure S2

Volcano plots of significant MXE, A5SS, and A3SS in different fungal treatments (from top to bottom). Each dot represents one AS event.

Supplementary Figure S3.

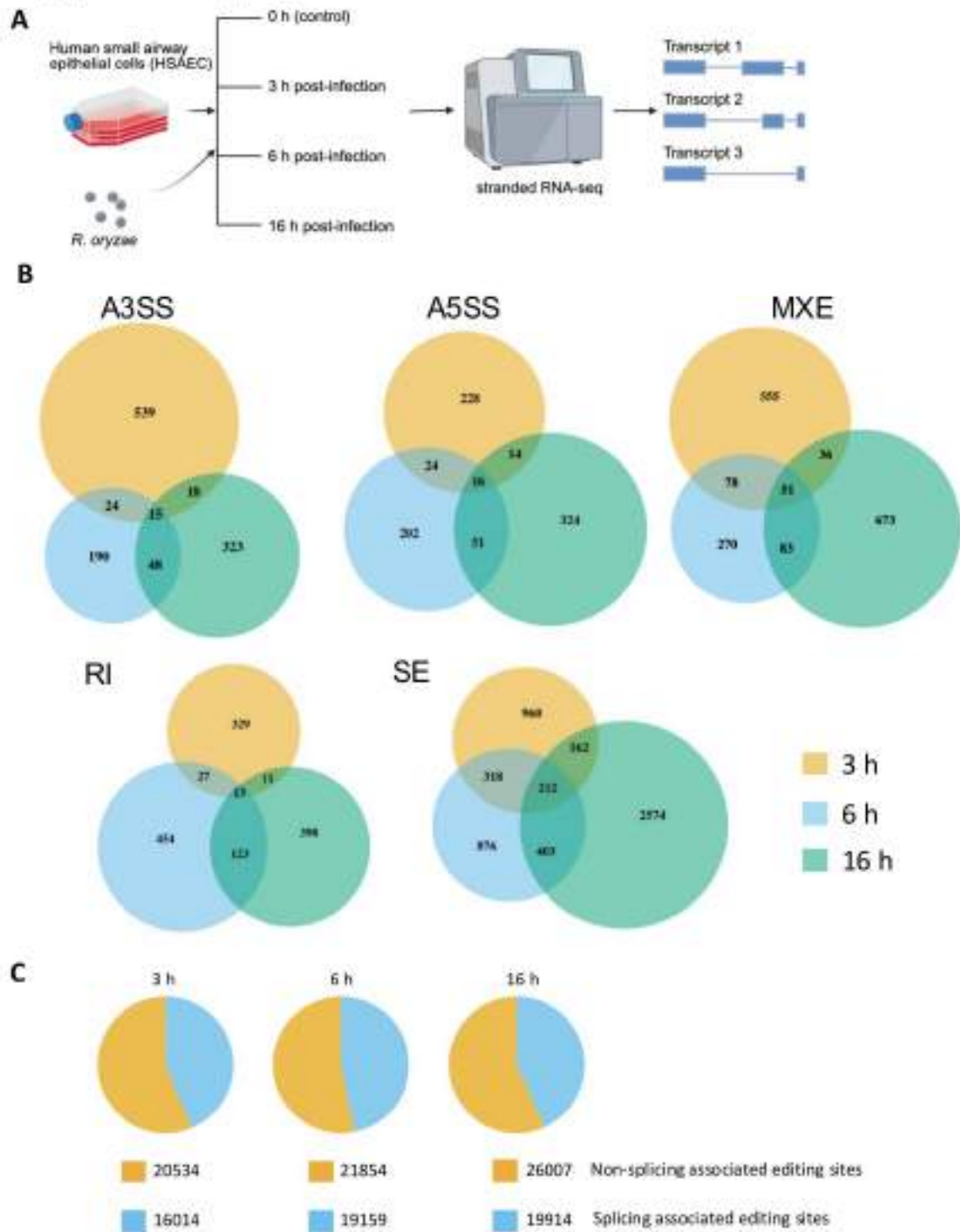


Figure S3

(A) Workflow of HSAECs infected by *R. oryzae* for 0 h, 3 h, 6 h, and 16 h. (B) Venn diagram of alternative spliced events in 3 h, 6 h, and 16 h post-infection time stages when comparing with the control. (C) AS-associated editing events. Colored by non-editing associated AS events (yellow) and editing associated AS events (blue).

5 Discussion

dsRNA is a molecule with complex regulation across biological systems. It serves as a pathogen-associated molecular pattern to trigger immune responses, facilitates interactions with RNA binding proteins, and even serves as the starting material for regulatory mechanisms like the RNAi pathway. Diverse RNAi-related functions have been studied in fungi, but its regulatory role in the critical human pathogen *A. fumigatus* remains ambiguous. We characterized the canonical RNAi pathway of this fungus using genetic manipulation and bioinformatic analysis to uncover an altered transcriptome in the conidia morphotype (**manuscript A**), DclB-dependent sRNAs after overexpression of an IRT gene, and the surprising growth inhibitory potential of excess dsRNA in *A. fumigatus* (**manuscript B**). By comparison, dsRNA metabolism is more intricate in the human host given that humans have a more complicated network of dsRBPs to regulate dsRNA. ADARs, as a type of human dsRBP, play a key role in infectious diseases through their A-to-I deaminating ability. **Manuscript C** first broadly highlights RNA modification-dependent regulation (including A-to-I RNA editing) and then discusses various examples during infection of the host. **Manuscript D** focuses on the ADAR1-mediated dsRNA editing landscape in the host upon fungal stimulations. ADAR1, as the main enzyme conferring A-to-I RNA editing, was differentially expressed upon stimulation of PBMCs with *C. albicans* and *R. oryzae*, the primary source of mucormycosis in humans. In comparison to *A. fumigatus* and *C. albicans*, the stimulation of PBMCs with *R. oryzae* resulted in increased level of hyper editing upon co-incubation. Furthermore, almost 6,000 significant AS events were observed upon *R. oryzae* stimulation but less than 100 significant AS events were found during two other fungal stimulations. The following chapter highlights the major findings from this PhD work and discusses their wider implications on the field.

5.1 Competition between dsRNAs and dsRBPs

A. fumigatus, as a prevalent pathogen causing 1.8 million deaths annually (156), has developed various strategies for masking host recognition during infection (157); however, the roles of RNAi or small RNAs in coordination of virulence of this pathogen are unknown. *A. fumigatus* has been confirmed to possess a functional RNAi pathway, as evidenced by the reduction of mRNA levels for the *ALB1/PKSP* and *FKS1* genes upon the introduction of a complementary repetitive dsRNA-forming transgene (152). This suppression of genes leads to phenotypic effects resembling those seen in gene disruption experiments, but the specifics of the underlying RNAi mechanism, its regulation, importance of the different involved components and ultimately relevance to the fungus remained unstudied. More importantly, exploiting RNAi for gene silencing could be one potential therapeutic avenue for future treatments against fungal infections.

5.1.1 RNAi components contribute to ribosomal processing in *A. fumigatus*

Fungal RNAi was first and extensively studied in the filamentous fungus *N. crassa*, where it was shown to be involved in quelling and MSUD (Meiotic silencing by unpaired DNA) against DNA damage, transposons, and viral sequences (158,159). The associated small RNAs produced by the RNAi machinery in *N. crassa* were classified as sRNA, masiRNA (MSUD-associated small interfering RNA), and qiRNA (79,160). Many other fungi have also confirmed functional RNAi-mediated gene silencing capacity, including *A. fumigatus* (152). When we used genetic methods to make RNAi double-knockout strains ($\Delta ppdA/B$, $\Delta dclA/B$ and $\Delta rrpA/B$) linked to argonaute proteins, dicer-like enzymes, and RNA-dependent RNA polymerases, respectively, results revealed that the RNAi genes (*ppdA*, *ppdB*, *dclA*, *dclB*, *rrpA*, *rrpB*) displayed differential stage-specific expression patterns, which implied co-regulatory or compensatory mechanisms to ensure pathway functionality (**manuscript A; Figure 5**). One explanation might be offered by the known role of RNAi contributing to asexual or sexual reproduction in multiple other fungi. For example, the head blight plant pathogen *Fusarium graminearum* requires RNAi for asexual ascosporeogenesis (161), whereas RNAi components in other fungi (e.g. *Sordaria macrospora*) are essential for the sexual cycle (162).

Moreover, many differentially expressed genes were observed when comparing wild-type and RNAi knockout strains, in particular genes associated with ribosome biogenesis showed significant up- and down-regulation when comparing $\Delta dclA/B$ and $\Delta rrpA/B$ strains with the wild type (**manuscript A; Figure 3**). This finding aligns well with the established role of the RNAi pathway in rRNA and other non-coding RNA processing. For example, Dicer1 in *C. albicans* (CaDcr1) serves additional roles other than

in the canonical RNAi pathway, including cleavage of the 3' external transcribed spacer from pre-rRNA and processing snRNA U4 (163). Similar evidence of Dicer-associated rRNA processing was also observed in mammals, where depletion of Drosha or Dicer in human cells led to the accumulation of 5.8S rRNA precursors, slowed the production of 28S rRNA, and led to structural alterations in the nucleolus (164). We observed significant downregulation of ribosomal biogenesis genes in the $\Delta rrpA/B$ strain *A. fumigatus*, in line with other reported evidence of RdRP being involved in ribosomal processing. For example, a study in plant *Brassica oleracea* demonstrated that RdRP was accumulated in the BoU3 snoRNA ribonucleoprotein complex (snoRNP) fraction and involved in the processing of pre-rRNA (165). While GO enrichment analysis (**manuscript A; Figure 3B**) provided preliminary evidence that the depletion of Dicer and RdRP proteins in *A. fumigatus* linked to ribosome biogenesis at the transcriptome level, no strong difference was observed when we performed LC-MS/MS-based proteomics on conidia (**manuscript A; Supplemental Dataset S4**), suggesting that the conidial transcriptome may not be indicative of the conidial proteome. Another reason could be that LC-MS/MS is not as sensitive as RNA-seq and therefore cannot detect as many low-abundance proteins. Taken together, these findings suggest that RNAi components may play a more pivotal role in ribosome assembly and potentially rRNA processing in *A. fumigatus*. Future studies should focus on elucidating the specific mechanisms by which these proteins contribute to ribosome biogenesis and their potential implications for fungal growth, development, and pathogenicity.

5.1.2 Endogenous sRNA species and potential functions in *A. fumigatus*

Non-coding small RNA molecules are typically 18 to 200 nucleotides in length and function in RNA silencing, rRNA processing, and translation. In the fungal kingdom, identified sRNA species include siRNA, miRNA, qiRNA, tDRs, and masiRNA (76,78,166–168). Many of these sRNA species have been reported to associate with the RNAi pathway or RNAi components. For example, canonical siRNAs and miRNAs are derived from dsRNAs with multiple steps of cleaving by Dicer and associated proteins and are typically incorporated into RISC consisting of AGO and other regulatory proteins (60). When we overexpressed a transgene that can form dsRNA in *A. fumigatus*, DclB-dependent 20-nt, 5'U containing reads accumulated (**manuscript B; Figure 1B, 1E**). Similar features of sRNAs were also identified in *F. graminearum* under *Cryphonectria hypovirus 2* (CHV2) infection (169). Fungi are known to employ RNAi-dependent silencing in order to defend against mycoviruses (170). Additionally, various differentially expressed miRNAs and siRNAs were detected when comparing mycovirus-free and mycovirus-infected *A. fumigatus* strains from sRNA-seq and northern blot assay (170). This indicates that mycovirus infection can alter fungal sRNA landscapes, and potentially fungal virulence, resulting in either in hyper- or hypovirulence depending on the type of virus (171,172).

To further examine the source and function of these sRNAs, RNAi components were deleted individually ($\Delta ppdA/B$, $\Delta dclA/B$, and $\Delta rrpA/B$) and sRNAs were sequenced. Depletion of the RNAi machinery in *A. fumigatus* didn't change the population of sRNAs (**manuscript A; Table 1 and manuscript B; Figure 2B**). Accordingly, it was postulated that *A. fumigatus* only produces high levels of sRNAs under specific stress conditions (e.g. mycovirus infection), or sRNA biogenesis in *A. fumigatus* might primarily rely on redundant or RNAi-independent mechanisms.

We next evaluated the dynamics of ncRNAs across different developmental stages of *A. fumigatus*, and identified expected classes of sRNAs, rRNAs, and tDRs (**manuscript B; Figure 2B**). tDR biogenesis has been reported to be associated with RNAi components like the Dicer proteins (173). Furthermore, tDRs were confirmed to be loaded into RISC and function in gene silencing with argonaute proteins in multiple organisms (174–176). A series of 5'- or 3'-derived tRNA halves were identified in *A. fumigatus* nearly two decades ago (167), but it was unknown if these tRNA halves were generated through a DCL-dependent pathway or if they could be associated with RISC to participate in RNA silencing in *A. fumigatus*. The comparative sRNA-seq analysis of RNAi knockout strains and the wild-type *A. fumigatus* (**manuscript B; Figure 2**) revealed only slight alterations in tDR expression patterns, which implies a minor role of the RNAi machinery in tDR biogenesis under standard growth conditions in *A. fumigatus*. The minimal variation aligns with findings in other organisms, where the biogenesis of tDRs can occur independently of the canonical RNAi pathways. For instance, sRNA-seq analysis of the human B cell lymphoma BCP1 cell line showed that tDR biogenesis was Dicer-independent (177). Given the limited role of RNAi in tDR production under standard growth conditions, it seems possible that tDRs in *A. fumigatus* could be induced in response to exogenous stresses (e.g., nutrient deprivation, viral infection, or host immune challenges) to function as canonical sRNA molecules, targeting specific mRNAs and trigger RNAi-dependent post-transcriptional gene silencing. In this direction, it was revealed that *F. graminearum* infected wheat showed high levels of tRNAs, which were subsequently degraded into tRNA fragments and later loaded into an AGO protein as part of RISC (178). Similar mechanisms were described in studies involving human HEK293T cells, which displayed the observation of gene silencing in the presence of tRF-3 guided AGO2 (175). We (and others (167)) detected tDRs in *A. fumigatus*, although more work is required to understand their function in the fungus.

In alignment with Jöchl, *et al* (167), we extracted low amounts of snoRNAs from sequencing data. As a class of housekeeping RNAs, snoRNAs guide the post-transcriptional modification of rRNAs, snRNAs,

tRNAs, and sometimes mRNAs (179). The majority of snoRNAs are derived from pre-mRNA introns (180). While no direct evidence of RNAi components involvement in the biogenesis of snoRNA exists, several snRNAs and snoRNA-derived RNAs (sdRNAs) have been shown to be substrates of Dicer (181–183) and can be processed into micro-like RNAs in mammals, plants, and yeast (184). Some of these sRNAs can be loaded into AGO proteins, as validated by immunoprecipitation and deep sequencing (183). Though confirmed in mammals, the presence and regulation of biogenesis and function of sdRNAs in pathogenic fungi remains unknown. Based on the observation presented in **manuscript B** in the sRNA-seq results, reads mapped to snoRNAs displayed low abundance and no alterations to expression in the comparisons between wild-type and RNAi knockout strains. This suggests that sdRNA biogenesis in *A. fumigatus* may occur independently of Dicer and AGO proteins, or that snoRNAs in this fungus are poorly annotated. This also encourages further investigation into their roles in fungal biology.

Another class of housekeeping RNA is rRNA. The involvement of RNAi components in ribosomal biogenesis was discussed previously. Here, I will focus on rRNA-derived fragments (rRFs) and their potential connection to the RNAi machinery. rRFs have been reported in diverse organisms, including mammals, plants, fungi, insects (185–187). These rRFs are not merely byproducts of RNA degradation, as evidenced by their accumulation in immunoprecipitated AGO (AGO-IP) complexes (185), suggesting their potential involvement in RNAi pathways or other underappreciated regulatory mechanisms. Studies have shown rRFs contributing to host-microbe interactions; for example, rRFs were reported to be secreted by primary human airway epithelial cells through EVs in response to infection by antibiotic-resistant strains of *Pseudomonas aeruginosa* (188). In **manuscript B; Figure 2B**, we also investigated the expression patterns of reads mapped to rRNA in *A. fumigatus* under different conditions. The absence of RNAi components didn't result in altered rRF expression patterns, indicating that the biogenesis of rRFs in *A. fumigatus* might primarily be independent of RNAi. Furthermore, wild-type and RNAi knockout strains all down-regulated rRFs when in conidia compared to hyphae. These findings raise several possibilities: 1) the observed variations could be derived from the poor annotation of rRNA in *A. fumigatus*, complicating the accurate mapping and interpretation of rRFs, 2) rRFs in *A. fumigatus* may be morphotype regulated, or 3) the biogenesis and regulation of rRFs may differ from those in other organisms, potentially involving unique pathways or mechanisms.

Taken together, our results revealed that *A. fumigatus* doesn't produce many sRNAs under standard conditions, DclB-dependent sRNAs were observed when overexpressing a dsRNA gene, and the biogenesis of tDRs, rRFs, and sdRNAs are DCL-independent in *A. fumigatus* under standard conditions.

Future studies should aim to address the function of DclA in fungal biology and identify the proteins responsible for the biogenesis of sRNA speices like tDRs, rRFs, and sdRNAs.

5.1.3 dsRNA introduced as inverted-repeat transgene influences fungal growth

The RNAi machinery is known for defending the genome against exogenous sequences like retrotransposons or viral infections (189). Many mycoviruses rely on dsRNA genomes and can alter fungal phenotypes or virulence (190). As part of our study, we hypothesized that overexpression of dsRNA in *A. fumigatus* was akin to a retrotransposon or mycovirus infection, thus resulting in a reduction of fungal growth that may give an advantage to unaffected fungal cells. Clearly, mycovirus infections present a far more complex scenario than a single dsRNA, and notable exceptions to this proposed mechanism have already been documented. For example, A78 virus-infected *A. fumigatus* isolates grew faster than A78 virus-free isolates and cause mild hypervirulence in infection assays (191). But the mechanism of A78 infection in influencing fungal growth and hypervirulence remains unknown. Alternatively, no obvious pathogenicity change was observed when infecting with a chrysovirus or a partitivirus (191). Early in the 1970s, *A. foetidus* (strain IMI 41871) and *A. niger* (strain IMI 146819) were reported to harbor virus particles, which were later isolated as dsRNA. These dsRNA-containing fungi were confirmed to promote interferon production in mice upon infection (192). The *A. fumigatus* reference strain Af293 contains *tetramycovirus 1* (*AfuTmV-1*), which was the first virus to be transformed into protoplasts by delivery of its four dsRNAs (193). *AfuTmV-1M*, in contrast to *AfuTmV-1*, has an additional open reading frame 5 (ORF5) and was shown to reduce virulence in a mouse-*A. fumigatus* infection model (194). Collectively, mycovirus-fungi interactions can result in different consequences for altering fungal virulence and growth. Future studies should focus on examining if IRT overexpression strains in our system alter fungal virulence or if the same phenotypes are also observed when the IRT is overexpressed in other strains.

The first indications of a functional RNAi machinery in *A. fumigatus* came from inserting a repetitive transgene onto the chromosome. Mouyna, *et al.* were able to show decreased *ALB1/PKSP* and *FKS1* mRNA levels with the presence of corresponding dsRNA, which phenotypes aligned with the $\Delta PKSP$ and $\Delta FKS1$ knockout strains (152). Likewise, Chen, *et al.* also observed *PKS* gene silencing in *F. graminearum* when using a similar strategy (195). Our results revealed that overexpression of an IRT gene appeared to silence in wild-type and $\Delta dclA$ strains, but no transgene dependent silencing was observed in $\Delta dclB$ or $\Delta dclA/B$ (**manuscript A; Figure 1**), indicating DclB to be necessary for hairpin-induced RNAi functionality in *A. fumigatus*. Interestingly, our results revealed that overexpression of an IRT gene reduced fungal biomass in the wild-type, $\Delta dclA/B$, $\Delta dclA$, and $\Delta dclB$ strains (**manuscript D; Figure 4A, 4B**). Surprisingly, growth inhibition was particularly strong in a $\Delta rrpA/B$ -IRT strain (**data not**

shown), but this result is beyond the scope of this thesis. These results suggest that excess dsRNA may inhibit fungal growth through a RNAi-independent manner. I hypothesize that excess dsRNA may compete for limited dsRBPs (e.g., Rnt1) resulting in fungal homeostasis dysregulation and growth inhibition. This arms race between dsRNA and dsRBPs has been discussed in many organisms, including humans. The idea is that dsRBPs bind to dsRNA in a sequence-independent manner, allowing different pathways to intersect and affect each other when in competition for dsRNA substrates, thus dysregulating the homeostasis of the organism (196,197).

5.2 Host RNA processing for infectious diseases

In humans, the competition between dsRBPs for dsRNA has been largely discussed in the context of immunology, particularly due to the increased complexity of the human system. Compared with 10 annotated dsRBPs in *A. fumigatus* (FungiDB.org), humans have at least 20 dsRBPs (198), of which the ADAR family is just one of the key regulators linked to immunity. ADAR1 competes with dsRNA sensors for dsRNA, influencing IFN production during viral infections. However, the outcome of such competitions and the breadth of ADAR1-dependent RNA editing are poorly understood during fungal infections. The aim of this work was to characterize ADAR1-mediated A-to-I RNA editing landscapes and their associated post-transcriptional regulators upon fungal stimulation.

5.2.1 ADAR1 expression and A-to-I RNA editing change during fungal stimulations

ADAR1-dependent A-to-I RNA editing is an important mechanism regulating the transcriptome in metazoans, associated with a wide range of human diseases, and effective in preventing MDA5- and PKR-mediated activation by endogenous dsRNA (112). Previous studies have shown that ADAR proteins respond to various infections (199), though host A-to-I editing in response to fungal infections has not been extensively characterized, leaving a significant gap in our understanding of this process. A study conducted by Huang *et al.* in 2018 shed some light on this topic by investigating the effects of *C. albicans* infection in human epithelial and endothelial cells (139). An increase of A-to-I RNA editing number was observed, which differed from the pattern seen in viral infections. However, when the authors subsequently performed the *in vivo* study in mouse tongue and kidney tissues upon the same fungal infection, the editing levels were reduced, likely due to the lack of *Alu* elements in the mouse genome, which are repetitive sequences that comprise ~11% of the human genome and are a primary substrate of ADAR1 editing (104). Nevertheless, it is still unknown if ADAR proteins also play a role in immune defense upon other fungal infections, particularly regarding hyper-editing.

To understand the alteration of the A-to-I RNA editing landscape in the context of three different host-fungal stimulated models, we re-analyzed mRNA-seq data from Bruno, *et al* (200). The authors characterized transcriptome dynamics in human PBMCs co-incubated with fungal pathogens, revealing that the PBMC-*C. albicans* interaction enriched type I IFN responses in comparison to *A. fumigatus* and *R. oryzae*. Unsurprisingly, we observed an up-regulation of *ADAR1* upon *C. albicans* stimulation in our reanalysis (**manuscript D; Figure 1D**). It is well-characterized that *ADAR1*, more specifically the long *ADAR1-p150* isoform, can be induced by type I IFN (201), and so seems logical that *C. albicans* triggered IFNs upregulate *ADAR1* expression.

Since we observed changes in *ADAR1* expression, we next assessed the A-to-I RNA editing landscape. The overall level of RNA editing detected from REDITools (202) in the three fungal stimulated samples showed decreased numbers of A-to-I editing sites (**manuscript D; Figure 1B left**). It is well known that ADAR editing is dynamically regulated due to different kinds of factors. For example, a study of more than 7,000 RNA-seq datasets revealed host A-to-I RNA editing to be differentially regulated by SARS-CoV-2 infections in diverse sample types (203). In contrast to the observed decrease in editing in PBMCs after fungal co-incubation, *ADAR1*-dependent hyper-editing appeared to be up-regulated upon *R. oryzae* stimulation (**manuscript D; Figure 1B right**), indicating that the fungus may specifically dysregulate the host RNA epitranscriptome.

We next focused on genes that were only found edited in stimulated samples from either the hyper-editing or REDITools results. Of note, these edited genes likely represent the current status of a given host transcriptome in response to stimulation, and do not indicate that editing is necessarily directly linked to the immune response. GO analysis (**manuscript D; Fig S1D**) showed that *R. oryzae* stimulation enriched host hyper-editing sites in “regulation of mRNA metabolic process” and “telomere localization” pathways. Although only speculation, perhaps PBMCs no longer maintain telomeres as a type of programmed cell death influenced by hyper-editing (204). Notably, as PBMCs consist of monocytes, macrophages, neutrophils, and other cell types (205), it is still unclear whether all cell types respond similarly or whether the observed phenotypes are driven by a subset of cells. REDITools revealed that 121 genes, 128 genes, and 172 genes were found edited only in *A. fumigatus*-, *C. albicans*-, and *R. oryzae*- stimulated PBMCs, respectively, compared to the control. Further applying GO enrichment analysis on these genes found only in stimulated samples uncovered a variety of shared and specific pathways (**manuscript D; Figure 1E**). Several GO categories were shared between all three infections related to “proliferation”, which is likely just associated with changes to cell growth during infection. PBMCs stimulated with *C. albicans* edited genes enriched for IFN-related pathways. As mentioned previously, *C. albicans* significantly induces the type I IFN pathway in humans as a central

part of the host defense mechanism (206), but so far no study has reported these transcripts to also be deaminated by ADARs. The deamination of IFN-related transcripts could result from the induction of IFNs triggered by *C. albicans*, which in turn upregulates ADAR1-p150 expression and potentially benefits fungal invasion/immune evasion. Interestingly, *R. oryzae* stimulation led to changed editing patterns of genes involved in “RNA splicing” and “mRNA processing”. Altered A-to-I editing of genes involved in RNA splicing and mRNA processing suggests a potential novel mechanism by which the fungus may dysregulate host dsRNA metabolism.

5.2.2 Host RNA splicing dysregulation upon fungal stimulation

The GO enrichment of RNA splicing pathways upon *R. oryzae* stimulation led us to analyze gene expression patterns of splicing genes. Differentially expressed gene (DEG) analysis suggested that splicing factors were dysregulated, with 45 splicing genes upregulated and 7 downregulated compared to the unstimulated sample (**manuscript D; Figure 2A**). Further detection of AS events during fungal coinfections confirmed that *R. oryzae* stimulated induction of differential splicing events in PBMCs, among which skipped exon events (SE) were the most prolific type detected (**manuscript D; Figure 2B**). These findings align with other studies reporting a changing AS landscape in host cells upon viral or bacterial infections, dependent on the pathogen, cell type, and other factors (207,208).

By analyzing RNA binding motifs (RBMs), we observed that YBX1 (Y-box binding protein 1) and SRSF2 (Serine/arginine-rich splicing factor 2) motifs associate with RI events (**manuscript D; Figure 3B**). Acting as both a transcription factor and a translational regulator, YBX1 plays a pivotal role in numerous cellular processes including proliferation, differentiation, drug resistance, and stress response (209). A handful of studies have shown YBX1 associating with AS. For example, YBX1 knockdown embryos in mammals exhibited decreased RNA/mRNA stability and downregulation of spliceosome-associated genes. Additionally, splicing factors and mRNA decay-related genes were observed to be aberrantly expressed (210). Interestingly, it was also found in our reanalysis results that six *HNRNP* genes (*HNRNPH1*, *HNRNPH3*, *HNRNPC*, *HNRNPA2B*, *HNRNPD*, and *HNRNPK*), a family of RNA binding proteins that are involved in multiple stages of AS, mRNA stabilization, and pre-mRNA processing (211), showed significant alternative splicing (**data not shown**). We thus hypothesized that *R. oryzae* stimulation induces splicing factor YBX1, which then binds to its associated motif in the RNA, and results in AS of heterogeneous nuclear ribonucleoprotein genes to further dysregulate RNA splicing machinery in the host.

The concept that pathogens alter host RNA splicing is not new, and some hypotheses are listed in the following paragraphs based on these previous studies. It was reported that *Mucor* species produce

mucorin and CotH proteins to facilitate hyphal and conidial invasion. A well-studied example is CotH3, a conidial surface protein that activates glucose-regulated proteins 78 (GRP78, also denoted as HSPA5 or BiP) on nasal epithelial cells for example (18,212). Bruno, *et al.* showed an upregulation of *GRP78* transcript in PBMCs upon *R. oryzae* stimulation (200), but our reanalysis provided no indication of it being differentially spliced (**data not shown**). It is possible that the transcriptome reprogramming that occurs upon GRP78 binding to CotH leads to the AS observed, but other possibilities also exist. For example, *R. oryzae* may produce an effector during the interaction that dysregulates the host splicing pathway.

Another intriguing hypothesis involves fungal-bacterial endosymbiosis, as certain *Rhizopus* species are known to carry bacterial endosymbionts. Endosymbiosis describes an interdependent relationship where one organism (endosymbiont) resides inside another one (host), to the benefit of both endosymbiont and host (216). A groundbreaking study two decades ago revealed that rice seedling blight, caused by genus *Rhizopus*, was linked to rhizoxin, a toxin synthesized not by the fungi but by the intracellular bacteria of the genus *Burkholderia* that lives within the fungus (217). While this highlights the role of endosymbionts in fungal pathogenicity, it remains unclear whether all *Rhizopus* isolates harbor such bacterial partners. Notably, some plant-pathogenic *Rhizopus* strains associated with endosymbionts have also been isolated from human patients (218). This raises the possibility that *Rhizopus* species, and potentially their endosymbionts could adapt to infect diverse hosts (e.g., plant and animal hosts). Future work could focus on 1) testing if *R. oryzae* harbors an endobacterium; and 2) if so, can the endobacterium produce secondary metabolites involved in controlling host splicing factors. This could hint to a previously unrecognized layer of complexity in tri-kingdom interactions, and result in a better understanding of how fungi and their bacterial partners synergize to promote virulence.

5.2.3 *R. oryzae* may induce hyper-editing through dysregulation of host RNA splicing

We observed an up-regulation of A-to-I hyper editing, an up-regulation of AS events, and an up-regulation of splicing factors during *R. oryzae* stimulation, but the link between global A-to-I RNA editing patterns and host RNA splicing still remains unclear. Both RNA editing and RNA splicing occur co-transcriptionally and are thought to be functionally interdependent. One study confirmed that A-to-I editing is linked to pre-mRNA splicing, as the depletion of splicing factors NOVA1 and NOVA2 changed global editing levels in mouse brain (150). In contrast, it was also demonstrated that the deficiency of ADAR1 and ADAR2 impacted RNA splicing in mice (219). In human PBMCs stimulated with *R. oryzae*, a higher level of hyper-editing and AS events was detected. Introns often contain repetitive sequences like *Alu* capable of forming dsRNA, meaning that the retained introns may serve as a source

of dsRNA deaminated by ADAR1. Interestingly, our analysis also revealed that the mature transcripts showed lower minimum free energy (**manuscript D; Figure 4C**) upon intron retention, indicating that more structural dsRNA could be formed. Significant intron retention may also indicate that *R. oryzae* manipulates the host to generate nonfunctional or aberrant transcripts (220), which could explain the increase in transcript abundance compared to *A. fumigatus* and *C. albicans* stimulations.

In addition to RI, SE was found to have the most differentially alternative spliced numbers among five tested AS events. Surprisingly, NLRC5 as a PRR in the host cytoplasm, was identified to differentially splice out exon 27, resulting in loss of the substrate binding site (**manuscript D; Figure 5**). NLRC5 belongs to the nucleotide-binding domain leucine-rich repeat (NLR) family and is associated with innate immunity to viruses and activation of inflammatory cell death (221). While it has been studied as a PRR against viruses, it is not well-characterized in response to fungal infection. We hypothesize that *R. oryzae* stimulation may disrupt antifungal immunity by alternatively splicing *NLRC5* to cause loss of the substrate recognition domain. Taken together, our results showed that *R. oryzae* dysregulated host RNA splicing, suppressed innate immunity, and increased levels of hyper-editing. These actions likely have major consequences to antifungal immunity and will require additional study in the future to determine the exact mechanism of action in place.

5.3 Summary and future directions

The metabolism of dsRNA is regulated through various cellular activities, such as synthesis, degradation, and editing. However, in the context of human-fungal infections, such mechanisms remain largely unknown. In this thesis, I explored dsRNA-mediated RNAi in the human fungal pathogen *A. fumigatus*, which causes 1.8 million death annually (156). In the fungal kingdom, RNAi has been reported to function in genome defense, heterochromatin formation, and gene regulation, among others (57). The first study on RNAi in *A. fumigatus* used a repetitive transgene to demonstrate that RNAi is indeed active in gene regulation in the fungus (152). To further explore this mechanism, we generated RNAi knockout strains, which were analyzed with different methods of genetic manipulation, high-throughput sequencing, and bioinformatic predictions. The presented data demonstrated that RNAi in *A. fumigatus* plays a role in the regulation of the conidial transcriptome, specifically of ribosome biogenesis genes (**manuscript A**). Using multiple sequencing approaches, little evidence was observed supporting the generation of sRNAs in dependence on the canonical RNAi machinery under laboratory growth conditions, but analysis of the nuclear- and mitochondrial-expression patterns of tDRs suggested differential expression across fungal morphotypes. Overexpression of an established repetitive transgene that forms a dsRNA structure resulted in the biogenesis of predominantly 20-nt, 5' uridine containing sRNAs, which heavily relied on DclB in *A. fumigatus*. Surprisingly, the overexpression of the repetitive transgene inhibited fungal growth in a dsRNA-dependent, RNAi-independent manner (**manuscript B**). Ultimately, these results provide an improved description of sRNA biogenesis and the role of RNAi in *A. fumigatus*, uncovering intriguing links between RNAi and ribosome biogenesis pathways as well as dsRNA metabolism and fungal growth. Future research should investigate multiple avenues. For example, how does RNAi mechanistically contribute to ribosome biogenesis or how are tDRs, rRF, and sdRNAs produced in *A. fumigatus*? The molecular mechanism of how dsRNA inhibits fungal growth may be another avenue to uncover previously overlooked fungal biology.

Humans have more dsRBPs that not only associate with gene regulation, but also relate to immune response. One representative immune-related dsRBP family are the ADARs, which harbor an A-to-I RNA editing function. **Manuscript C** includes a review of the host-pathogen arms race in the context of A-to-I RNA editing and other commonly studied RNA modifications. **Manuscript D** explores the idea of how host dsRNA is regulated by ADAR-mediated A-to-I RNA editing during the host response to fungal stimulations. To investigate this idea, various bioinformatic tools (e.g., REDITools (202) for A-to-I RNA editing site prediction, *hyper-editing* (134) for A-to-I hyper-editing prediction, and RNAEditingIndexer (222) for RNA editing index prediction in *Alu* elements) were employed to analyze

different publicly available high-throughput RNA-seq datasets, including a dataset of human PBMCs stimulated with three pathogenic fungi (*A. fumigatus*, *C. albicans*, and *R. oryzae var delemar*) (200). The studies presented in this thesis revealed significant up-regulation in A-to-I hyper-editing levels and down-regulation of *ADAR1* transcript in response to *R. oryzae* stimulation but not two other tested fungi. To better elucidate the underlying mechanism of the opposite regulations of hyper-editing and *ADAR1* expression, we further analyzed GO term of edited genes present in only stimulated samples. Interestingly, GO analysis of genes being edited only upon *R. oryzae* stimulation revealed the involvement of A-to-I editing in various biological processes, particularly RNA splicing. DEG analysis revealed many splicing factor genes to be up- and down-regulated upon *R. oryzae* stimulation, including *YBX-1* and *SRSF-2*. AS detection using rMATS-turbo (223) confirmed *R. oryzae* stimulation induced a substantial number of AS events in PBMCs. Moreover, when analyzing another RNA-seq dataset of human small airway epithelial cells (HSAECs) infected with live *R. delemar* (224), different A-to-I editing patterns were observed, indicating that the ADAR1-mediated RNA editing patterns are dynamic and vary based on cell type, pathogen stimulation and time. These intriguing observations will direct the future research focus on the upstream cause of AS events in the host, or more broadly, AS during host-fungal interactions.

6 References

1. Denning DW. Global incidence and mortality of severe fungal disease. *Lancet Infect Dis*. 2024 July 1;24(7):e428–38.
2. WHO Fungal Priority Pathogens List to Guide Research, Development and Public Health Action. 1st ed. Geneva: World Health Organization; 2022. 1 p.
3. Zhao Y, Ye L, Zhao F, Zhang L, Lu Z, Chu T, *et al*. *Cryptococcus neoformans*, a global threat to human health. *Infect Dis Poverty*. 2023 Mar 17;12(1):20.
4. Hope W, Natarajan P, Goodwin L. Invasive fungal infections. *Clin Med*. 2013 Oct;13(5):507–10.
5. Huang J, Hu P, Ye L, Shen Z, Chen X, Liu F, *et al*. Pan-drug resistance and hypervirulence in a human fungal pathogen are enabled by mutagenesis induced by mammalian body temperature. *Nat Microbiol*. 2024 July;9(7):1686–99.
6. Garcia-Solache MA, Casadevall A. Global Warming Will Bring New Fungal Diseases for Mammals. *mBio*. 2010 May 18;1(1):10.1128/mbio.00061-10.
7. Castrejón-Pérez AD, Welsh EC, Miranda I, Ocampo-Candiani J, Welsh O. Cutaneous mucormycosis. *An Bras Dermatol*. 2017;92(3):304–11.
8. Ibrahim AS, Spellberg B, Walsh TJ, Kontoyiannis DP. Pathogenesis of Mucormycosis. *Clin Infect Dis Off Publ Infect Dis Soc Am*. 2012 Feb 1;54(Suppl 1):S16–22.
9. Jia LJ, González K, Orasch T, Schmidt F, Brakhage AA. Manipulation of host phagocytosis by fungal pathogens and therapeutic opportunities. *Nat Microbiol*. 2024 Sept;9(9):2216–31.
10. Ghannoum MA, Rice LB. Antifungal Agents: Mode of Action, Mechanisms of Resistance, and Correlation of These Mechanisms with Bacterial Resistance. *Clin Microbiol Rev*. 1999 Oct;12(4):501–17.
11. Zhang Z, Bills GF, An Z. Advances in the treatment of invasive fungal disease. *PLOS Pathog*. 2023 May 4;19(5):e1011322.
12. Kim JY. Human fungal pathogens: Why should we learn? *J Microbiol*. 2016 Mar 1;54(3):145–8.
13. de Jesus Carrion S, Leal SM, Ghannoum MA, Pearlman E. The RodA hydrophobin on *Aspergillus fumigatus* spores masks Dectin-1 and Dectin-2 dependent responses and enhances fungal survival in vivo. *J Immunol Baltim Md 1950*. 2013 Sept 1;191(5):2581–8.
14. Díaz-Jiménez DF, Pérez-García LA, Martínez-Álvarez JA, Mora-Montes HM. Role of the Fungal Cell Wall in Pathogenesis and Antifungal Resistance. *Curr Fungal Infect Rep*. 2012 Dec 1;6(4):275–82.
15. Tang D, Kang R, Coyne CB, Zeh HJ, Lotze MT. PAMPs and DAMPs: signal 0s that spur autophagy and immunity. *Immunol Rev*. 2012;249(1):158–75.
16. Silva-Gomes S, Decout A, Nigou J. Pathogen-Associated Molecular Patterns (PAMPs). In: *Encyclopedia of Inflammatory Diseases*. Birkhäuser, Basel; 2014. p. 1–16.
17. Moyes DL, Wilson D, Richardson JP, Mogavero S, *et al*. Candidalysin is a fungal peptide toxin critical for mucosal infection. *Nature*. 2016 Mar 1;64–68(532).

REFERENCES

18. Soliman SSM, Baldin C, Gu Y, Singh S, Gebremariam T, Swidergall M, *et al.* Mucoricin is a ricin-like toxin that is critical for the pathogenesis of mucormycosis. *Nat Microbiol.* 2021 Mar;6(3):313–26.
19. Alqarihi A, Gebremariam T, Gu Y, Swidergall M, Alkhazraji S, Soliman SSM, *et al.* GRP78 and Integrins Play Different Roles in Host Cell Invasion during Mucormycosis. *mBio.* 2020 June 2;11(3):e01087-20.
20. Gebremariam T, Liu M, Luo G, Bruno V, Phan QT, Waring AJ, *et al.* CoH3 mediates fungal invasion of host cells during mucormycosis. *J Clin Invest.* 2014 Jan 2;124(1):237–50.
21. Soliman SSM, El-Labbad EM, Abu-Qiyas A, Fayed B, Hamoda AM, Al-Rawi AM, *et al.* Novel Secreted Peptides From *Rhizopus arrhizus* var. *delemar* With Immunomodulatory Effects That Enhance Fungal Pathogenesis. *Front Microbiol.* 2022 Mar 21;13.
22. Jia LJ, Rafiq M, Radosa L, Hortschansky P, *et al.* *Aspergillus fumigatus* hijacks human p11 to redirect fungal-containing phagosomes to non-degradative pathway. *Cell Host & Microbe.* 2023 Mar 8;31(1):373–88.
23. Howlett BJ. Secondary metabolite toxins and nutrition of plant pathogenic fungi. *Curr Opin Plant Biol.* 2006 Aug 1;9(4):371–5.
24. Kiriya H, Kinoshita SN, Hayashi Y, Honda R, Kasuga S, Kinoshita T, *et al.* Fungal toxin fusaric acid enhances plant growth by upregulating 14-3-3 interaction with plasma membrane H⁺-ATPase. *Sci Rep.* 2024 Oct 8;14(1):23431.
25. Tsug T, Harimoto Y, Akimitsu K, Ohtani K. Host-selective toxins produced by the plant pathogenic fungus *Alternaria alternata*. *FEMS Microbiology Reviews.* 2013 Jan 1;37(1):44–66.
26. Casadevall A, Coelho C, Cordero RJB, Dragotakes Q, Jung E, Vij R, *et al.* The capsule of *Cryptococcus neoformans*. *Virulence.* 2018 Aug 1;10(1):822–31.
27. Rodrigues ML, Janbon G, O'Connell RJ, Chu TTH, May RC, Jin H, *et al.* Characterizing extracellular vesicles of human fungal pathogens. *Nat Microbiol.* 2025 Apr;10(4):825–35.
28. Robbins PD, Morelli AE. Regulation of immune responses by extracellular vesicles. *Nat Rev Immunol.* 2014 Mar;14(3):195–208.
29. Vargas G, Rocha JDB, Oliveira DL. Compositional and immunobiological analyses of extracellular vesicles released by *Candida albicans*. *Cell Microbiol.* 2014 Oct 6;17:389–407.
30. Borges BM, Santana MG, Preite NW, Kaminski V de L. Extracellular vesicles from virulent *P. brasiliensis* induce TLR4 and dectin-1 expression in innate cells and promote enhanced Th1/Th17 response - PMC. *Virulence.* 2024 Mar 21;15(1):2329573.
31. Delves PJ, Roitt IM. The Immune System. *N Engl J Med.* 2000 July 6;343(1):37–49.
32. Cai X, Chiu YH, Chen ZJ. The cGAS-cGAMP-STING Pathway of Cytosolic DNA Sensing and Signaling. *Mol Cell.* 2014 Apr 24;54(2):289–96.
33. Lai HC, Ho UY, James A, De Souza P, Roberts TL. RNA metabolism and links to inflammatory regulation and disease. *Cell Mol Life Sci.* 2021 Dec 31;79(1):21.

REFERENCES

34. Lemaitre B, Nicolas E, Michaut L, Reichhart JM, Hoffmann JA. The Dorsoventral Regulatory Gene Cassette *spätzle/Toll/cactus* Controls the Potent Antifungal Response in *Drosophila* Adults. *Cell*. 1996 Sept 20;86(6):973–83.
35. Yauch LE, Mansour MK, Shoham S, Rottman JB, Levitz SM. Involvement of CD14, Toll-Like Receptors 2 and 4, and MyD88 in the Host Response to the Fungal Pathogen *Cryptococcus neoformans* In Vivo. *Infect Immun*. 2004 Sept;72(9):5373–82.
36. Biondo C, Midiri A, Messina L, Tomasello F, Garufi G, Catania MR, *et al*. MyD88 and TLR2, but not TLR4, are required for host defense against *Cryptococcus neoformans*. *Eur J Immunol*. 2005;35(3):870–8.
37. Steele C, Rapaka RR, Metz A, Pop SM, Williams DL, Gordon S, *et al*. The Beta-Glucan Receptor Dectin-1 Recognizes Specific Morphologies of *Aspergillus fumigatus*. *PLOS Pathog*. 2005 Dec 9;1(4):e42.
38. Verma A, Wüthrich M, Deepe G, Klein B. Adaptive Immunity to Fungi. *Cold Spring Harb Perspect Med*. 2014 Nov 6;5(3):a019612.
39. Schimmel P, Pouplana L. Transfer RNA: From Minihelix to Genetic Code. *Cell*. 1995 June 30;81(1):983–6.
40. Hur S. Double-Stranded RNA Sensors and Modulators in Innate Immunity. 2019;
41. Cottrell KA, Andrews RJ, Bass BL. The competitive landscape of the dsRNA world. *Mol Cell*. 2024 Jan 4;84(1):107–19.
42. Saunders LR, Barber GN. The dsRNA binding protein family: critical roles, diverse cellular functions. *FASEB J Off Publ Fed Am Soc Exp Biol*. 2003 June;17(9):961–83.
43. Bass BL, Weintraub H. A developmentally regulated activity that unwinds RNA duplexes. *Cell*. 1987 Feb 27;48(4):607–13.
44. Wedekind JE, Dance GSC, Sowden MP, Smith HC. Messenger RNA editing in mammals: new members of the APOBEC family seeking roles in the family business. *Trends Genet*. 2003 Apr 1;19(4):207–16.
45. Suresh Babu S, Joladarashi D, Jeyabal P, Thandavarayan RA, Krishnamurthy P. RNA-stabilizing proteins as molecular targets in cardiovascular pathologies. *Trends Cardiovasc Med*. 2015 Nov 1;25(8):676–83.
46. Roy K, Chanfreau GF. Chapter Ten - The Diverse Functions of Fungal RNase III Enzymes in RNA Metabolism. In: Chanfreau GF, Tamanoi F, editors. *The Enzymes*. Academic Press; 2012. p. 213–35. (Eukaryotic RNases and their Partners in RNA Degradation and Biogenesis, Part A; vol. 31).
47. NICHOLSON A. The ribonuclease III family: forms and functions in RNA maturation, decay, and gene silencing. *RNAi Guide Gene Silenc*. 2003;149–74.
48. Court DL, Gan J, Liang YH, Shaw GX, Tropea JE, Costantino N, *et al*. RNase III: Genetics and Function; Structure and Mechanism. *Annu Rev Genet*. 2013;47:405–31.

REFERENCES

49. Leulliot N, Quevillon-Cheruel S, Graille M, van Tilbeurgh H, Leeper TC, Godin KS, *et al.* A new alpha-helical extension promotes RNA binding by the dsRBD of Rnt1p RNase III. *EMBO J.* 2004 July 7;23(13):2468–77.
50. Abou Elela S, Ji X. Structure and function of Rnt1p: An alternative to RNAi for targeted RNA degradation. *WIREs RNA.* 2019;10(3):e1521.
51. Carmell MA, Hannon GJ. RNase III enzymes and the initiation of gene silencing. *Nat Struct Mol Biol.* 2004 Mar;11(3):214–8.
52. Conrad C, Rauhut R. Ribonuclease III: new sense from nuisance. *Int J Biochem Cell Biol.* 2002 Feb 1;34(2):116–29.
53. Catala M, Tremblay M, Samson É, Conconi A, Abou Elela S. Deletion of Rnt1p Alters the Proportion of Open versus Closed rRNA Gene Repeats in Yeast. *Mol Cell Biol.* 2008 Jan 1;28(2):619–29.
54. Giorgi C, Fatica A, Nagel R, Bozzoni I. Release of *U18 snoRNA* from its host intron requires interaction of Nop1p with the Rnt1p endonuclease. *EMBO J.* 2001 Dec 3;20(23):6856–65.
55. Wang Q, An B, Hou X, Guo Y, Luo H, He C. Dicer-like Proteins Regulate the Growth, Conidiation, and Pathogenicity of *Colletotrichum gloeosporioides* from *Hevea brasiliensis*. *Front Microbiol.* 2018 Jan 4;8:2621.
56. Weiberg A, Wang M, Lin FM, Zhao H, Zhang Z, Kaloshian I, *et al.* Fungal Small RNAs Suppress Plant Immunity by Hijacking Host RNA Interference Pathways. *Science.* 2013 Oct 4;342(6154):118–23.
57. Dang Y, Yang Q, Xue Z, Liu Y. RNA Interference in Fungi: Pathways, Functions, and Applications ▽ . *Eukaryot Cell.* 2011 Sept;10(9):1148–55.
58. Zeng W, Wang J, Wang Y, Lin J, Fu Y, Xie J, *et al.* Dicer-Like Proteins Regulate Sexual Development via the Biogenesis of Peritheciium-Specific MicroRNAs in a Plant Pathogenic Fungus *Fusarium graminearum*. *Front Microbiol.* 2018 Apr 26;9.
59. Werner BT, Koch A, Šečić E, Engelhardt J, Jelonek L, Steinbrenner J, *et al.* *Fusarium graminearum* DICER-like-dependent sRNAs are required for the suppression of host immune genes and full virulence. *PLoS ONE.* 2021 Aug 5;16(8):e0252365.
60. Tang G. siRNA and miRNA: an insight into RISCs. *Trends Biochem Sci.* 2005 Feb 1;30(2):106–14.
61. Vergani-Junior CA, Tonon-da-Silva G, Inan MD, Mori MA. DICER: structure, function, and regulation. *Biophys Rev.* 2021 Nov 16;13(6):1081–90.
62. Frank F, Sonenberg N, Nagar B. Structural basis for 5'-nucleotide base-specific recognition of guide RNA by human AGO2. *Nature.* 2010 June;465(7299):818–22.
63. Suzuki HI, Katsura A, Yasuda T, Ueno T, Mano H, Sugimoto K, *et al.* Small-RNA asymmetry is directly driven by mammalian Argonautes. *Nat Struct Mol Biol.* 2015 July;22(7):512–21.
64. Khvorova A, Reynolds A, Jayasena SD. Functional siRNAs and miRNAs Exhibit Strand Bias. *Cell.* 2003 Oct 17;115(2):209–16.
65. Schwarz DS, Hutvágner G, Du T, Xu Z, Aronin N, Zamore PD. Asymmetry in the Assembly of the RNAi Enzyme Complex. *Cell.* 2003 Oct 17;115(2):199–208.

REFERENCES

66. Bartel DP. Metazoan MicroRNAs. *Cell*. 2018 Mar 22;173(1):20–51.
67. Cogoni C, Macino G. Gene silencing in *Neurospora crassa* requires a protein homologous to RNA-dependent RNA polymerase. *Nature*. 1999 May;399(6732):166–9.
68. Calo S, Nicolás FE, Vila A, Torres-Martínez S, Ruiz-Vázquez RM, *et al*. Two distinct RNA-dependent RNA polymerases are required for initiation and amplification of RNA silencing in the basal fungus *Mucor circinelloides*. *Mol Microbiol*. 2012;83(2):379–94.
69. Nicolás FE, Ruiz-Vázquez RM. Functional Diversity of RNAi-Associated sRNAs in Fungi. *Int J Mol Sci*. 2013 July 24;14(8):15348–60.
70. Hammond SM, Bernstein E, Beach D, Hannon GJ. An RNA-directed nuclease mediates post-transcriptional gene silencing in *Drosophila* cells. *Nature*. 2000 Mar;404(6775):293–6.
71. Zamore PD, Tuschl T, Sharp PA, Bartel DP. RNAi: Double-Stranded RNA Directs the ATP-Dependent Cleavage of mRNA at 21 to 23 Nucleotide Intervals. *Cell*. 2000 Mar 31;101(1):25–33.
72. Ambros V, Bartel B, Bartel DP, Burge CB, Carrington JC, Chen X, *et al*. A uniform system for microRNA annotation. *RNA*. 2003 Jan 3;9(3):277–9.
73. Lagos-Quintana M, Rauhut R, Lendeckel W, Tuschl T. Identification of Novel Genes Coding for Small Expressed RNAs. *Science*. 2001 Oct 26;294(5543):853–8.
74. Lau NC, Lim LP, Weinstein EG, Bartel DP. An Abundant Class of Tiny RNAs with Probable Regulatory Roles in *Caenorhabditis elegans*. *Science*. 2001 Oct 26;294(5543):858–62.
75. Lee RC, Ambros V. An Extensive Class of Small RNAs in *Caenorhabditis elegans*. *Science*. 2001 Oct 26;294(5543):862–4.
76. Lee HC, Li L, Gu W, Xue Z, Crosthwaite SK, Pertsemlidis A, *et al*. Diverse Pathways Generate MicroRNA-like RNAs and Dicer-Independent Small Interfering RNAs in Fungi. *Mol Cell*. 2010 June;38(6):803–14.
77. Zeng Y, Yi R, Cullen BR. MicroRNAs and small interfering RNAs can inhibit mRNA expression by similar mechanisms. *Proc Natl Acad Sci*. 2003 Aug 19;100(17):9779–84.
78. Lee HC, Chang SS, Choudhary S, Aalto AP, Maiti M, Bamford DH, *et al*. qiRNA is a new type of small interfering RNA induced by DNA damage. *Nature*. 2009 May;459(7244):274–7.
79. Pickford A, Braccini L, Macino G, Cogoni C. The QDE-3 homologue RecQ-2 co-operates with QDE-3 in DNA repair in *Neurospora crassa*. *Curr Genet*. 2003 Jan;42(4):220–7.
80. Maiti M, Lee HC, Liu Y. QIP, a putative exonuclease, interacts with the *Neurospora* Argonaute protein and facilitates conversion of duplex siRNA into single strands. *Genes Dev*. 2007 Jan 3;21(5):590–600.
81. Forrest EC, Cogoni C, Macino G. The RNA-dependent RNA polymerase, QDE-1, is a rate-limiting factor in post-transcriptional gene silencing in *Neurospora crassa*. *Nucleic Acids Res*. 2004;32(7):2123–8.

82. Cozen AE, Quartley E, Holmes AD, Hrabeta-Robinson E, Phizicky EM, Lowe TM. ARM-seq: AlkB-facilitated RNA methylation sequencing reveals a complex landscape of modified tRNA fragments. *Nat Methods*. 2015 Sept;12(9):879–84.
83. Watkins CP, Zhang W, Wylder AC, Katanski CD, Pan T. A multiplex platform for small RNA sequencing elucidates multifaceted tRNA stress response and translational regulation. *Nat Commun*. 2022 May 5;13(1):2491.
84. Sun C, Fu Z, Wang S, Li J, Li Y, Zhang Y, *et al*. Roles of tRNA-derived fragments in human cancers. *Cancer Lett*. 2018 Feb 1;414:16–25.
85. Pliatsika V, Loher P, Telonis AG, Rigoutsos I. MINTbase: a framework for the interactive exploration of mitochondrial and nuclear tRNA fragments. *Bioinformatics*. 2016 Aug 15;32(16):2481–9.
86. Maute RL, Schneider C, Sumazin P, Holmes A, Califano A, Basso K, *et al*. tRNA-derived microRNA modulates proliferation and the DNA damage response and is down-regulated in B cell lymphoma. *Proc Natl Acad Sci*. 2013 Jan 22;110(4):1404–9.
87. Yamasaki S, Ivanov P, Hu G fu, Anderson P. Angiogenin cleaves tRNA and promotes stress-induced translational repression. *J Cell Biol*. 2009 Mar 30;185(1):35–42.
88. Fu M, Gu J, Wang M, Zhang J. Emerging roles of tRNA-derived fragments in cancer. *Molecular Cancer*. 2023 Feb 13;22(30).
89. Priest SJ, Yadav V, Roth C, Dahlmann TA, Kück U, Magwene PM, *et al*. Uncontrolled transposition following RNAi loss causes hypermutation and antifungal drug resistance in clinical isolates of *Cryptococcus neoformans*. *Nat Microbiol*. 2022 Aug;7(8):1239–51.
90. Navarro-Mendoza MI, Pérez-Arques C, Heitman J. Heterochromatin and RNAi act independently to ensure genome stability in *Mucorales* human fungal pathogens. *Proc Natl Acad Sci U S A*. 120(7):e2220475120.
91. Segers GC, Zhang X, Deng F, Sun Q, Nuss DL. Evidence that RNA silencing functions as an antiviral defense mechanism in fungi. *Proc Natl Acad Sci*. 2007 July 31;104(31):12902–6.
92. Martienssen R, Moazed D. RNAi and Heterochromatin Assembly. *Cold Spring Harb Perspect Biol*. 2015 Aug;7(8):a019323.
93. Volpe TA, Kidner C, Hall IM, Teng G, Grewal SIS, Martienssen RA, *et al*. Regulation of Heterochromatic Silencing and Histone H3 Lysine-9 Methylation by RNAi. *Science*. 2002 Sept 13;297(5588):1833–7.
94. Navarro E, Ruiz-Pérez VL, Torres-Martínez S, *et al*. Overexpression of the *crgA* gene abolishes light requirement for carotenoid biosynthesis in *Mucor circinelloides*. *Eur J Biochem*. 2000;267(3):800–7.
95. Welsh JD, Leibowitz MJ. Localization of genes for the double-stranded RNA killer virus of yeast. *Proc Natl Acad Sci*. 1982 Feb;79(3):786–9.
96. Schmitt MJ, Breinig F. Yeast viral killer toxins: lethality and self-protection. *Nat Rev Microbiol*. 2006 Mar;4(3):212–21.

REFERENCES

97. Drinnenberg IA, Fink GR, Bartel DP. Compatibility with Killer explains the Rise of RNAi-deficient fungi. *Science*. 2011 Sept 16;333(6049):1592.
98. Roberts SA, Gordenin DA. Hypermutation in human cancer genomes: footprints and mechanisms. *Nat Rev Cancer*. 2014 Dec;14(12):786–800.
99. Jolivet-Gougeon A, Kovacs B, Le Gall-David S, Le Bars H, Bousarghin L, Bonnaure-Mallet M, *et al*. Bacterial hypermutation: clinical implications. *J Med Microbiol*. 2011;60(5):563–73.
100. Chang Z, Billmyre RB, Lee SC, Heitman J. Broad antifungal resistance mediated by RNAi-dependent epimutation in the basal human fungal pathogen *Mucor circinelloides*. *PLOS Genet*. 2019 Feb 11;15(2):e1007957.
101. Alves LR, Peres da Silva R, Sanchez DA, Zamith-Miranda D, Rodrigues ML, Goldenberg S, *et al*. Extracellular Vesicle-Mediated RNA Release in *Histoplasma capsulatum*. *mSphere*. 2019 Mar 27;4(2):10.1128/msphere.00176-19.
102. Rayner S, Bruhn S, Vallhov H, Andersson A, Billmyre RB, Scheynius A. Identification of small RNAs in extracellular vesicles from the commensal yeast *Malassezia sympodialis*. *Sci Rep*. 2017 Jan 4;7(1):39742.
103. Peres da Silva R, Puccia R, Rodrigues ML, Oliveira DL, Joffe LS, César GV, *et al*. Extracellular vesicle-mediated export of fungal RNA. *Sci Rep*. 2015 Jan 14;5(1):7763.
104. Cai Q, He B, Weiberg A, Buck AH, Jin H. Small RNAs and extracellular vesicles: New mechanisms of cross-species communication and innovative tools for disease control. *PLOS Pathog*. 2019 Dec 30;15(12):e1008090.
105. Zand Karimi H, Innes RW. Molecular mechanisms underlying host-induced gene silencing. *Plant Cell*. 2022 Sept 1;34(9):3183–99.
106. He B, Wang H, Liu G, Chen A, Calvo A, Cai Q, *et al*. Fungal small RNAs ride in extracellular vesicles to enter plant cells through clathrin-mediated endocytosis. *Nat Commun*. 2023 July 20;14(1):4383.
107. Kato H, Sato S, Yoneyama M, Yamamoto M, Uematsu S, Matsui K, *et al*. Cell Type-Specific Involvement of RIG-I in Antiviral Response. *Immunity*. 2005 July 1;23(1):19–28.
108. Platanias LC. Mechanisms of type-I- and type-II-interferon-mediated signalling. *Nat Rev Immunol*. 2005 May;5(5):375–86.
109. Song B, Chen Y, Liu X, Yuan F, Tan EYJ, Lei Y, *et al*. Ordered assembly of the cytosolic RNA-sensing MDA5-MAVS signaling complex via binding to unanchored K63-linked poly-ubiquitin chains. *Immunity*. 2021 Oct 12;54(10):2218-2230.e5.
110. Wang Y, Swiecki M, McCartney SA, Colonna M. dsRNA sensors and plasmacytoid dendritic cells in host defense and autoimmunity. *Immunol Rev*. 2011 Sept;243(1):74–90.
111. Peisley A, Lin C, Wu B. Cooperative assembly and dynamic disassembly of MDA5 filaments for viral dsRNA recognition. *Proc Natl Acad Sci USA*. 2011 Dec 12;108(52):21010–5.

REFERENCES

112. Liddicoat BJ, Piskol R, Chalk AM, Ramaswami G, Higuchi M, Hartner JC, *et al.* RNA editing by ADAR1 prevents MDA5 sensing of endogenous dsRNA as nonself. *Science*. 2015 Sept 4;349(6252):1115–20.
113. Grice LF, Degnan BM. The origin of the ADAR gene family and animal RNA editing. *BMC Evol Biol*. 2015 Jan 2;15(5).
114. Jin Y, Zhang W, Li Q. Origins and evolution of ADAR-mediated RNA editing. *IUBMB Life*. 2009;61(6):572–8.
115. Keegan LP, McGurk L, Palavicini JP, Brindle J, Paro S, Li X, *et al.* Functional conservation in human and *Drosophila* of Metazoan ADAR2 involved in RNA editing: loss of ADAR1 in insects. *Nucleic Acids Res*. 2011 Sept 1;39(16):7249–62.
116. Moroz LL, Kocot KM, Citarella MR, Dosung S, Norekian TP, Povolotskaya IS, *et al.* The ctenophore genome and the evolutionary origins of neural systems. *Nature*. 2014 June;510(7503):109–14.
117. Teichert I, Dahlmann TA, Kück U, Nowrousian M. RNA Editing During Sexual Development Occurs in Distantly Related Filamentous Ascomycetes. *Genome Biol Evol*. 2017 Apr 1;9(4):855–68.
118. Liu H, Wang Q, He Y, Chen L, Hao C, Jiang C, *et al.* Genome-wide A-to-I RNA editing in fungi independent of ADAR enzymes. *Genome Res*. 2016 Jan 4;26(4):499–509.
119. Desterro JMP, Keegan LP, Lafarga M, Berciano MT, O’Connell M, Carmo-Fonseca M. Dynamic association of RNA-editing enzymes with the nucleolus. *J Cell Sci*. 2003 May 1;116(9):1805–18.
120. Yanai M, Kojima S, Sakai M, Komorizono R, Tomonaga K, Makino A. ADAR2 Is Involved in Self and Nonself Recognition of Bornavirus Genomic RNA in the Nucleus. *J Virol*. 2020 Feb 28;94(6):e01513-19.
121. George CX, Samuel CE. Human RNA-specific adenosine deaminase ADAR1 transcripts possess alternative exon 1 structures that initiate from different promoters, one constitutively active and the other interferon inducible. *Proc Natl Acad Sci*. 1999 Apr 13;96(8):4621–6.
122. Mannion NM, Greenwood SM, Young R, Cox S, Brindle J, Read D, *et al.* The RNA-Editing Enzyme ADAR1 Controls Innate Immune Responses to RNA. *Cell Rep*. 2014 Nov 20;9(4):1482–94.
123. Yee PS, Chai AWY, Yee SM, Ooi S, Tan YH, Garnett MJ, *et al.* Interferon-Inducible ADAR1 p150 Is Essential for the Survival of Oral Squamous Cell Carcinoma. *Mol Carcinog*. 2025;64(6):1066–77.
124. Hu SB, Heraud-Farlow J, Sun T, Liang Z, Goradia A, Taylor S, *et al.* ADAR1p150 prevents MDA5 and PKR activation via distinct mechanisms to avert fatal autoinflammation. *Mol Cell*. 2023 Nov 2;83(21):3869-3884.e7.
125. Sun T, Yu Y, Wu X, Acevedo A, Luo JD, Wang J, *et al.* Decoupling expression and editing preferences of *ADAR1 p150* and *p110* isoforms. *Proc Natl Acad Sci*. 2021 Mar 23;118(12):e2021757118.
126. Deininger P. Alu elements: know the SINEs. *Genome Biol*. 2011 Dec 28;12(236).
127. Melcher T, Maas S, Herb A, Sprengel R, Seeburg PH, Higuchi M. A mammalian RNA editing enzyme. *Nature*. 1996 Feb;379(6564):460–4.

128. Chen CX, Cho DSC, Wang Q, Lai F, Carter KC, Nishikura K. A third member of the RNA-specific adenosine deaminase gene family, ADAR3, contains both single- and double-stranded RNA binding domains. *RNA*. 2000 May;6(5):755–67.
129. Yuan J, Xu L, Bao HJ, Wang J lin, Zhao Y, Chen S. Biological roles of A-to-I editing: implications in innate immunity, cell death, and cancer immunotherapy. *J Exp Clin Cancer Res*. 2023 June 17;42(1):149.
130. Goldstein B, Agranat-Tamir L, Light D, Ben-Naim Zgayer O, Fishman A, Lamm AT. A-to-I RNA editing promotes developmental stage-specific gene and lncRNA expression. *Genome Res*. 2017 Mar;27(3):462–70.
131. Lin Z, Zhao S, Li X, Miao Z, Cao J, Chen Y, *et al*. Cathepsin B S-nitrosylation promotes ADAR1-mediated editing of its own mRNA transcript via an ADD1/MATR3 regulatory axis. *Cell Res*. 2023 July;33(7):546–61.
132. Chattopadhyay P, Mehta P, Kanika, Mishra P, Chen Liu CS, Tarai B, *et al*. RNA editing in host lncRNAs as potential modulator in SARS-CoV-2 variants-host immune response dynamics. *iScience*. 2024 Apr 29;27(6):109846.
133. Heale BS, Eulalio A, Schulte L, Vogel J, O’Connell MA. Analysis of A to I editing of miRNA in macrophages exposed to *salmonella*. *RNA Biol*. 2010;7(5):621–7.
134. Porath HT, Carmi S, Levanon EY. A genome-wide map of hyper-edited RNA reveals numerous new sites. *Nat Commun*. 2014 Aug 27;5(1):4726.
135. Porath HT, Knisbacher BA, Eisenberg E, Levanon EY. Massive A-to-I RNA editing is common across the Metazoa and correlates with dsRNA abundance. *Genome Biol*. 2017 Oct 2;18(1):185.
136. Ashley CN, Broni E, Miller WA. ADAR Family Proteins: A Structural Review. *Curr Issues Mol Biol*. 2024 May;46(5):3919–45.
137. Chung H, Calis JJA, Wu X, Sun T, Yu Y, Sarbanes SL, *et al*. Human ADAR1 Prevents Endogenous RNA from Triggering Translational Shutdown. *Cell*. 2018 Feb 8;172(4):811-824.e14.
138. Pfaller CK, George CX, Samuel CE. Adenosine Deaminases Acting on RNA (ADARs) and Viral Infections. *Annu Rev Virol*. 2021 Sept 29;8(1):239–64.
139. Huang Y, Cao Y, Li J, Liu Y, Zhong W, Li X, *et al*. A survey on cellular RNA editing activity in response to *Candida albicans* infections. *BMC Genomics*. 2018 Jan 19;19(1):43.
140. Wei ZY, Wang ZX, Li JH, Wen YS, Gao D, Xia SY, *et al*. Host A-to-I RNA editing signatures in intracellular bacterial and single-strand RNA viral infections. *Front Immunol*. 2023 Apr 4;14:1121096.
141. Ringlander J, Fingal J, Kann H, Prakash K, Rydell G, Andersson M, *et al*. Impact of ADAR-induced editing of minor viral RNA populations on replication and transmission of SARS-CoV-2. *Proc Natl Acad Sci*. 2022 Feb 8;119(6):e2112663119.
142. Piontkivska H, Frederick M, Miyamoto MM, Wayne ML. RNA editing by the host ADAR system affects the molecular evolution of the Zika virus. *Ecol Evol*. 2017;7(12):4475–85.

REFERENCES

143. Sommer B, Köhler M, Sprengel R, Seeburg PH. RNA editing in brain controls a determinant of ion flow in glutamate-gated channels. *Cell*. 1991 Oct 4;67(1):11–9.
144. Köhler M, Burnashev N, Sakmann B, Seeburg PH. Determinants of Ca^{2+} permeability in both TM1 and TM2 of high affinity kainate receptor channels: Diversity by RNA editing. *Neuron*. 1993 Mar 1;10(3):491–500.
145. Lomeli H, Mosbacher J, Melcher T, Höger T, Geiger JRP, Kuner T, *et al.* Control of Kinetic Properties of AMPA Receptor Channels by Nuclear RNA Editing. *Science*. 1994 Dec 9;266(5191):1709–13.
146. Wang Q, Miyakoda M, Yang W, Khillan J, Stachura DL, Weiss MJ, *et al.* Stress-induced Apoptosis Associated with Null Mutation of ADAR1 RNA Editing Deaminase Gene. *J Biol Chem*. 2004 Feb 6;279(6):4952–61.
147. Bentley DL. Coupling mRNA processing with transcription in time and space. *Nat Rev Genet*. 2014 Feb 11;15(3):163.
148. Hsiao YHE, Bahn JH, Yang Y, Lin X, Tran S, Yang EW, *et al.* RNA editing in nascent RNA affects pre-mRNA splicing. *Genome Res*. 2018 June;28(6):812–23.
149. Tang SJ, Shen H, An O, Hong H, Li J, Song Y, *et al.* Cis- and trans-regulations of pre-mRNA splicing by RNA editing enzymes influence cancer development. *Nat Commun*. 2020 Feb 7;11(1):799.
150. Licht K, Kapoor U, Amman F, Picardi E, Martin D, Bajad P, *et al.* A high resolution A-to-I editing map in the mouse identifies editing events controlled by pre-mRNA splicing. *Genome Res*. 2019 Sept;29(9):1453–63.
151. Saldi T, Riemondy K, Erickson B, Bentley DL. Alternative RNA structures formed during transcription depend on elongation rate and modify RNA processing. *Mol Cell*. 2021 Apr 15;81(8):1789-1801.e5.
152. Mouyna I, Henry C, Doering TL, Latgé JP. Gene silencing with RNA interference in the human pathogenic fungus *Aspergillus fumigatus*. *FEMS Microbiol Lett*. 2004 Aug 1;237(2):317–24.
153. Chen Y, Jiang Q, Qing F, Xue J, Xiao Q, He W, *et al.* MDA5 Enhances Invasive *Candida albicans* Infection by Regulating Macrophage Apoptosis and Phagocytosis/Killing Functions. *Inflammation*. 2024 Feb 1;47(1):191–208.
154. Wang X, Caffrey-Carr AK, Liu KW, Espinosa V, Croteau W, Dhingra S, *et al.* MDA5 Is an Essential Sensor of a Pathogen-Associated Molecular Pattern Associated with Vitality That Is Necessary for Host Resistance against *Aspergillus fumigatus*. *J Immunol Baltim Md 1950*. 2020 Dec 1;205(11):3058–70.
155. Hoenigl M, Seidel D, Sprute R. COVID-19-associated fungal infections. *Nat Microbiol*. 2022;7:1127–40.
156. van Rhijn N, Storer ISR, Birch M, Oliver JD, Bottery MJ, Bromley MJ. *Aspergillus fumigatus* strains that evolve resistance to the agrochemical fungicide ipflufenquin in vitro are also resistant to olorofim. *Nat Microbiol*. 2024 Jan;9(1):29–34.
157. Liu F, Zeng M, Zhou X, Huang F, Song Z. *Aspergillus fumigatus* escape mechanisms from its harsh survival environments. *Appl Microbiol Biotechnol*. 2024 Jan 4;108(1):53.

REFERENCES

158. Napoli C, Lemieux C, Jorgensen R. Introduction of a Chimeric Chalcone Synthase Gene into *Petunia* Results in Reversible Co-Suppression of Homologous Genes in trans. *Plant Cell*. 1990 Apr 1;2(4):279–89.
159. Shiu PKT, Metzenberg RL. Meiotic Silencing by Unpaired DNA: Properties, Regulation and Suppression. *Genetics*. 2002 Aug 1;161(4):1483–95.
160. Hammond TM, Xiao H, Boone EC, Perdue TD, Pukkila PJ, Shiu PKT. SAD-3, a Putative Helicase Required for Meiotic Silencing by Unpaired DNA, Interacts with Other Components of the Silencing Machinery. *G3 GenesGenomesGenetics*. 2011 Oct 1;1(5):369–76.
161. Gaffar FY, Imani J, Karlovsky P, Koch A, Kogel KH. Different Components of the RNA Interference Machinery Are Required for Conidiation, Ascosporeogenesis, Virulence, Deoxynivalenol Production, and Fungal Inhibition by Exogenous Double-Stranded RNA in the Head Blight Pathogen *Fusarium graminearum*. *Front Microbiol*. 2019 Aug 7;10.
162. Girard C, Budin K, Boisnard S, Zhang L, Debuchy R, Zickler D, *et al*. RNAi-Related Dicer and Argonaute Proteins Play Critical Roles for Meiocyte Formation, Chromosome-Axes Lengths and Crossover Patterning in the Fungus *Sordaria macrospora*. *Front Cell Dev Biol*. 2021 June 28;9.
163. Bernstein DA, Vyas VK, Weinberg DE. *Candida albicans* Dicer (CaDcr1) is required for efficient ribosomal and spliceosomal RNA maturation. *Proc Natl Acad Sci USA*. 2011 Dec 15;109(2):523–8.
164. Liang X hai, Crooke ST. Depletion of key protein components of the RISC pathway impairs pre-ribosomal RNA processing, *Nucleic Acids Research*. *Nucleic Acids Research*. 39(11):4875–89.
165. Samaha H, Delorme V, Pontvianne F, Cooke R, Delalande F, Van Dorsselaer A, *et al*. Identification of protein factors and *U3 snoRNAs* from a *Brassica oleracea* RNP complex involved in the processing of pre-rRNA. *Plant J*. 2010;61(3):383–98.
166. Hammond TM, Xiao H, Boone EC, Decker LM, Lee SA, Perdue TD, *et al*. Novel Proteins Required for Meiotic Silencing by Unpaired DNA and siRNA Generation in *Neurospora crassa*. *Genetics*. 2013 May 1;194(1):91–100.
167. Jöchl C, Rederstorff M, Hertel J, Stadler PF, Hofacker IL, Schrettl M, *et al*. Small ncRNA transcriptome analysis from *Aspergillus fumigatus* suggests a novel mechanism for regulation of protein synthesis. *Nucleic Acids Res*. 2008 May;36(8):2677–89.
168. Torres-Martínez S, Ruiz-Vázquez RM. The RNAi Universe in Fungi: A Varied Landscape of Small RNAs and Biological Functions. *Annu Rev Microbiol*. 2017 Sept 8;71:371–91.
169. Wang S, Li P, Zhang J, Qiu D, Guo L. Generation of a high resolution map of sRNAs from *Fusarium graminearum* and analysis of responses to viral infection. *Sci Rep*. 2016 May 18;6(1):26151.
170. Özkan S, Mohorianu I, Xu P. Profile and functional analysis of small RNAs derived from *Aspergillus fumigatus* infected with double-stranded RNA mycoviruses. *BMC Genomics*. 2017 May 30;18(416).
171. Kotta-Loizou I. Mycoviruses and their role in fungal pathogenesis. *Curr Opin Microbiol*. 2021 Oct 1;63:10–8.

REFERENCES

172. Takahashi-Nakaguchi AT, Shishido E, Yahara M. Analysis of an Intrinsic Mycovirus Associated With Reduced Virulence of the Human Pathogenic Fungus *Aspergillus fumigatus*. *Front Microbiol*. 2020 Jan 17;10.
173. Di Fazio A, Schlackow M, Pong SK, Alagia A, Gullerova M. Dicer dependent tRNA derived small RNAs promote nascent RNA silencing. *Nucleic Acids Res*. 2022 Jan 20;50(3):1734–52.
174. Cole C, Sobala A, Lu C, Thatcher SR, Bowman A, Brown JWS, *et al*. Filtering of deep sequencing data reveals the existence of abundant Dicer-dependent small RNAs derived from tRNAs. *RNA*. 2009 Jan 12;15(12):2147–60.
175. Kuscü C, Kumar P, Kiran M, Su Z, Malik A, Dutta A. tRNA fragments (tRFs) guide Ago to regulate gene expression post-transcriptionally in a Dicer-independent manner. *RNA*. 2018 Jan 8;24(8):1093–105.
176. Alves CS, Vicentini R, Duarte GT, Pinoti VF, Vincentz M, Nogueira FTS. Genome-wide identification and characterization of tRNA-derived RNA fragments in land plants. *Plant Mol Biol*. 2017 Jan 1;93(1):35–48.
177. Li Z, Ender C, Meister G, Moore PS, Chang Y, John B. Extensive terminal and asymmetric processing of small RNAs from rRNAs, snoRNAs, snRNAs, and tRNAs. *Nucleic Acids Res*. 2012 Aug;40(14):6787–99.
178. Sun Z, Hu Y, Zhou Y, Jiang N, Hu S, Li L, *et al*. tRNA-derived fragments from wheat are potentially involved in susceptibility to Fusarium head blight. *BMC Plant Biol*. 2022 Jan 3;22(1):3.
179. Bachellerie JP, Cavaillé J, Hüttenhofer A. The expanding snoRNA world. *Biochimie*. 2002 Aug 1;84(8):775–90.
180. Kiss T. SnoRNP Biogenesis Meets Pre-mRNA Splicing. *Mol Cell*. 2006 Sept 15;23(6):775–6.
181. Rybak-Wolf A, Jens M, Murakawa Y, Herzog M, Landthaler M, Rajewsky N. A Variety of Dicer Substrates in Human and *C. elegans*. *Cell*. 2014 Nov 20;159(5):1153–67.
182. Saraiya AA, Wang CC. snoRNA, a Novel Precursor of microRNA in *Giardia lamblia*. *PLOS Pathogens*. 2008 Nov 28;4(11):e1000224.
183. Ender C, Krek A, Friedländer MR, Beitzinger M, Weinmann L, Chen W, *et al*. A Human snoRNA with MicroRNA-Like Functions. *Mol Cell*. 2008 Nov 21;32(4):519–28.
184. Taft RJ, Glazov EA, Lassmann T, Hayashizaki Y, Carninci P, Mattick JS. Small RNAs derived from snoRNAs. *RNA*. 2009 Jan 7;15(7):1233–40.
185. Guan L, Grigoriev A. Computational meta-analysis of ribosomal RNA fragments: potential targets and interaction mechanisms. *Nucleic Acids Res*. 2021 Apr 19;49(7):4085–103.
186. Chen Z, Sun Y, Yang X, Wu Z, Guo K, Niu X, *et al*. Two featured series of rRNA-derived RNA fragments (rRFs) constitute a novel class of small RNAs. *PLOS ONE*. 2017 Apr 25;12(4):e0176458.
187. Nunes CC, Sailsbery JK, Dean RA. Characterization and application of small RNAs and RNA silencing mechanisms in fungi. *Fungal Biol Rev*. 2011 Dec 1;25(4):172–80.

188. Koeppen K, Hampton TH, Barnaby R, Roche C, Gerber SA, Goo YA, *et al.* An rRNA fragment in extracellular vesicles secreted by human airway epithelial cells increases the fluoroquinolone sensitivity of *P. aeruginosa*. *Am J Physiol Lung Cell Mol Physiol*. 2023 July 1;325(1):L54–65.
189. Chang SS, Zhang Z, Liu Y. RNA Interference Pathways in Fungi: Mechanisms and Functions. *Annu Rev Microbiol*. 2012 Oct 13;66(1):305–23.
190. Refos JM, Vonk AG, Eadie K, Lo-Ten-Foe JR, Verbrugh HA, Diepeningen AD van, *et al.* Double-Stranded RNA Mycovirus Infection of *Aspergillus fumigatus* Is Not Dependent on the Genetic Make-Up of the Host. *PLOS ONE*. 2013 Oct 22;8(10):e77381.
191. Özkan S, Coutts RHA. *Aspergillus fumigatus* mycovirus causes mild hypervirulent effect on pathogenicity when tested on *Galleria mellonella*. *Fungal Genet Biol*. 2015 Mar 1;76:20–6.
192. Banks GT, Buck KW, Chain EB, Darbyshire JE, Himmelweit F, Ratti G, *et al.* Antiviral Activity of Double Stranded RNA from a Virus isolated from *Aspergillus foetidus*. *Nature*. 1970 Aug;227(5257):505–7.
193. Kanhayuwa L, Kotta-Loizou I, Özkan S, Gunning AP, Coutts RHA. A novel mycovirus from *Aspergillus fumigatus* contains four unique dsRNAs as its genome and is infectious as dsRNA. *Proc Natl Acad Sci*. 2015 July 21;112(29):9100–5.
194. Takahashi-Nakaguchi A, Shishido E, Yahara M, Urayama SI, Ninomiya A, Chiba Y, *et al.* Phenotypic and Molecular Biological Analysis of Polymycovirus *AfuPmV-1M* From *Aspergillus fumigatus*: Reduced Fungal Virulence in a Mouse Infection Model. *Front Microbiol*. 2020;11.
195. Chen Y, Gao Q, Huang M, Liu Y, Liu Z, Liu X, *et al.* Characterization of RNA silencing components in the plant pathogenic fungus *Fusarium graminearum*. *Sci Rep*. 2015 July 27;5(1):12500.
196. Bass B. How Does RNA Editing Affect dsRNA-mediated Gene Silencing? *Cold Spring Harb Symp Quant Biol*. 2006;71:285–92.
197. Reich DP, Bass BL. Mapping the dsRNA World. *Cold Spring Harb Perspect Biol*. 2019 Jan 3;11(3):a035352.
198. Wang X, Vukovic L, Koh HR, Schulten K, Myong S. Dynamic profiling of double-stranded RNA binding proteins. *Nucleic Acids Res*. 2015 Sept 3;43(15):7566–76.
199. Pan X, Bruch A, Blango MG. Past, Present, and Future of RNA Modifications in Infectious Disease Research. *ACS Infect Dis*. 2024 Dec 13;10(12):4017–29.
200. Bruno M, Dewi IMW, Matzaraki V, ter Horst R, Pekmezovic M, Rösler B, *et al.* Comparative host transcriptome in response to pathogenic fungi identifies common and species-specific transcriptional antifungal host response pathways. *Comput Struct Biotechnol J*. 2020 Dec 26;19:647–63.
201. Vlachogiannis NI, Tual-Chalot S, Zormpas E, Bonini F, Ntouros PA, Pappa M, *et al.* Adenosine-to-inosine RNA editing contributes to type I interferon responses in systemic sclerosis. *J Autoimmun*. 2021 Dec 1;125:102755.
202. Flati T, Gioiosa S, Spallanzani N, Tagliaferri I, Diroma MA, Pesole G, *et al.* HPC-REDIttools: a novel HPC-aware tool for improved large scale RNA-editing analysis. *BMC Bioinformatics*. 2020 Aug 25;21(10):353.

203. Song Y, He X, Yang W, Wu Y, Cui J, Tang T, *et al.* Virus-specific editing identification approach reveals the landscape of A-to-I editing and its impacts on SARS-CoV-2 characteristics and evolution. *Nucleic Acids Res.* 2022 Mar 21;50(5):2509–21.
204. Ichihashi E, Kubota M, Shiromoto Y, Sakurai M. ADAR1 RNA-Editing Enzyme Regulates Telomeric R-Loop Formation. In: Hamada M, Fukunaga T, Zeng C, editors. *RNA-RNA and RNA-DNA Interactions*. New York, NY: Springer US; 2021. p. 109–27.
205. Kleiveland CR. Peripheral Blood Mononuclear Cells. In: Verhoeckx K, Cotter P, López-Expósito I, Kleiveland C, Lea T, Mackie A, *et al.* *The Impact of Food Bioactives on Health: in vitro and ex vivo models*. Cham (CH): Springer; 2015.
206. Smeeckens SP, Ng A, Kumar V, Johnson MD, Plantinga TS, van Diemen C, *et al.* Functional genomics identifies type I interferon pathway as central for host defense against *Candida albicans*. *Nat Commun.* 2013;4:1342.
207. Zhang W, Niu C, Fu RY, Peng ZY. *Mycobacterium tuberculosis H37Rv* infection regulates alternative splicing in Macrophages. *Bioengineered.* 2018 Jan 1;9(1):203–8.
208. Boudreault S, Roy P, Lemay G. Viral modulation of cellular RNA alternative splicing: A new key player in virus–host interactions? *WIREs RNA.* 2019;10(e1543).
209. Alkrekshi A, Wang W, Rana PS, Markovic V, Sossey-Alaoui K. A comprehensive review of the functions of YB-1 in cancer stemness, metastasis and drug resistance. *Cell Signal.* 2021 Sept 1;85:110073.
210. Deng M, Chen B, Liu Z, Wan Y, Li D, Yang Y, *et al.* YBX1 mediates alternative splicing and maternal mRNA decay during pre-implantation development. *Cell Biosci.* 2022 Feb 2;12(1):12.
211. Geuens T, Bouhy D, Timmerman V. The hnRNP family: insights into their role in health and disease. *Hum Genet.* 2016;135:851–67.
212. Szebenyi C, Gu Y, Gebremariam T, Kocsubé S, Kiss-Vetráb S, Jáger O, *et al.* *coth* Genes Are Necessary for Normal Spore Formation and Virulence in *Mucor lusitanicus*. *mBio.* 2023 Jan 10;14(1):e03386-22.
213. Giger GH, Ernst C, Richter I, Gassler T, Field CM, Sintsova A, *et al.* Inducing novel endosymbioses by implanting bacteria in fungi. *Nature.* 2024 Nov;635(8038):415–22.
214. Partida-Martinez LP, Hertweck C. Pathogenic fungus harbours endosymbiotic bacteria for toxin production. *Nature.* 2005 Oct;437(7060):884–8.
215. Pandey M, Singh G, Agarwal R, Dabas Y, Jyotsna VP, Kumar R, *et al.* Emerging *Rhizopus microsporus* Infections in India. *J Clin Microbiol.* 2018 May 25;56(6):10.1128/jcm.00433-18.
216. Kapoor U, Licht K, Amman F, Jakobi T, Martin D, Dieterich C, *et al.* ADAR-deficiency perturbs the global splicing landscape in mouse tissues. *Genome Res.* 2020 Aug;30(8):1107–18.
217. Zheng JT, Lin CX, Fang ZY, Li HD. Intron Retention as a Mode for RNA-Seq Data Analysis. *Front Genet.* 2020 July 7;11.
218. Neerincx A, Castro W, Guarda G, Kufer TA. NLRC5, at the Heart of Antigen Presentation. *Front Immunol.* 2013 Nov 22;4.

REFERENCES

219. Roth SH, Levanon EY, Eisenberg E. Genome-wide quantification of ADAR adenosine-to-inosine RNA editing activity. *Nat Methods*. 2019 Nov;16(11):1131–8.
220. Wang Y, Xie Z, Kutschera E, Adams JI, Kadash-Edmondson KE, Xing Y. rMATS-turbo: an efficient and flexible computational tool for alternative splicing analysis of large-scale RNA-seq data. *Nat Protoc*. 2024 Apr;19(4):1083–104.
221. Kavaliauskas P, Gu Y, Hasin N, Graf KT, Alqarihi A, Shetty AC, *et al*. Multiple roles for hypoxia inducible factor 1-alpha in airway epithelial cells during mucormycosis. *Nat Commun*. 2024 June 20;15(1):5282.

7 Appendix

7.1 Contribution to figures

7.1.1 Manuscript A

Short reference: Kelani *et al.* (2023), *RNA*.

Contribution of the doctoral candidate

Contribution of the doctoral candidate to figures reflecting experimental data (only for original articles):

Figure: 3, 5, S5, S6, S7	☐ 100% (the data presented in this figure come entirely from experimental work carried out by the candidate) ☐ 0% (the data presented in this figure are based exclusively on the work of other co-authors) ☒ Approximate contribution of the doctoral candidate to the figure: 10% I contributed to the sRNA-seq analysis
Figure: 1, 2, 4, 6, S1, S2, S3, S4, S8	☐ 100% (the data presented in this figure come entirely from experimental work carried out by the candidate) ☒ 0% (the data presented in this figure are based exclusively on the work of other co-authors) ☐ Approximate contribution of the doctoral candidate to the figure: 0%

Signature candidate

Signature supervisor (member of the Faculty)

7.1.2 Manuscript B

Short reference: Pan and Kelani *et al.* (2026). *Mol. Microbiol.*

Contribution of the doctoral candidate

Contribution of the doctoral candidate to figures reflecting experimental data (only for original articles):

Figure: 1, 2, 3, 4	☐ 100% (the data presented in this figure come entirely from experimental work carried out by the candidate)
	☐ 0% (the data presented in this figure are based exclusively on the work of other co-authors)
	☒ Approximate contribution of the doctoral candidate to the figure: 50% I analyzed the RNA-seq data and image visualization.

Figure: S1, S2, S3, S4, S5, S6, S7, S8, S9	☐ 100% (the data presented in this figure come entirely from experimental work carried out by the candidate)
	☐ 0% (the data presented in this figure are based exclusively on the work of other co-authors)
	☒ Approximate contribution of the doctoral candidate to the figure: 40% I helped with RNA isolation (Figure S2A), I did the data visualization (Figure S2D-E, Figure S4B-C, Figure S8B-C). I did the experiment (Figure S9C).

Signature candidate

Signature supervisor (member of the Faculty)

7.1.3 Manuscript C

Short reference: Pan and Bruch *et al.* (2024), *ACS Infect. Dis.*

Contribution of the doctoral candidate

Contribution of the doctoral candidate to figures reflecting experimental data (only for original articles):

Figures: 1, 2	☐	100% (the data presented in this figure come entirely from experimental work carried out by the candidate)
	☐	0% (the data presented in this figure are based exclusively on the work of other co-authors)
	☒	Approximate contribution of the doctoral candidate to the figure: 33% I contributed to making the figures.

Signature candidate

Signature supervisor (member of the Faculty)

7.1.4 Manuscript D

Short reference: Pan *et al.* (2025), *in preparation*.

Contribution of the doctoral candidate

Contribution of the doctoral candidate to figures reflecting experimental data (only for original articles):

Figures: 2 -5, S1-S3	☒ 100% (the data presented in this figure come entirely from experimental work carried out by the candidate) ☐ 0% (the data presented in this figure are based exclusively on the work of other co-authors) ☐ Approximate contribution of the doctoral candidate to the figure: ____%
Figures: 1	☐ 100% (the data presented in this figure come entirely from experimental work carried out by the candidate) ☐ 0% (the data presented in this figure are based exclusively on the work of other co-authors) ☒ Approximate contribution of the doctoral candidate to the figure: 75% Brief description of the contribution: The model was made by me (Figure 1A). I analyzed RNA editing (Figure 1B). I analyzed <i>ADAR1</i> TPM (Figure 1C). Figure 1C and 1E were also carried out by me.

Signature candidate

Signature supervisor (member of the Faculty)

8 Curriculum vitae

PERSONAL INFORMATION

Name: Xiaoqing Pan
Date of birth: 25th of May 1994
Place of birth: Fuzhou, China
Nationality: Chinese
Address: Werner-Seelenbinder-Straße 3, 07747 Jena
ORCID: 0009-0009-0894-7366

EDUCATION

2021 – Present

Doctor of Philosophy: Microbiology

Friedrich-Schiller University, Jena, Germany

Dissertation: RNA regulation in host-fungal interactions: investigation of the host epitranscriptome and fungal small RNAs

Supervisors: Prof. Dr. Kai Papenfort

Dr. Matthew G. Blango

2018 – 2021

Master of Science: Biotechnology Engineering

University of Chinese Academy of Sciences, China

Master thesis: Functional Analysis of non-coding RNA *ciR133* and *miR168a* in rice

Supervisors: Associate Professor. Dr. Xia Kuaifei

Dr. Cheng Taihui

2012 – 2016

Bachelor of Science: Gardening

Hainan University, China

WORK EXPERIENCE

2021 – Present

Doctoral Researcher

Leibniz Institute for Natural Product Research and Infection Biology | Jena, Germany

Supervisor: Dr. Matthew G. Blango

- Took part in production of collaborative research reports and publications.
- Presented research findings at national and international conferences.
- Contributed to teaching by assisting in supervision of student projects and developing students' research skills.

PUBLICATIONS

- **Pan, X***, AA. Kelani*, L. Schrettenbrunner, S. Prabakar, B. Israni, and M. G. Blango. 2026. An Improved Description of the Small RNA Landscape of the Human Fungal Pathogen *Aspergillus fumigatus*. *Molecular Microbiology*. 2026 Jan; 1–12.
doi: 10.1111/mmi.70042.
(* equal contribution)
- **Pan X***, Bruch A*, Blango MG. Past, Present, and Future of RNA Modifications in Infectious Disease Research. *ACS Infect. Dis.* 2023 Dec;10(12): 4017-4029.
doi: 10.1021/acsinfecdis.4c00598.
(* equal contribution)
- Kelani AA, Bruch A, Riviuccio F, Visser C, Krüger T, Weaver D, **Pan X**, Schäuble S, Panagiotou G, Kniemeyer O, Bromley MJ, Bowyer P, Barber AE, Brakhage AA, Blango MG. Disruption of the *Aspergillus fumigatus* RNA interference machinery alters the conidial transcriptome. *RNA*. 2023 Jul;29(7):1033-1050.
doi: 10.1261/rna.079350.122.

- Xia K*, **Pan X***, Zeng X, Zhang M. *Xoo*-responsive transcriptome reveals the role of the circular *RNA133* in disease resistance by regulating expression of *OsARAB* in rice. *Phytopathol Res* 5, 33 (2023).
doi: 10.1186/s42483-023-00188-8.
(* equal contribution)
- Xia K*, **Pan X***, Chen H, Xu X, Zhang M. Rice *miR168a-5p* regulates seed length, nitrogen allocation and salt tolerance by targeting *OsOFP3*, *OsNPF2.4* and *OsAGO1a*, respectively. *Journal of Plant Physiology*, 280, 153905 (2023)
doi: 10.1016/j.jplph.2022.153905.
(* equal contribution)

CONFERENCES

- | | |
|------|---|
| 2025 | <p>Talk Title: An improved description of the small RNA landscape of the human fungal pathogen <i>Aspergillus fumigatus</i>.
Deutschsprachigen Mykologischen Gesellschaft. Germany</p> |
| 2025 | <p>Poster Title: Human RNA editing patterns reveal disrupted mRNA splicing upon <i>R. oryzae</i> stimulation.
Gordon Research Seminar - RNA Editing. Italy.</p> |
| 2024 | <p>Poster Title: Fungal infections lead to changes in A-to-I RNA editing and alternative splicing in host immune cells
Kiel Conference on Molecular Biology of Fungi. Germany</p> |
| 2024 | <p>Talk Title: Fungal infections lead to changes in A-to-I RNA editing in host immune cells.
Jena RNA Club Symposium. Germany.</p> |
| 2024 | <p>Poster Title: Fungal infections lead to changes in A-to-I RNA editing in host immune cells.
International Student Conference on Microbial Communication. Germany.</p> |

- 2024 **Poster Title:** Fungal infections lead to changes in A-to-I RNA editing in host immune cells.
RNA Society Meeting. UK.
- 2023 **Poster Title:** Unveiling the hidden world of tRNA fragments in *Aspergillus fumigatus*.
EMBL The Non-coding Genome Symposium. Germany
- 2022 **Poster Title:** The role of RNA structure in fungal pathogenesis.
International Joint Meeting Infection Biology and Antibiotics. Germany
-

TEACHING AND
SUPERVISION
2022 – 2024

- Training for MSc students // Microbiology and Molecular Biology - MMB005**
- Mentored students in the experiment section

Jena, 2026

PAN, Xiaoqing

9 Acknowledgments

I have come a long way to get here. I hail from China, where the culture, the language, everything is different. I still chose to start this journey not because it's easy, but because it's hard. Because this challenge is one that I am willing to accept, one that I am unwilling to postpone, and one I intend to overcome. Now is a good chance to say that it is only possible because of the abundant love and support from people who I would like to acknowledge and offer my sincerest thanks to.

Life was hard when I first came to this land, with the COVID-19 pandemic and everything. What was even harder was that I had to use English as my primary language, if you are aware it is neither my first nor my second language. And this journey involved my supervisor, Dr. Matthew Blango. As an American, he has never made fun of my accent. As a supervisor, he is always there to answer my questions, brainstorm ideas, try to move the projects forward as fast as he can. I can't remember how many times he replied my emails after 10 PM; How many times I angrily declared RNAi didn't exist in *A. fumigatus* and he responded with "we're getting there". Indeed, it took us some years to "get there", when we finally observed sRNAs just standing there and using their canonical "U"s to say hi to us. Another fun story is when I carefully suggested having a project of my own. That's how the editing story came to fruition, with Matt proposing to try with the PBMC paper. It was the project I used to find a collaborator. I have no idea how many PhD candidates establish their network and find their own collaborators, so I was quite surprised and grateful when Magnus offered the help.

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A special thank to ChatGPT, Gemini and Deepseek. Thank you for making my life better and easier.

To my motherland.

致谢

罗曼罗兰说过：

“世界上只有一种真正的英雄主义，那就是看清生活的真相之后，依然热爱生活。”

Il n'y a qu'un héroïsme au monde : c'est de voir le monde tel qu'il est, et de l'aimer.

博士旅程在此即将告一段落。故事始于盛夏终于耶拿的寒冬。四年半的时间，不长不短。我读了很多文章，虽没有到看清生活的真相，也没有热爱生活。普普通通，身体健康。博士学位的完成是生活的一部分，这期间我学会了面对很多困难和挑战。路总会继续，学海无涯，书囊无底。明早的太阳依旧会升起，希望我的未来也如那般灿烂无比。

希望爸爸妈妈身体健康。

潘晓清

乙巳年小寒

10 Ehrenwörtliche Erklärung

Ich erkläre hiermit, dass mir die Promotionsordnung der Fakultät für Biowissenschaften der Friedrich-Schiller-Universität Jena bekannt ist. Ich habe die vorliegende Arbeit selbstständig angefertigt und keine Textabschnitte Dritter oder eigener Prüfungsarbeiten ohne entsprechende Kennzeichnung übernommen. Alle benutzten Hilfsmittel, persönlichen Mitteilungen und Quellen sind in der Arbeit angegeben. Unterstützung bei der Auswahl und Auswertung des Materials sowie bei der Erstellung des Manuskriptes erhielt ich ausschließlich von den in der Danksagung genannten Personen.

Ich versichere, dass ich keine Hilfe kommerzieller Promotionsvermittlungen in Anspruch genommen habe und dass Dritte weder unmittelbar noch mittelbar geldwerte Leistungen von mir für Arbeiten erhalten haben, die im Zusammenhang mit dem Inhalt der vorgelegten Dissertation stehen. Diese Dissertation wurde von mir bisher weder für eine staatliche noch eine wissenschaftliche Prüfung eingereicht. Weder die gleiche, eine in wesentlichen Teilen ähnliche noch eine andere Abhandlung wurde bei einer anderen Hochschule oder Fakultät als Dissertation eingereicht.

Jena, 2026

PAN, Xiaoqing